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(54) **MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME**

(71) Applicant: **YANGTZE MEMORY TECHNOLOGIES CO., LTD.**, Wuhan (CN)

(72) Inventors: **Wei LIU**, Wuhan (CN); **Hongbin Zhu**, Wuhan (CN); **Ziqun Hua**, Wuhan (CN); **Ning Jiang**, Wuhan (CN); **Wenyu Hua**, Wuhan (CN)

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Related U.S. Application Data

(63) Continuation of application No. 17/553,776, filed on Dec. 16, 2021, now Pat. No. 12,349,341, which is a continuation of application No. PCT/CN2021/115775, filed on Aug. 31, 2021.

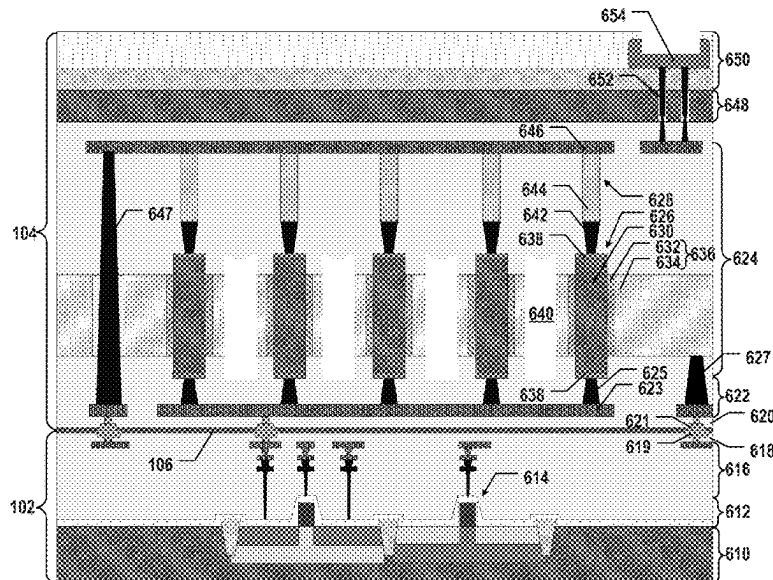
Publication Classification

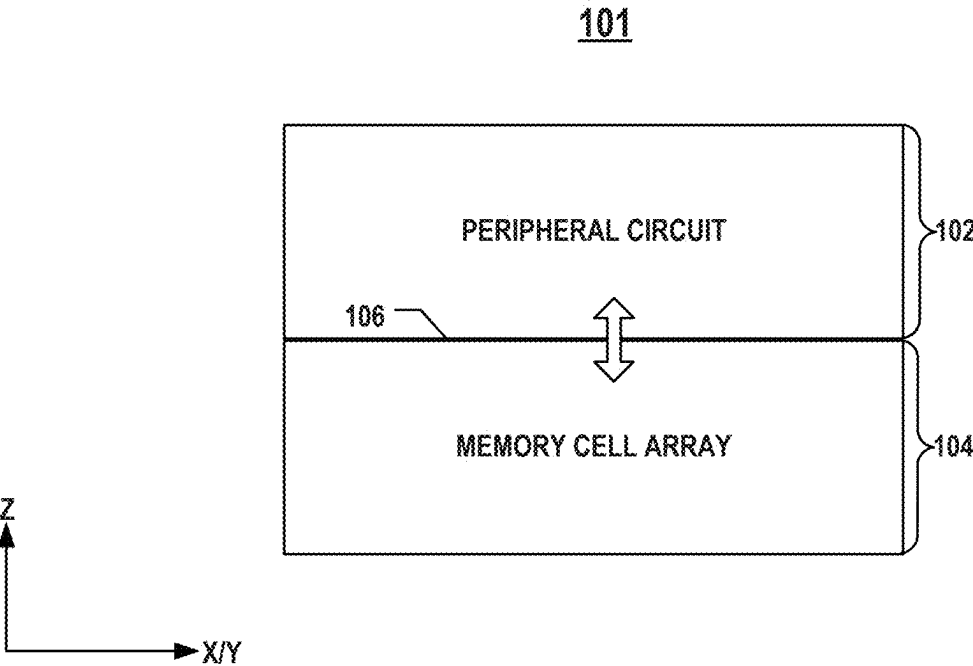
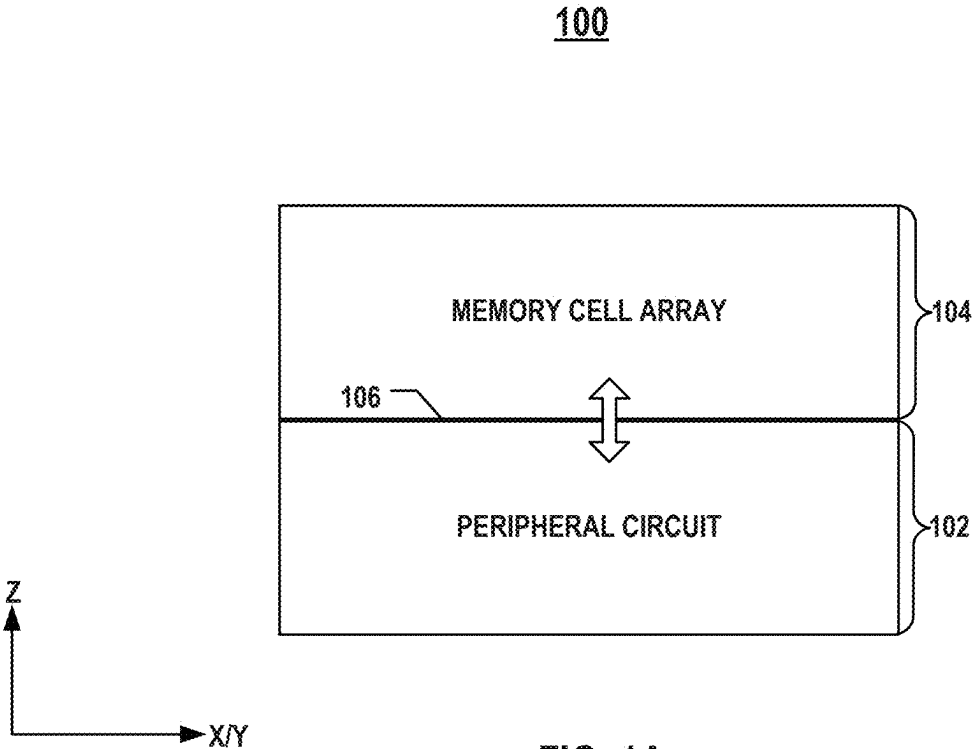
(51) **Int. Cl.**
H10B 12/00 (2023.01)
H10N 70/00 (2023.01)
H10N 70/20 (2023.01)
(52) **U.S. Cl.**
CPC *H10B 12/50* (2023.02); *H10B 12/395* (2023.02); *H10B 12/482* (2023.02); *H10N 70/231* (2023.02); *H10N 70/841* (2023.02)

(57) **ABSTRACT**

A memory device include a bit line extending in a first direction, a first transistor comprising a first semiconductor body extending in a second direction perpendicular to the first direction and a first gate structure located at one side of the first semiconductor body in the first direction, the first semiconductor body coupled to the bit line, a second transistor comprising a second semiconductor body extending in the second direction and a second gate structure located at one side of the second semiconductor body in the first direction, the second semiconductor body coupled to the bit line; and a first isolation structure located between the first semiconductor body and the second semiconductor body in the first direction. The first semiconductor body is in contact with the first isolation structure, and the second semiconductor body is in contact with the first isolation structure

600





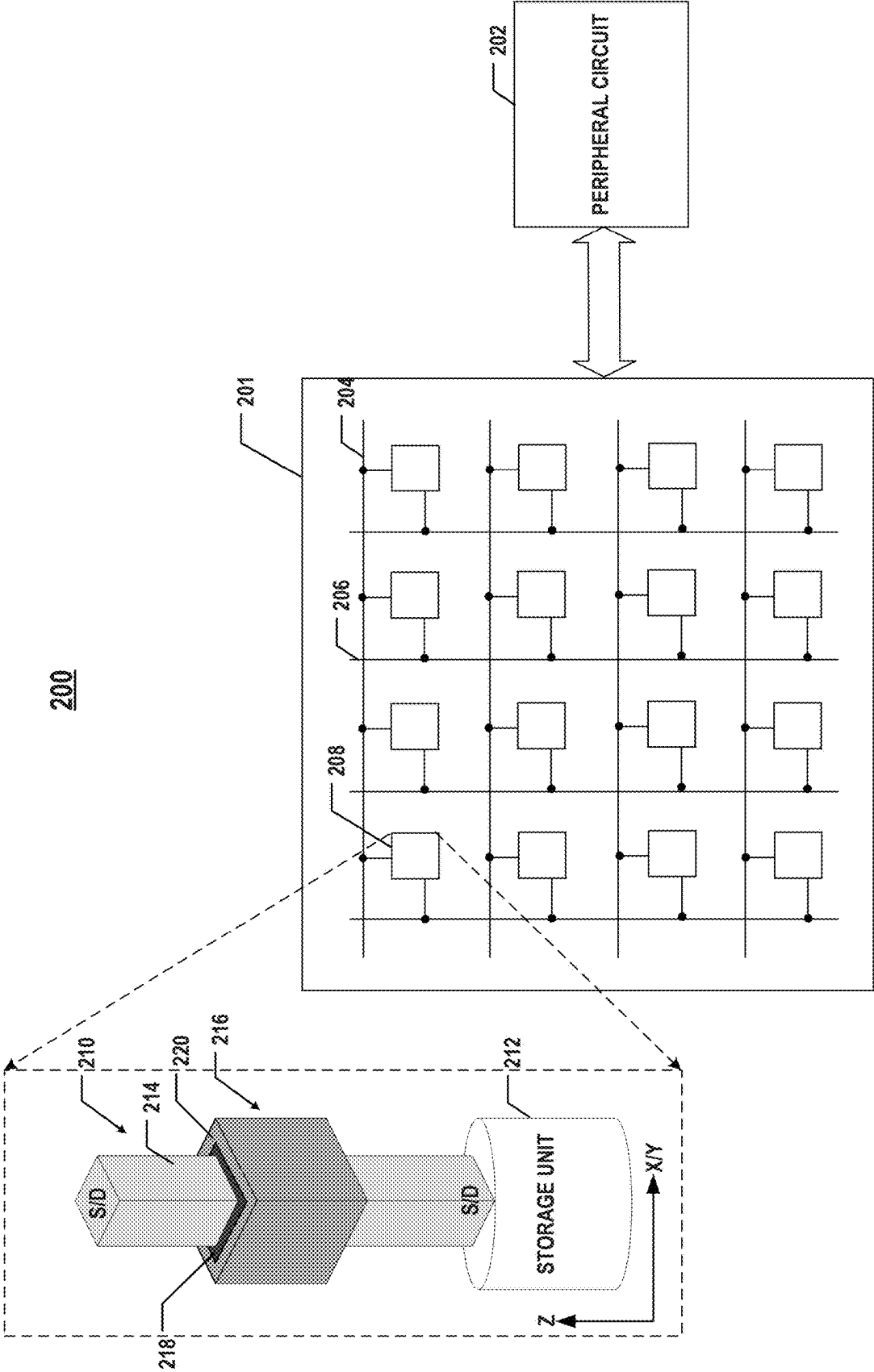


FIG. 2

200

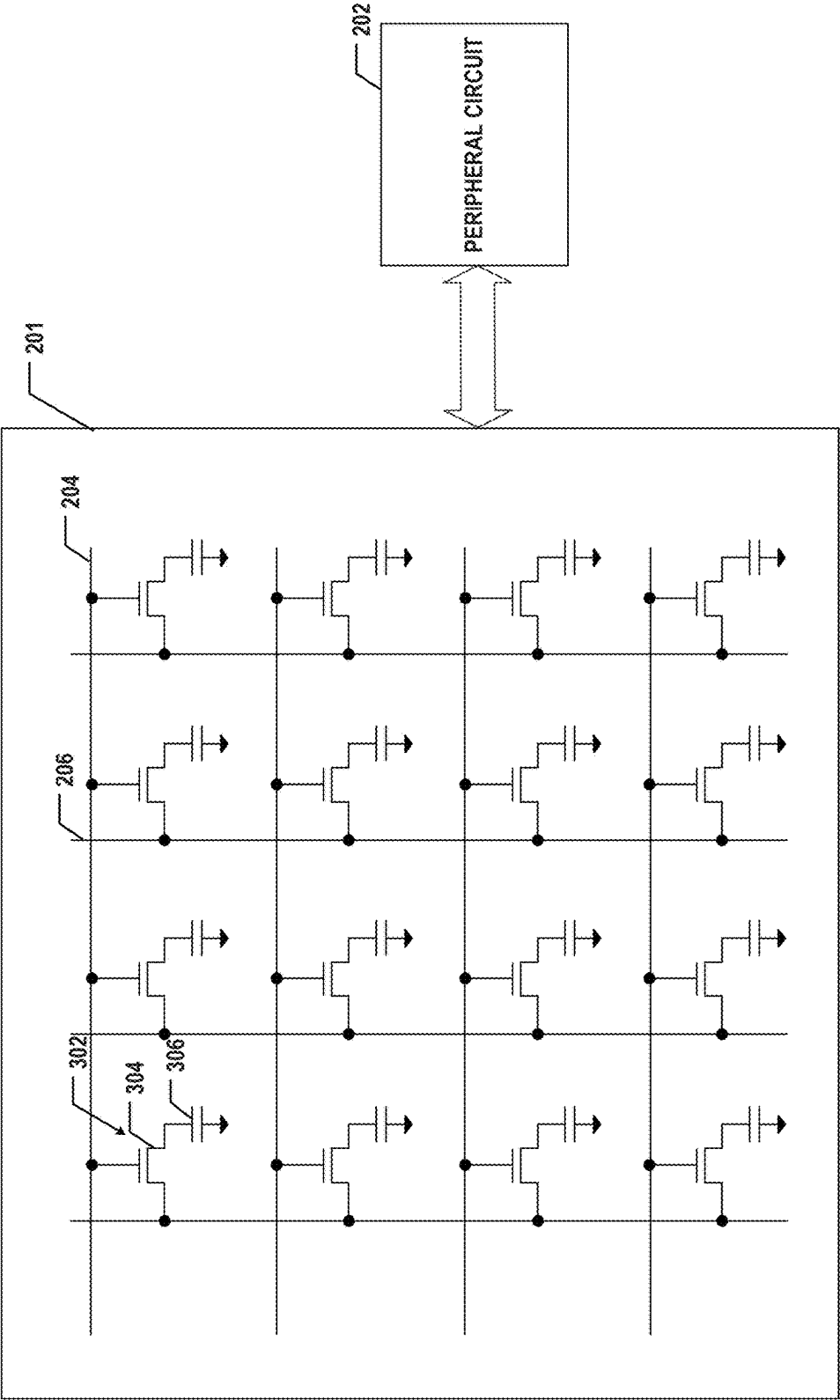


FIG. 3

200

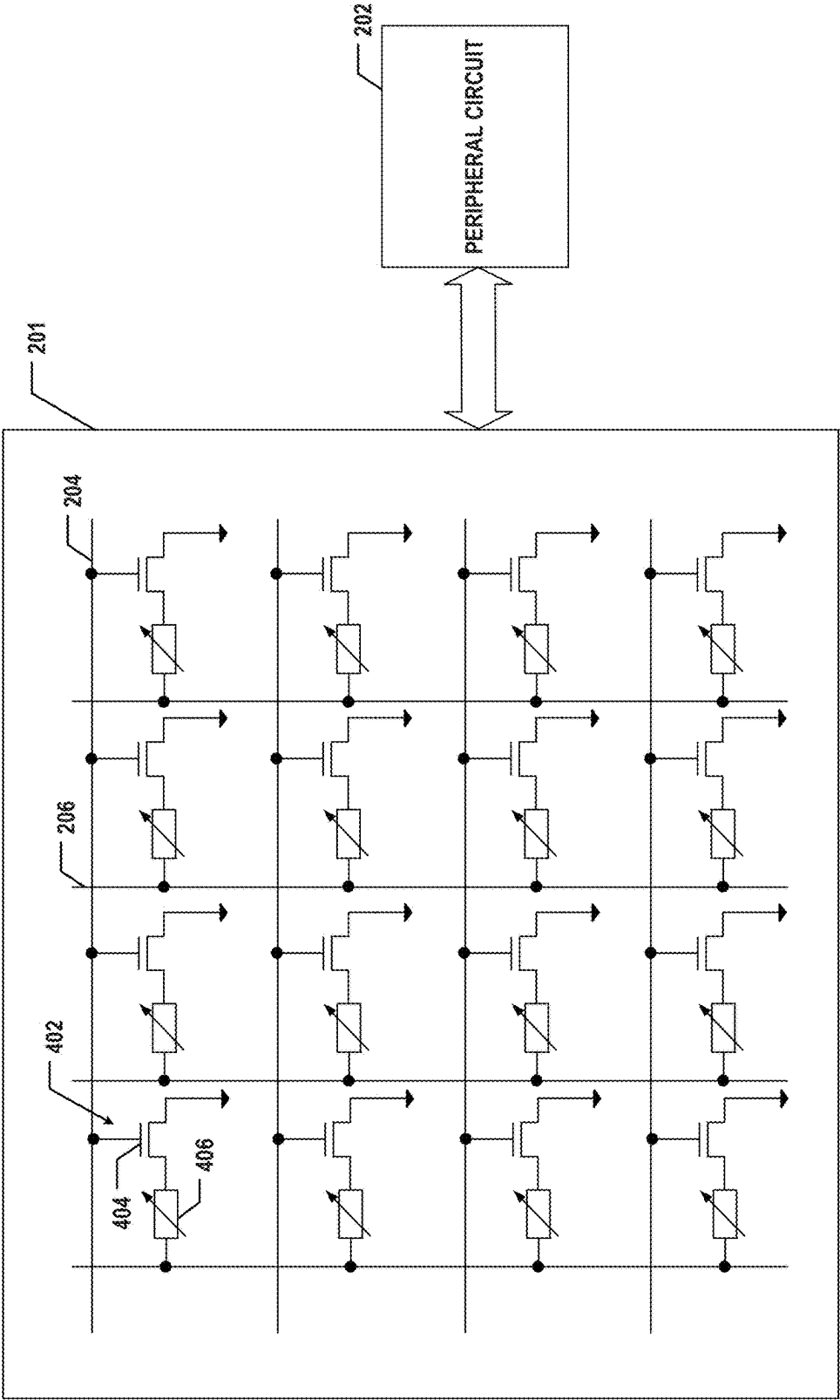


FIG. 4

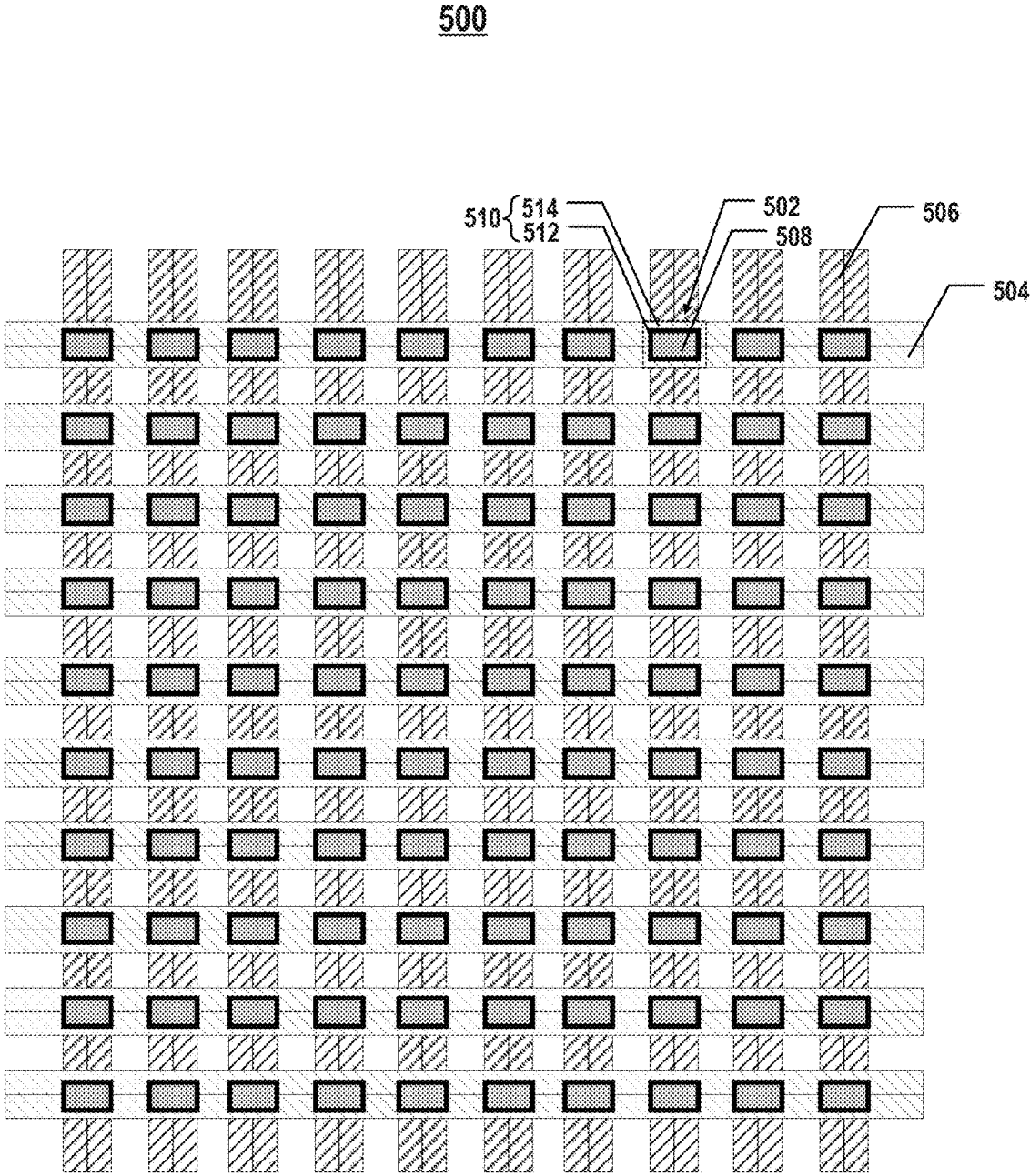


FIG. 5

600

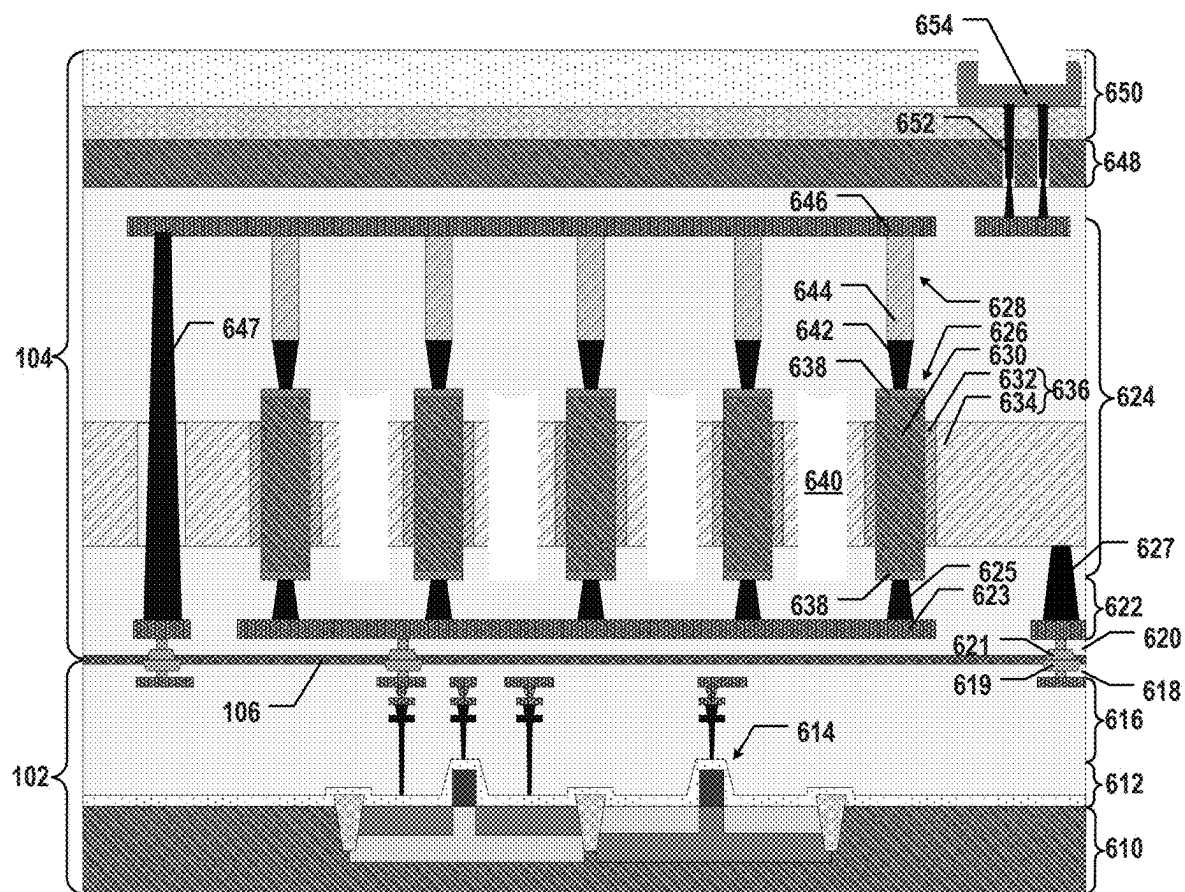


FIG. 6A

601

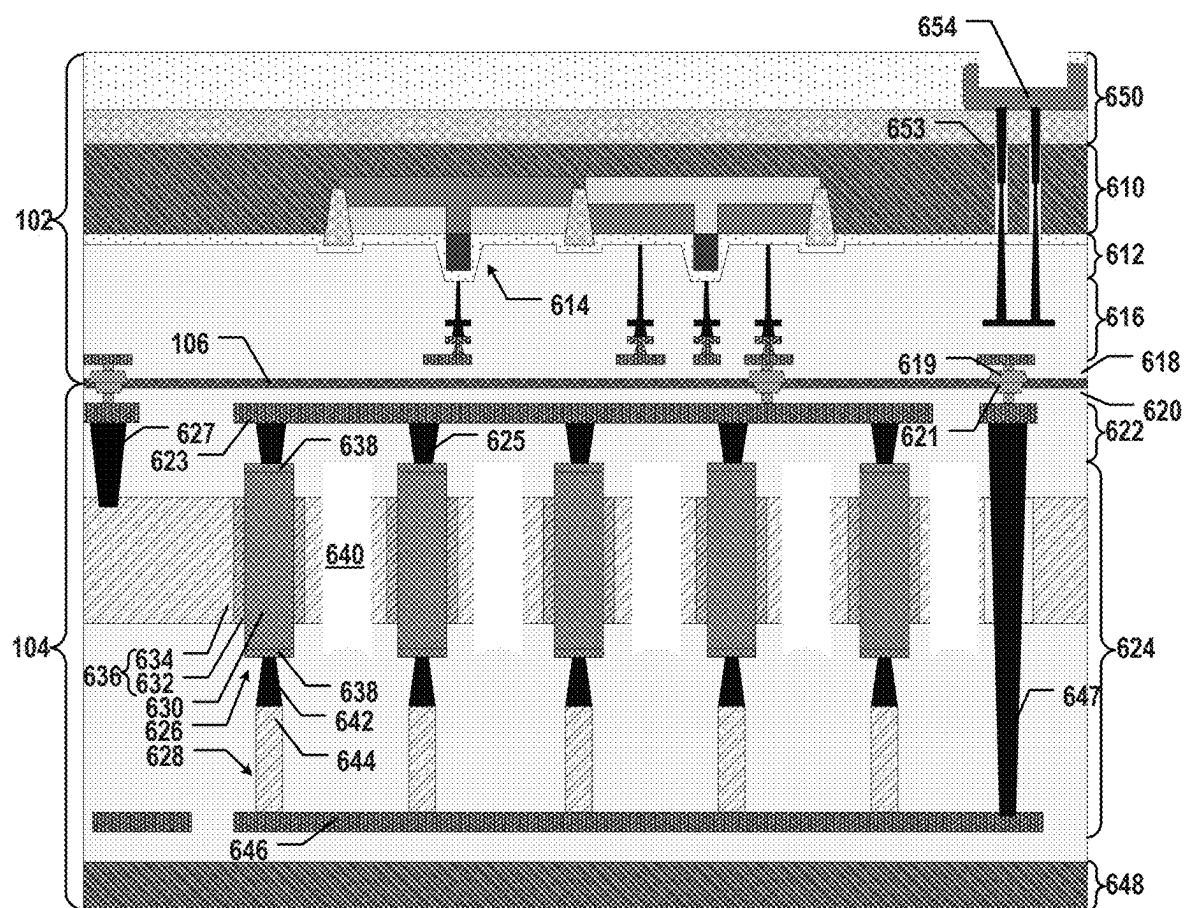


FIG. 6B

603

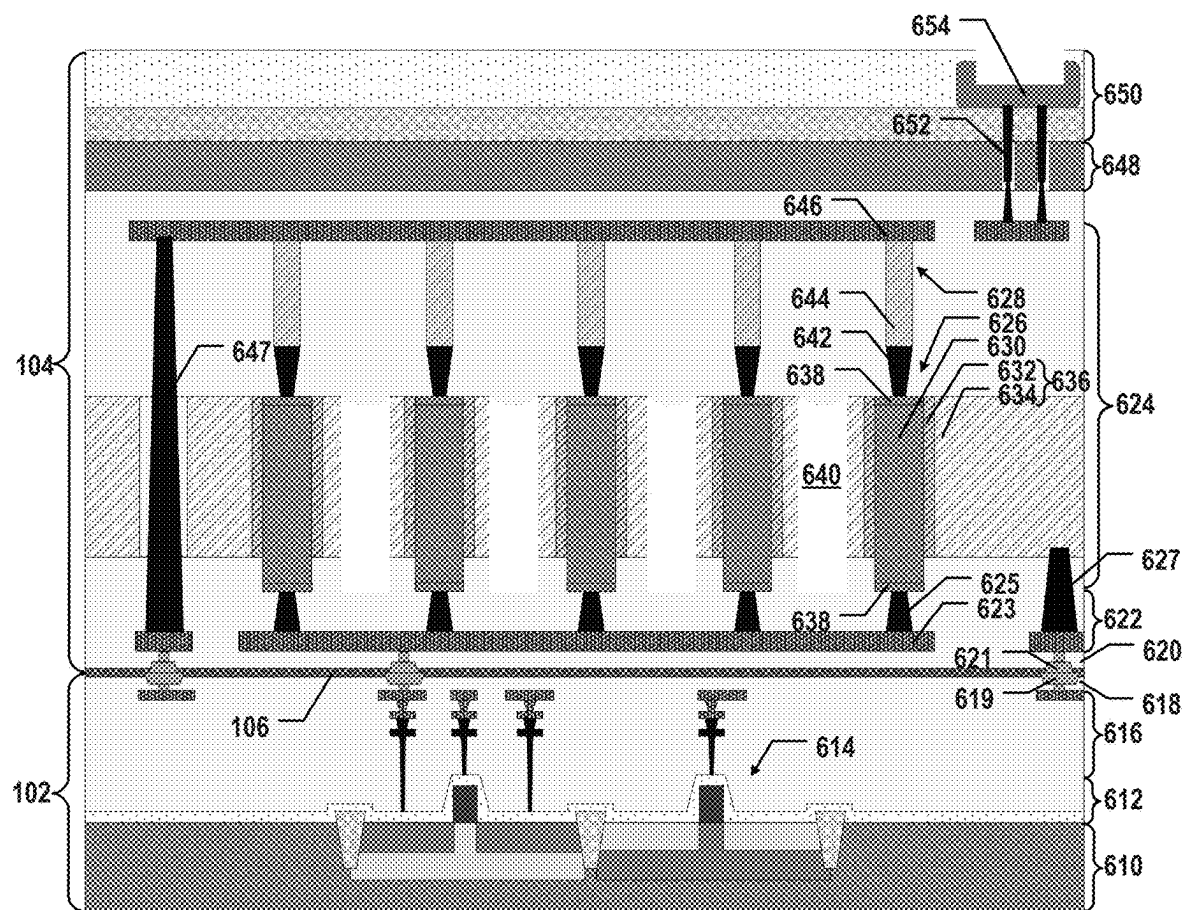


FIG. 6C

605

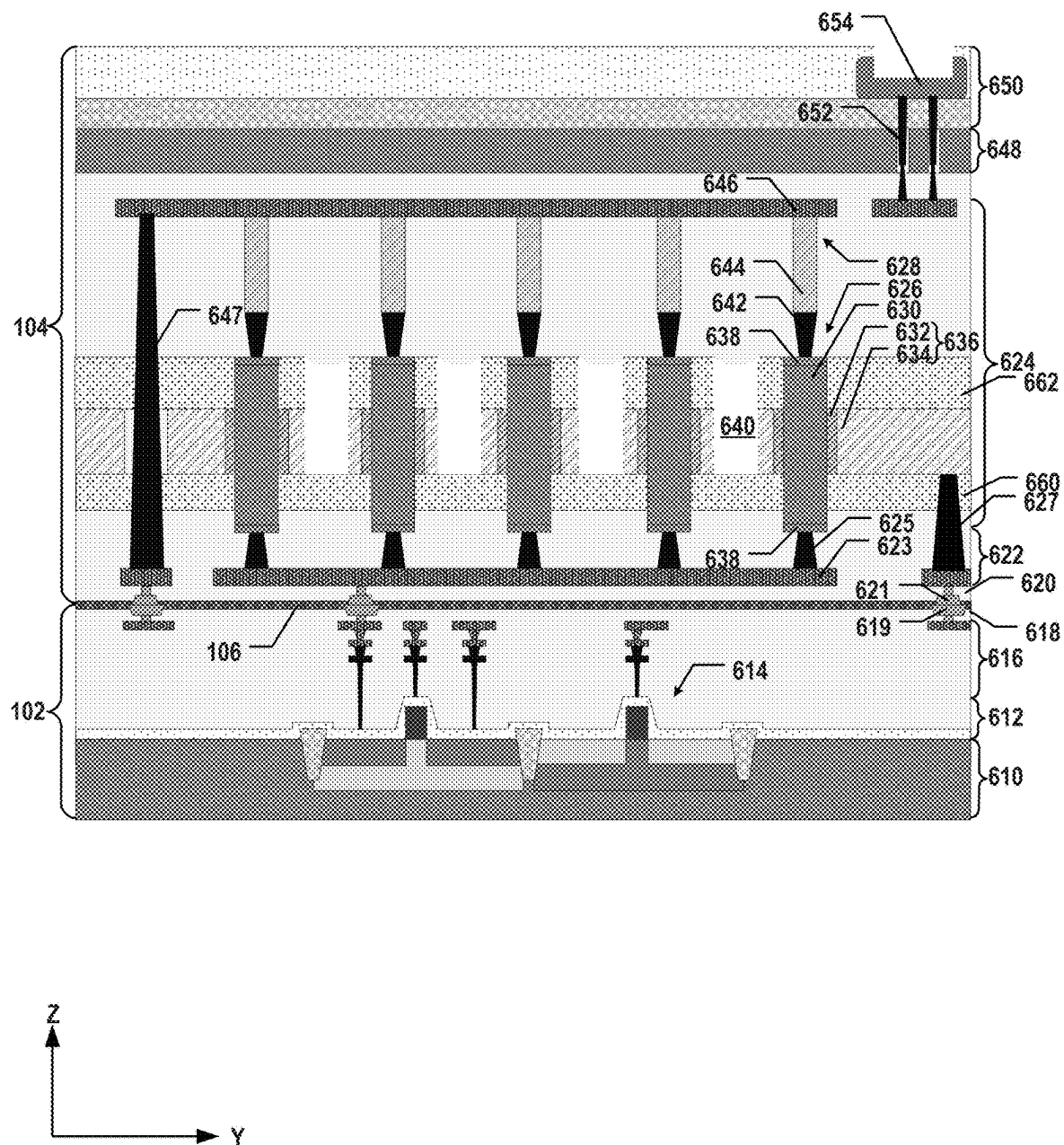


FIG. 6D

607

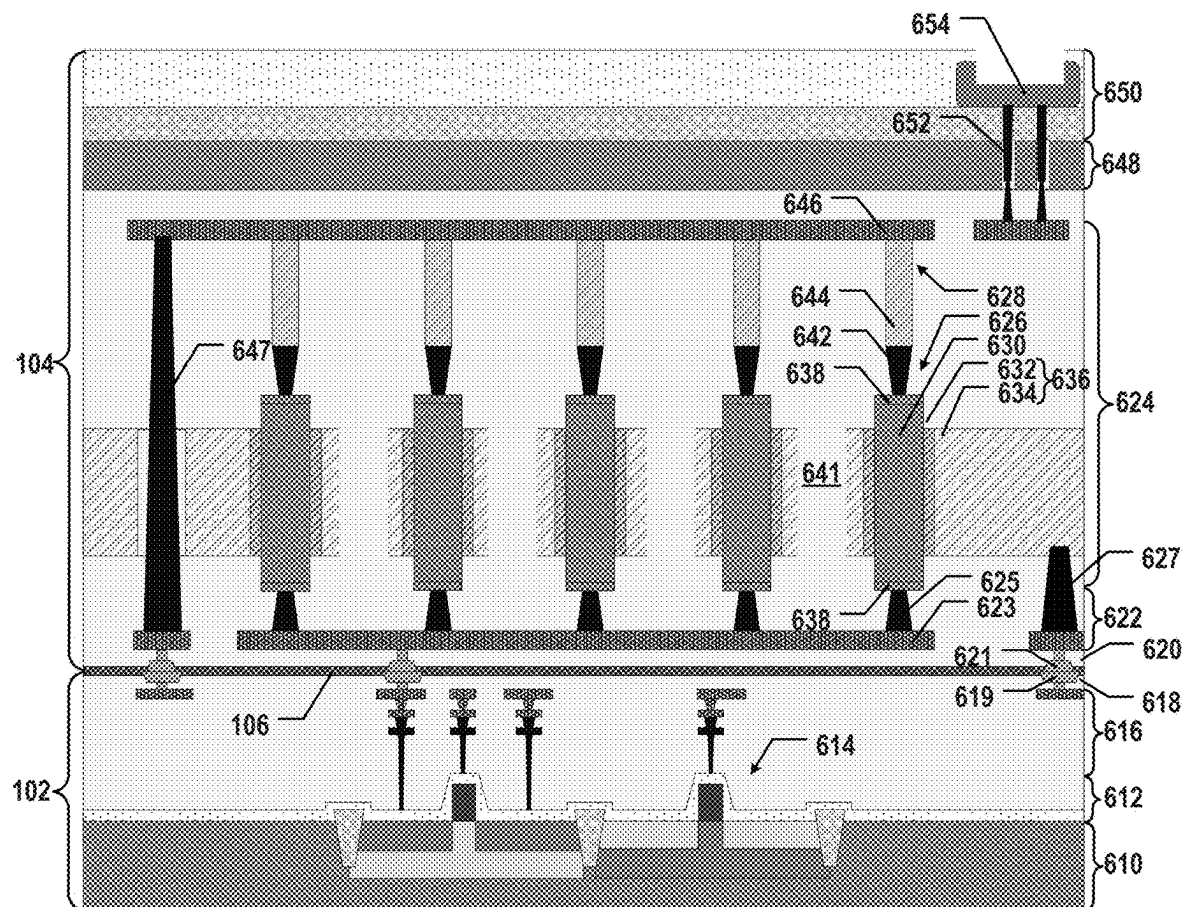


FIG. 6E

700

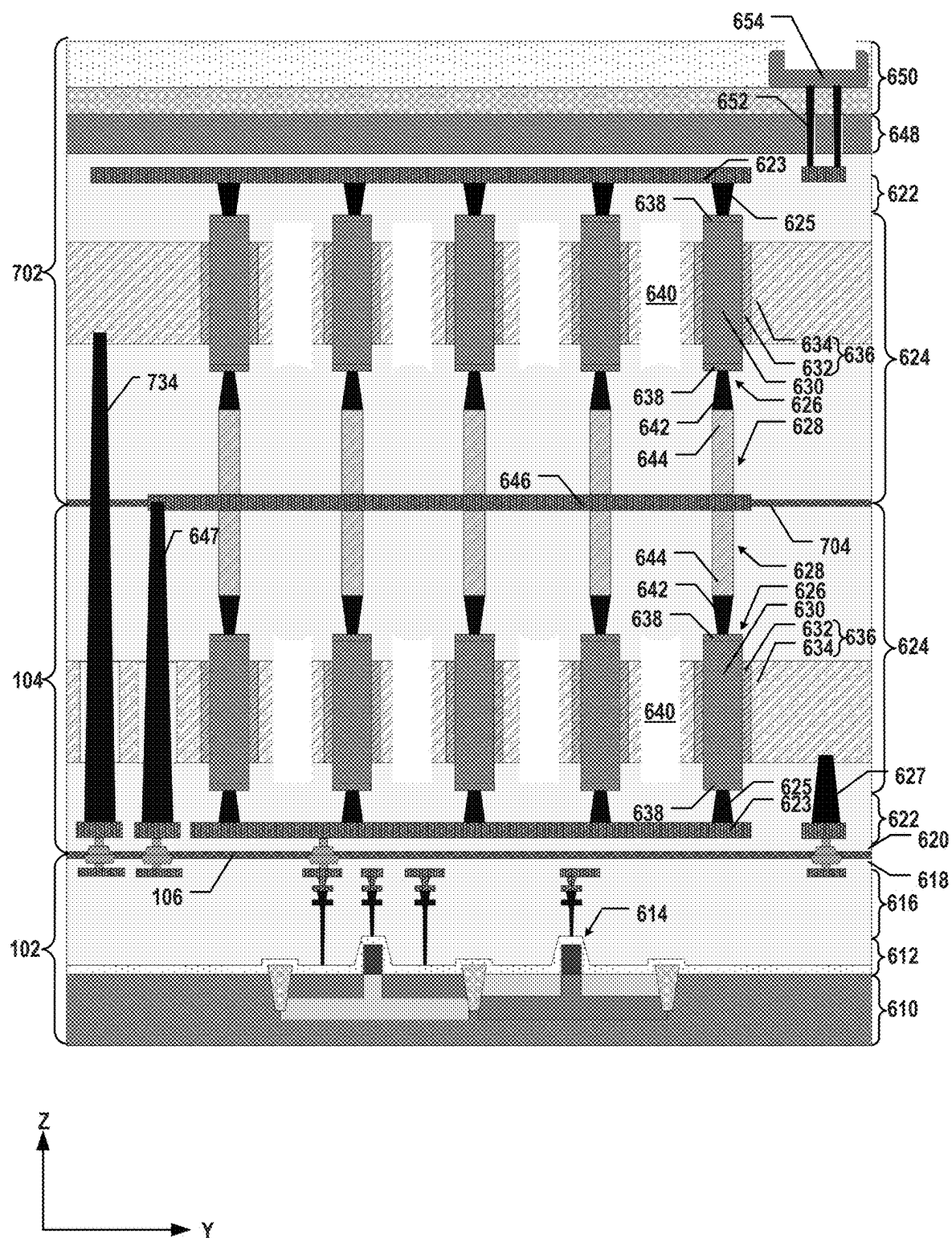


FIG. 7

800

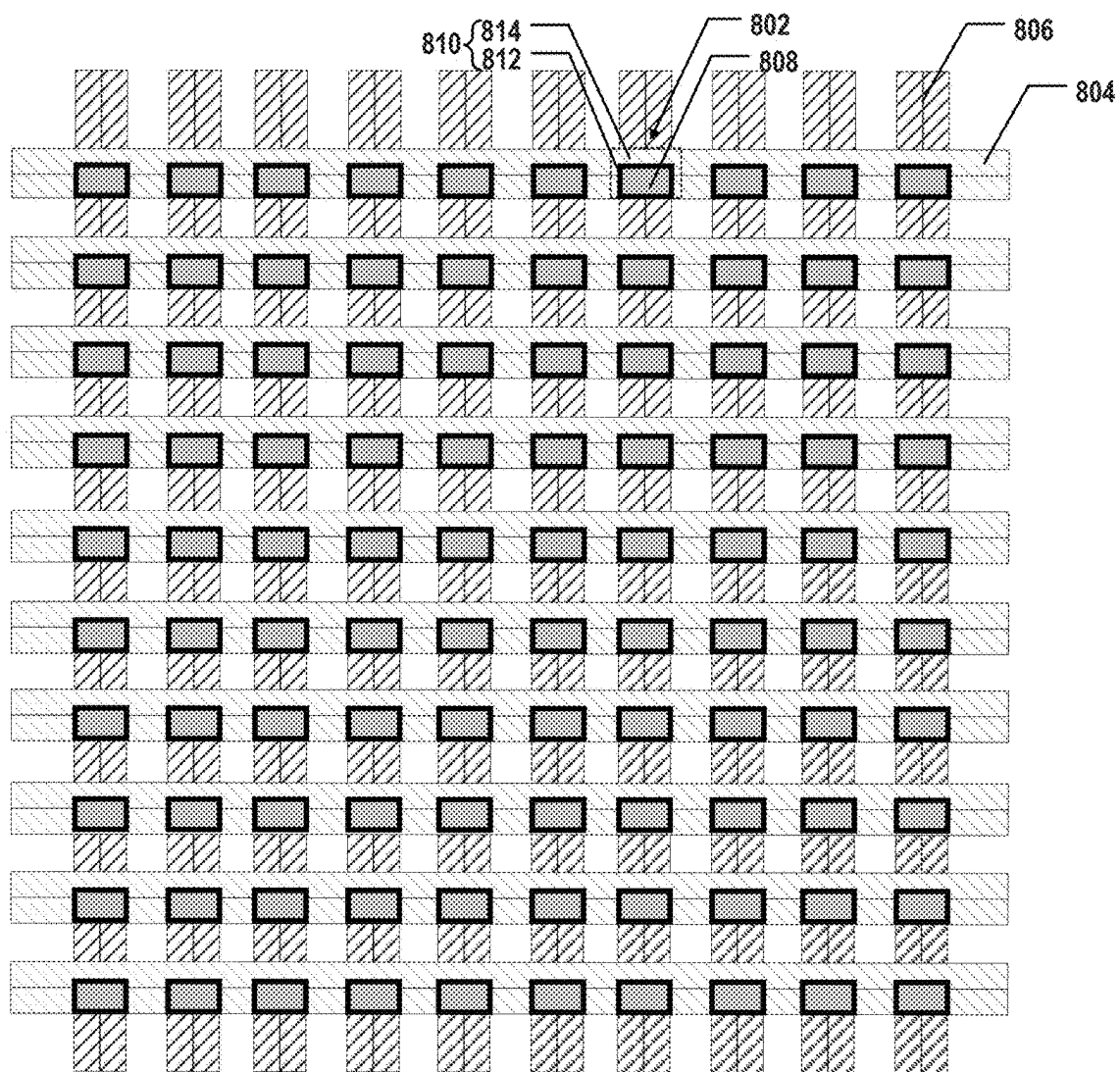


FIG. 8

900

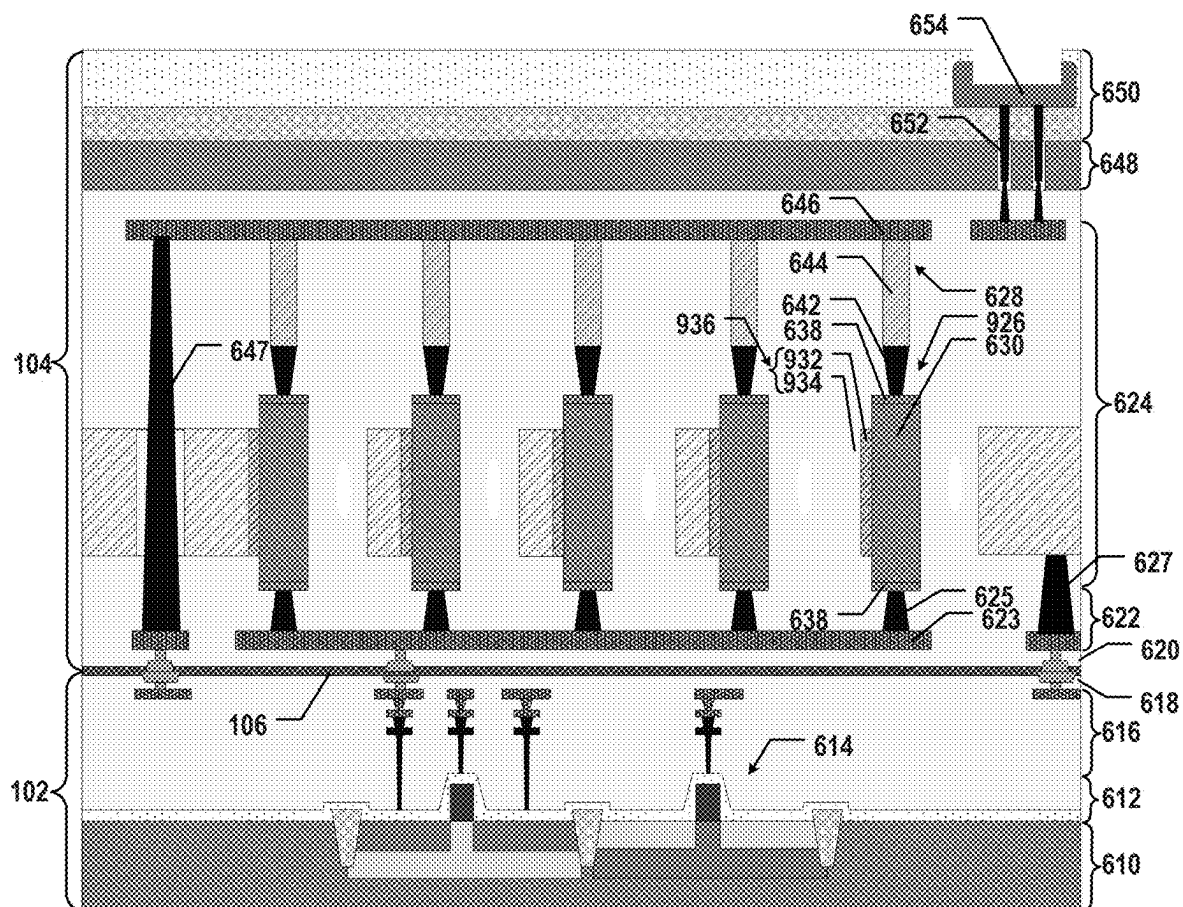


FIG. 9

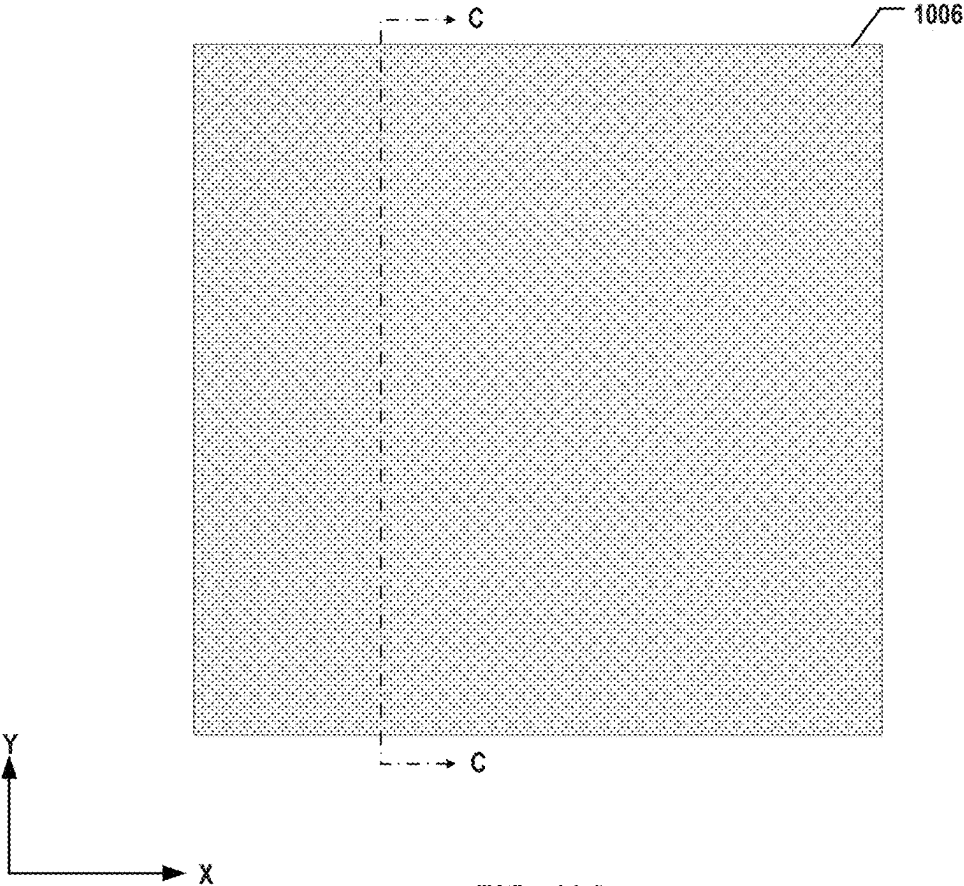
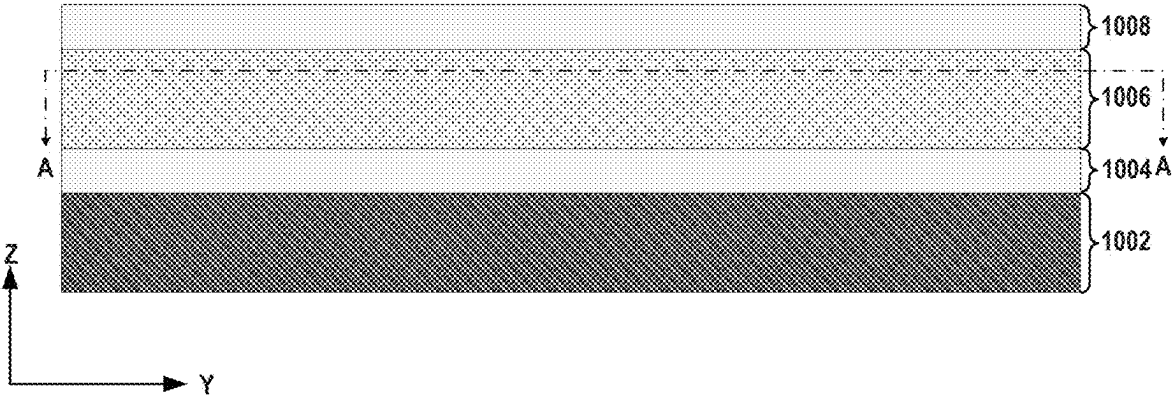


FIG. 10A

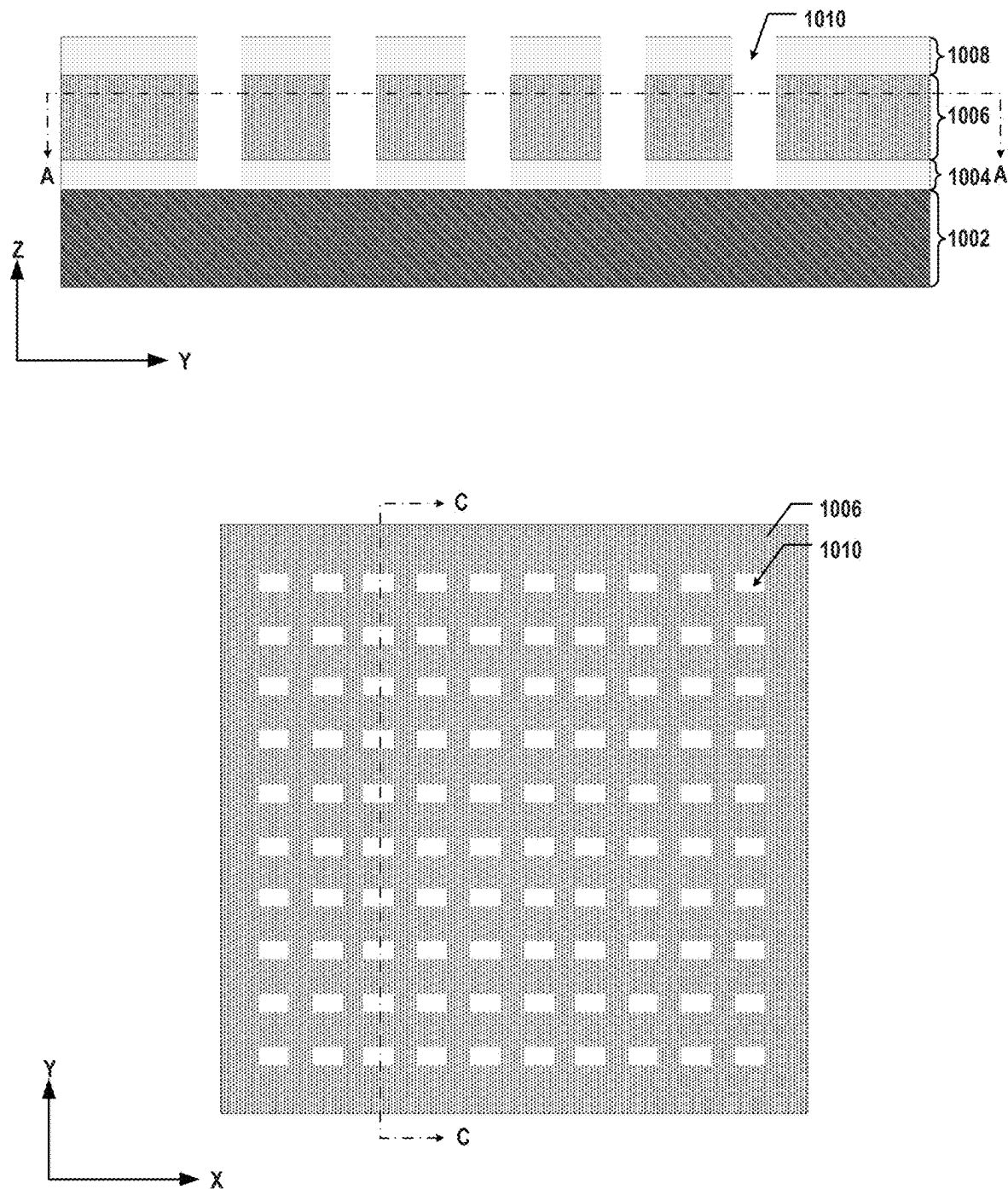


FIG. 10B

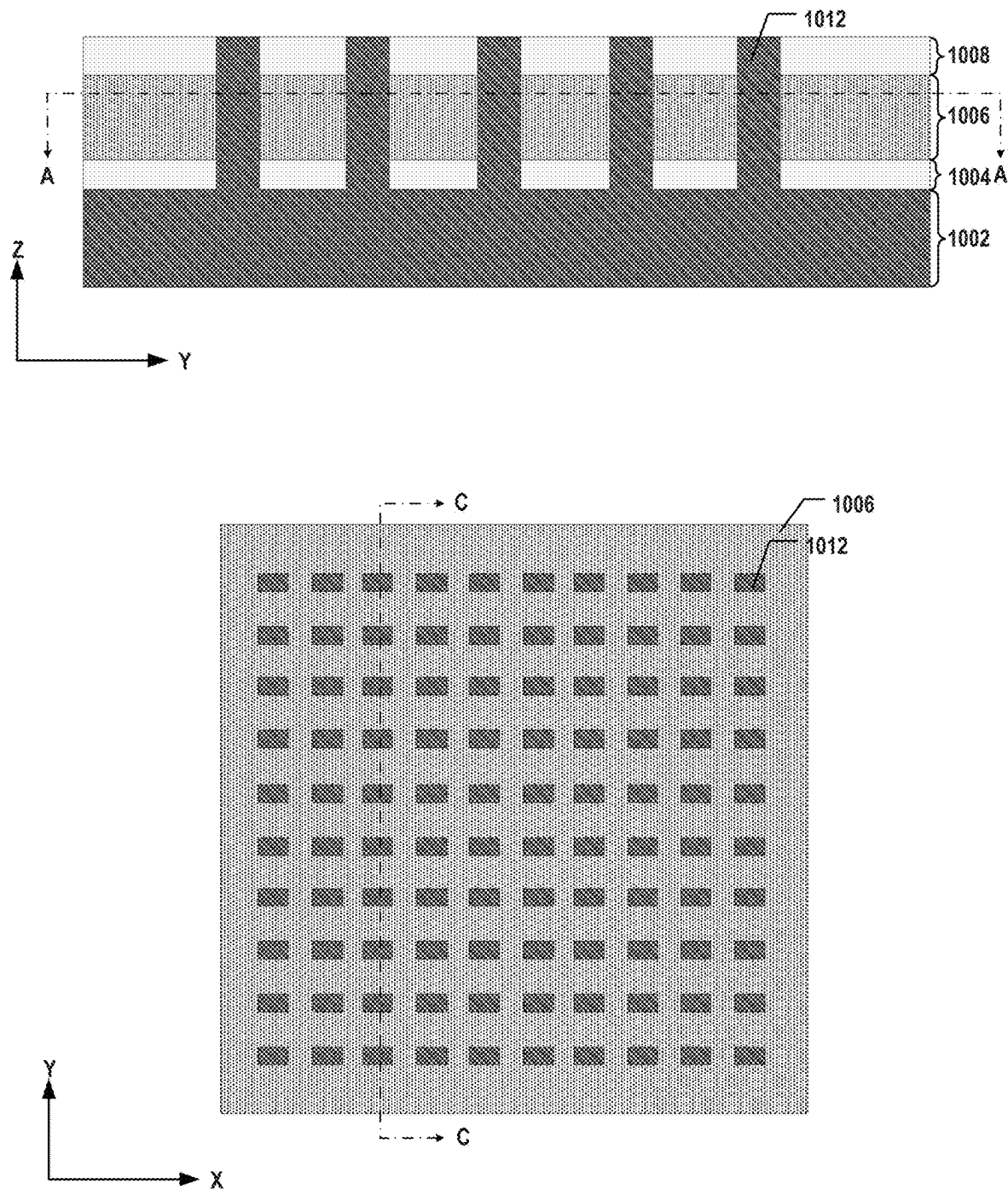


FIG. 10C

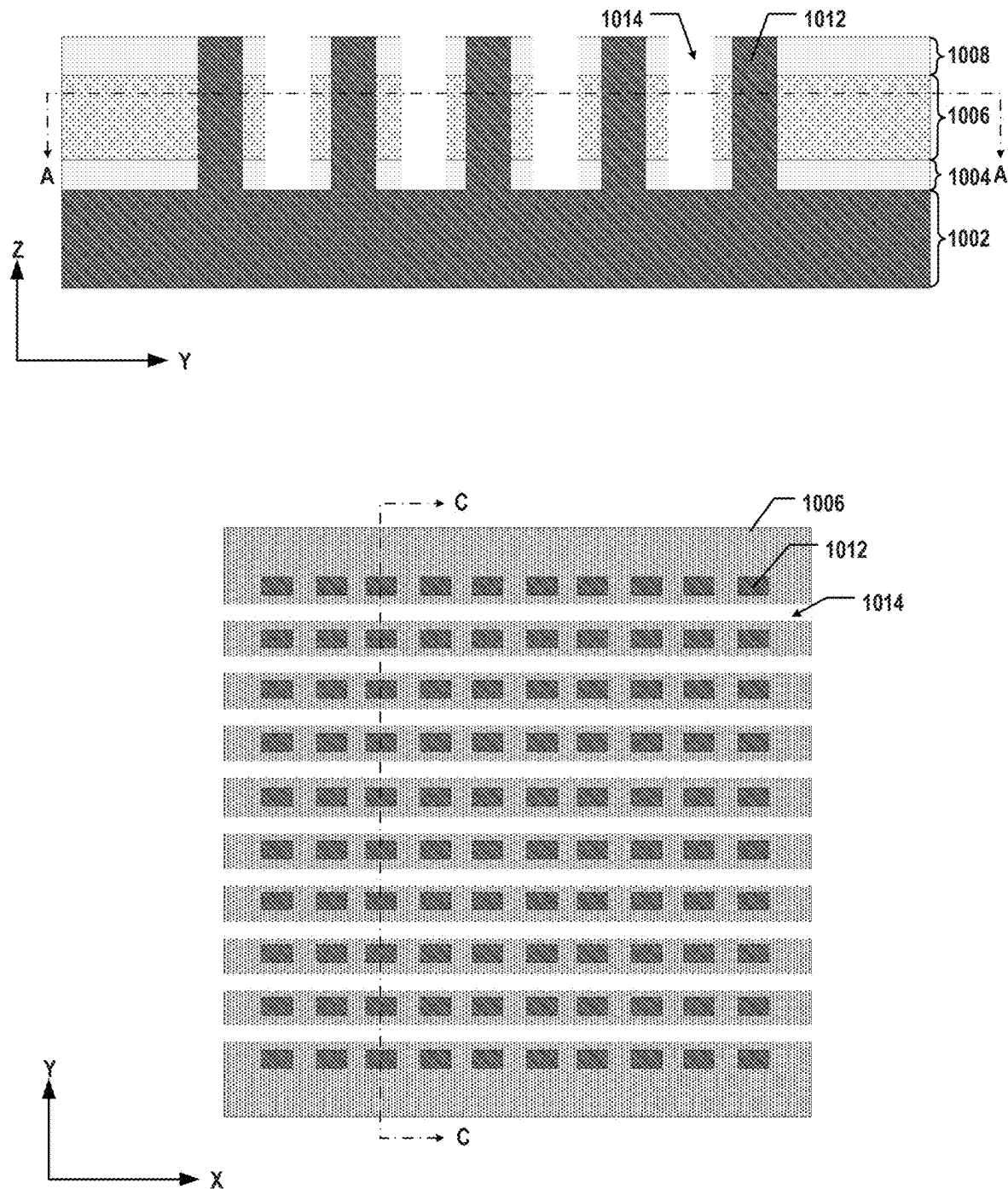


FIG. 10D

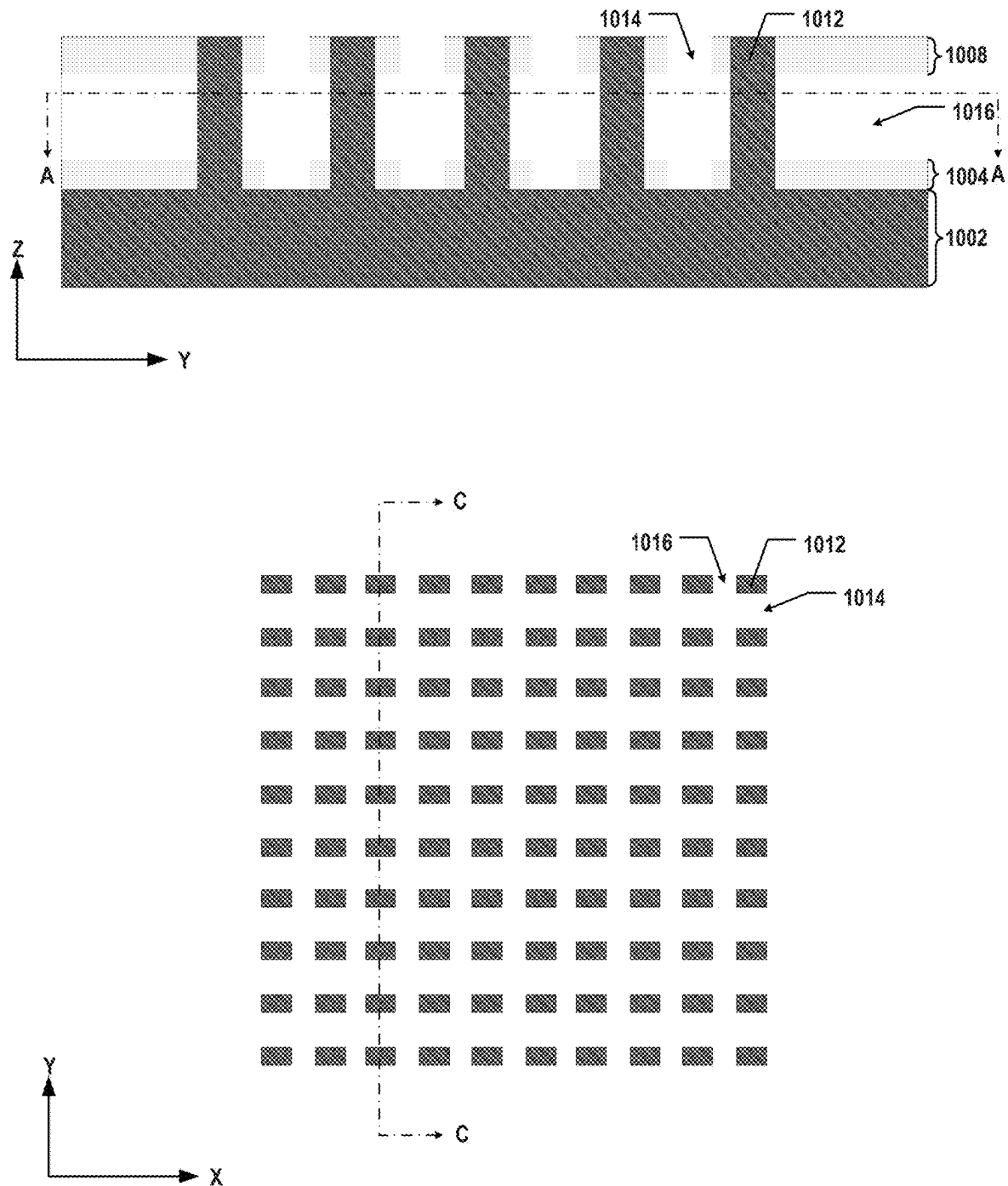


FIG. 10E

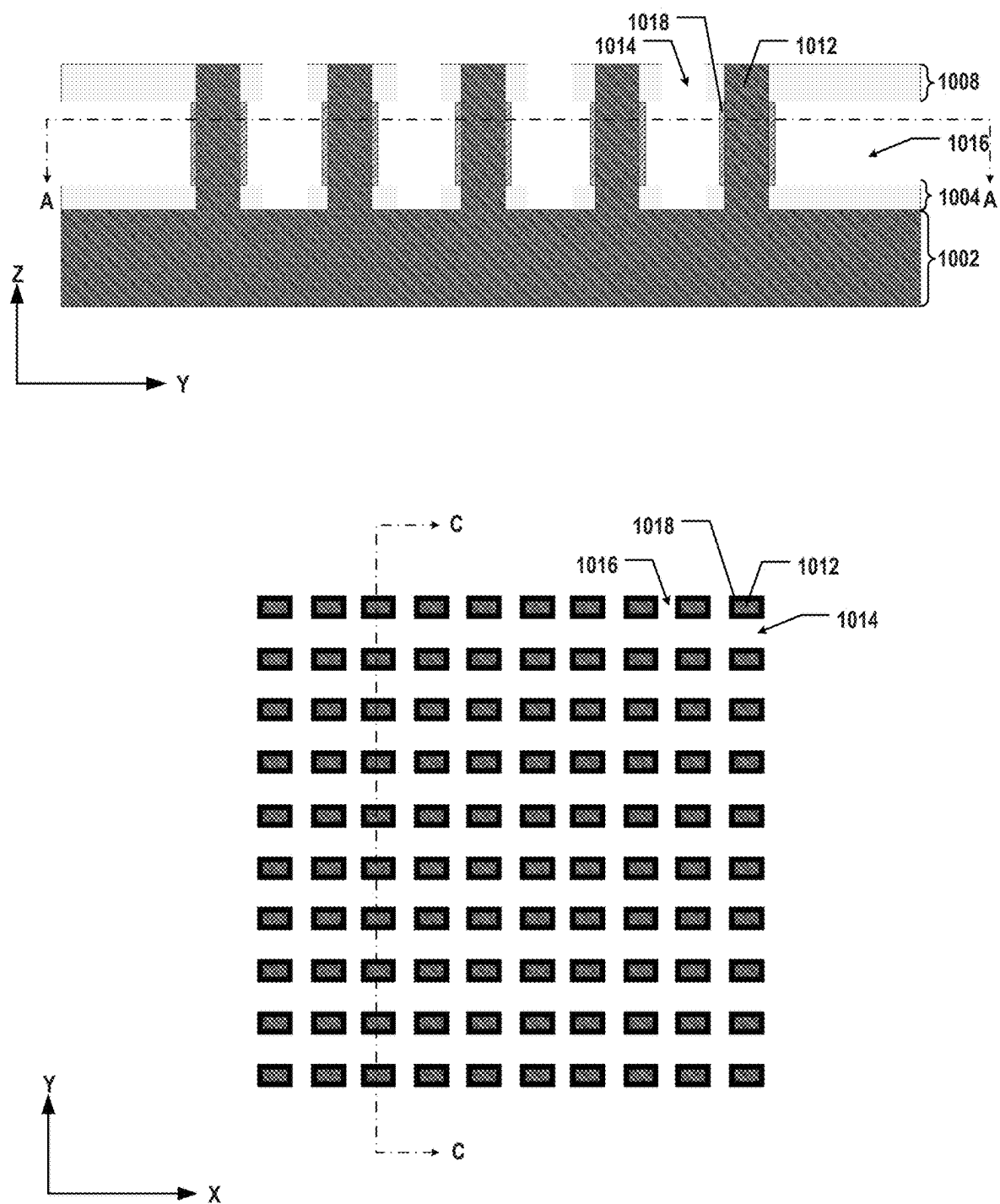


FIG. 10F

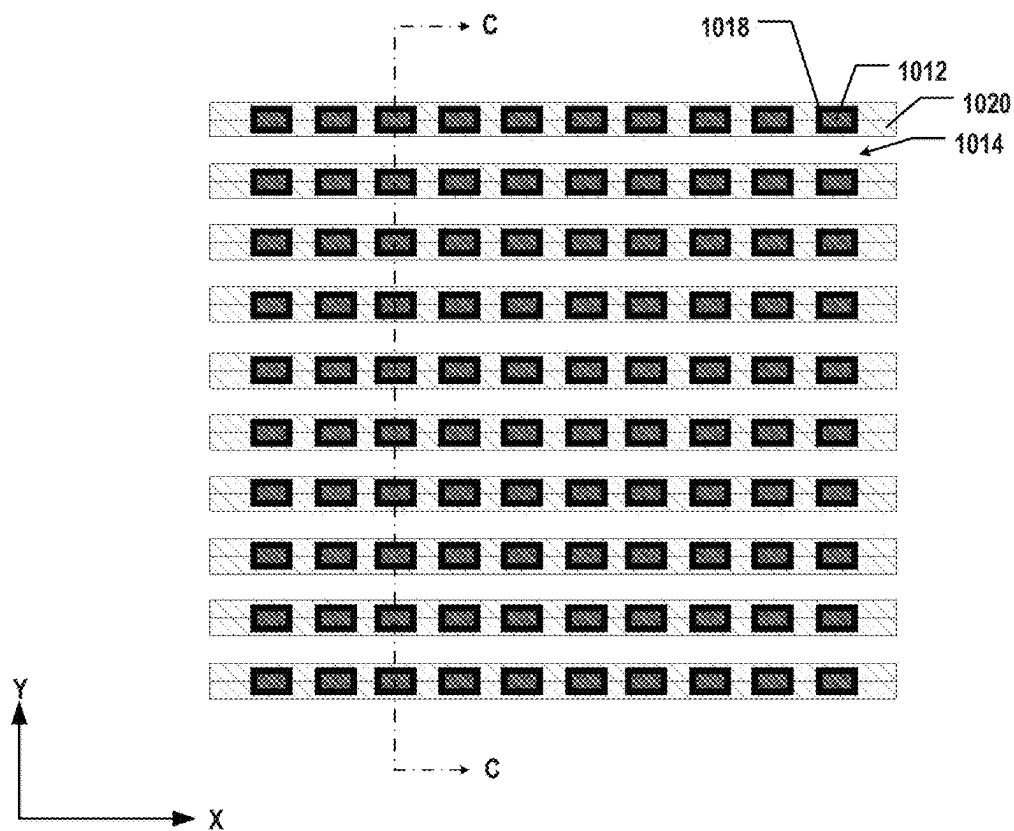
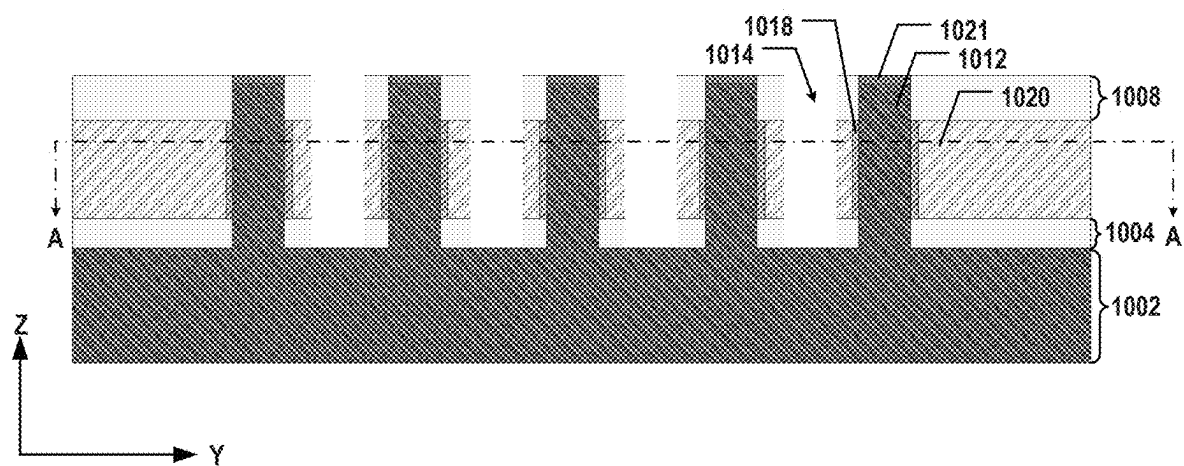


FIG. 10G

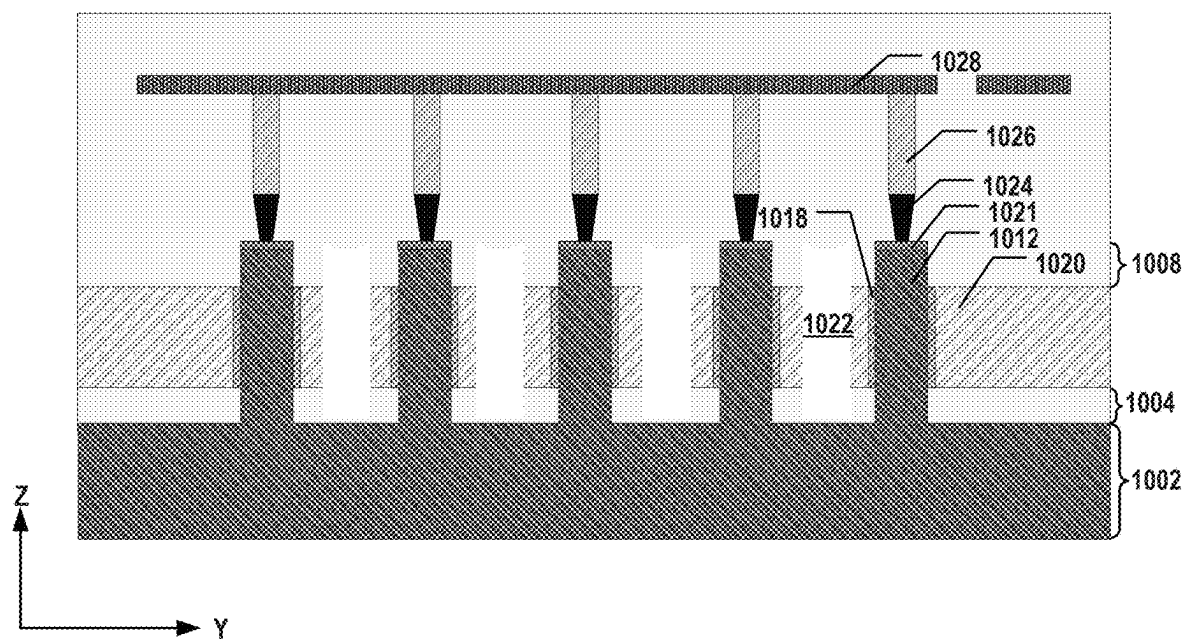


FIG. 10H

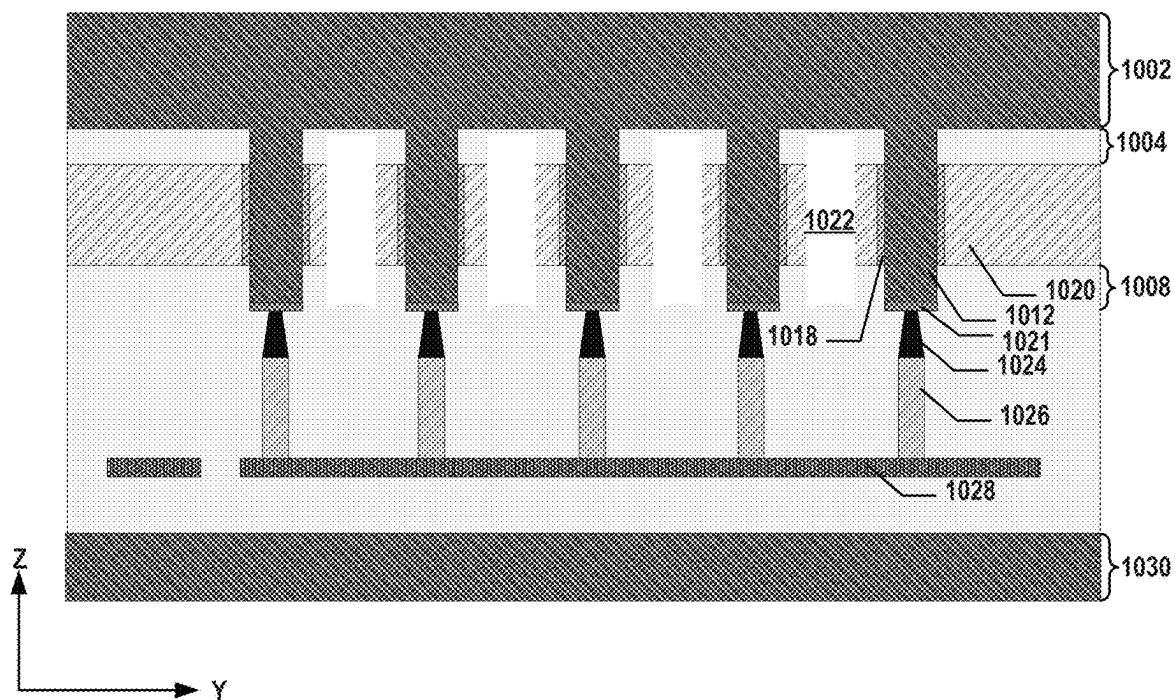


FIG. 10I

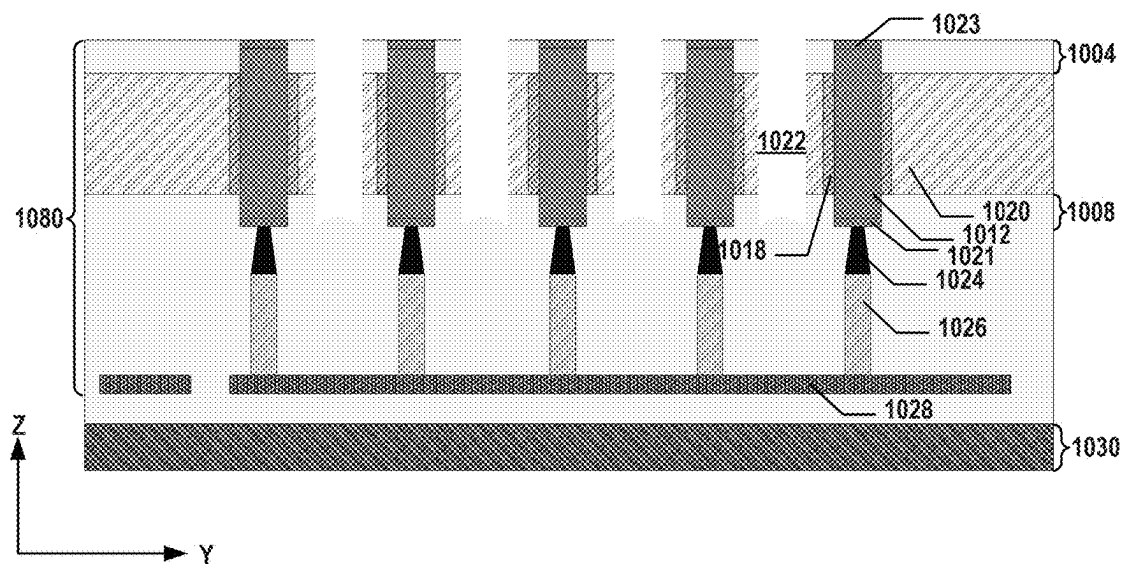


FIG. 10J

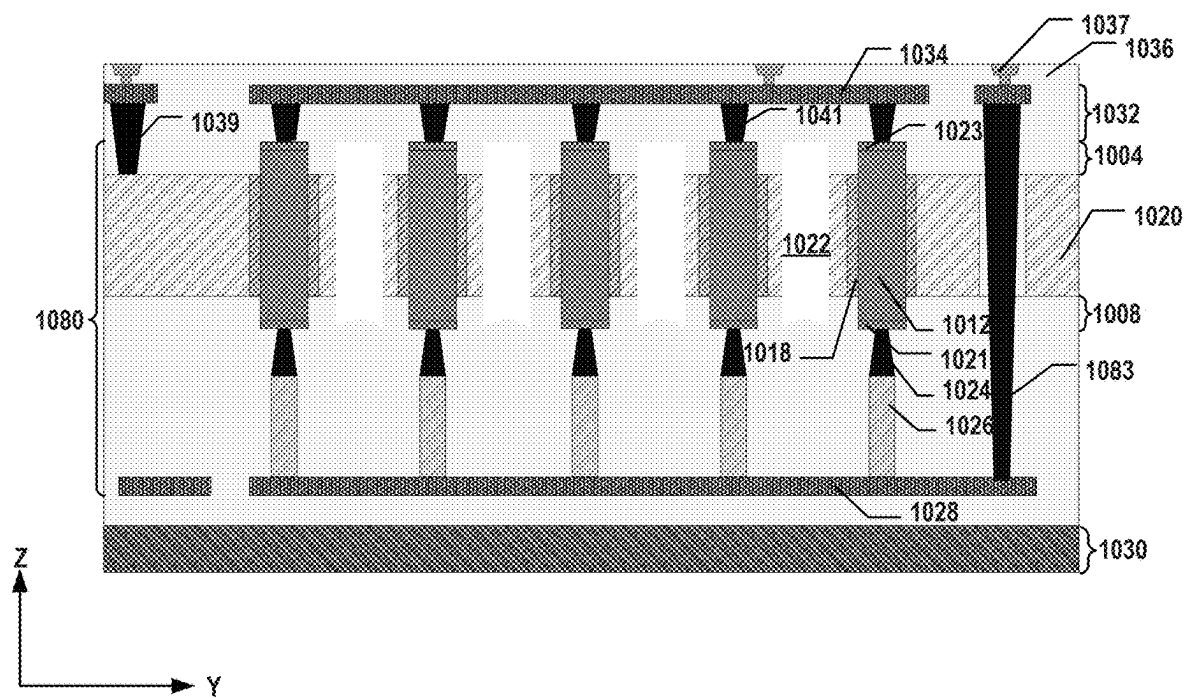


FIG. 10K

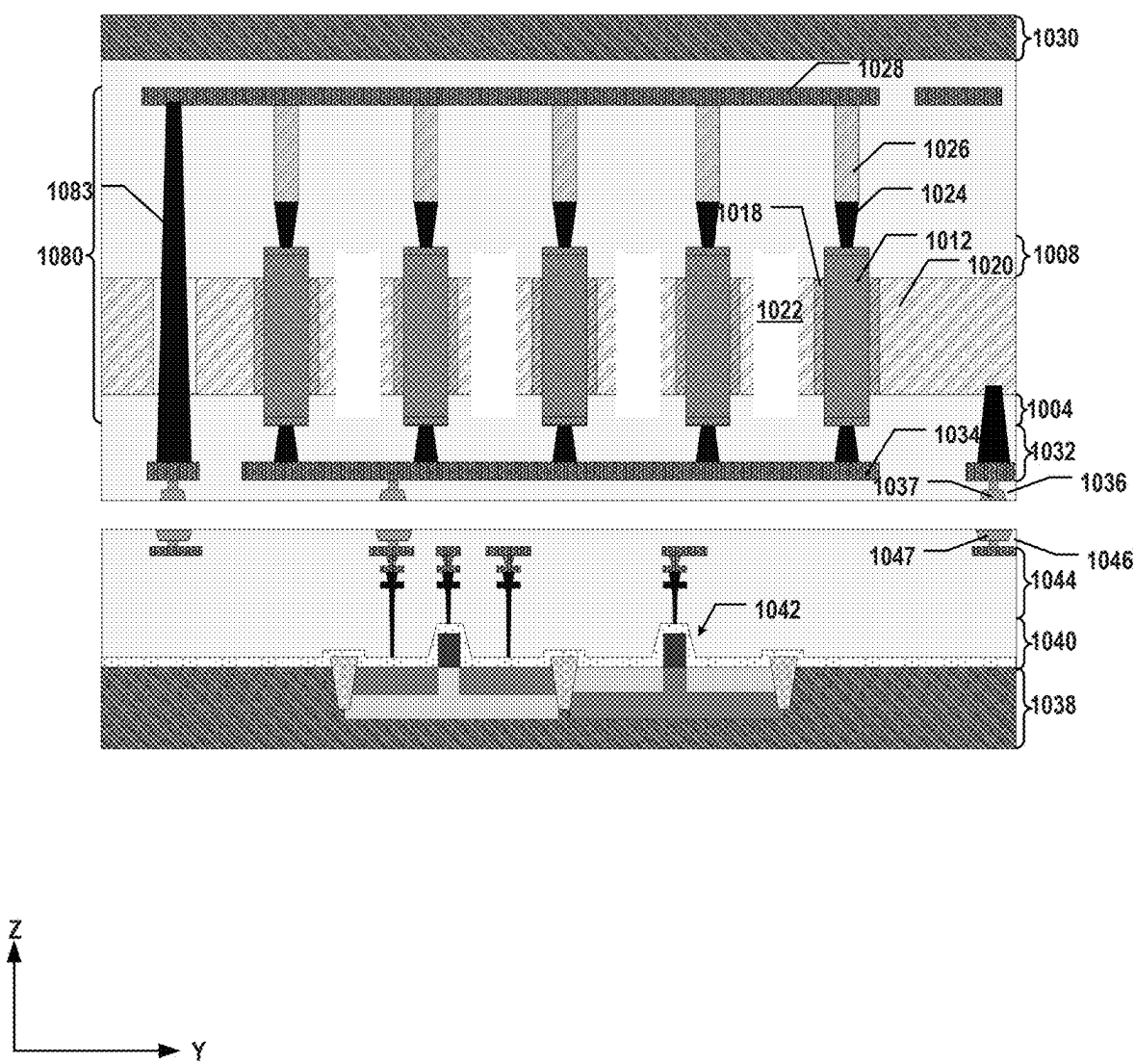


FIG. 10L

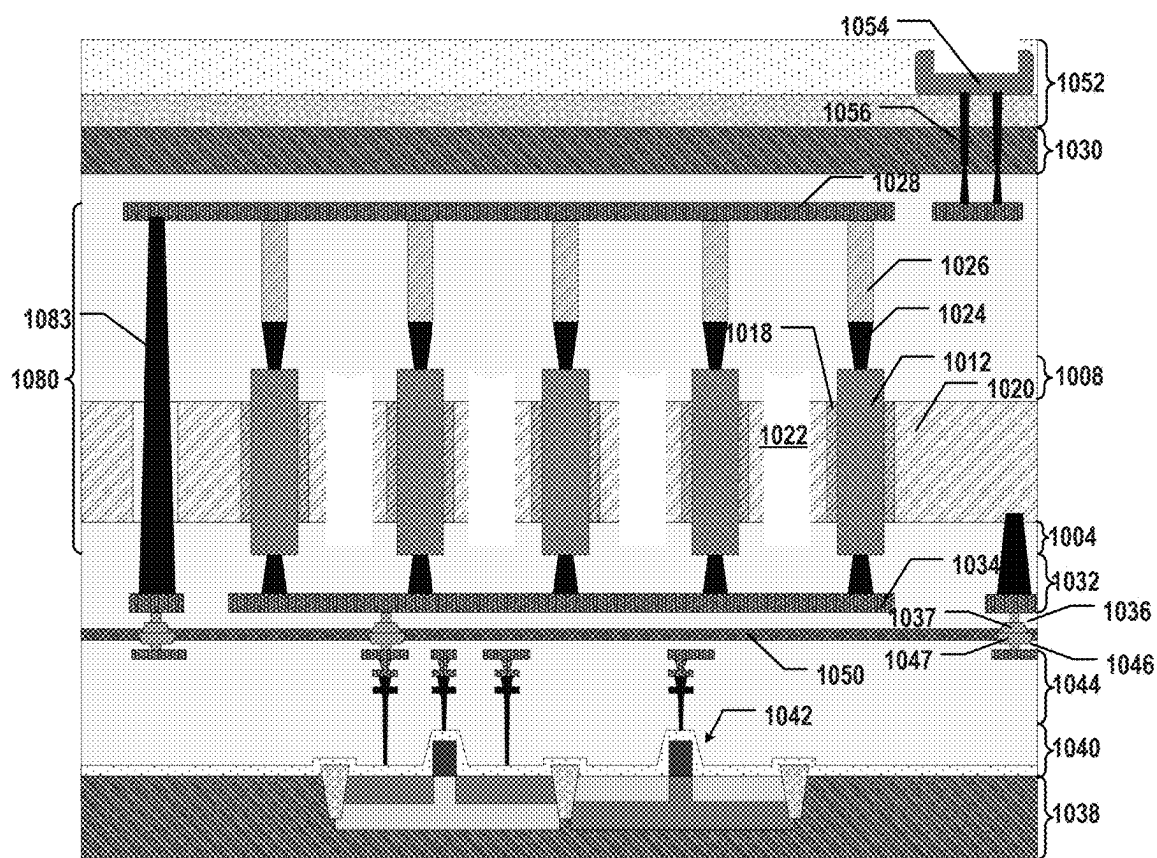


FIG. 10M

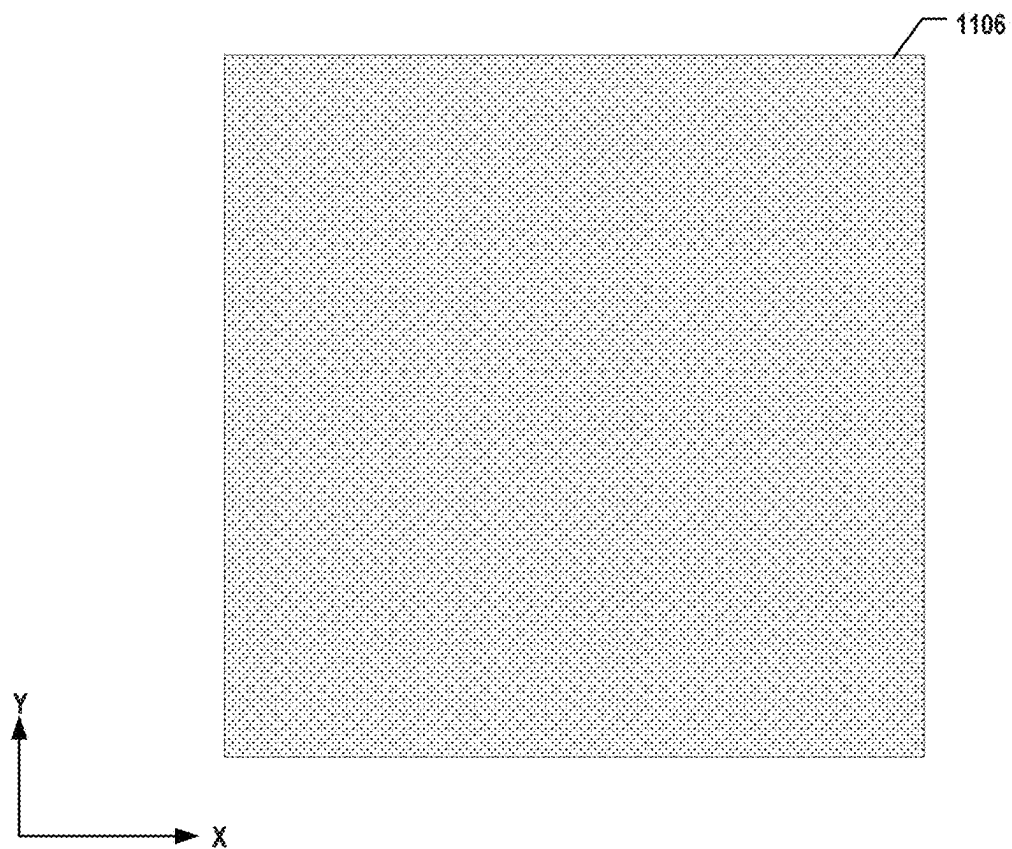
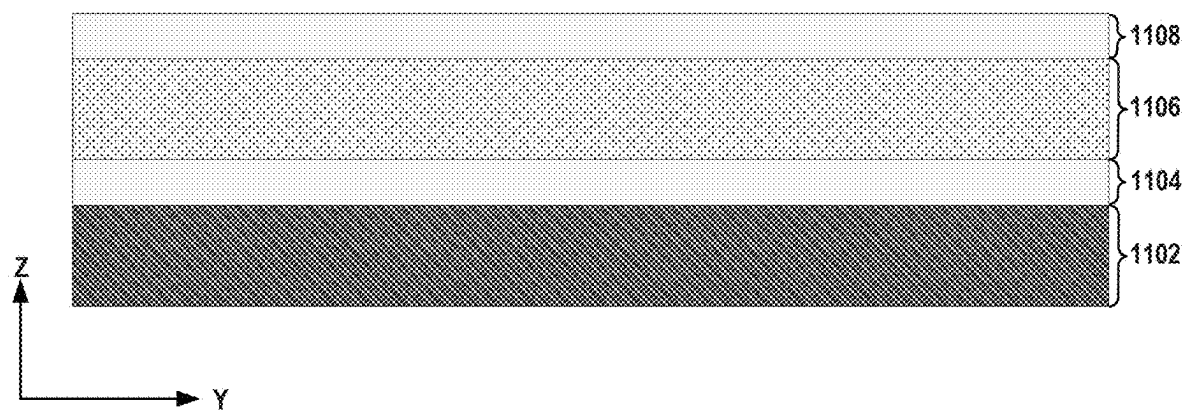


FIG. 11A

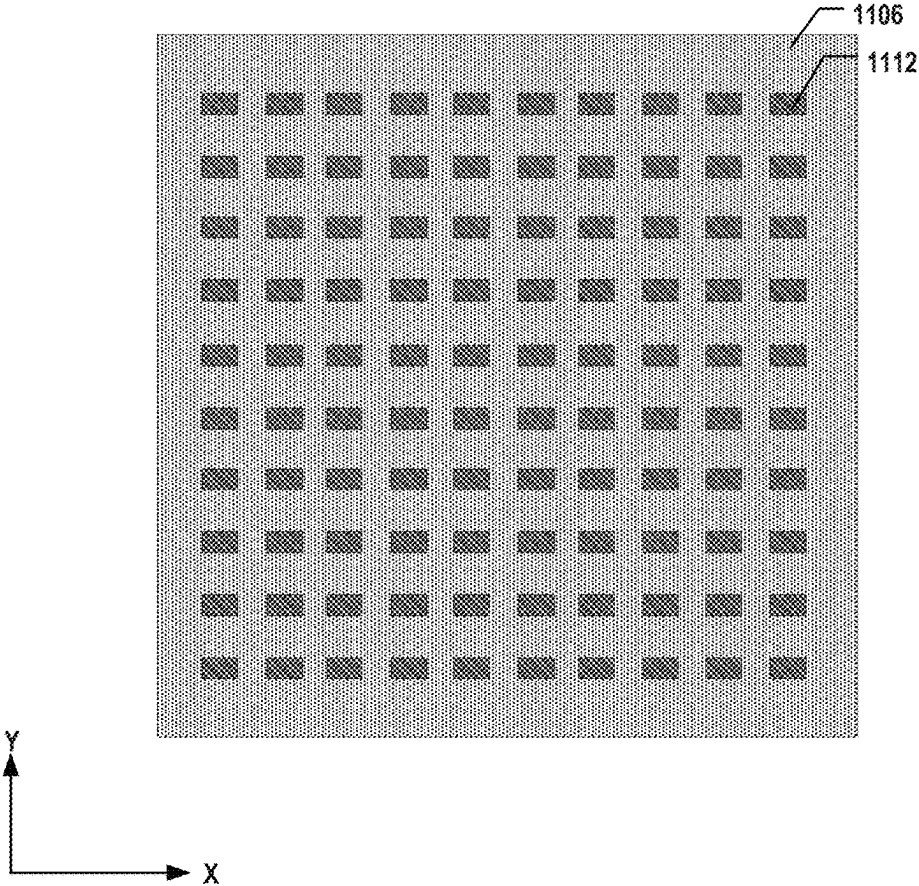
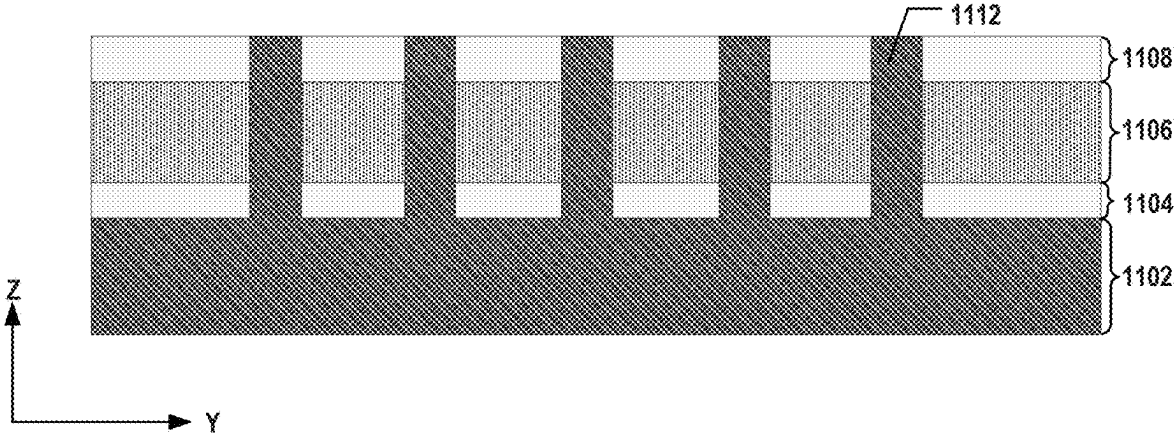


FIG. 11B

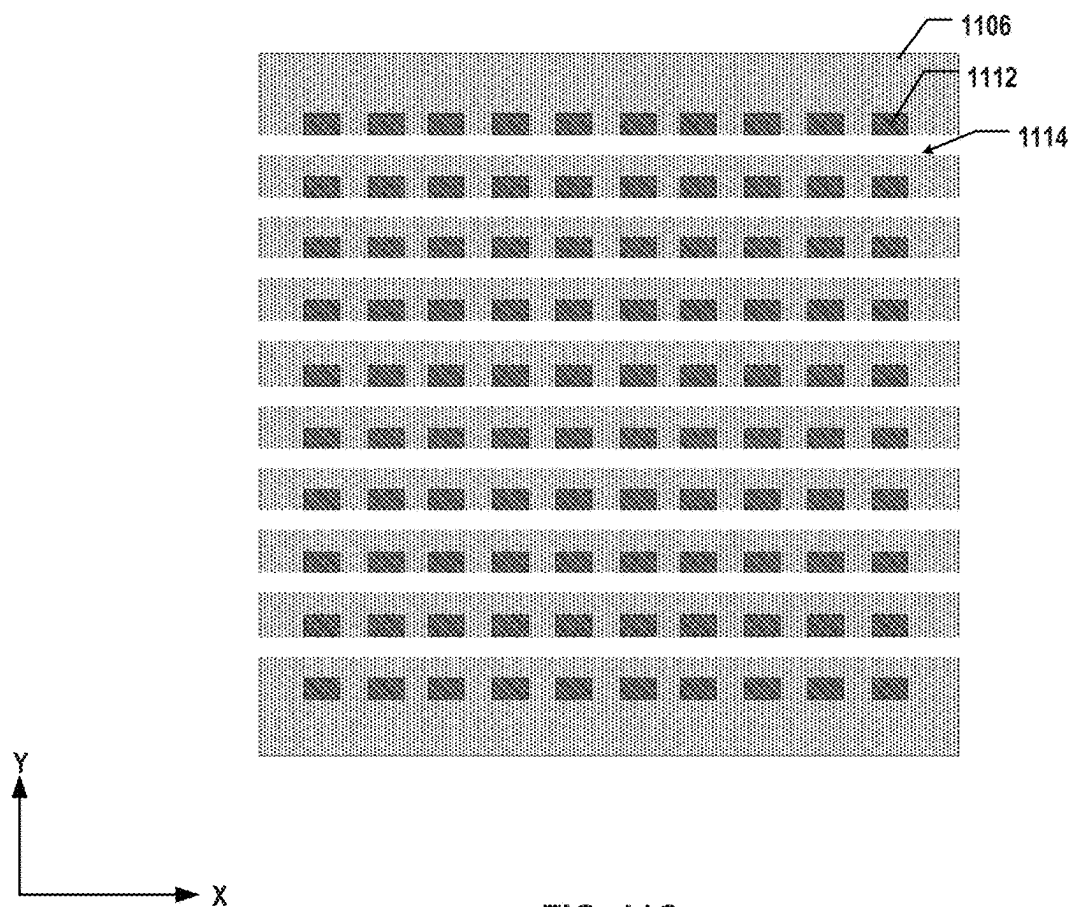
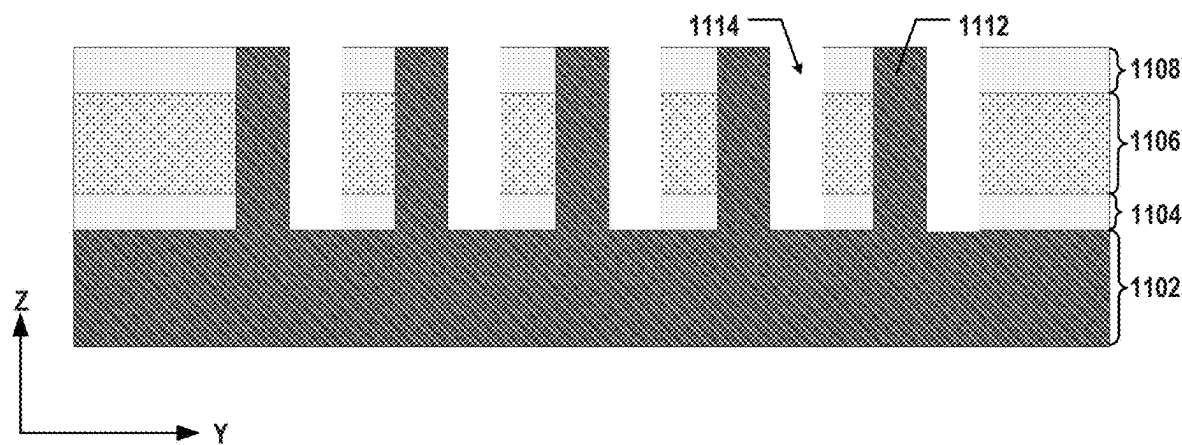


FIG. 11C

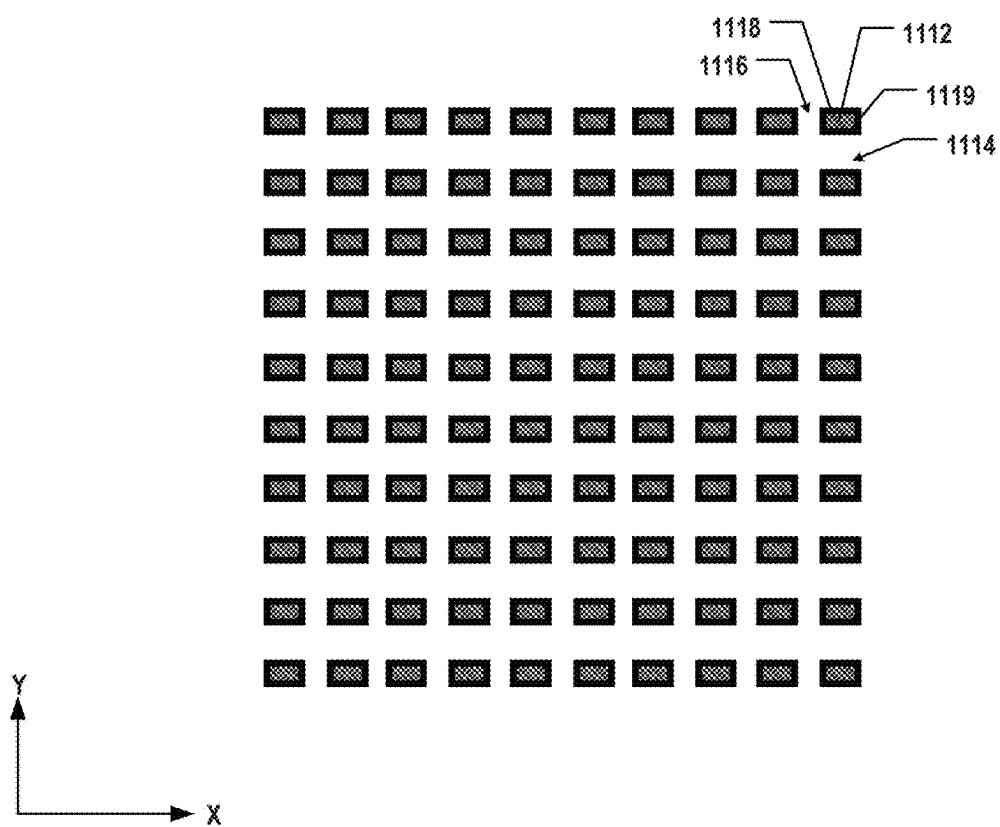
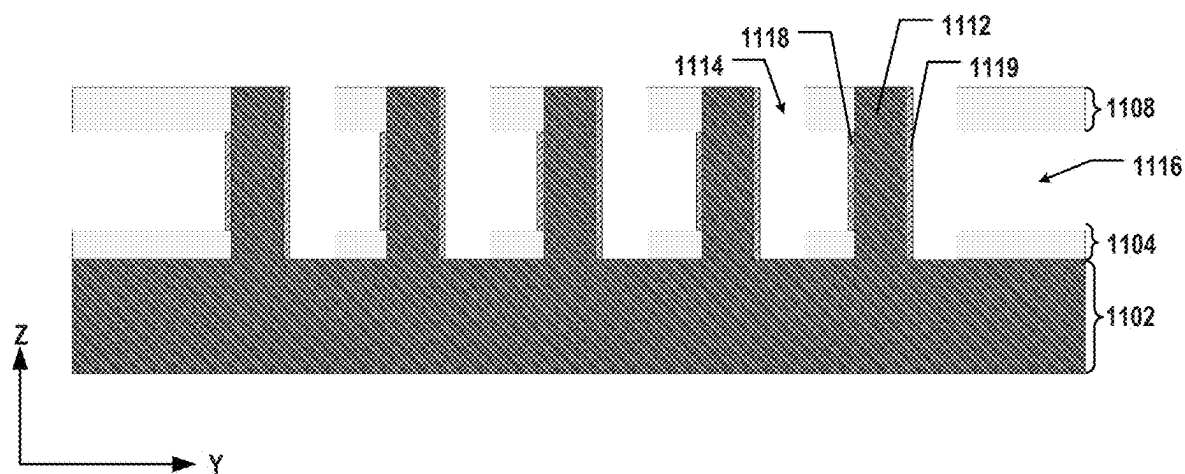


FIG. 11D

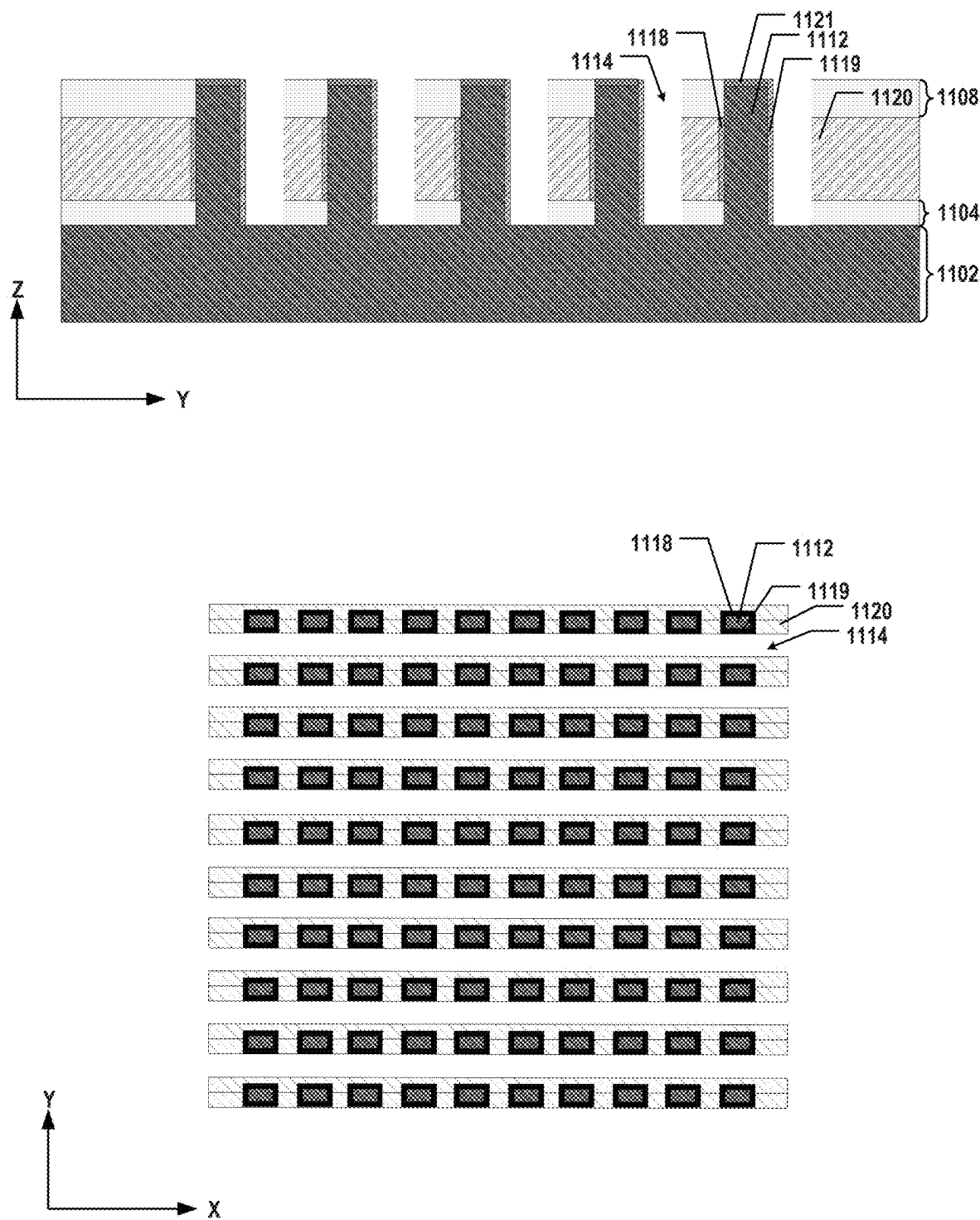


FIG. 11E

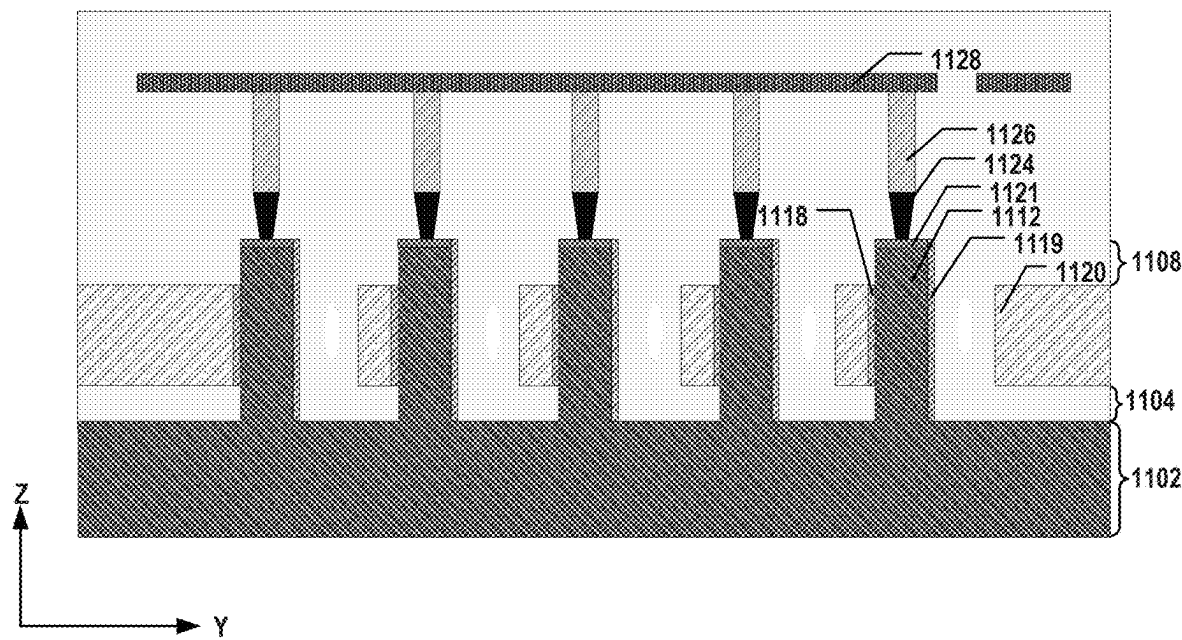


FIG. 11F

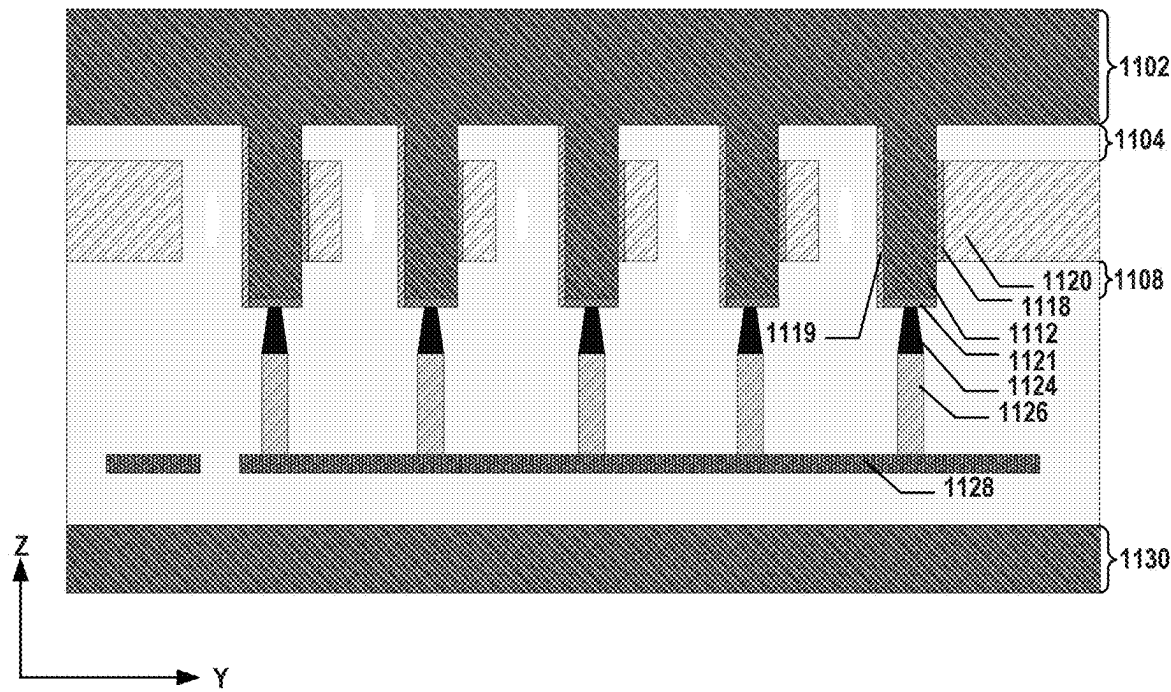


FIG. 11G

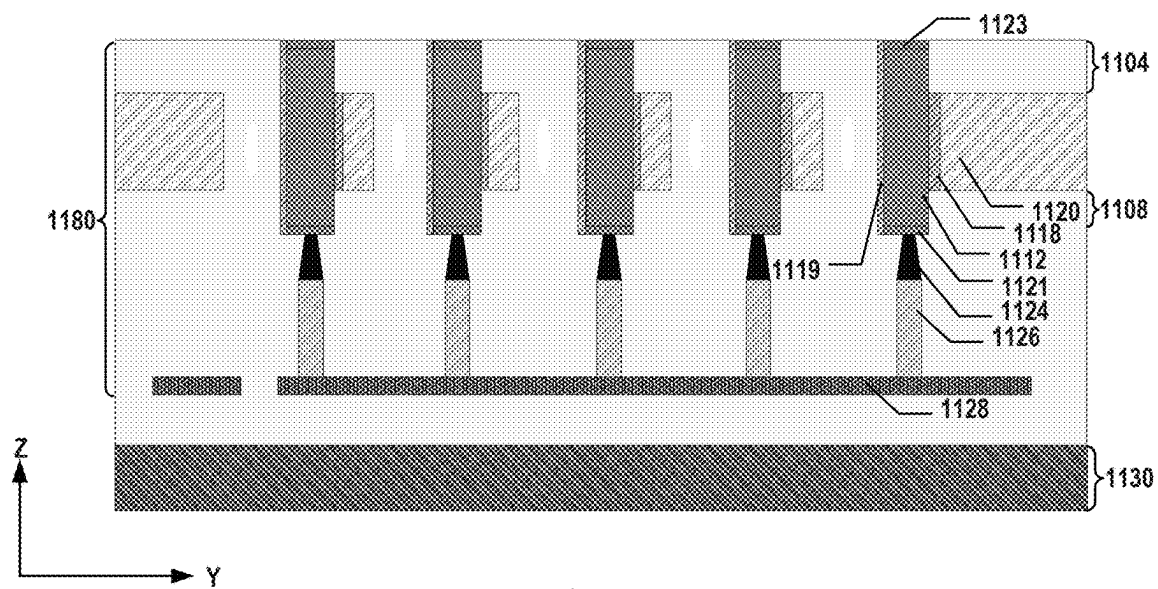


FIG. 11H

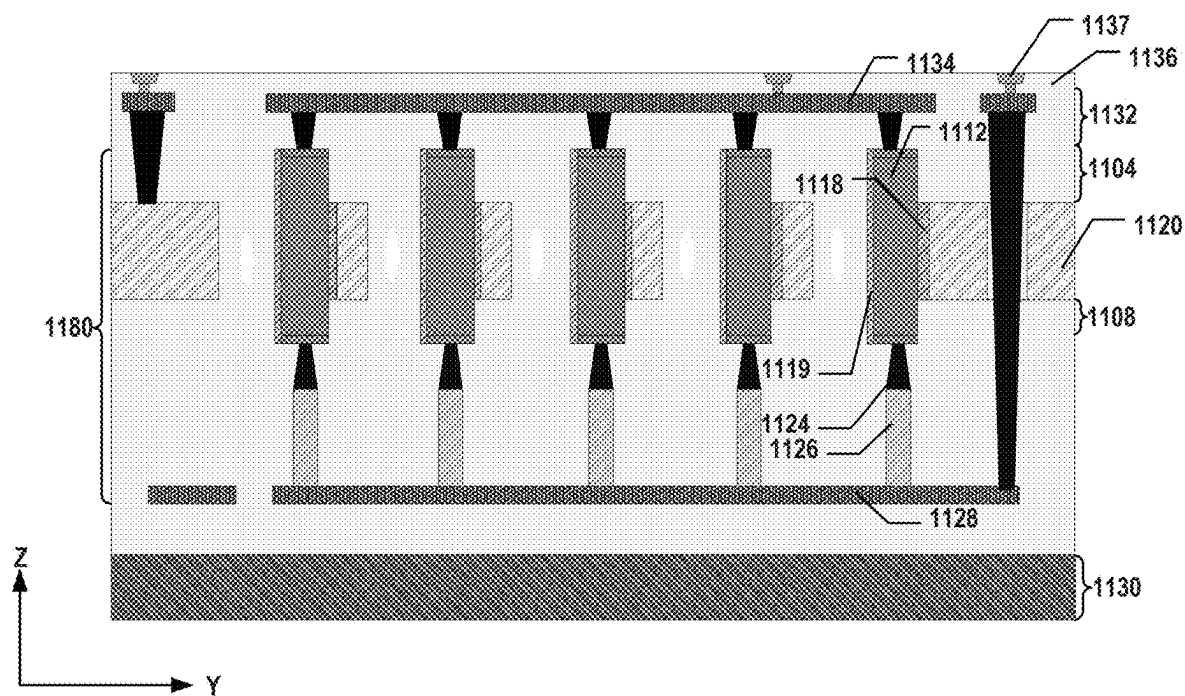


FIG. 11I

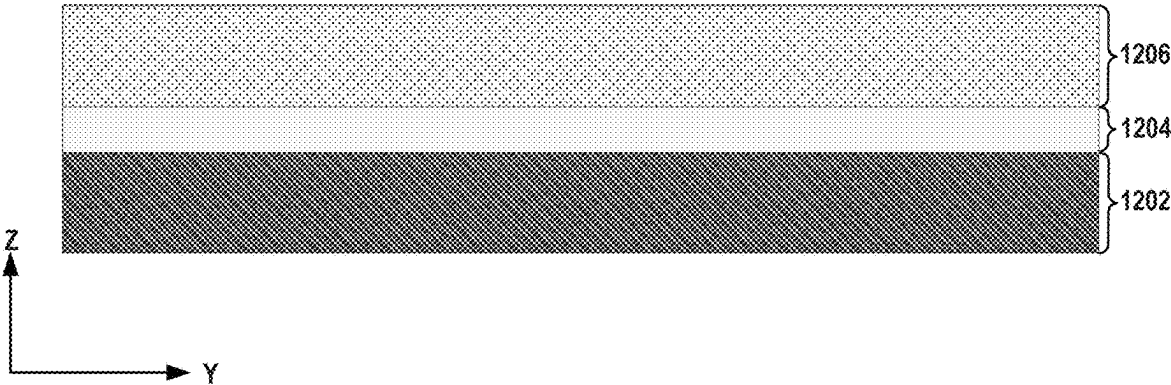


FIG. 12A

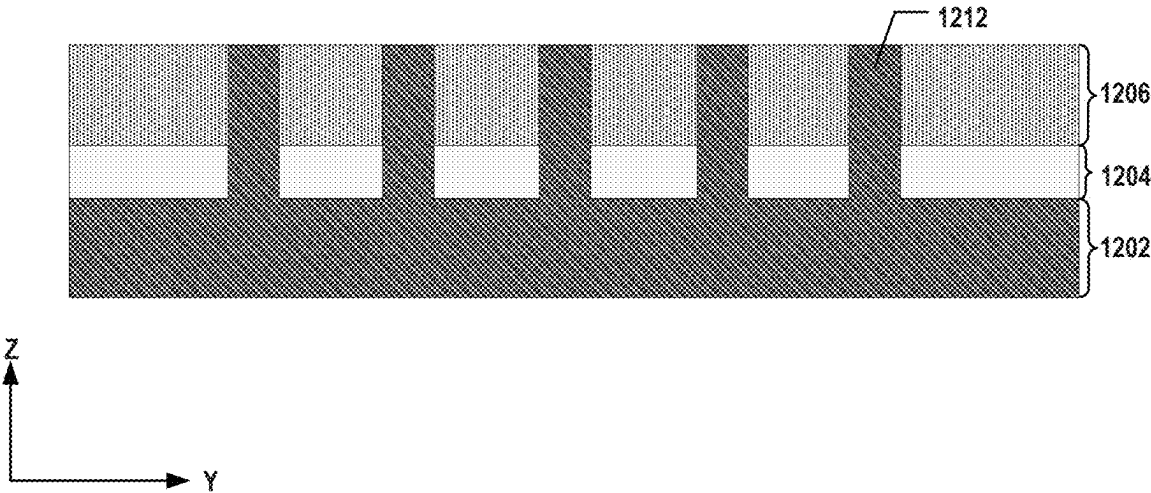


FIG. 12B

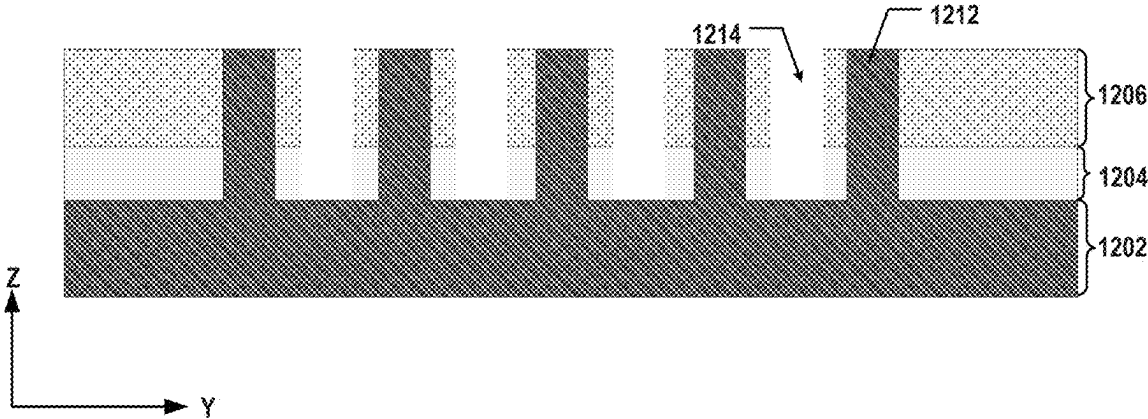


FIG. 12C

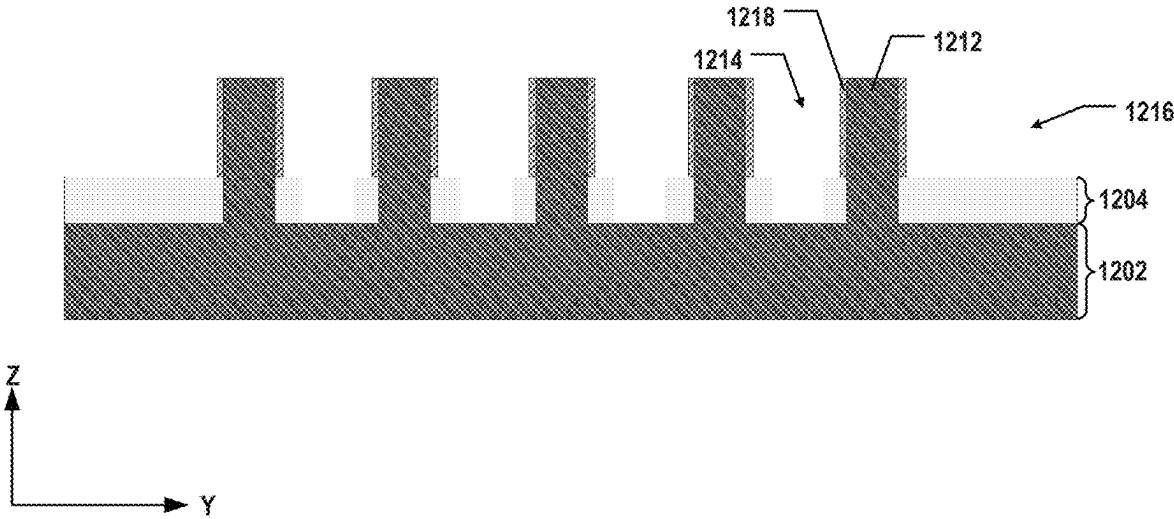


FIG. 12D

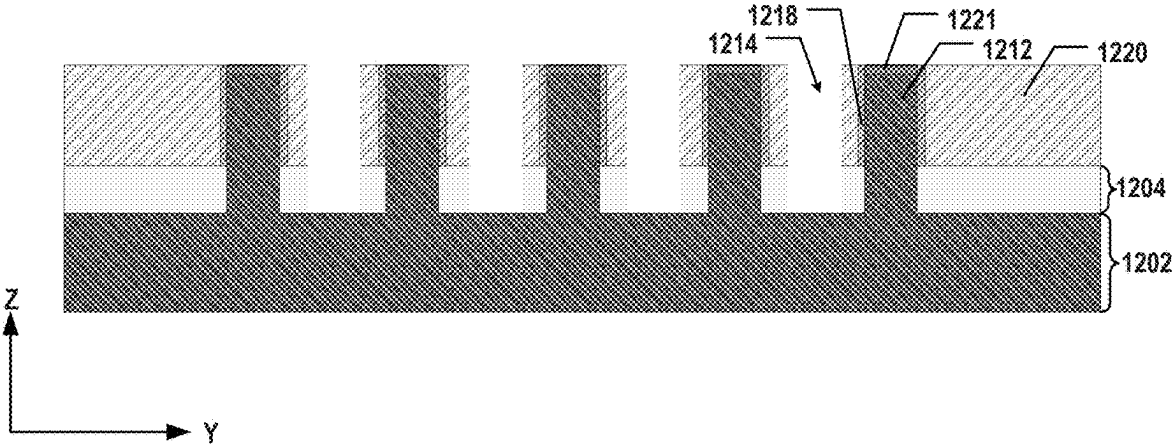


FIG. 12E

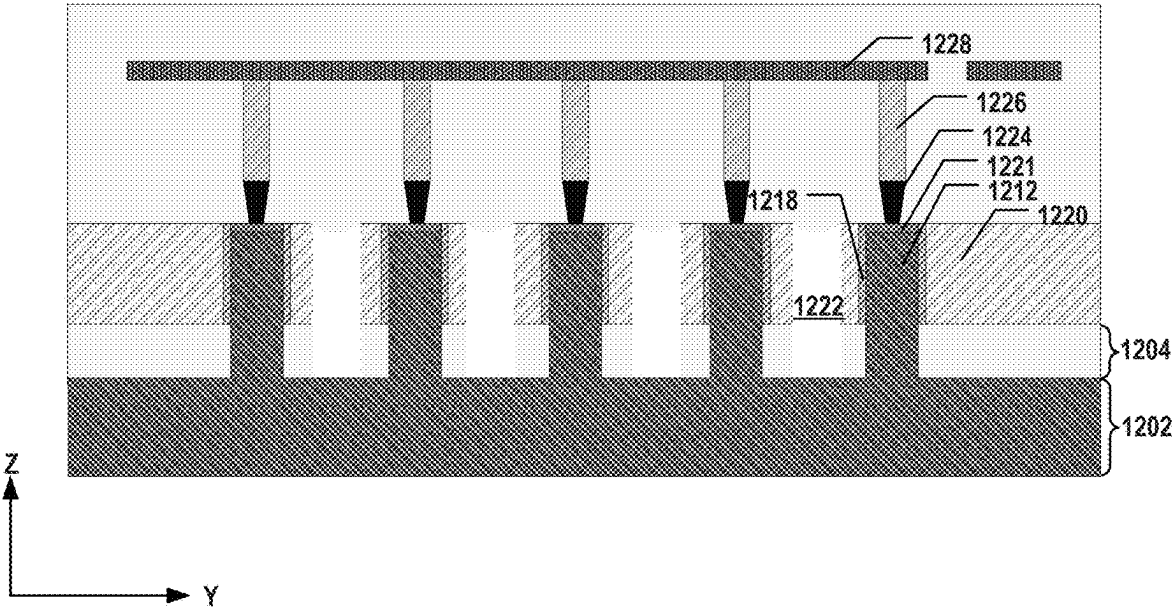


FIG. 12F

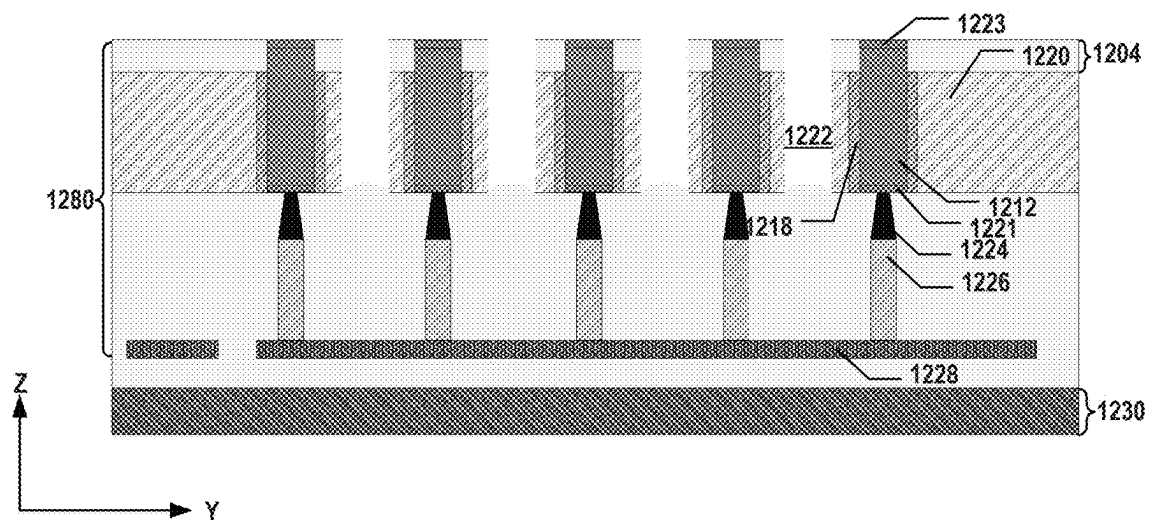


FIG. 12G

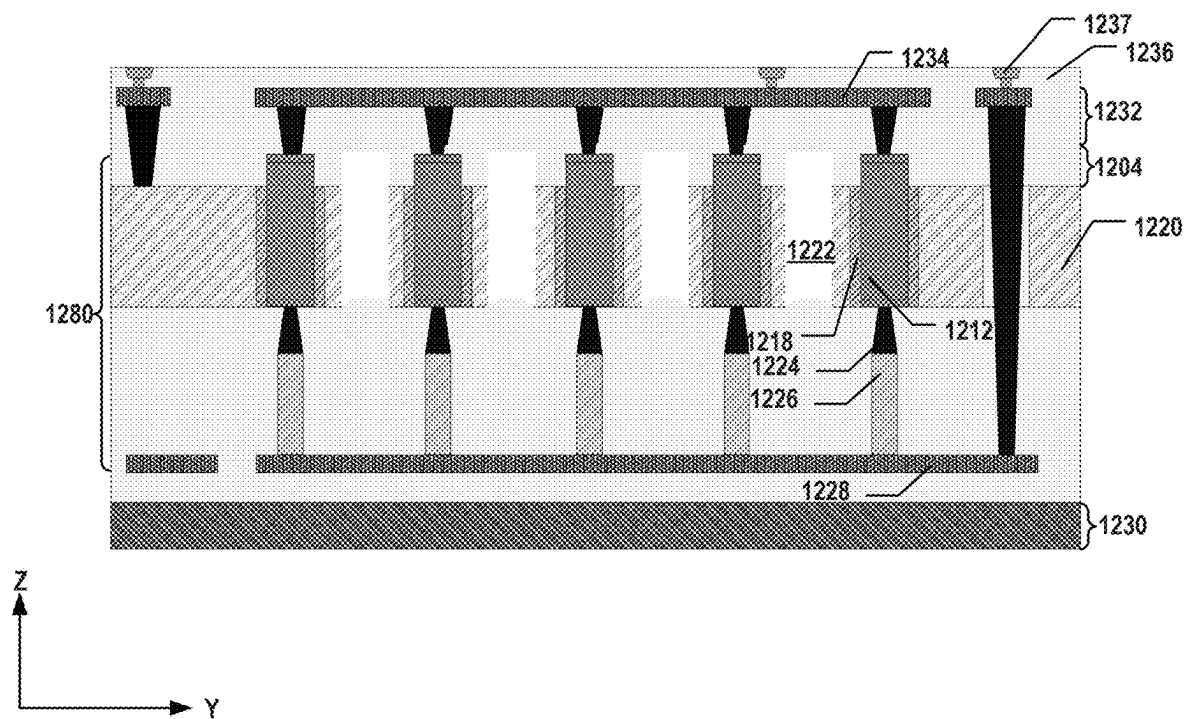


FIG. 12H

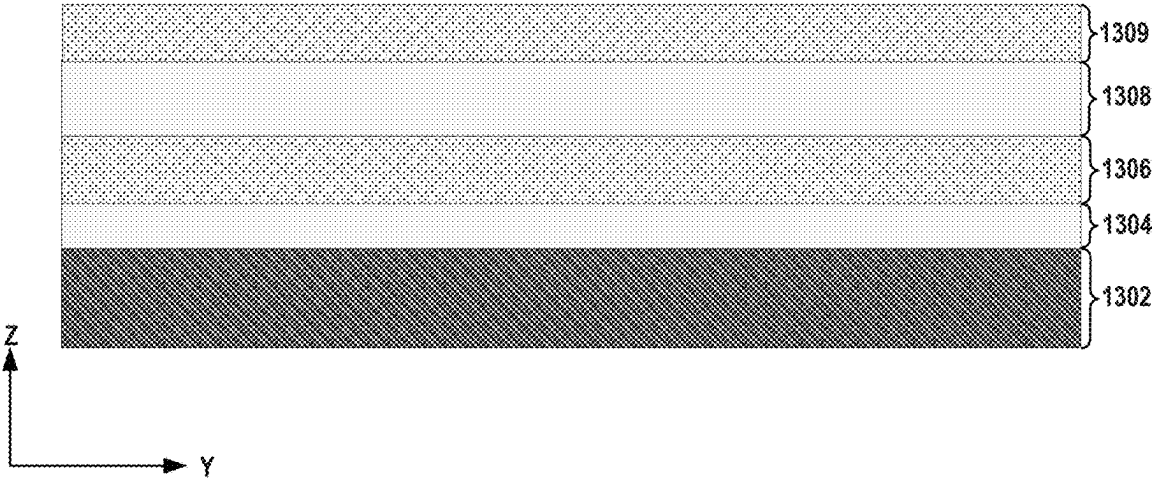


FIG. 13A

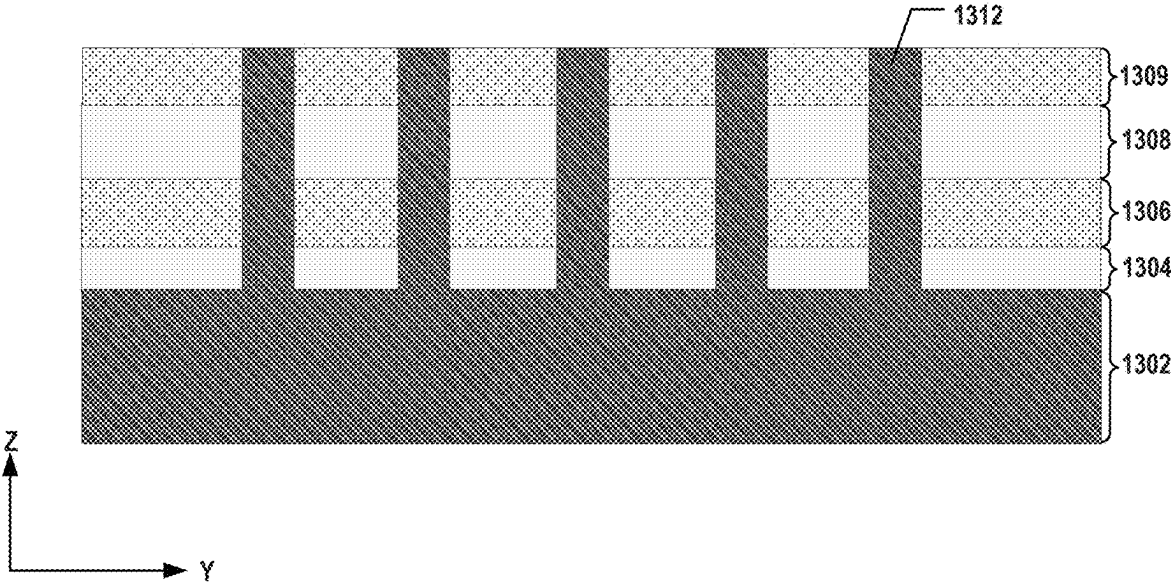


FIG. 13B

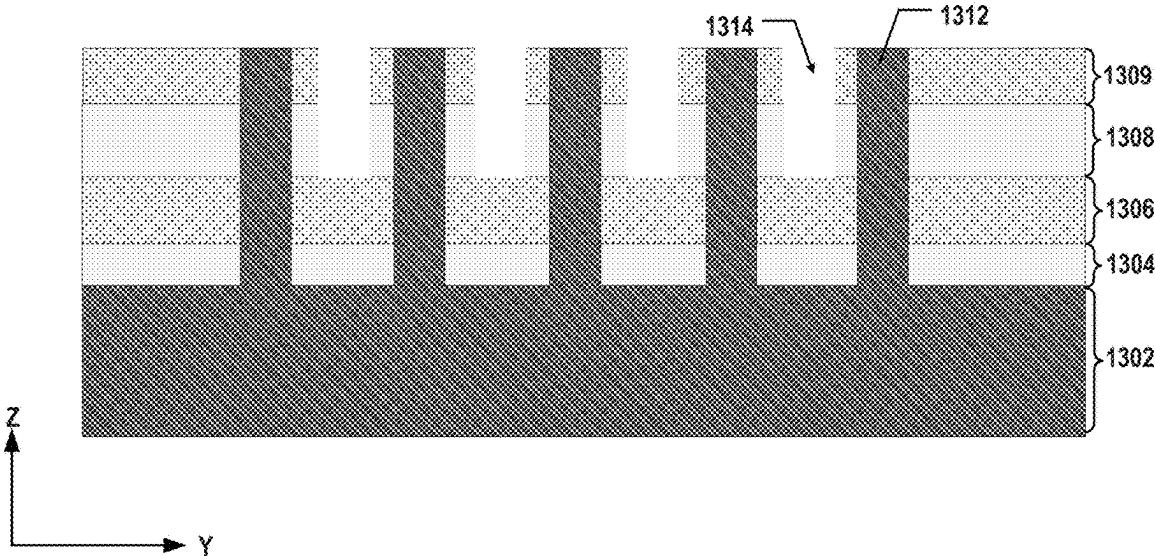


FIG. 13C

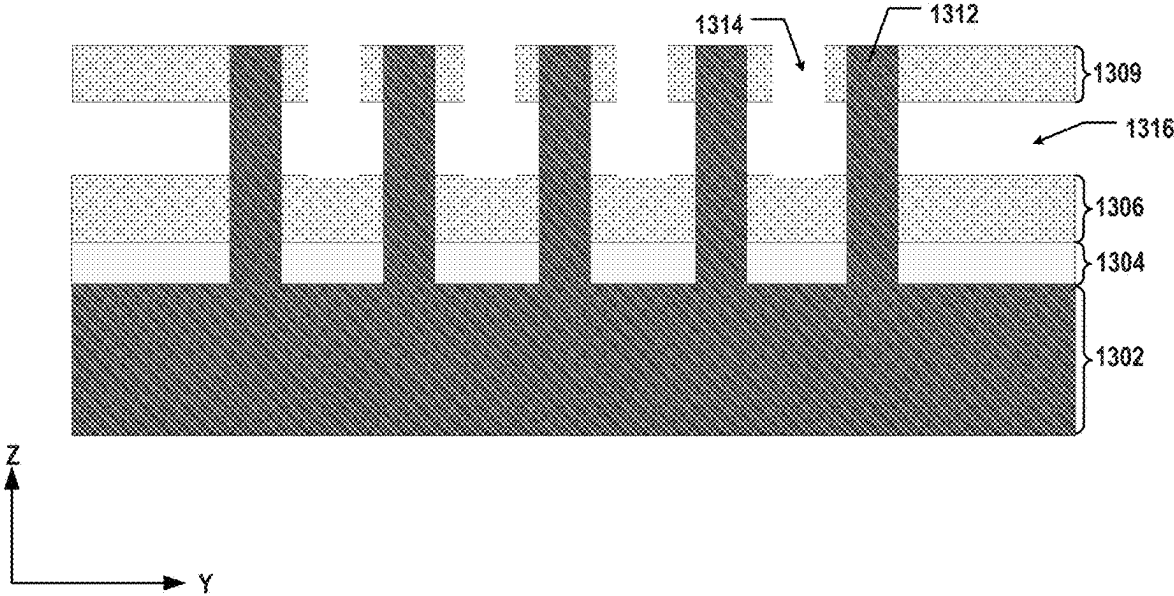


FIG. 13D

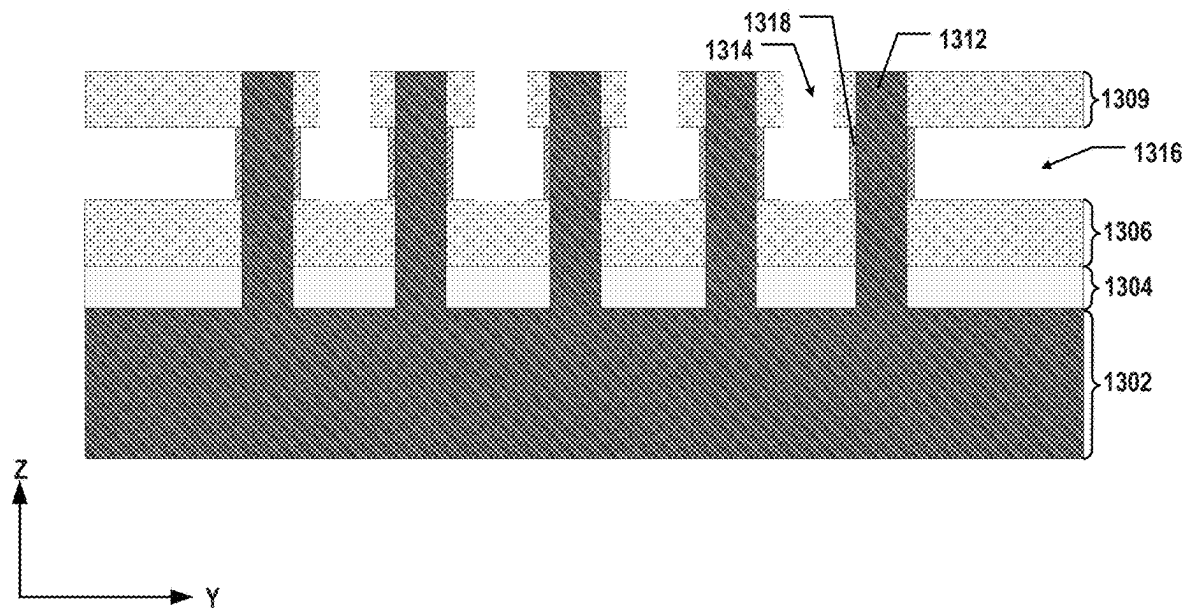


FIG. 13E

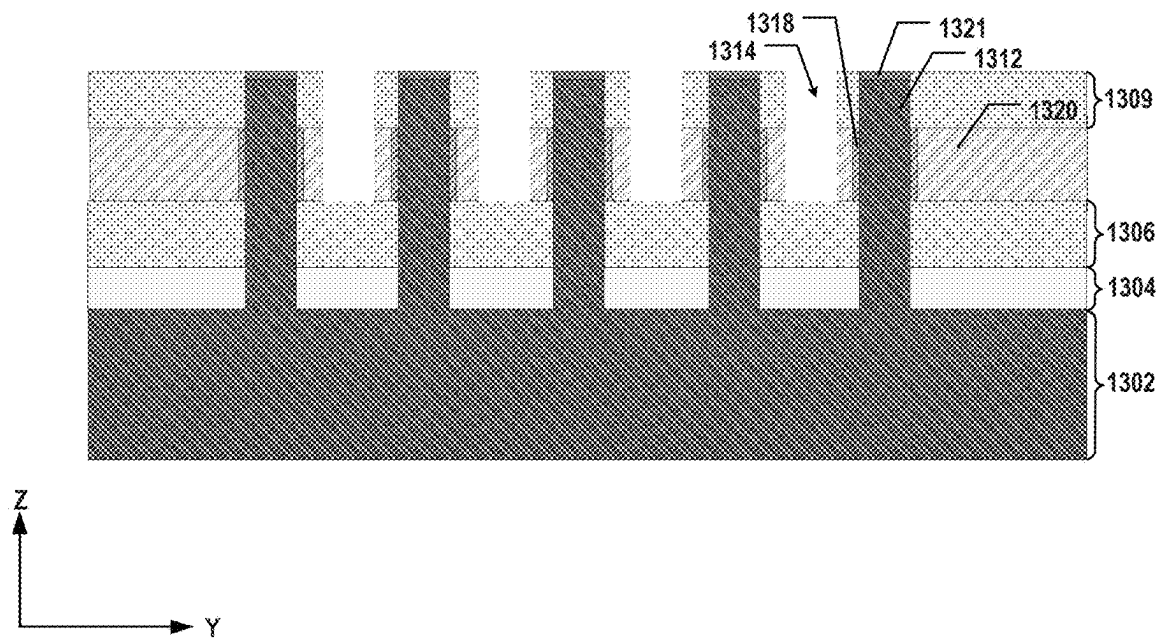


FIG. 13F

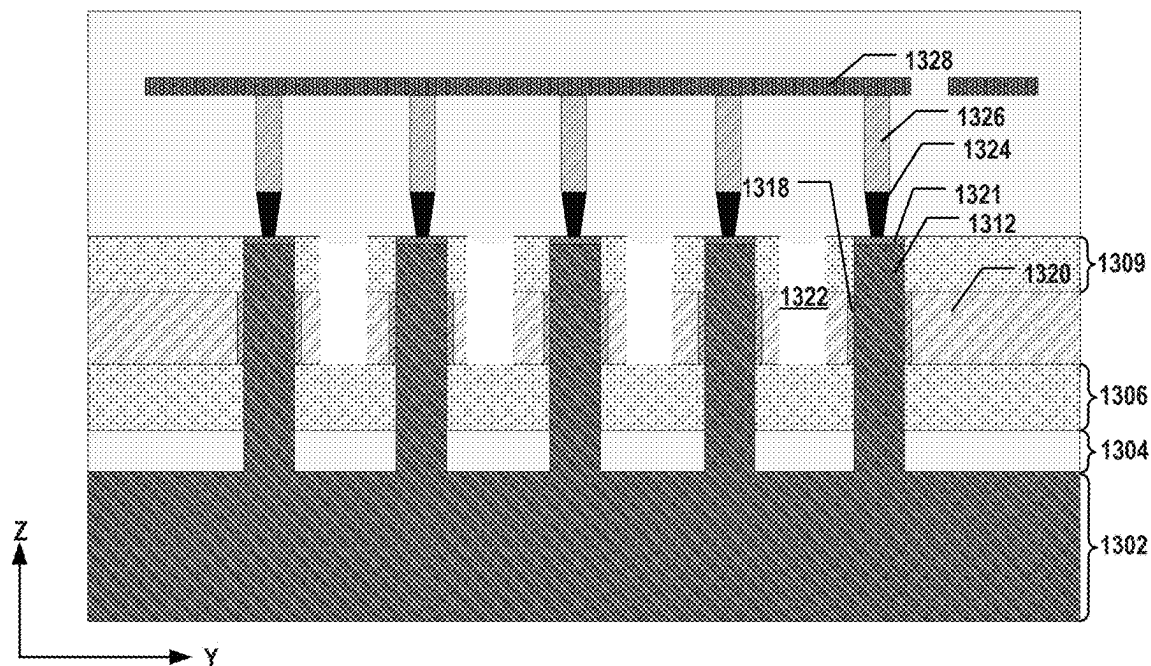


FIG. 13G

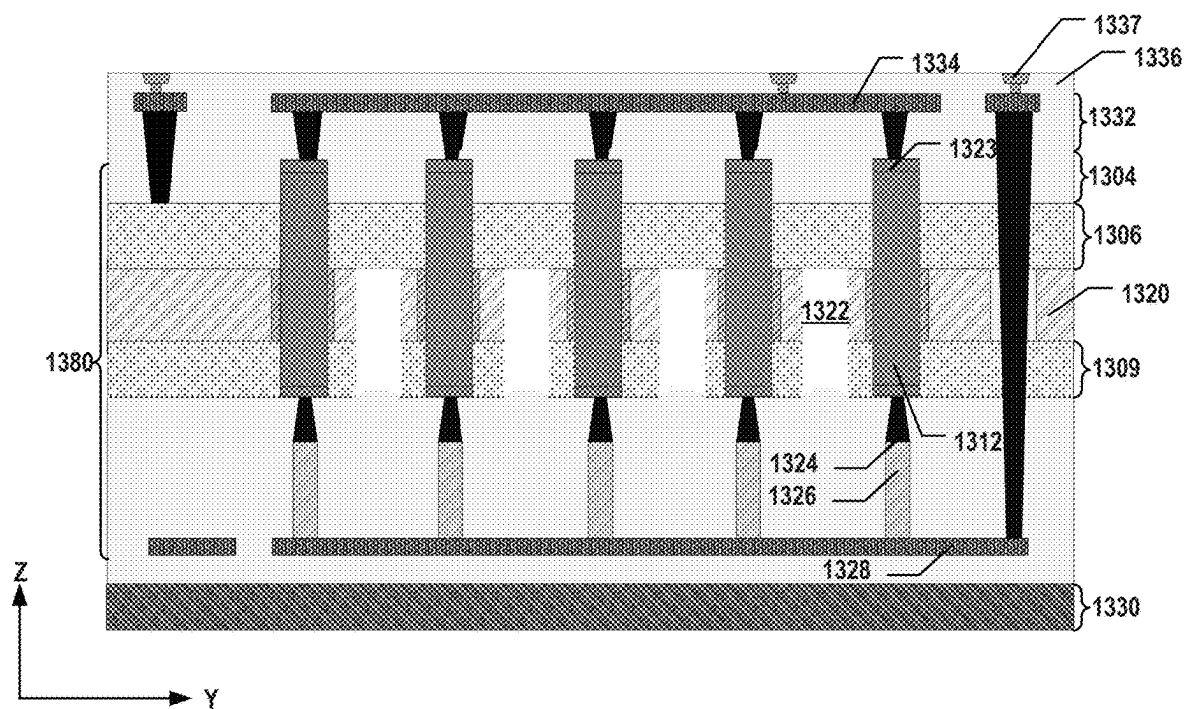


FIG. 13H

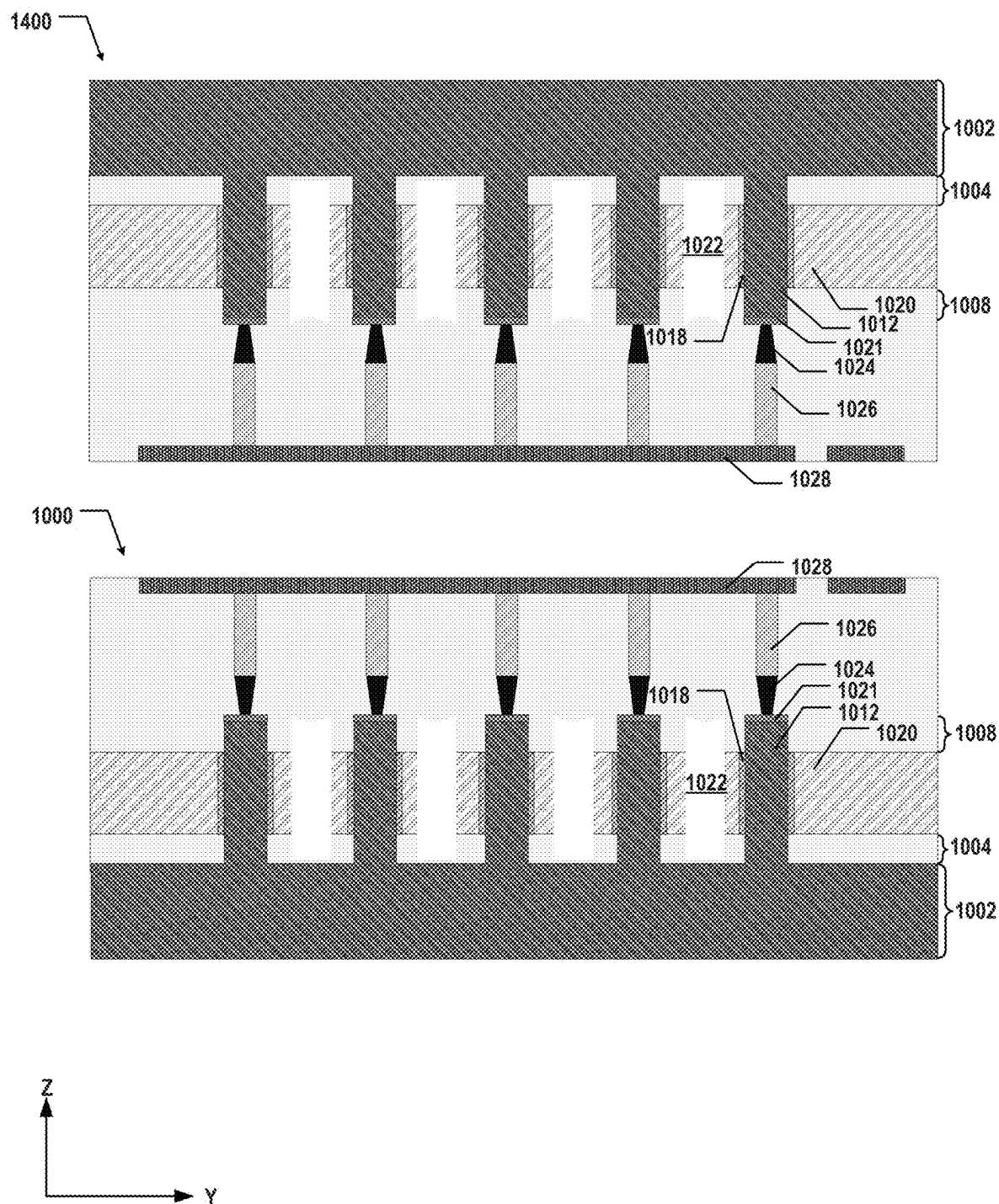


FIG. 14A

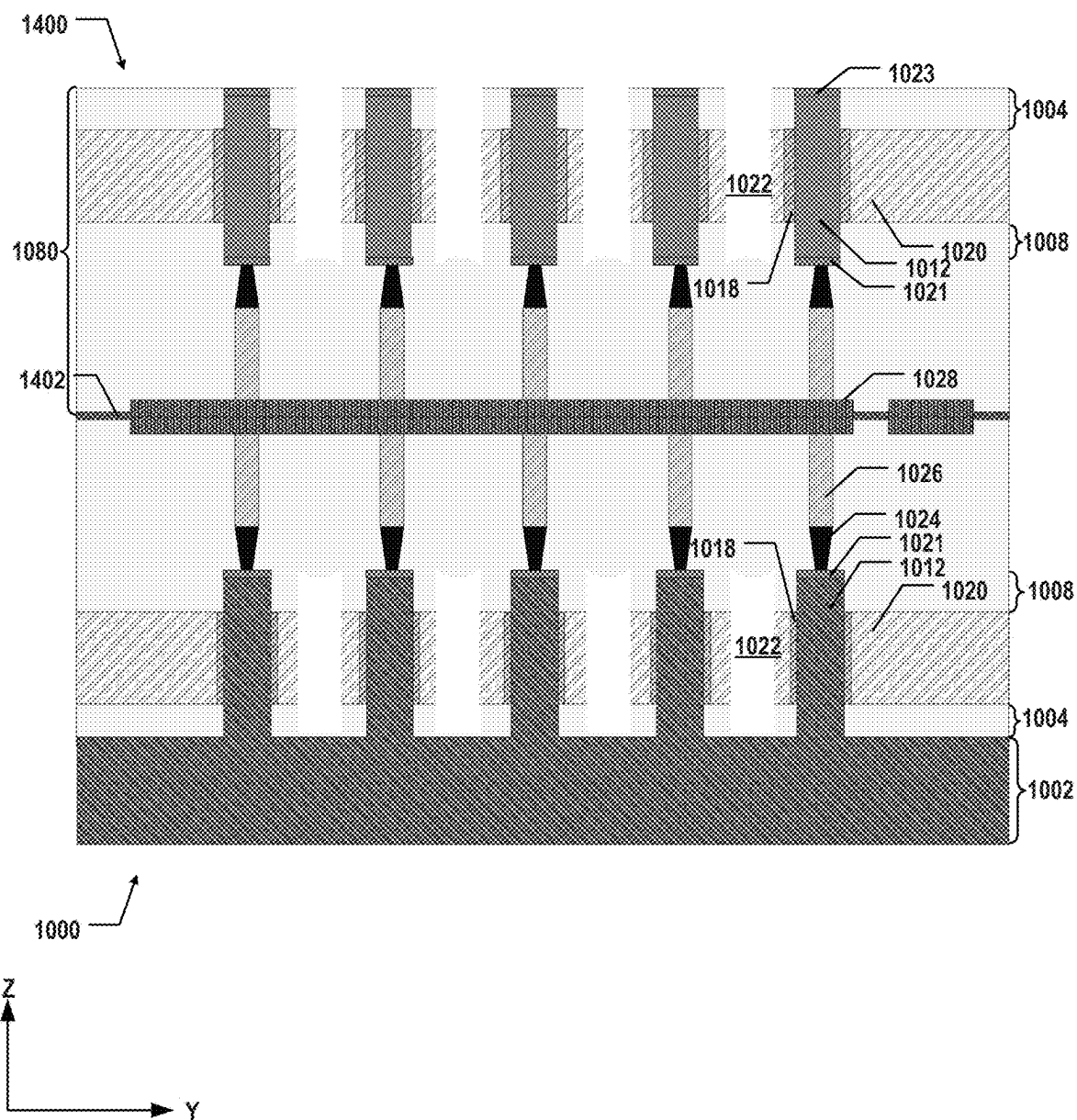


FIG. 14B

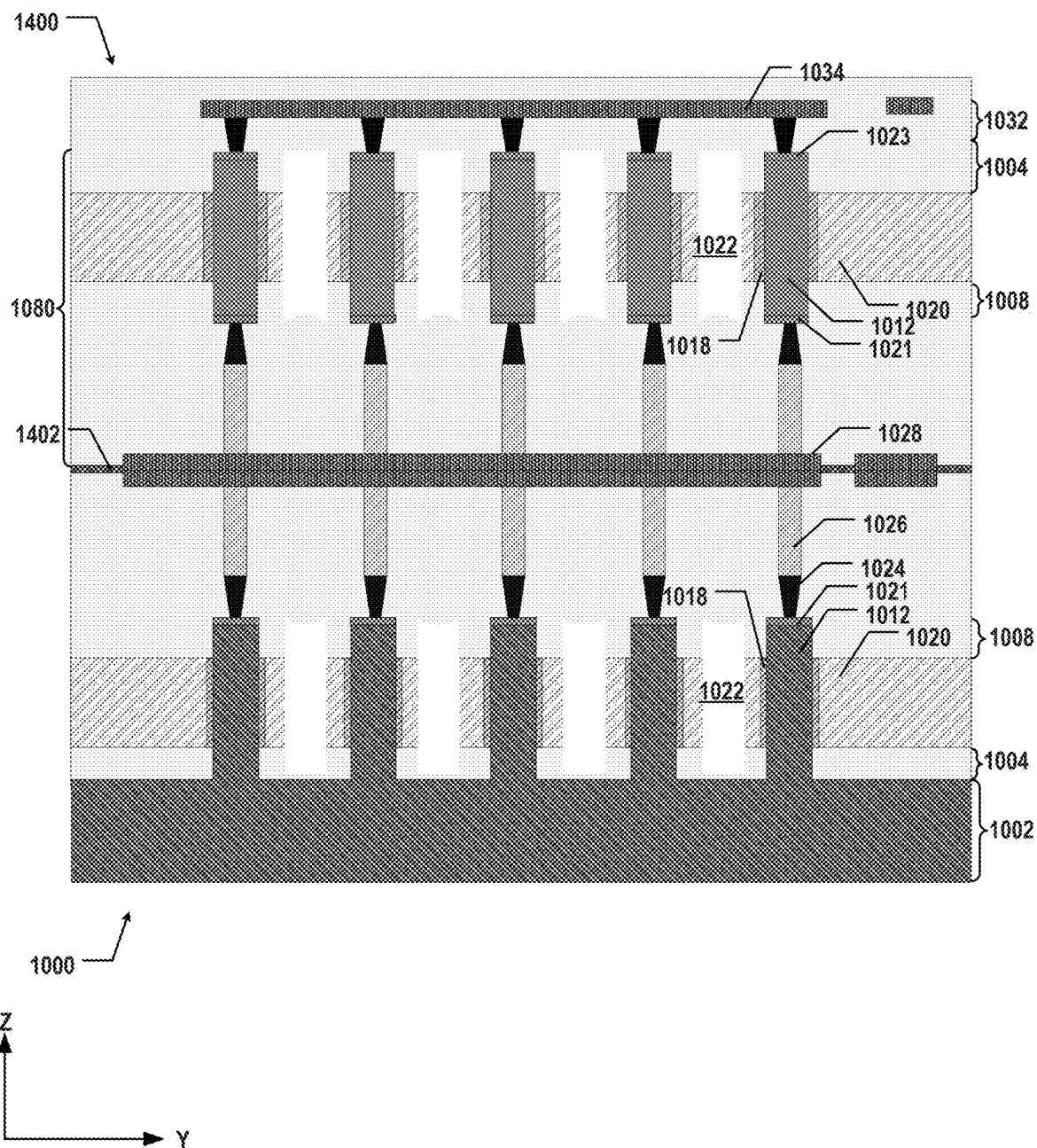


FIG. 14C

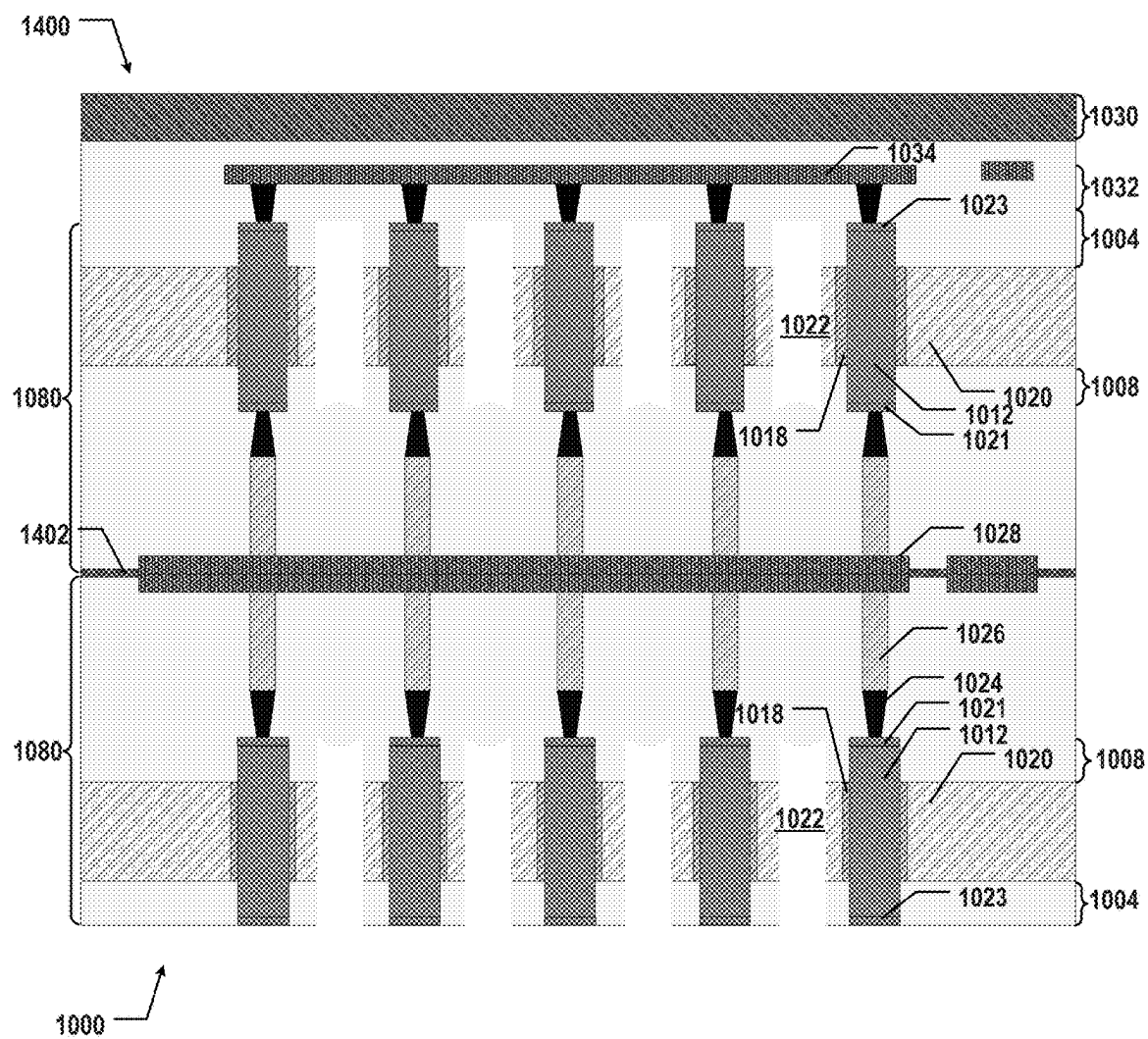


FIG. 14D

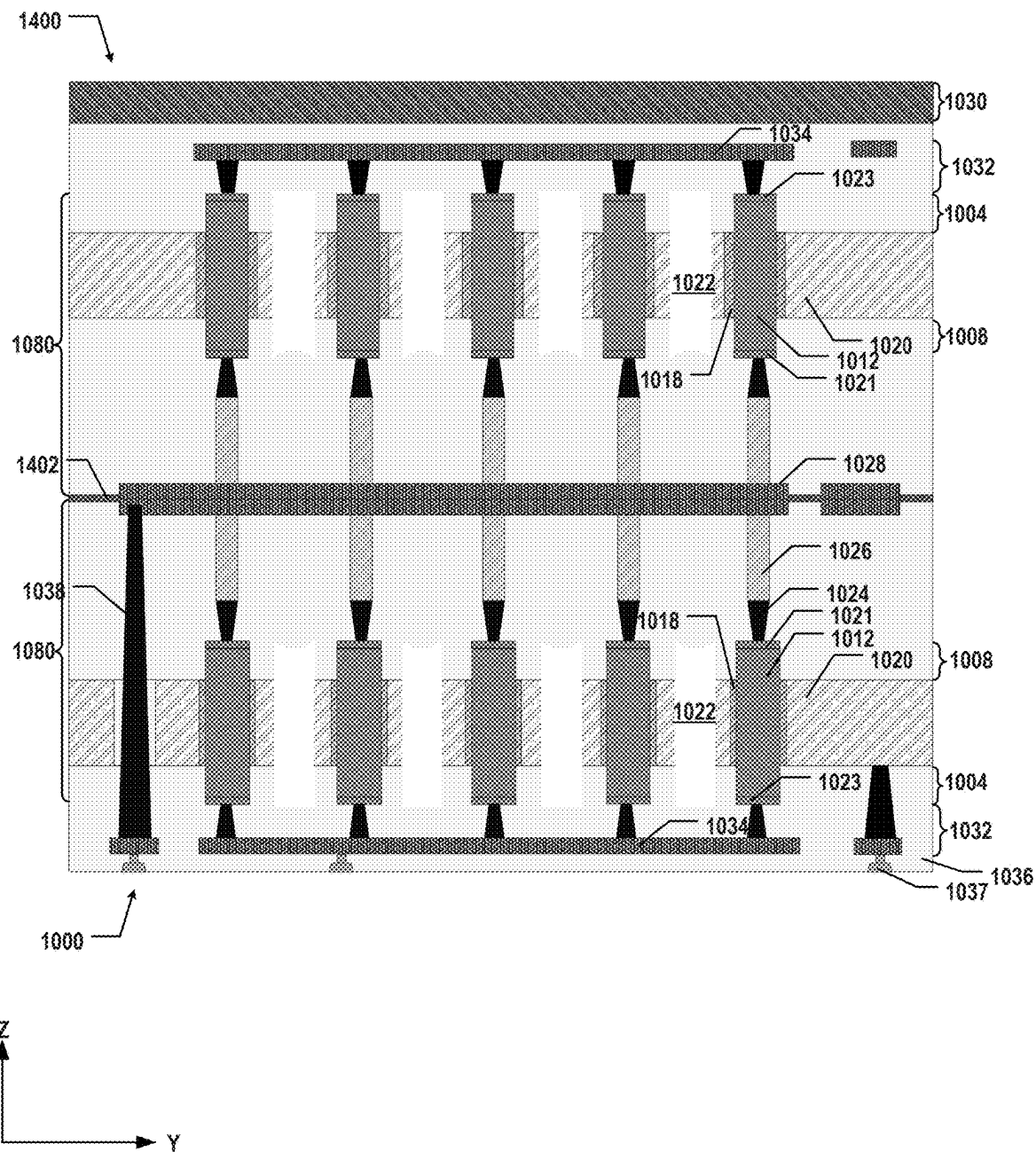


FIG. 14E

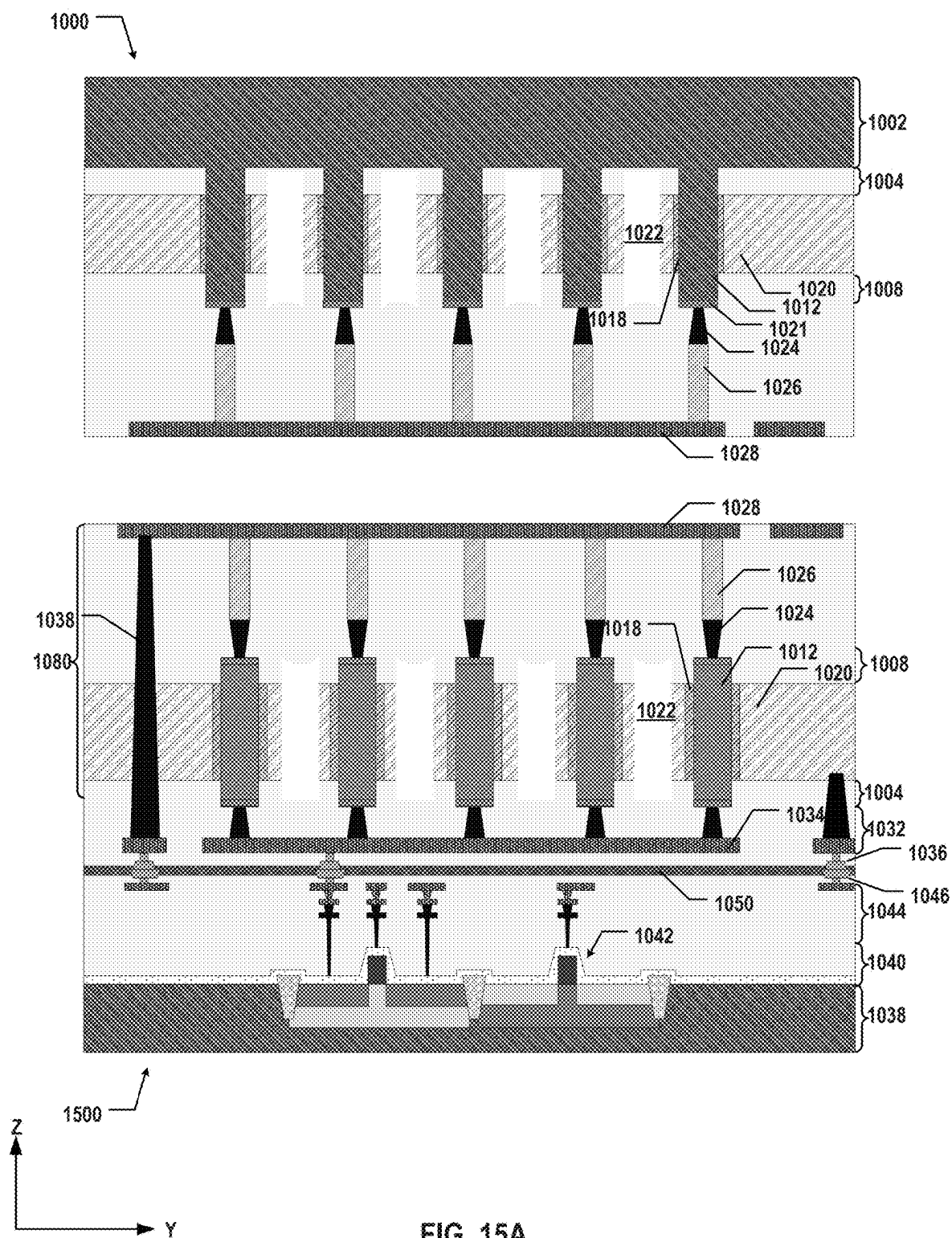


FIG. 15A

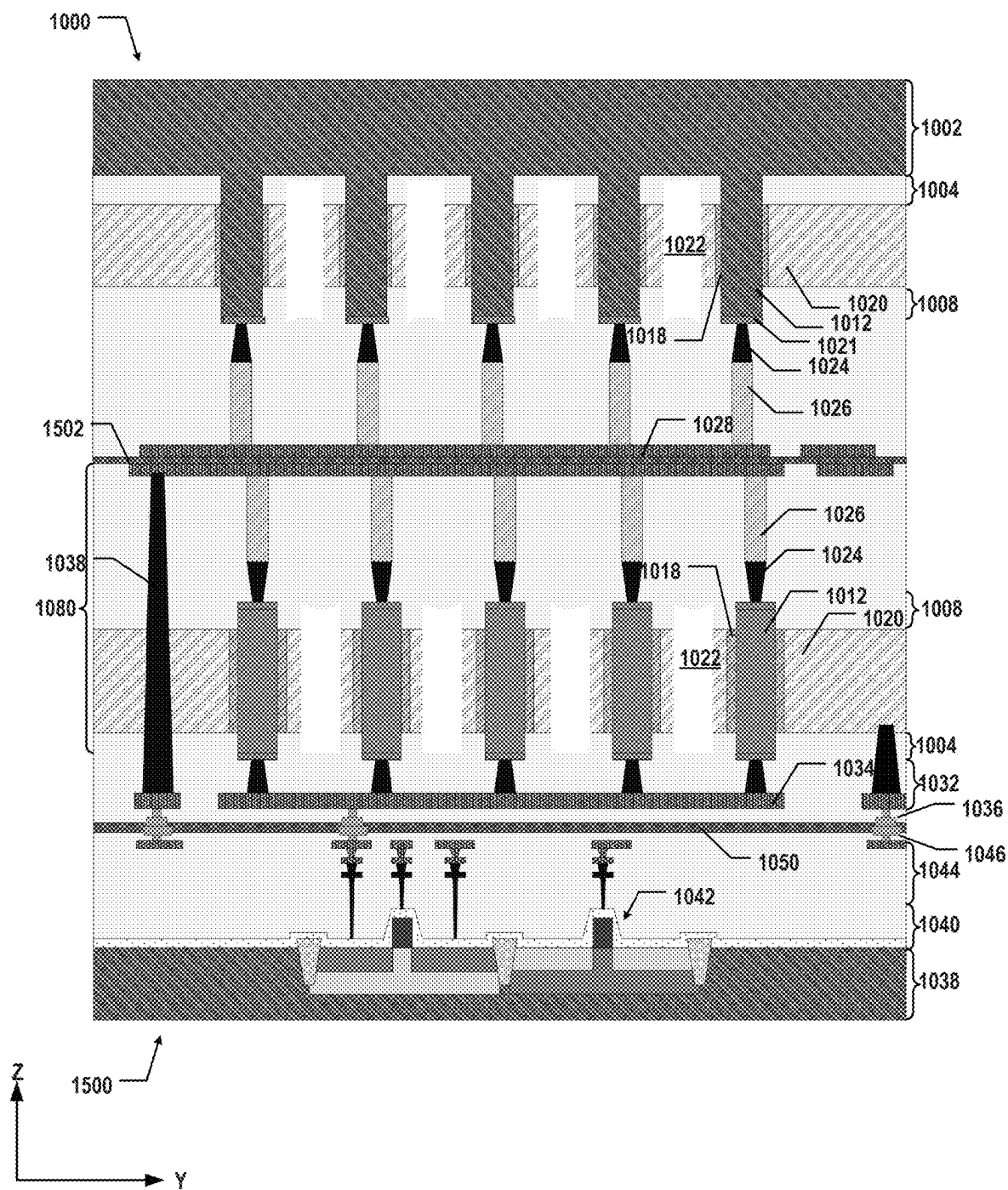


FIG. 15B

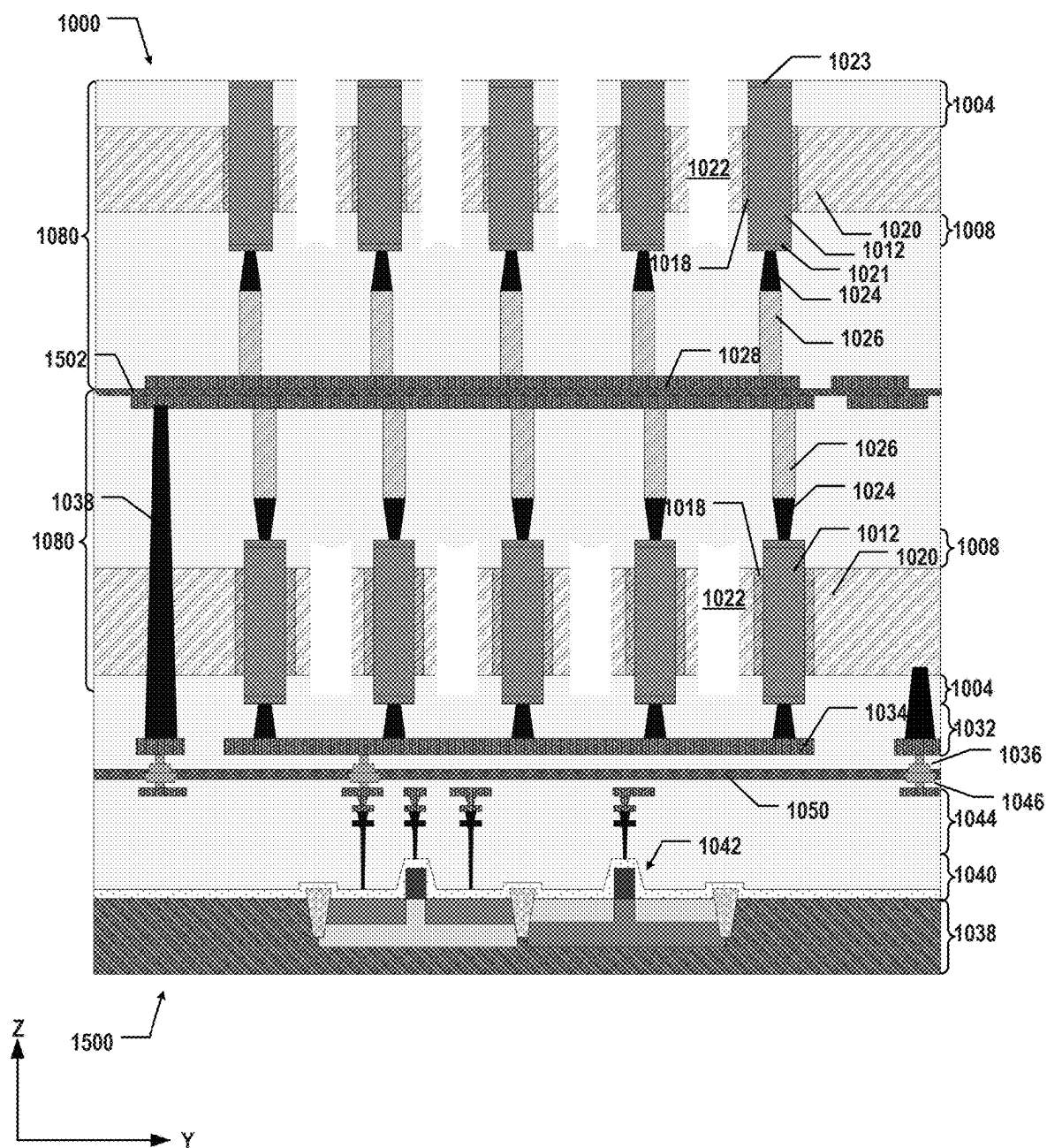


FIG. 15C

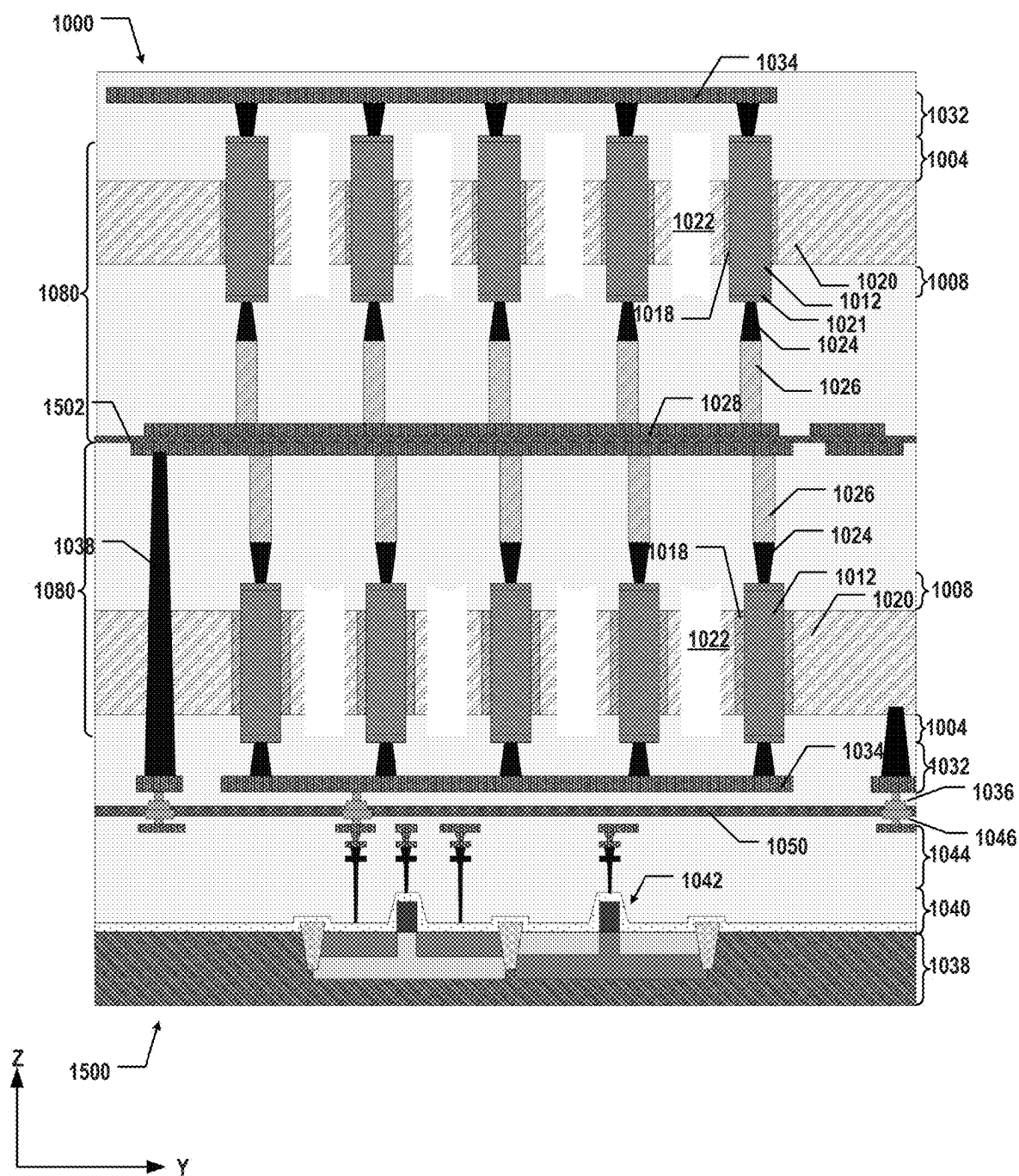


FIG. 15D

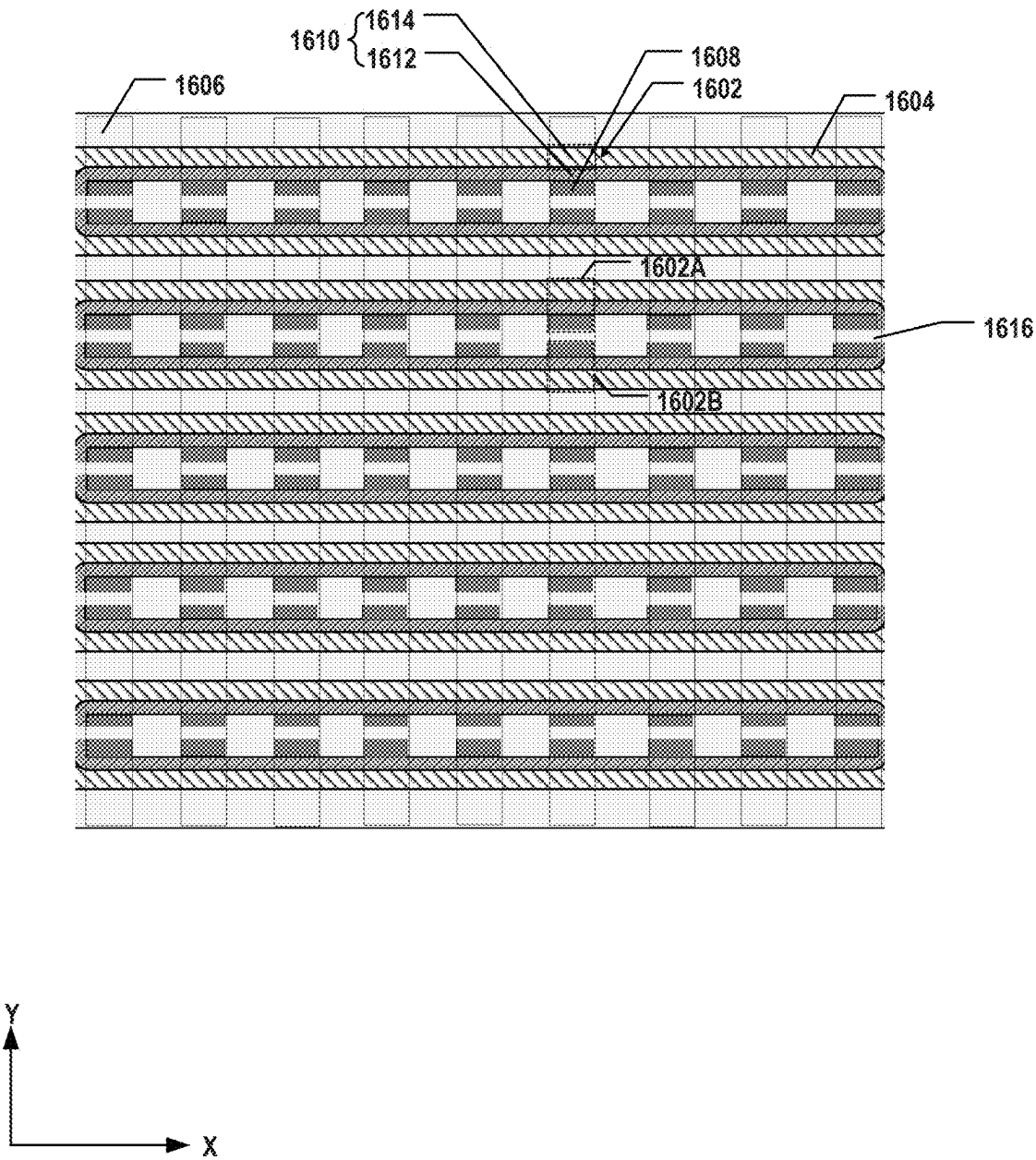


FIG. 16

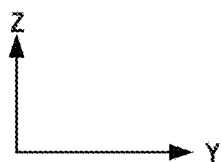
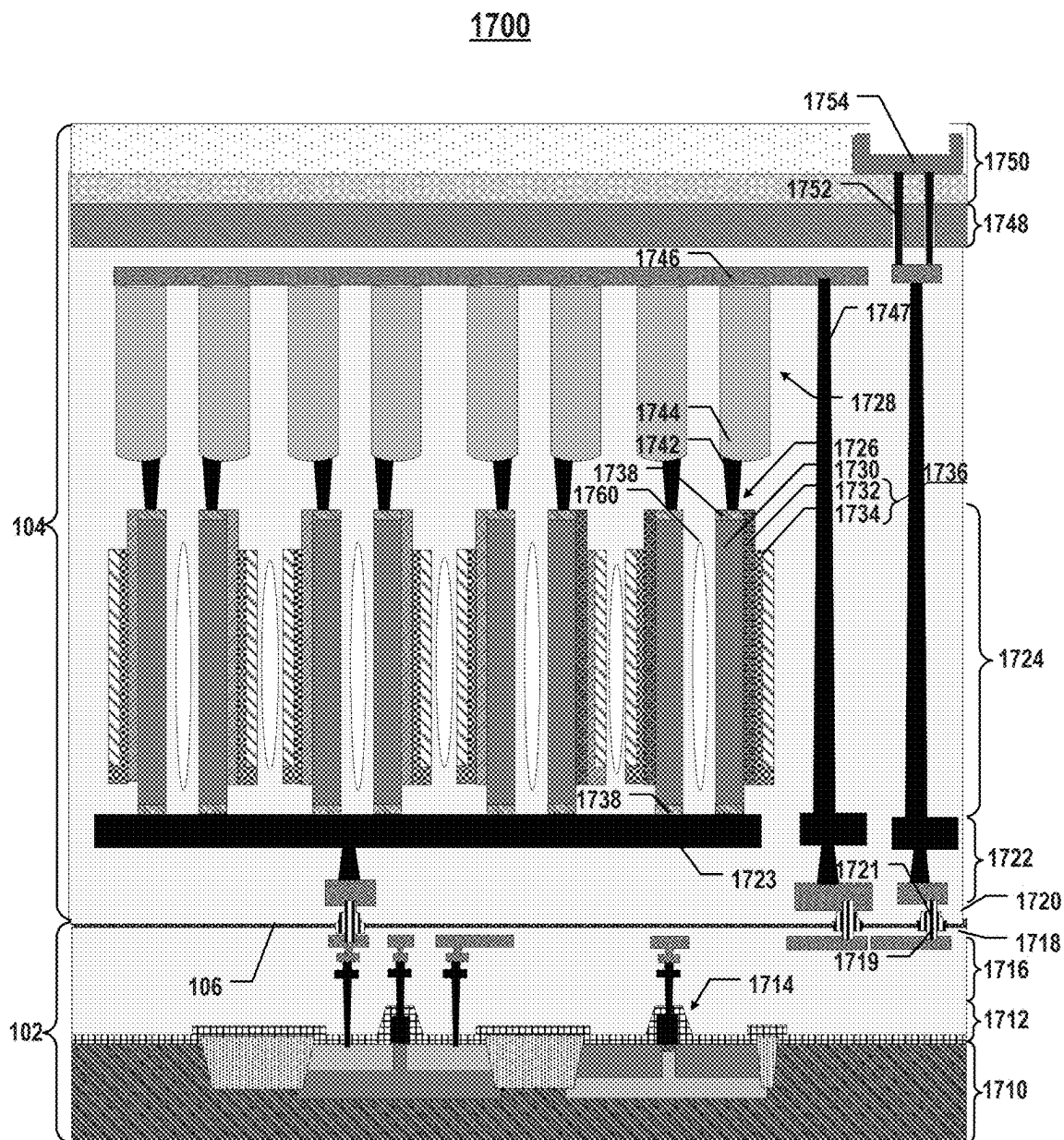


FIG. 17

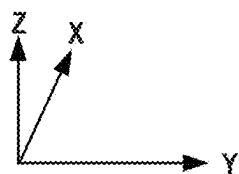
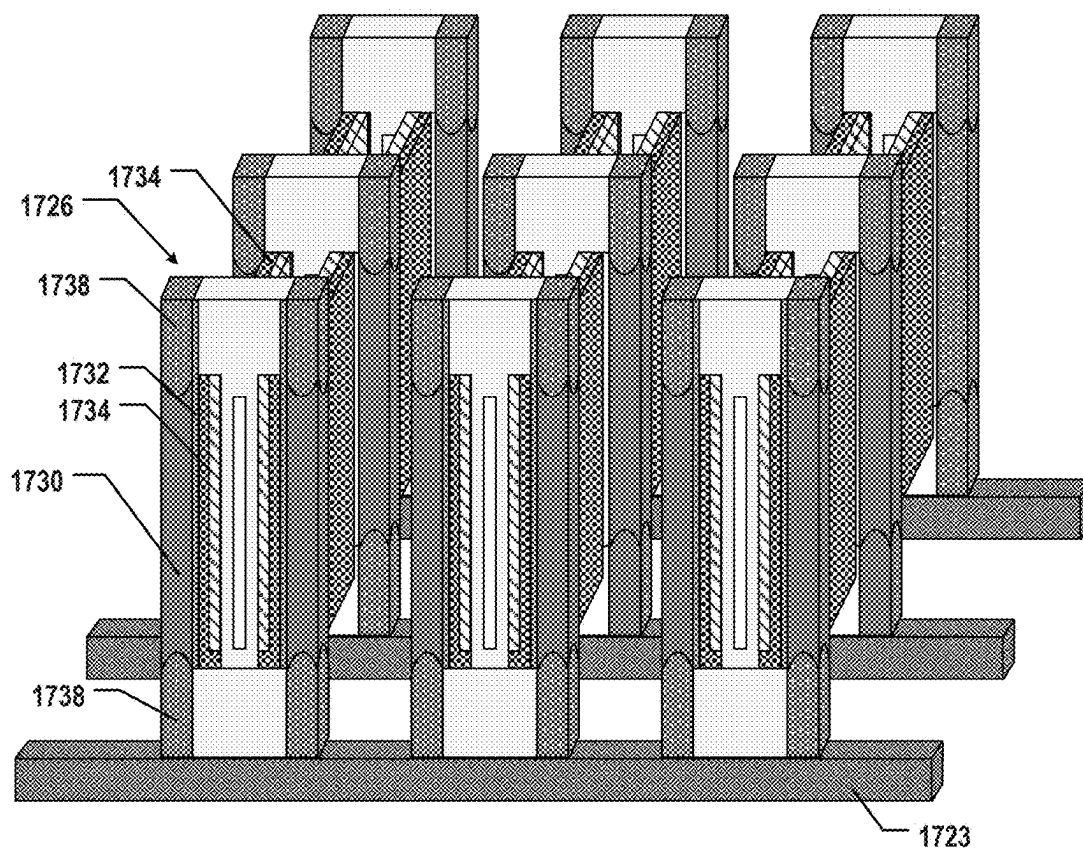


FIG. 18

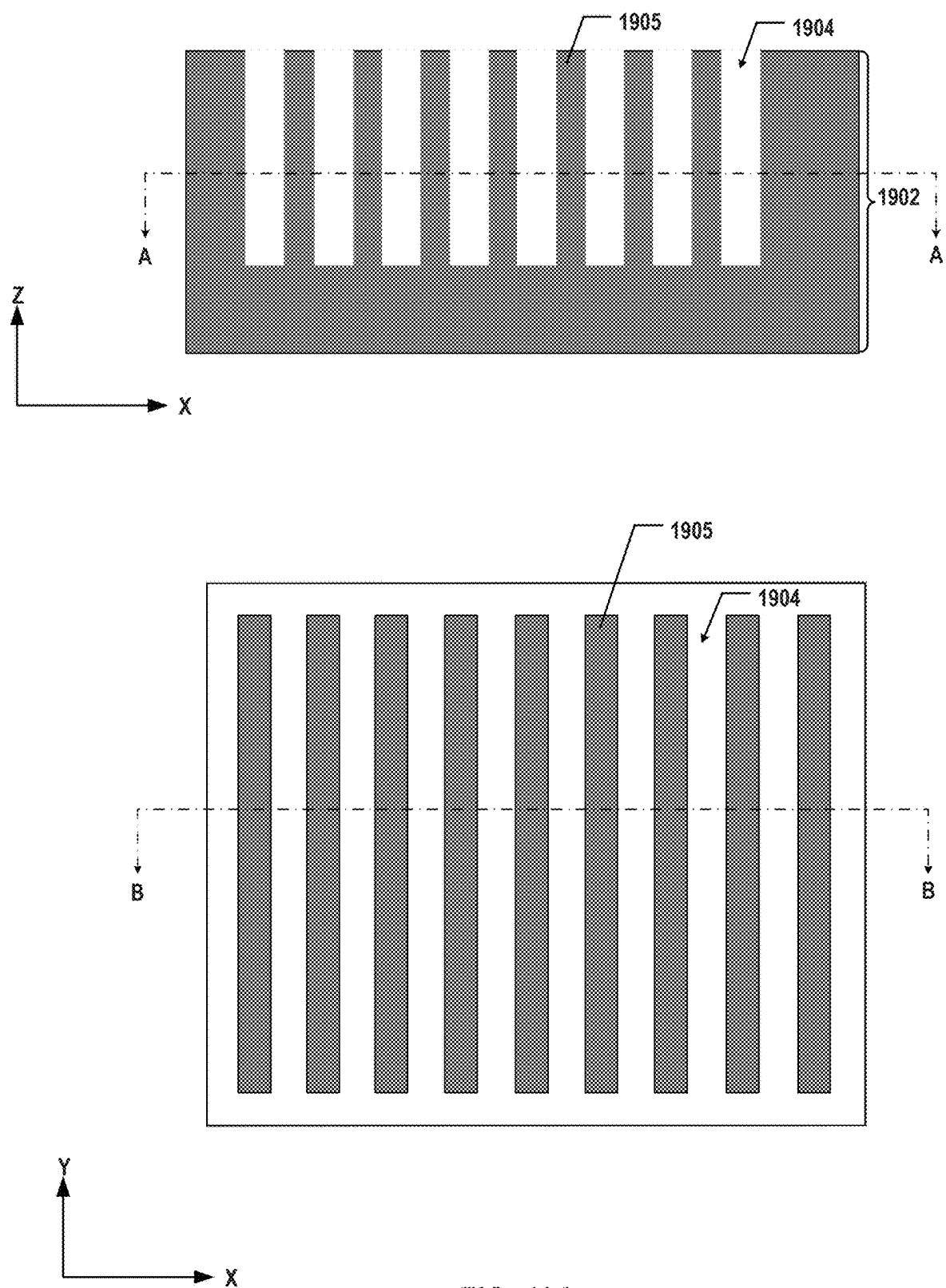


FIG. 19A

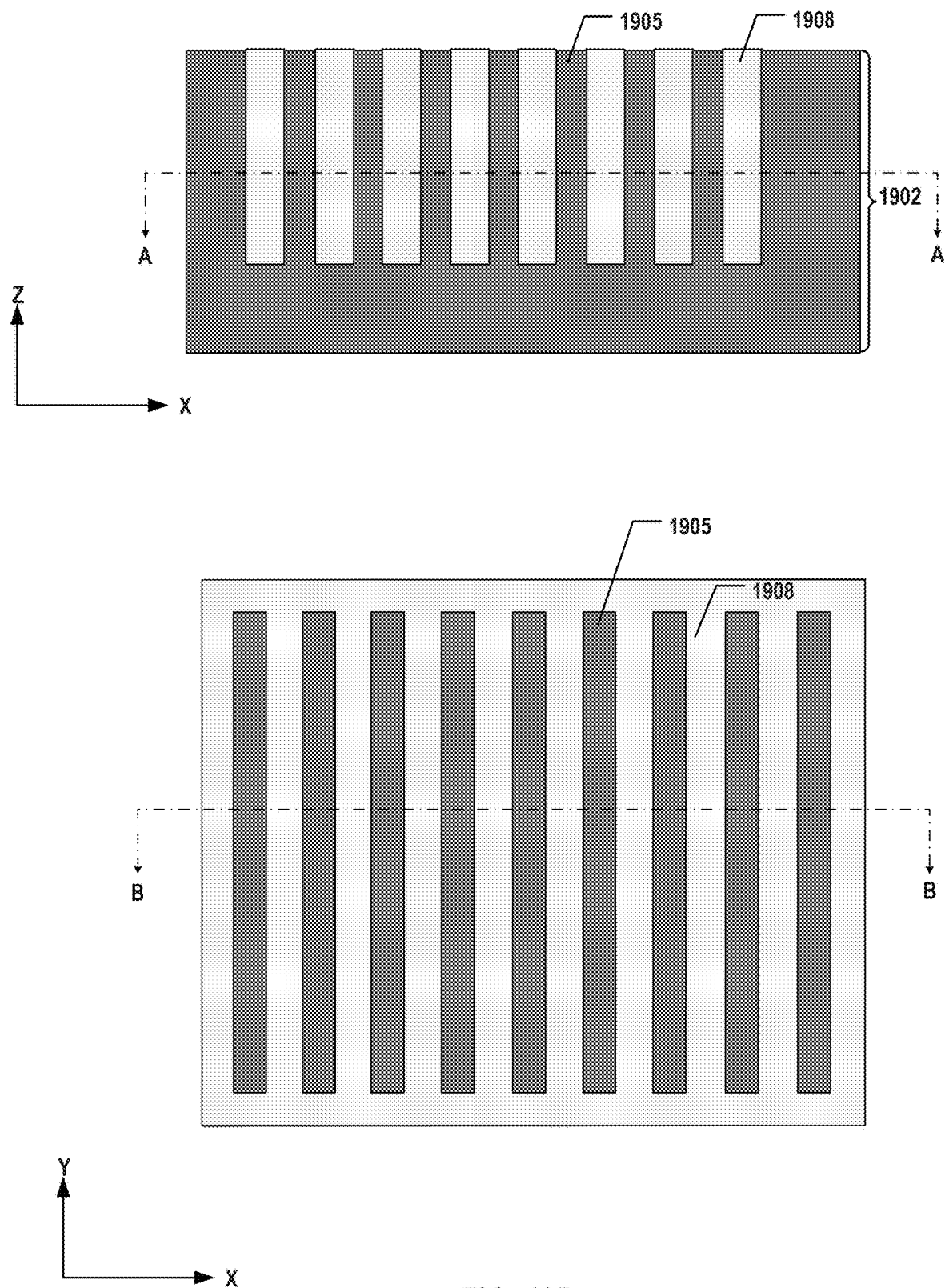


FIG. 19B

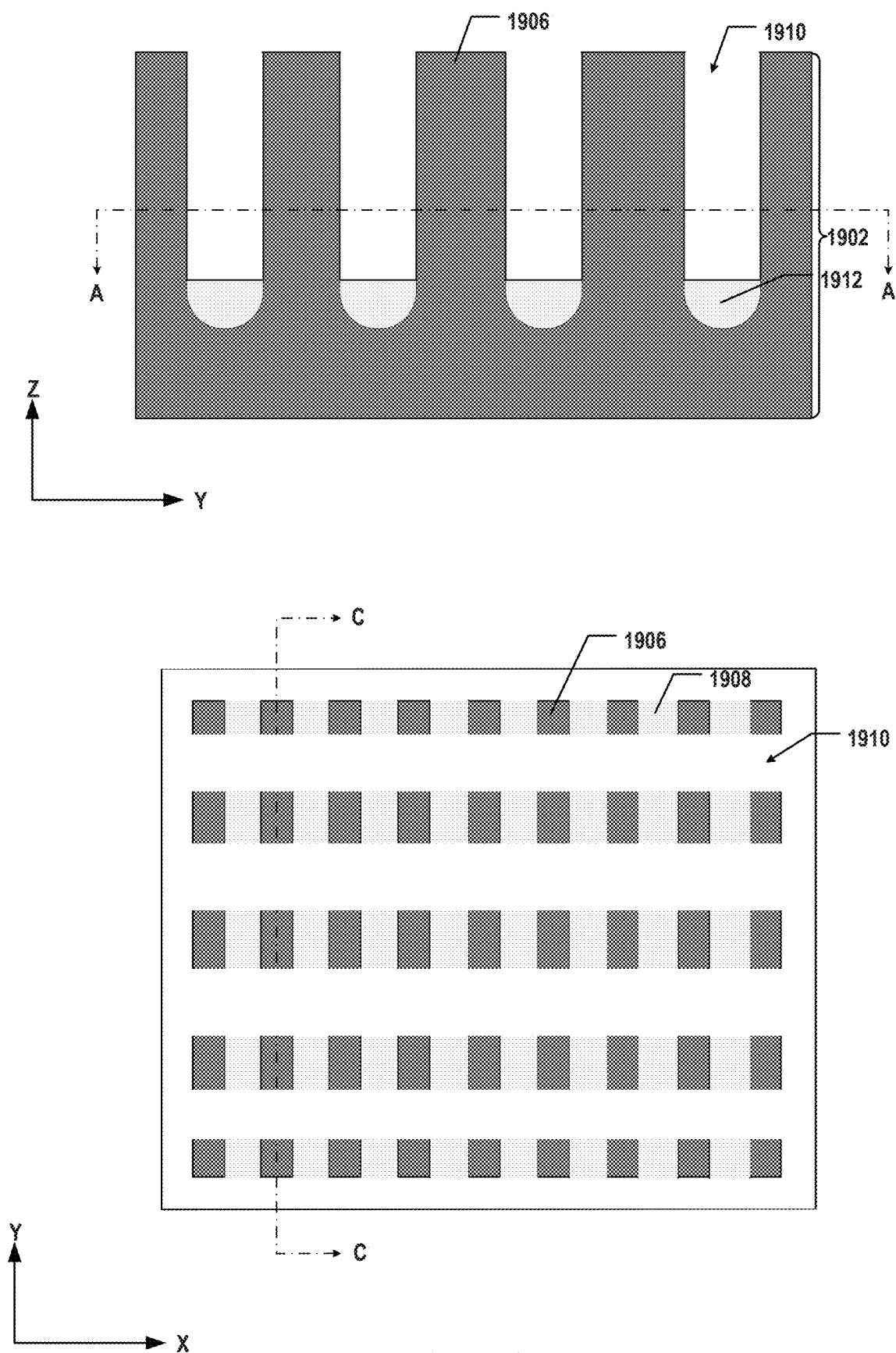


FIG. 19C

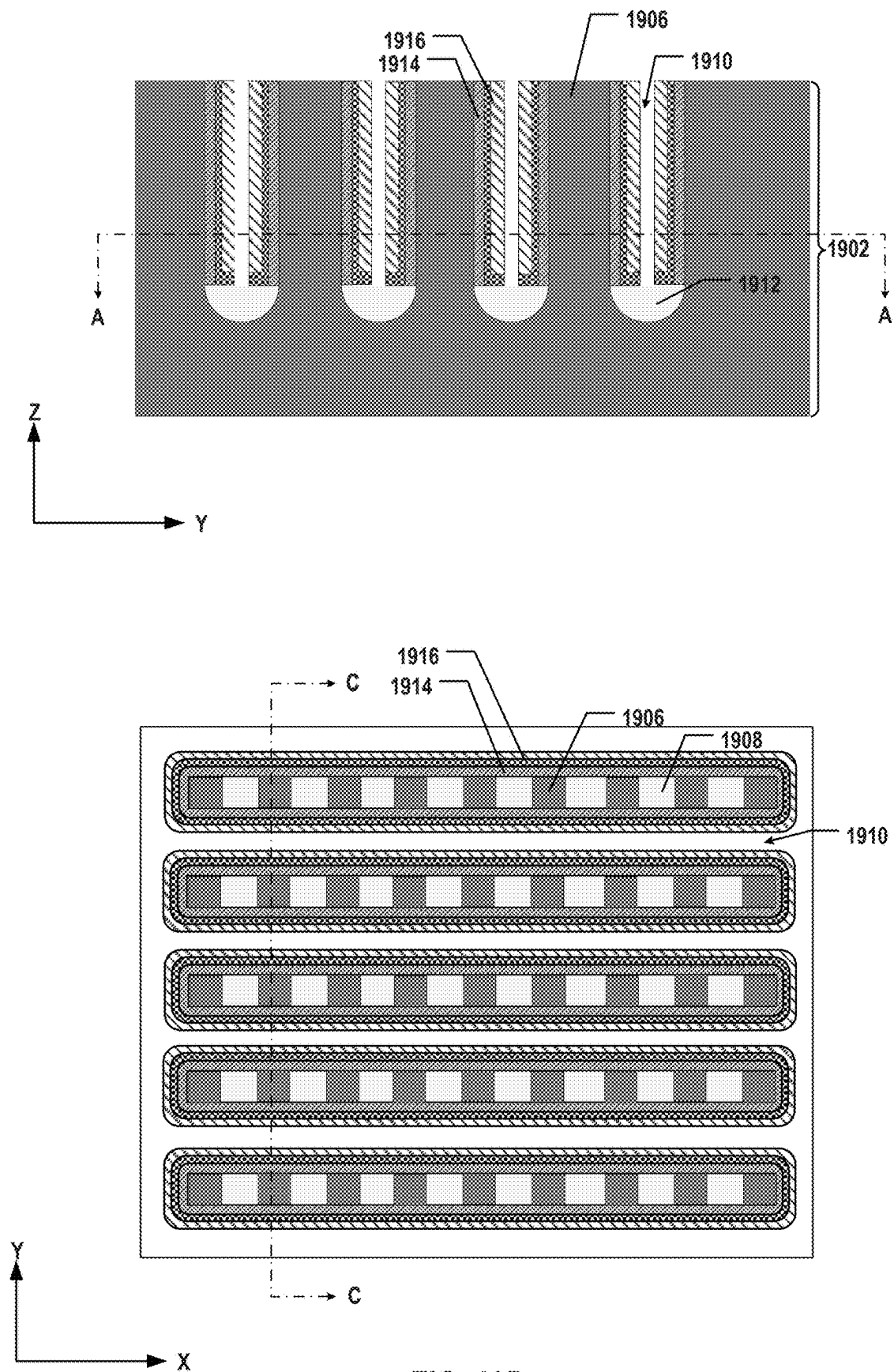


FIG. 19D

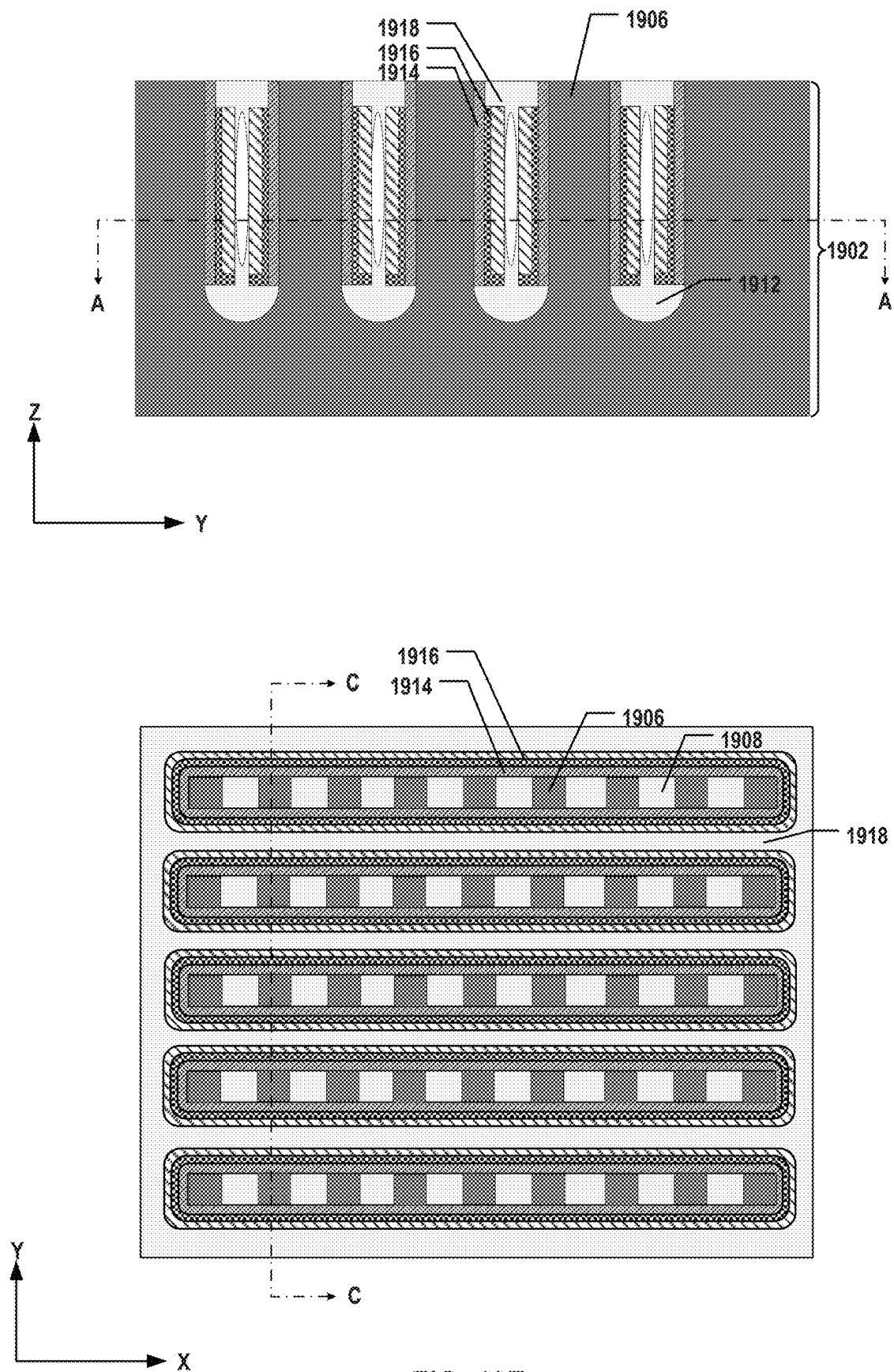


FIG. 19E

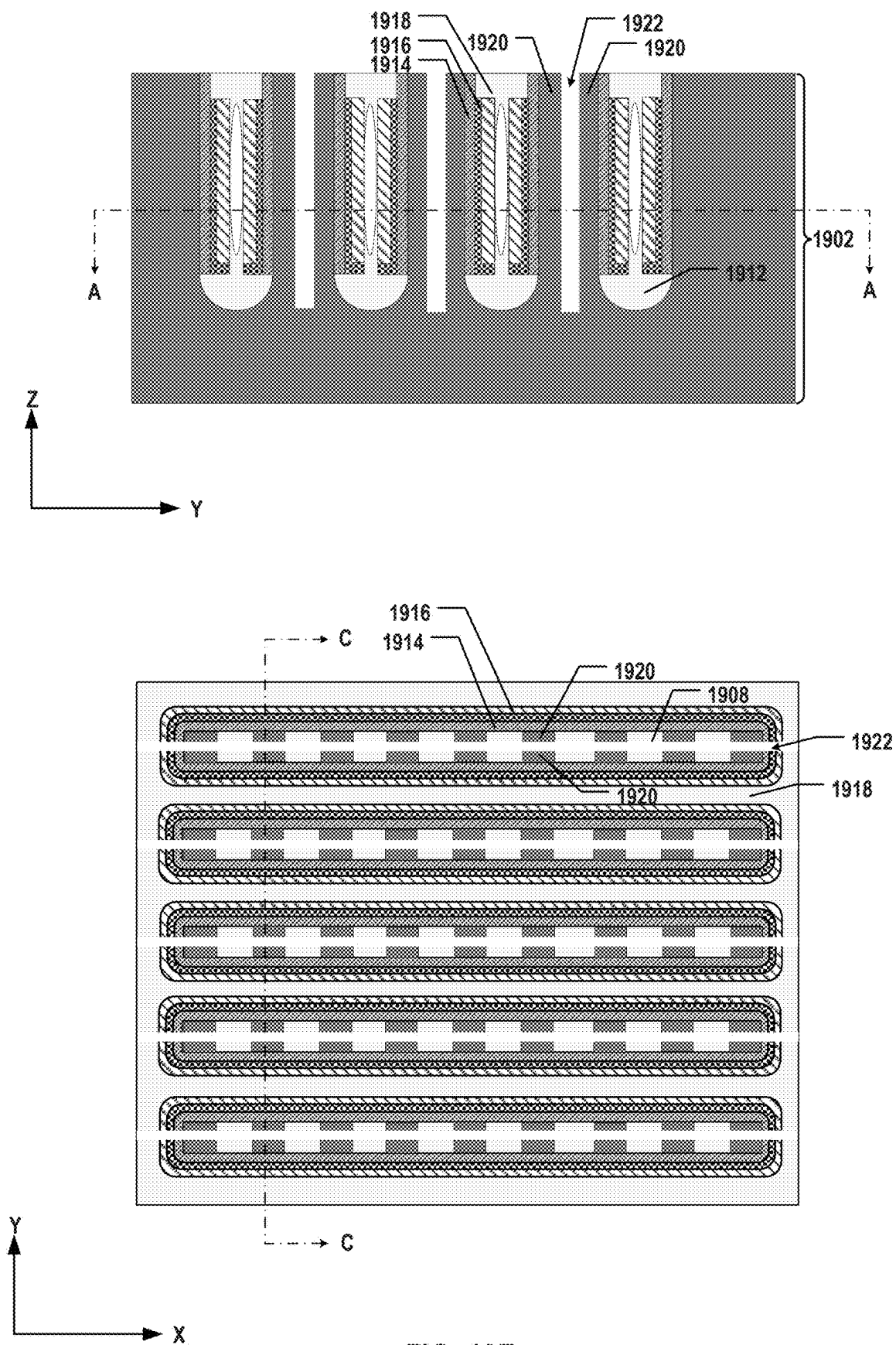


FIG. 19F

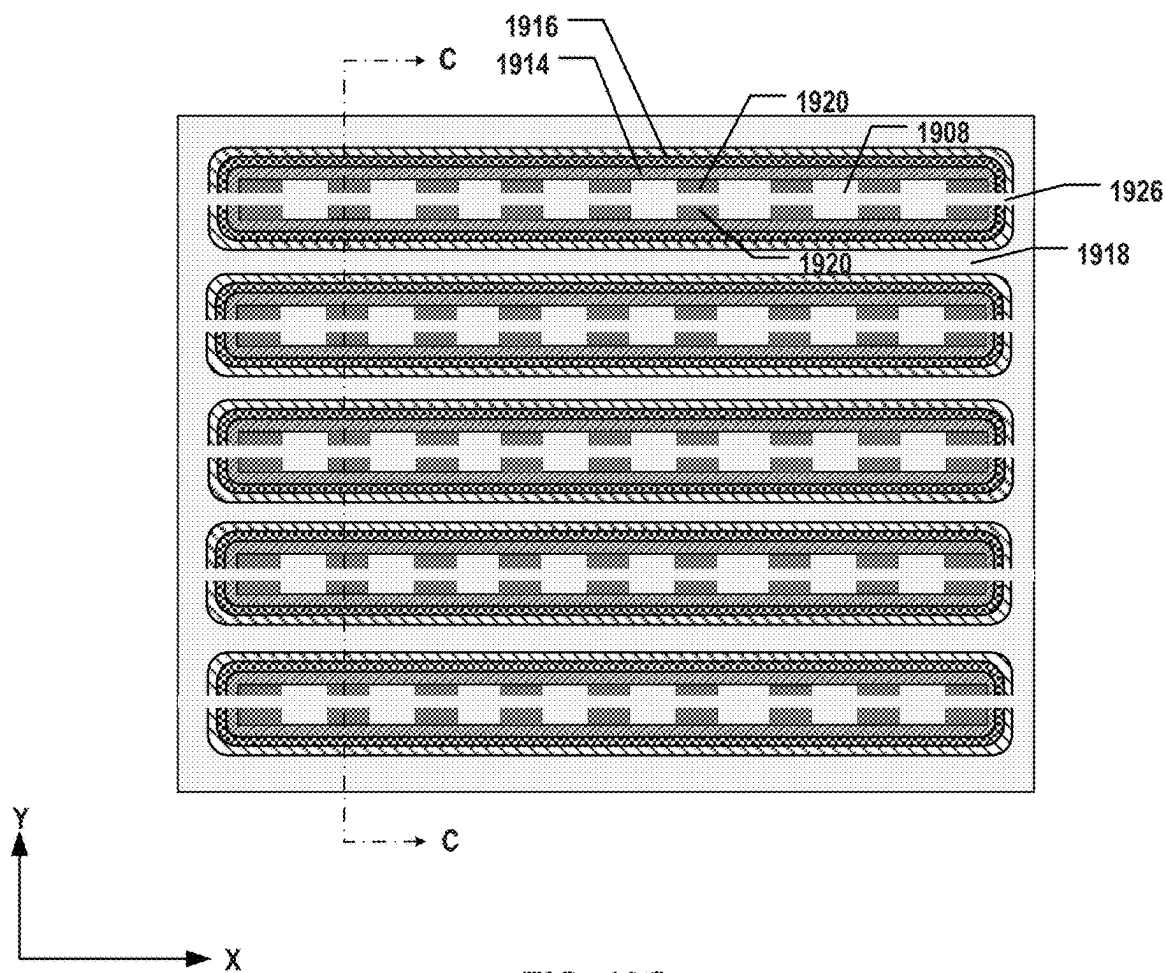
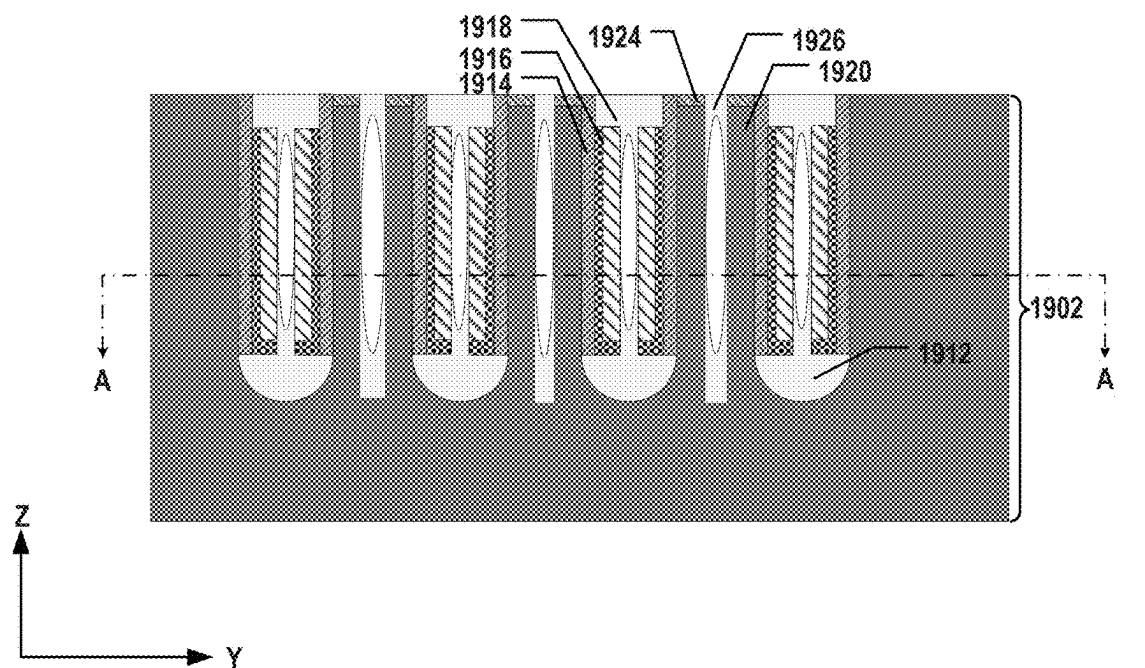


FIG. 19G

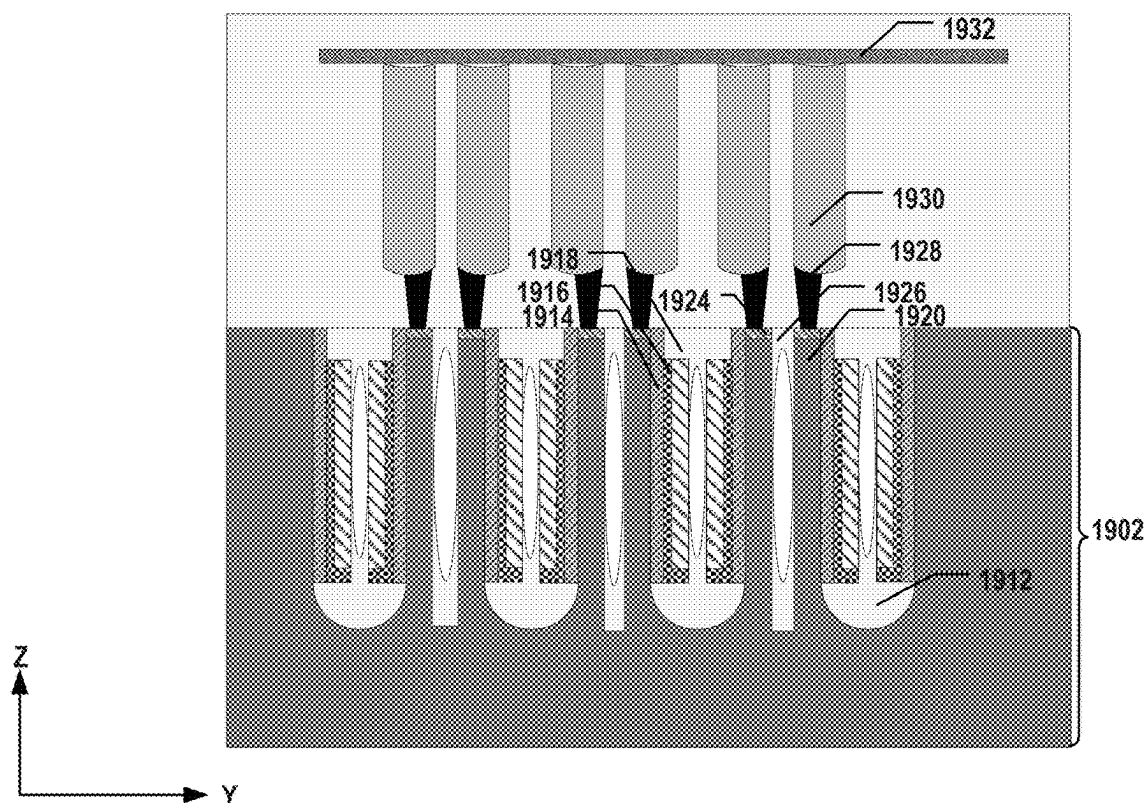


FIG. 19H

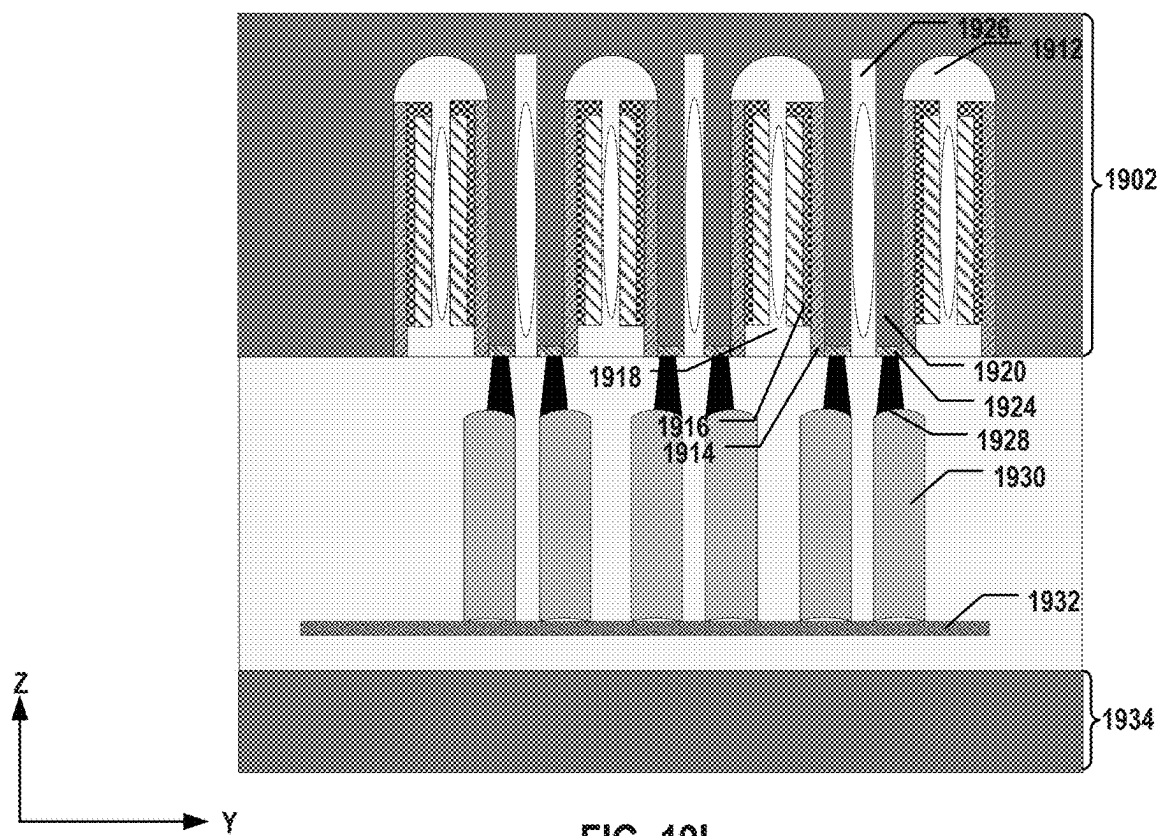
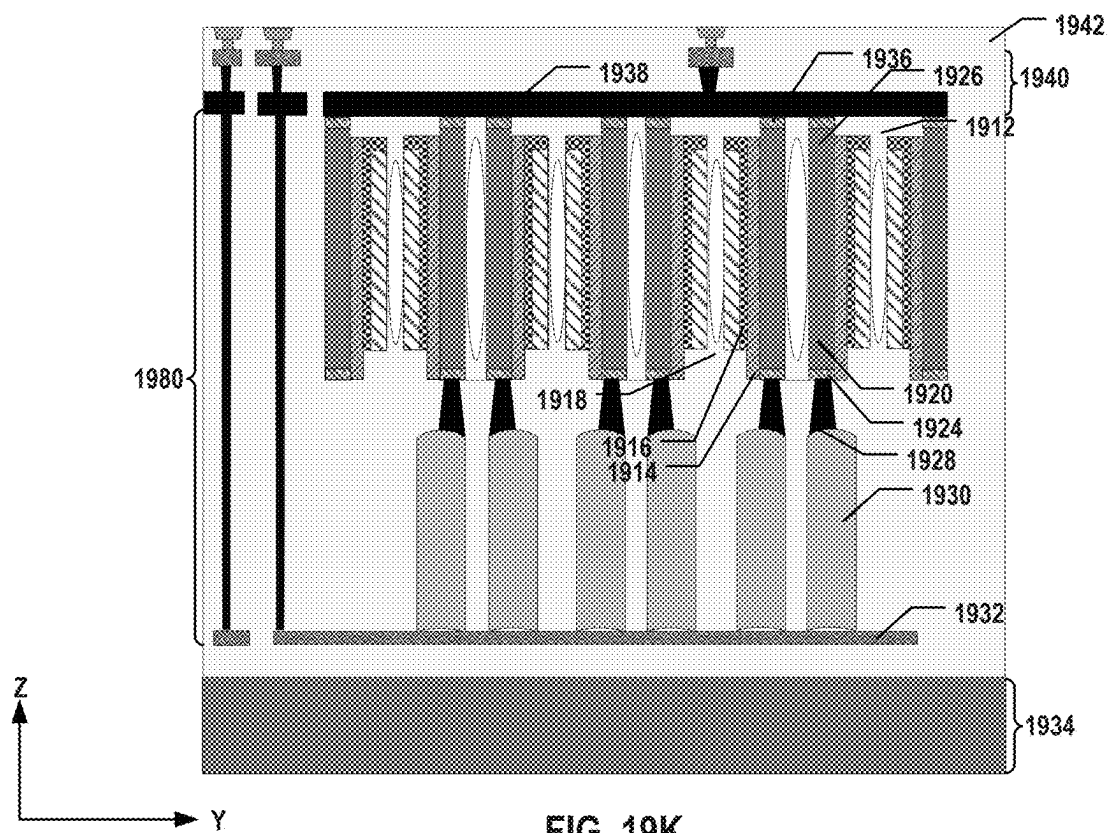
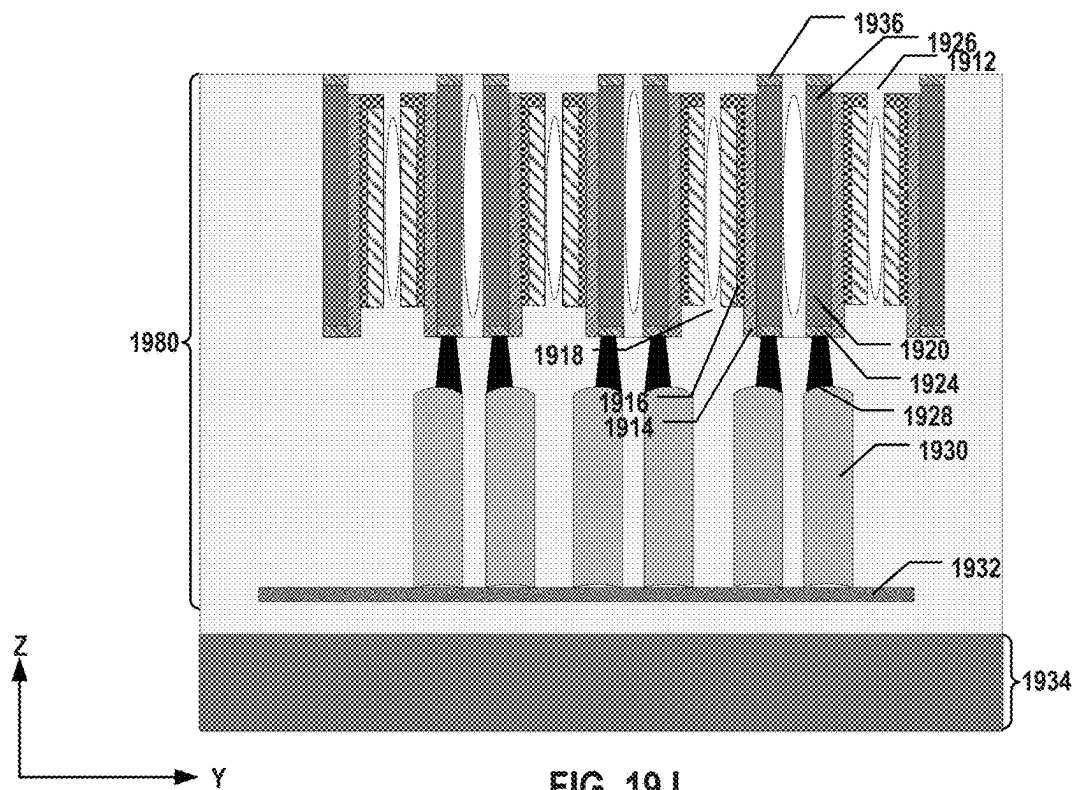


FIG. 19I



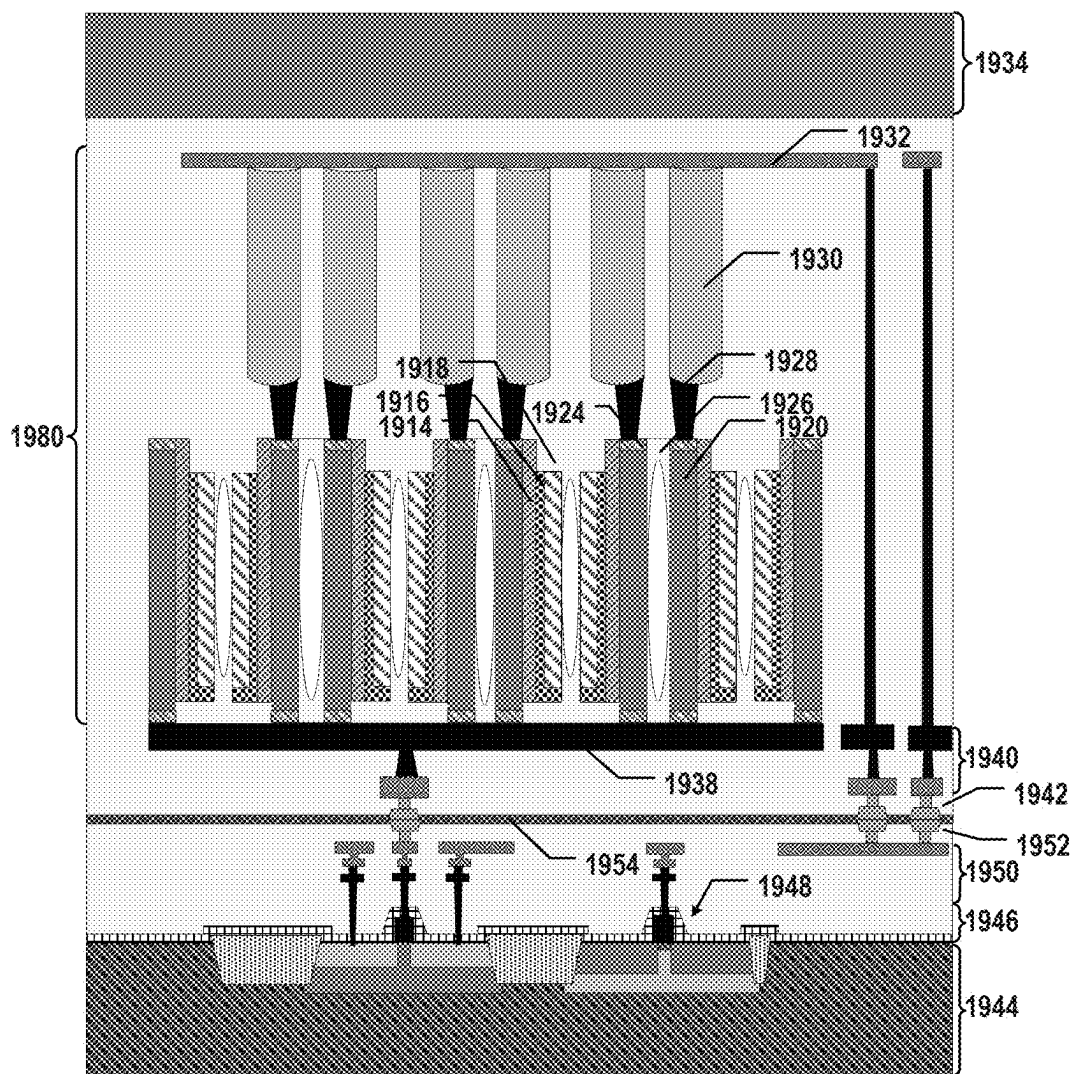


FIG. 19L

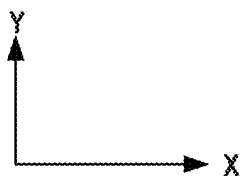
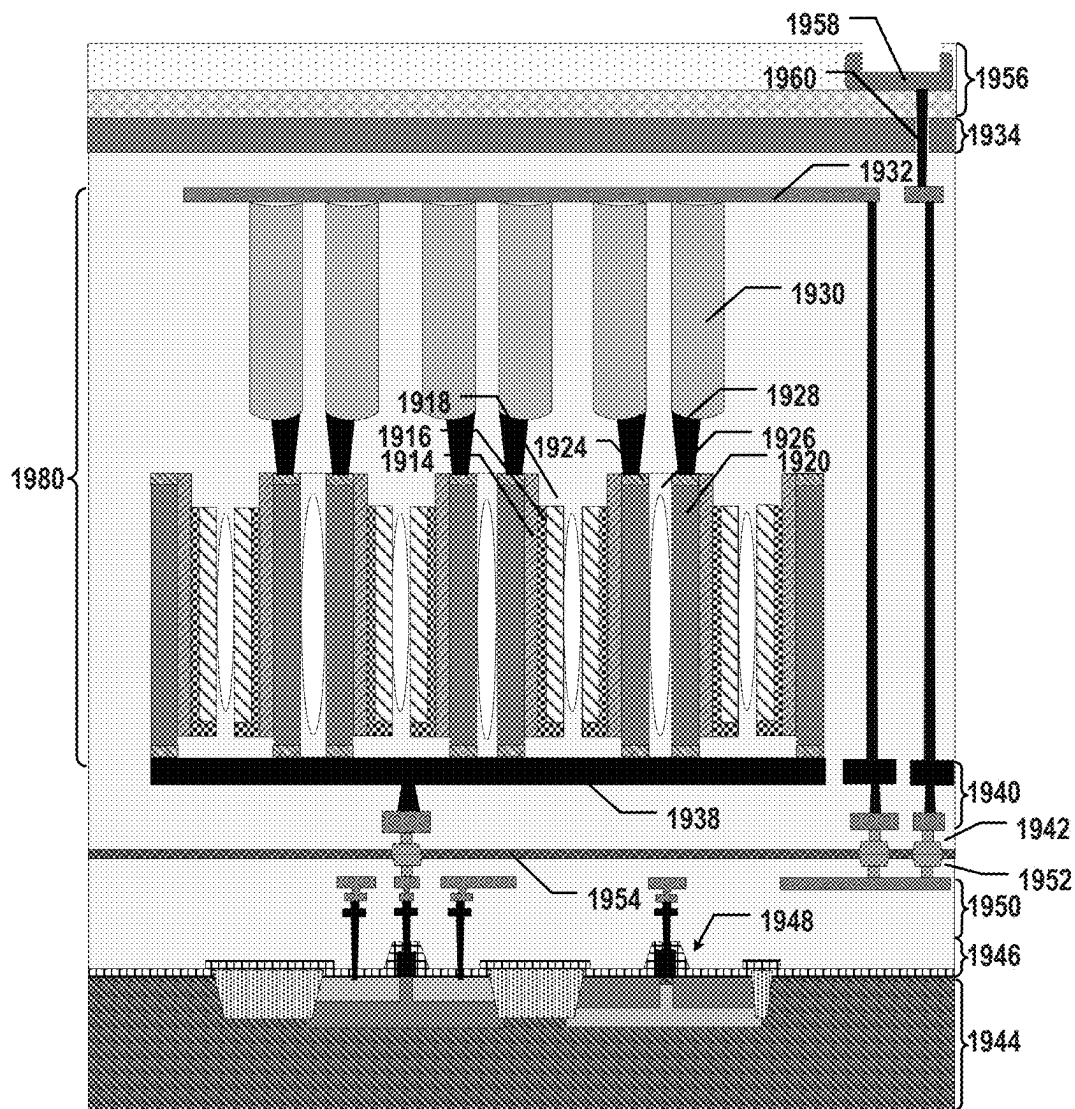


FIG. 19M

2000

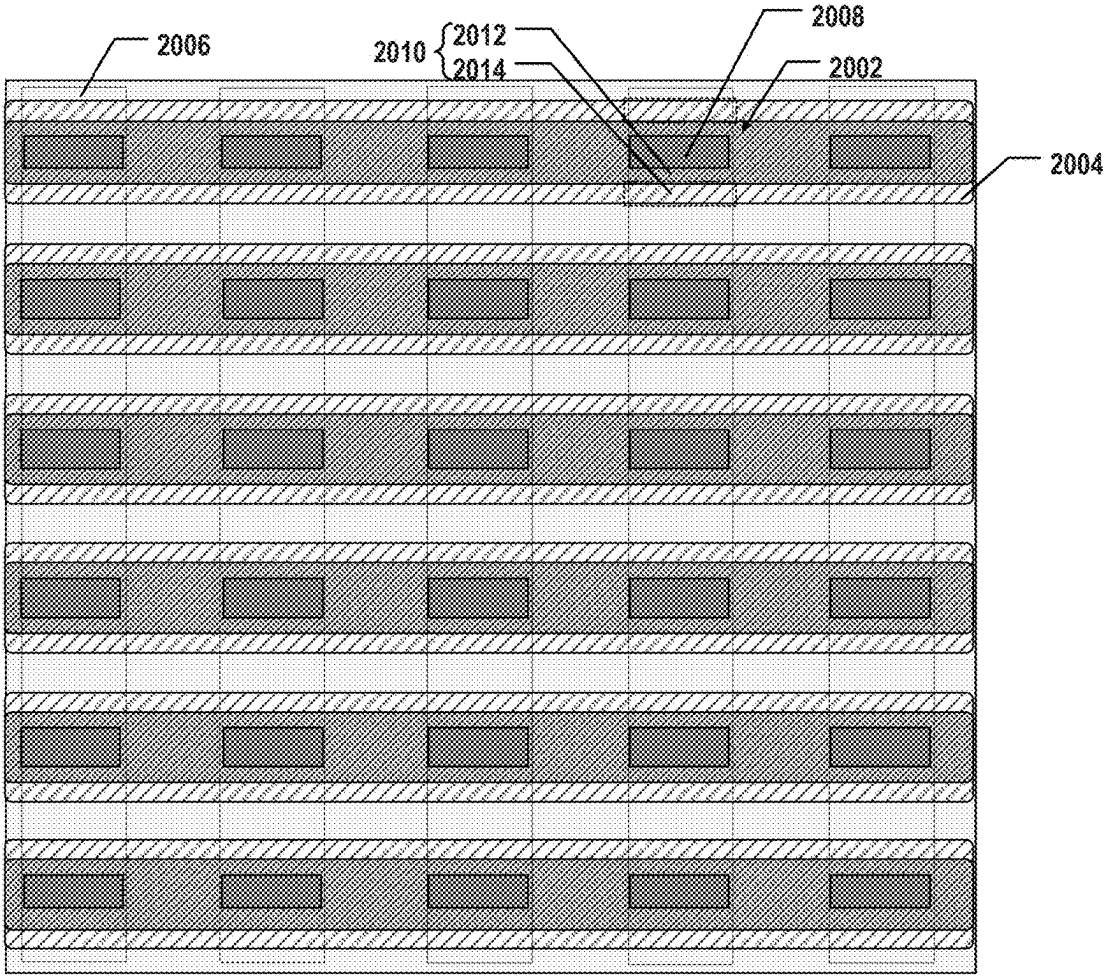


FIG. 20

2100

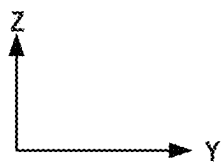
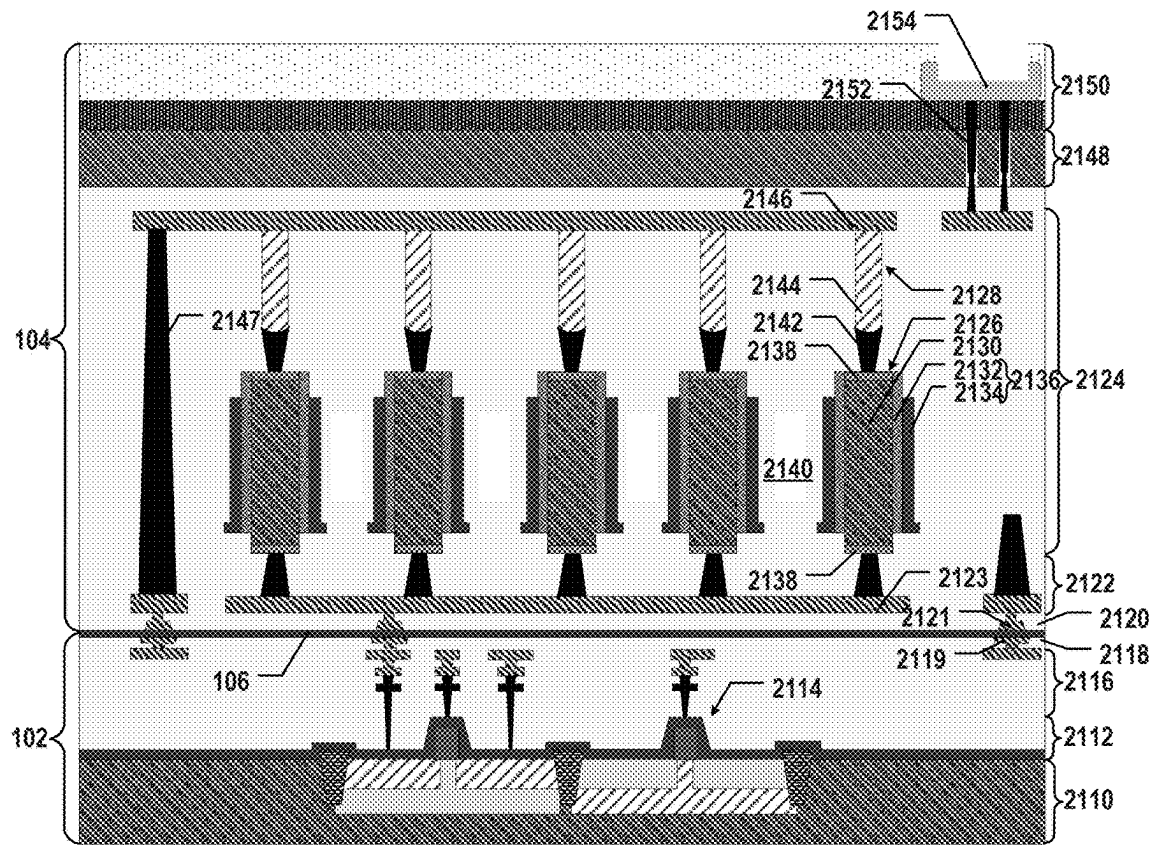


FIG. 21

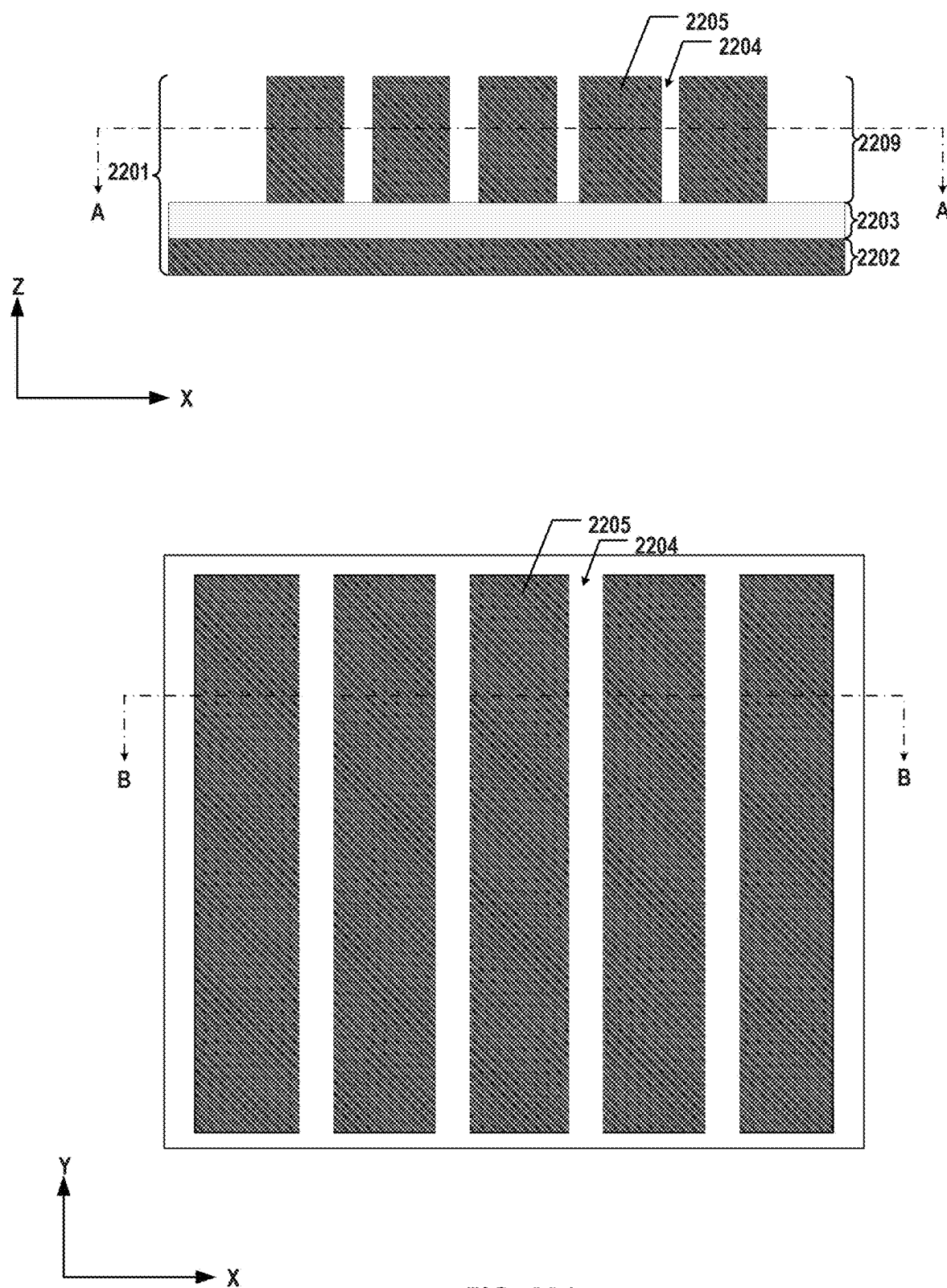


FIG. 22A

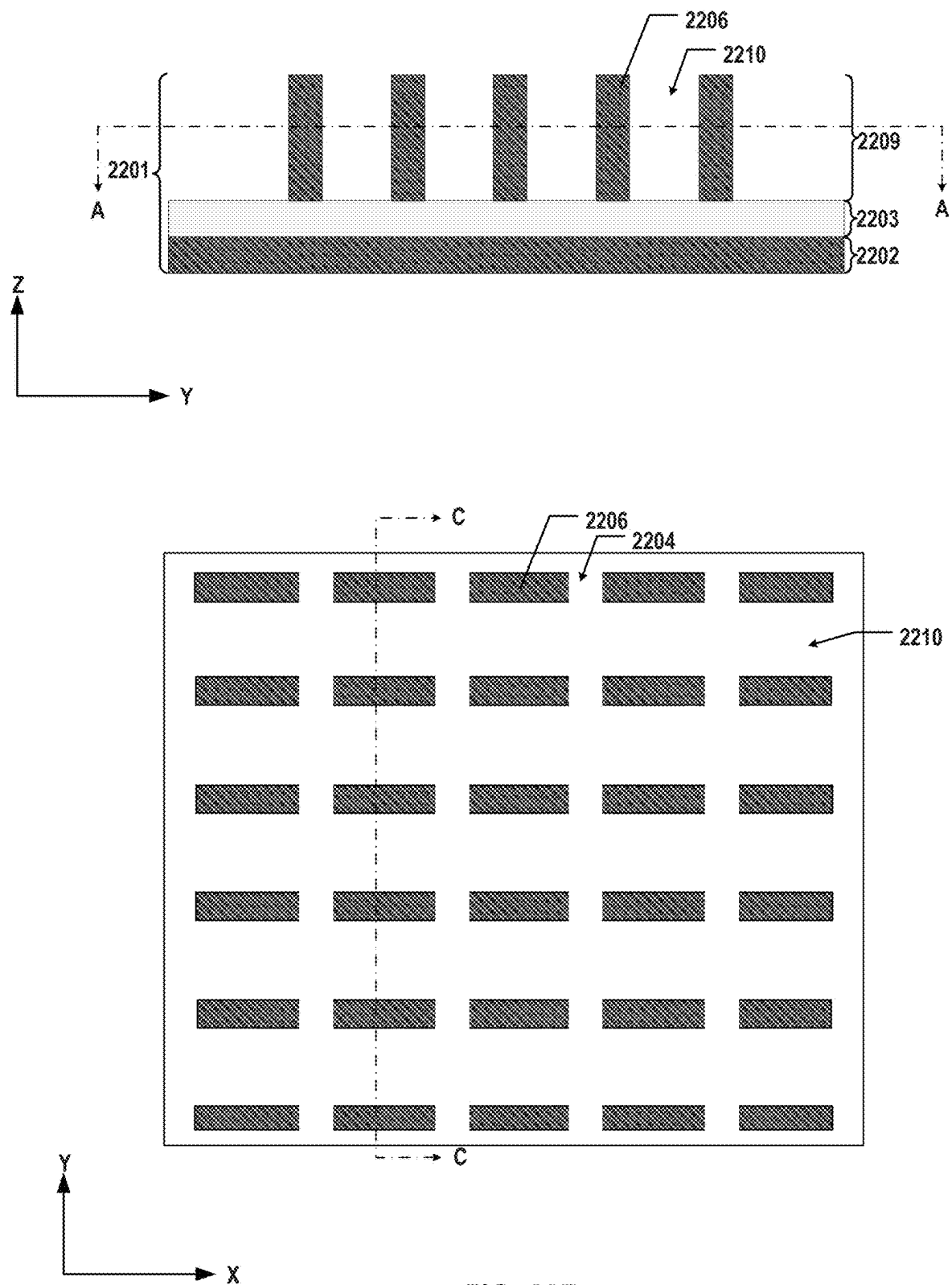


FIG. 22B

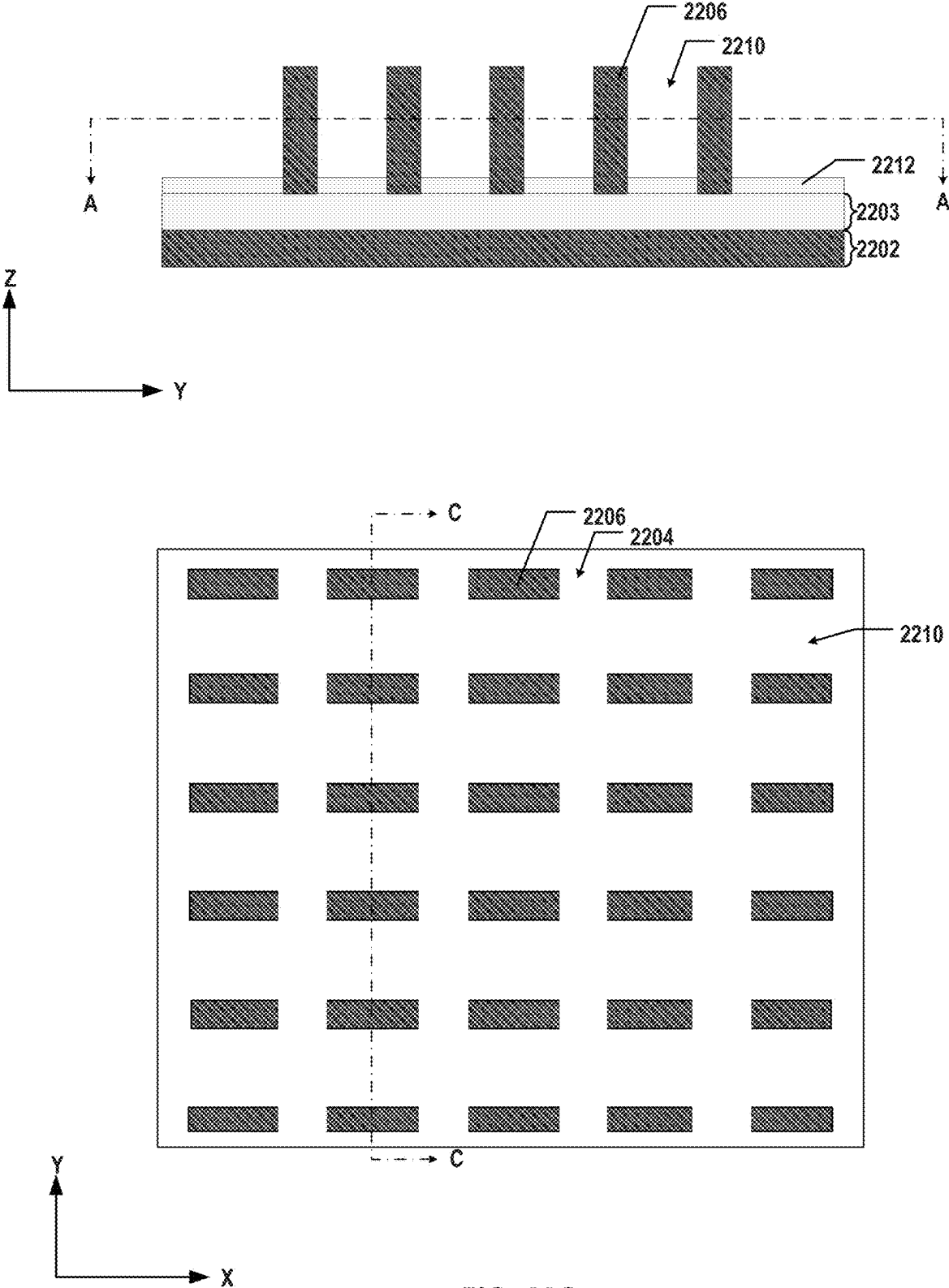


FIG. 22C

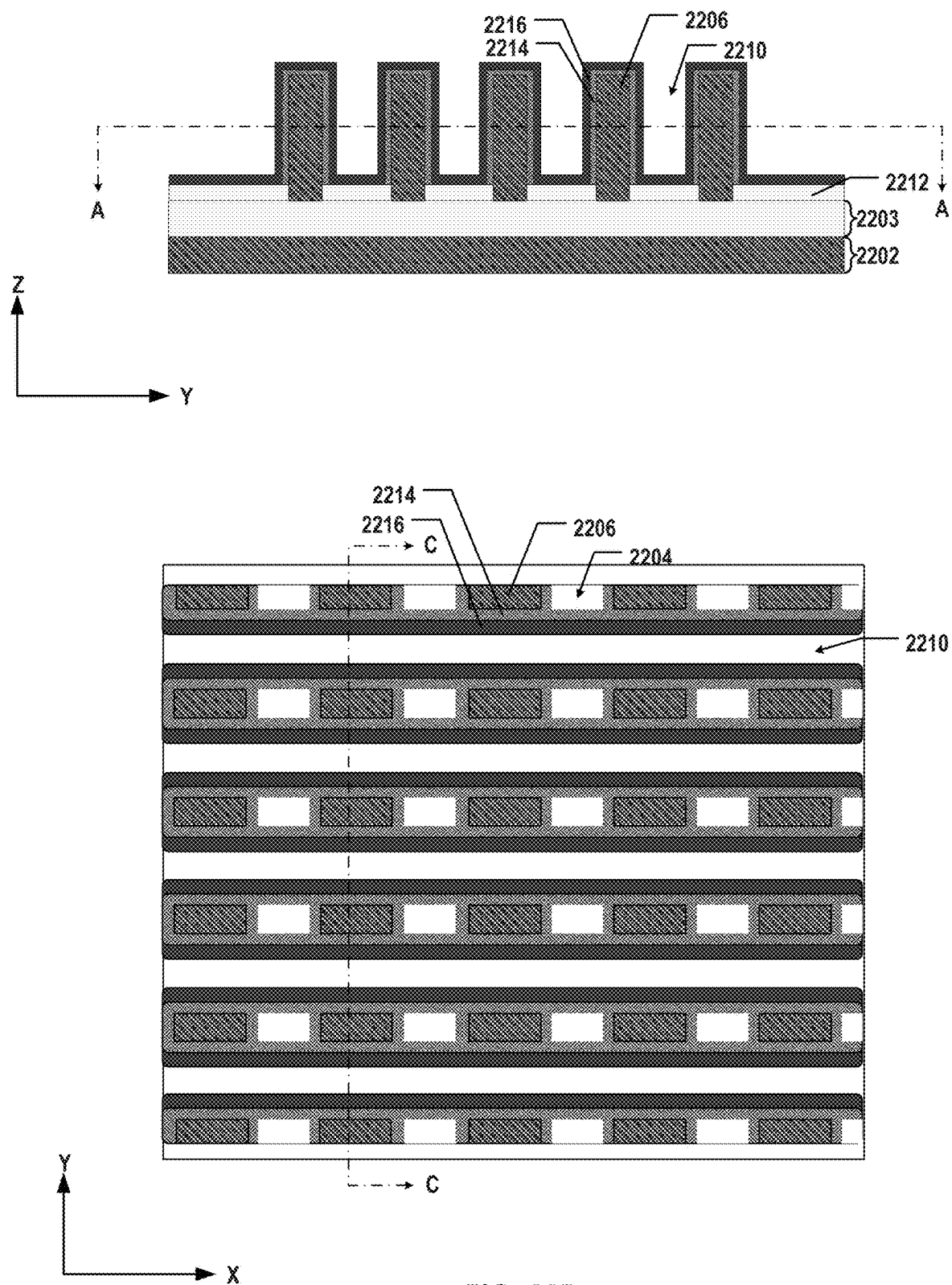


FIG. 22D

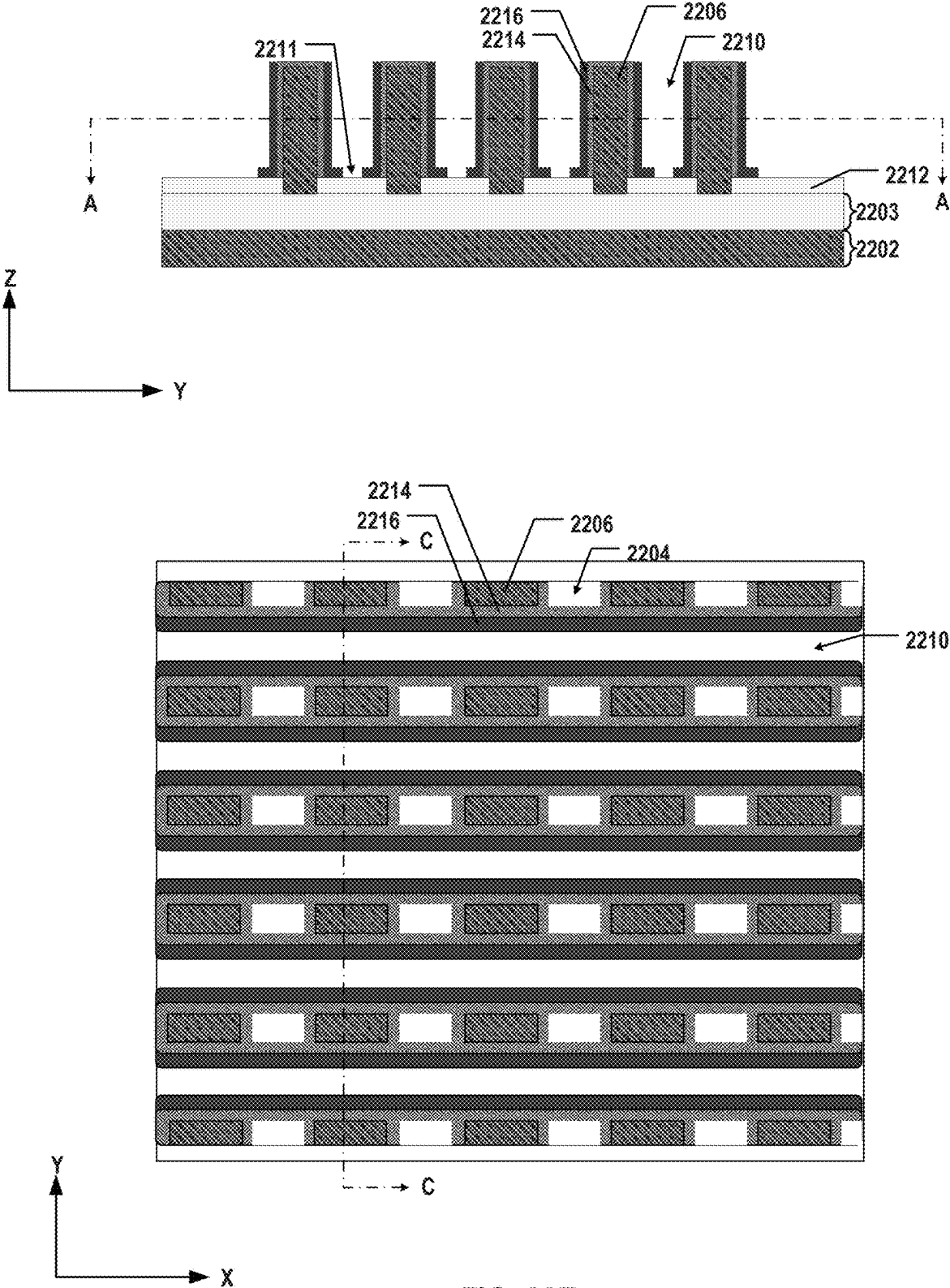


FIG. 22E

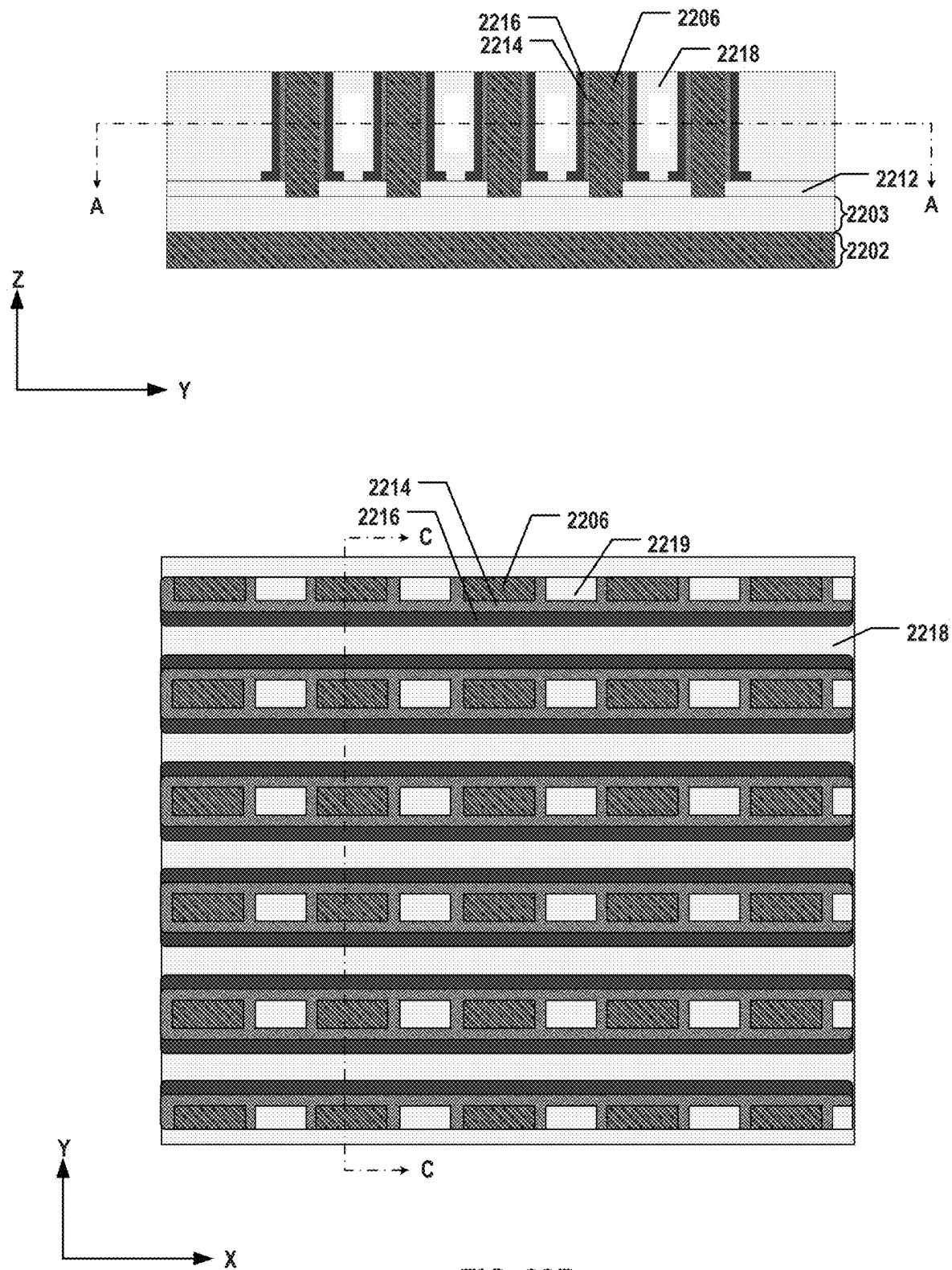


FIG. 22F

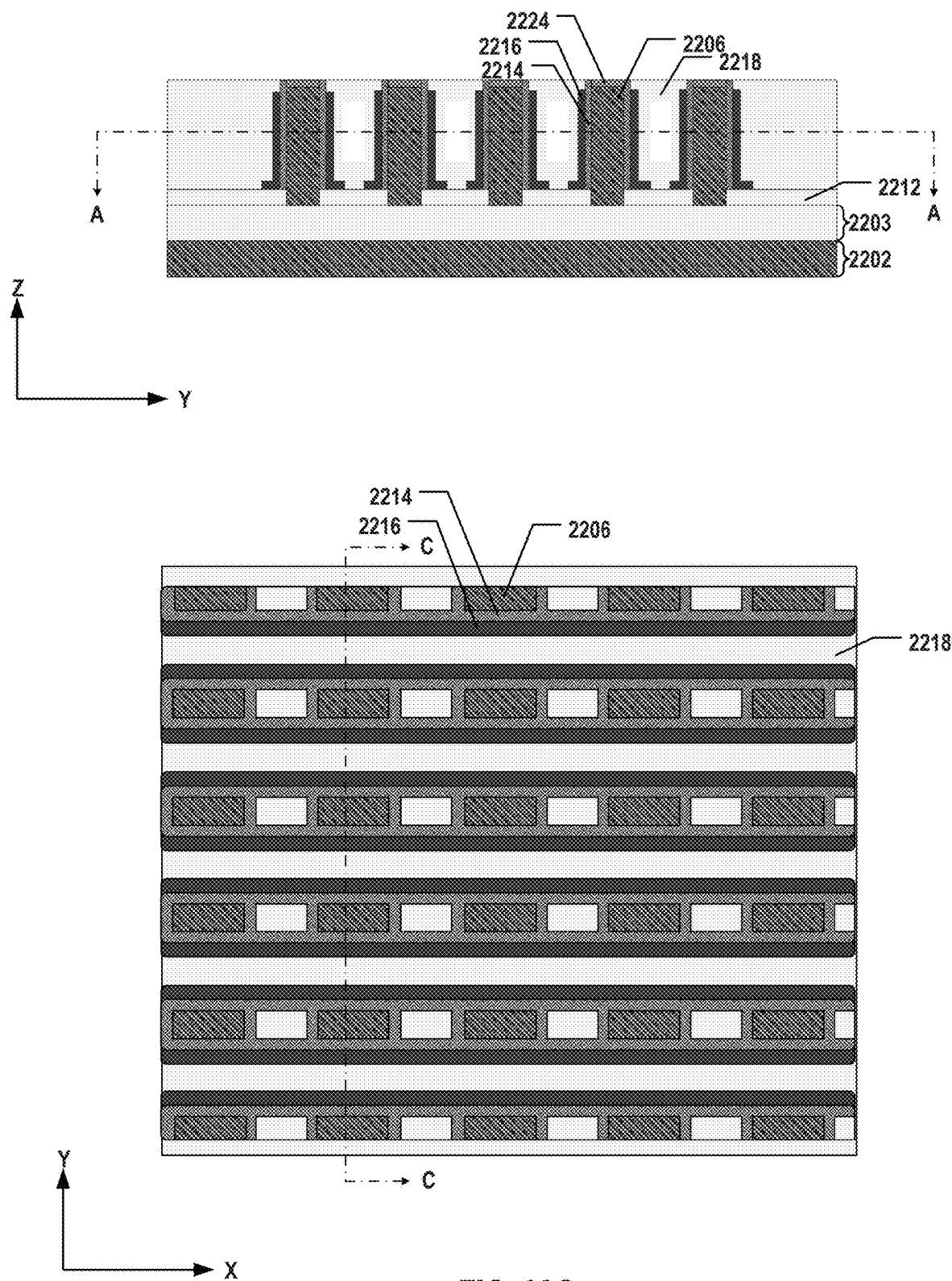


FIG. 22G

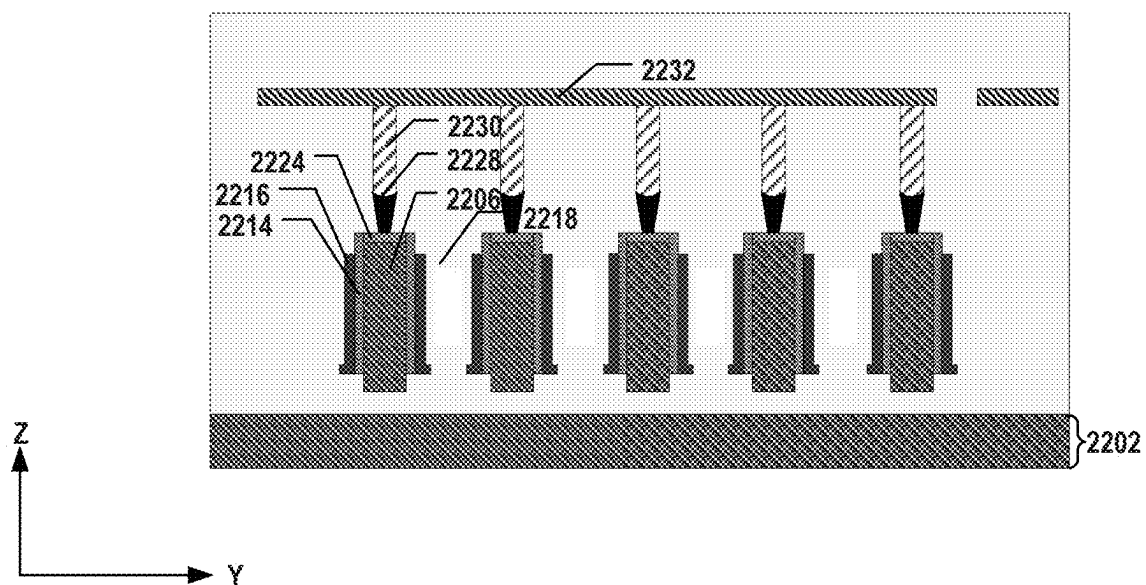


FIG. 22H

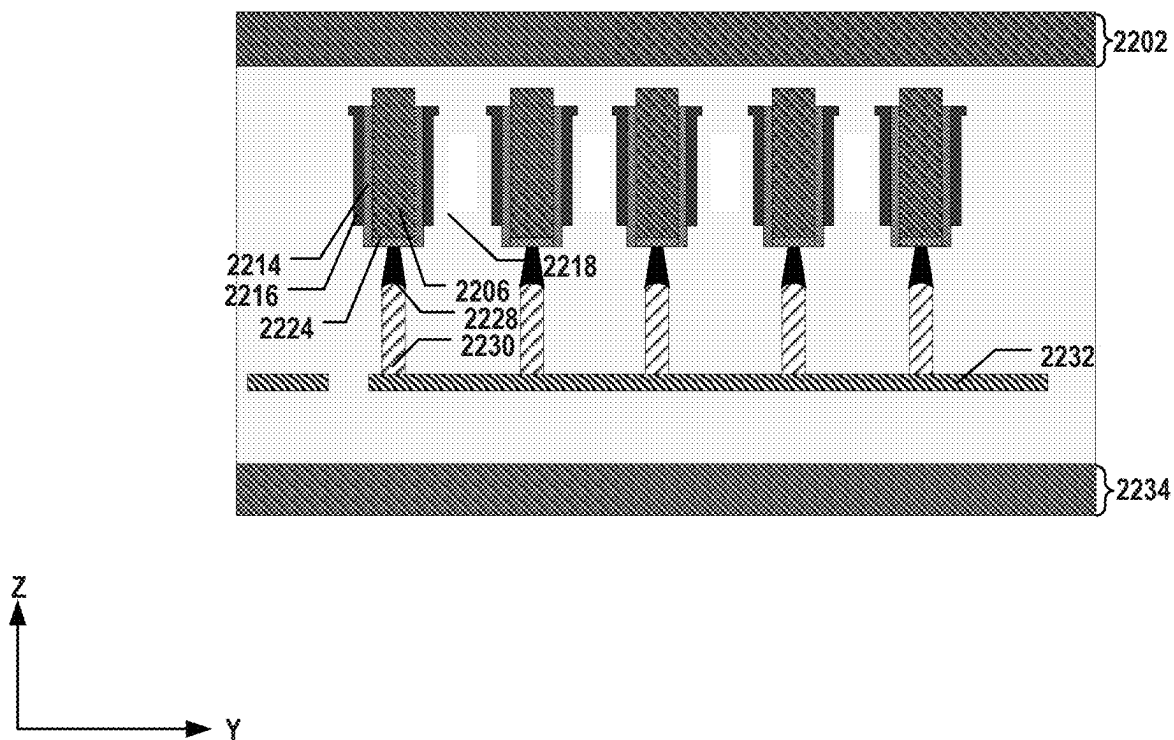


FIG. 22I

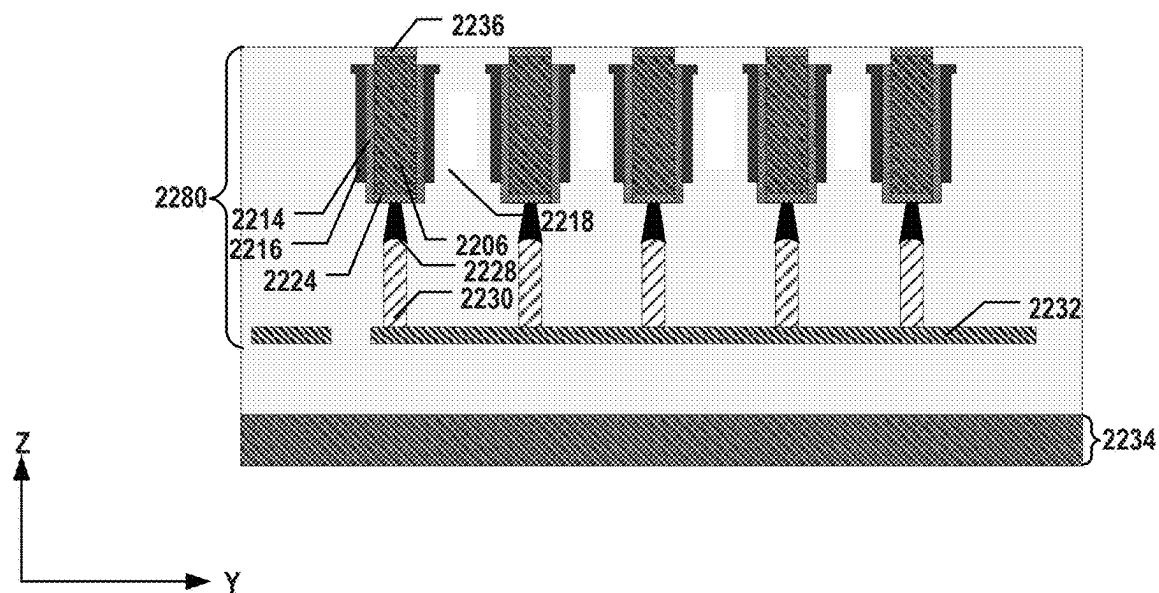


FIG. 22J

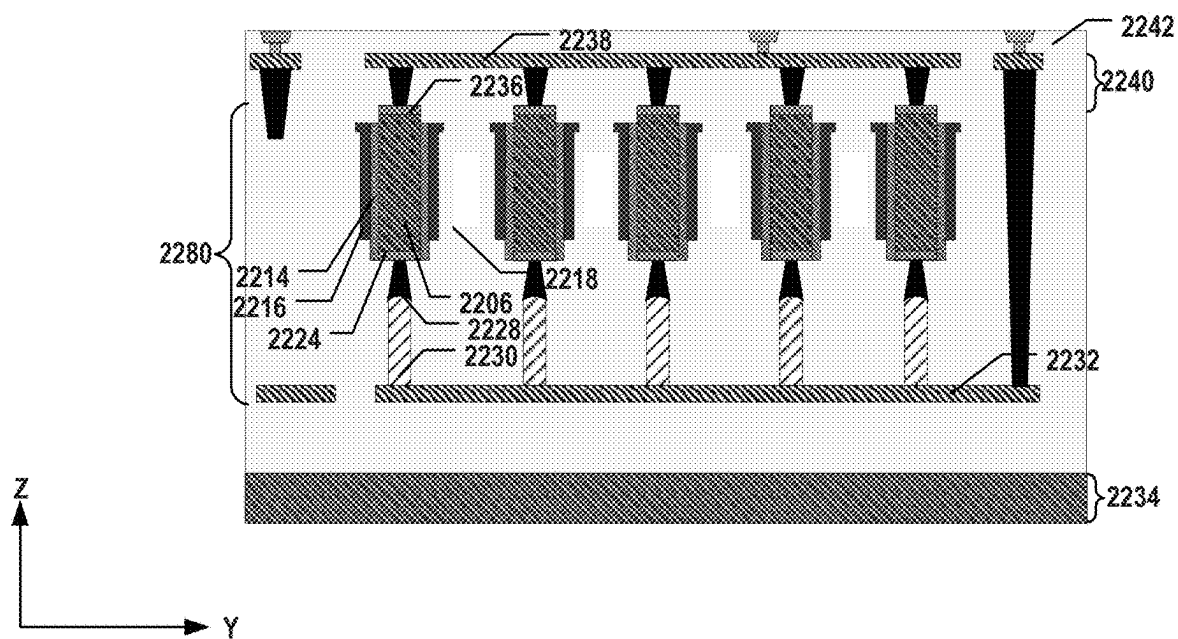


FIG. 22K

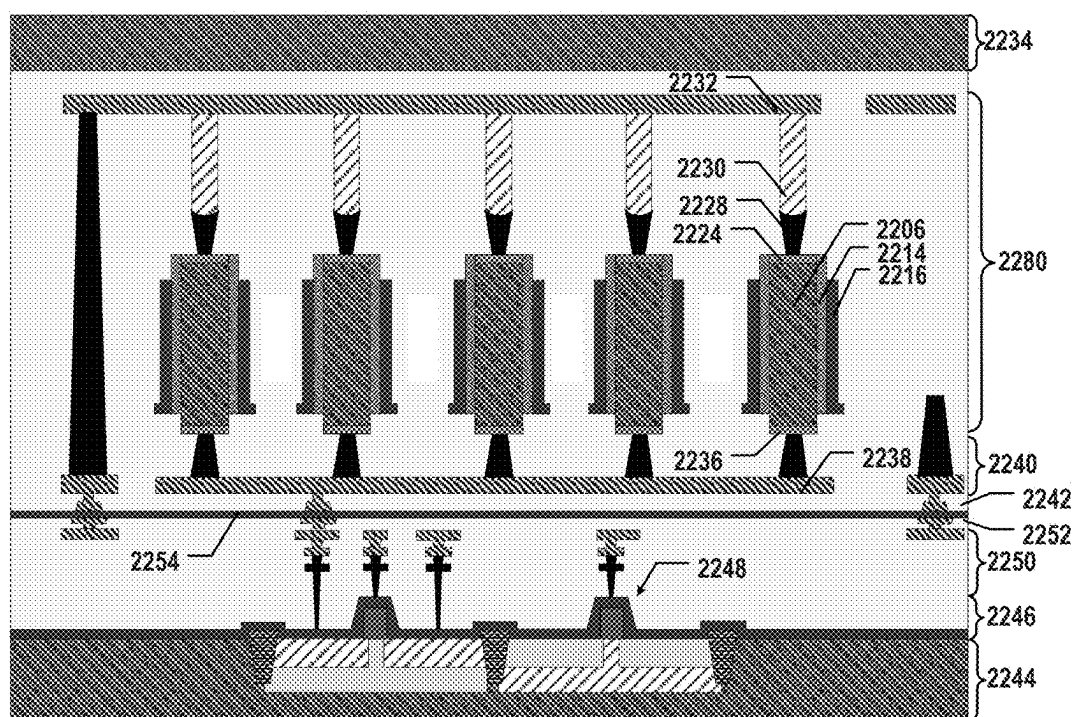


FIG. 22L

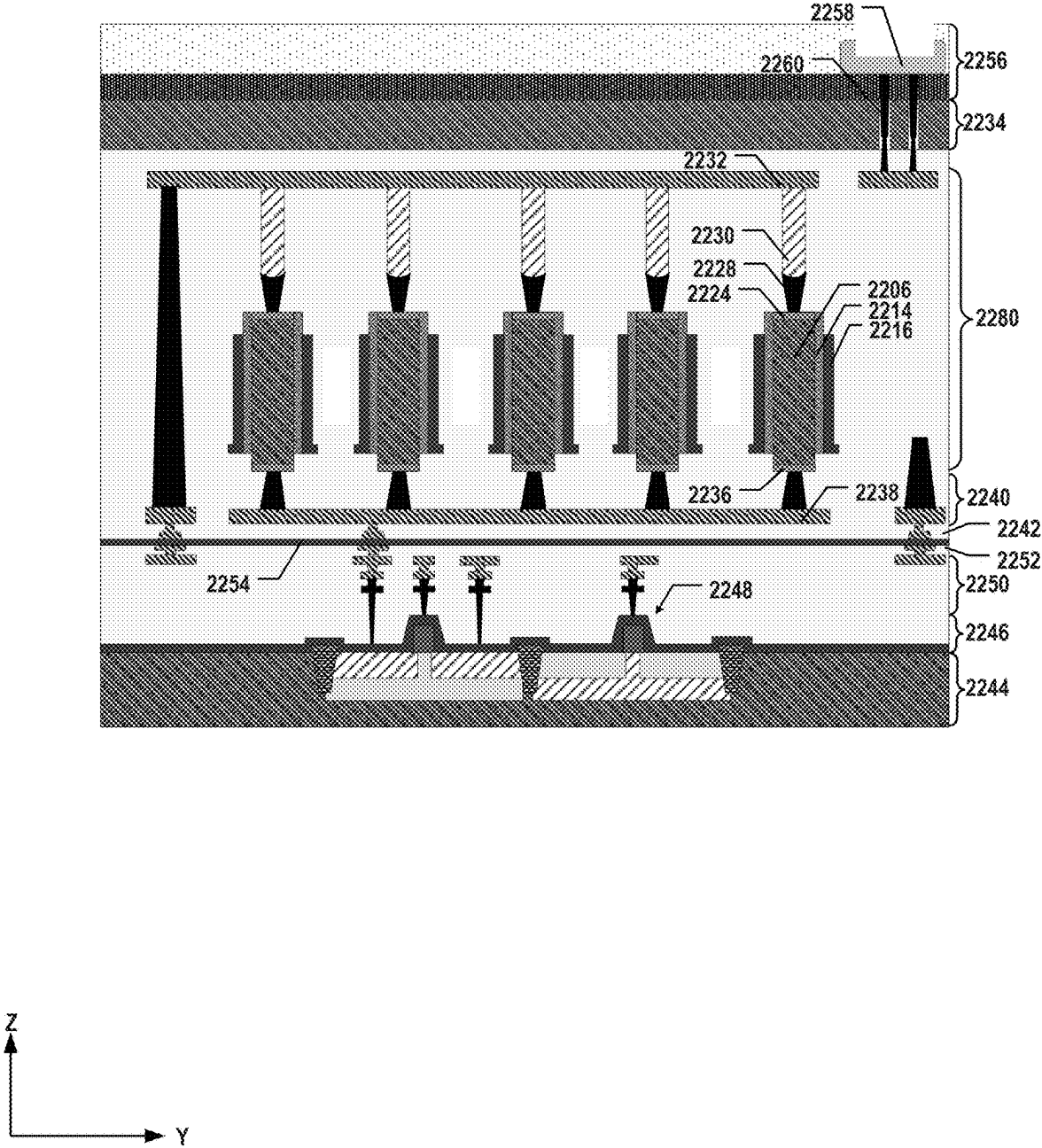


FIG. 22M

2300

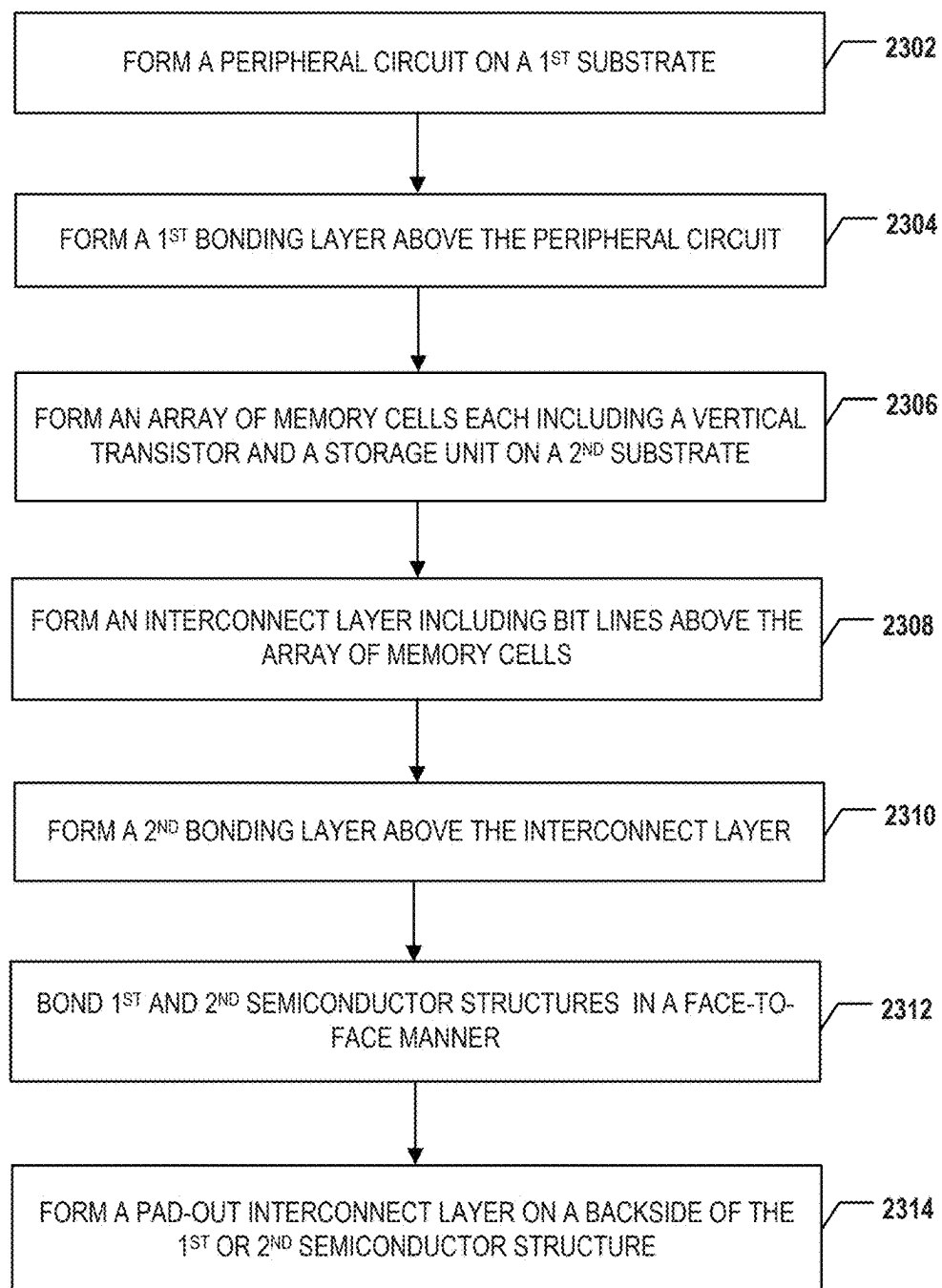


FIG. 23

2400

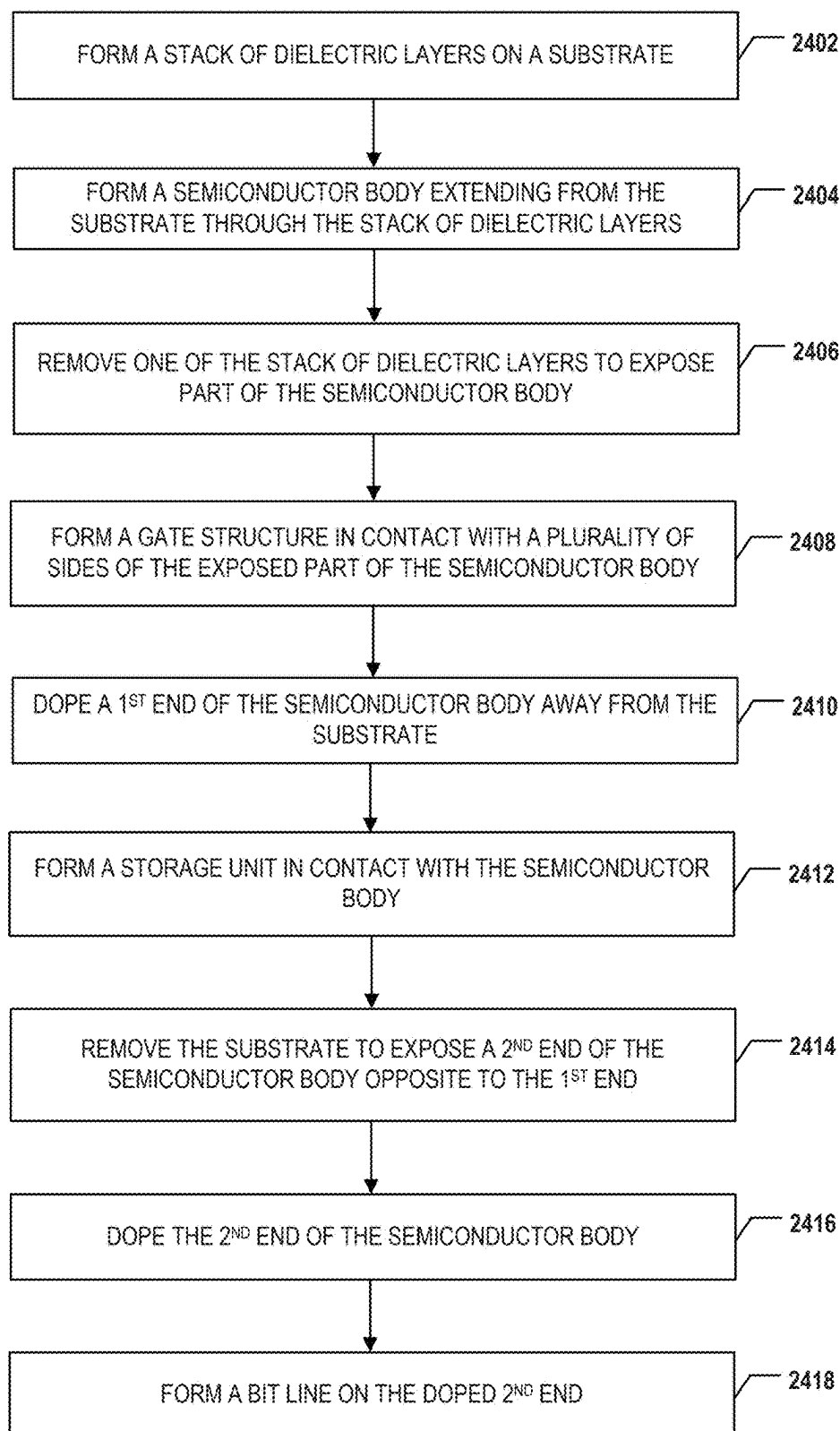


FIG. 24

2500

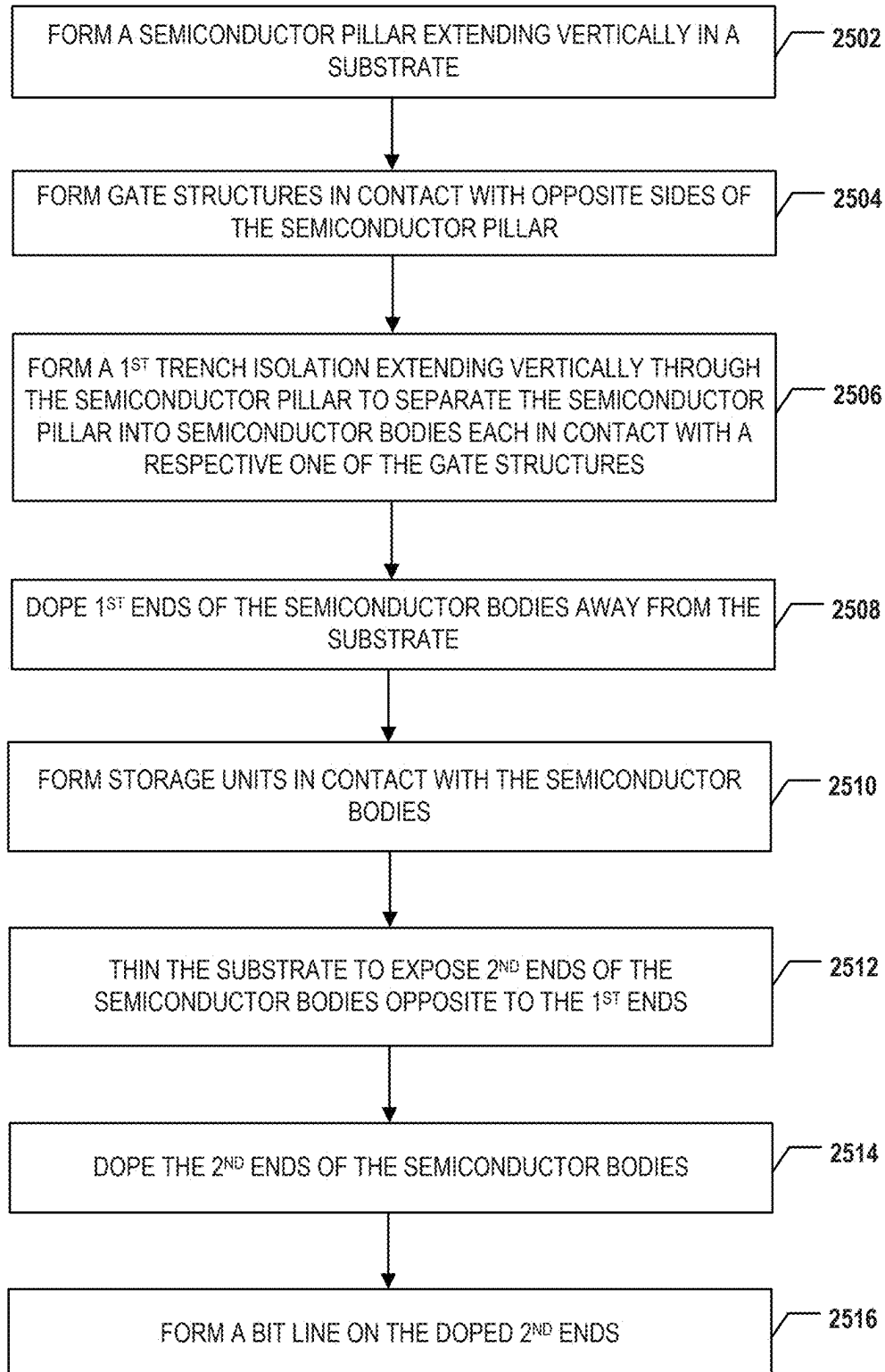


FIG. 25

2600

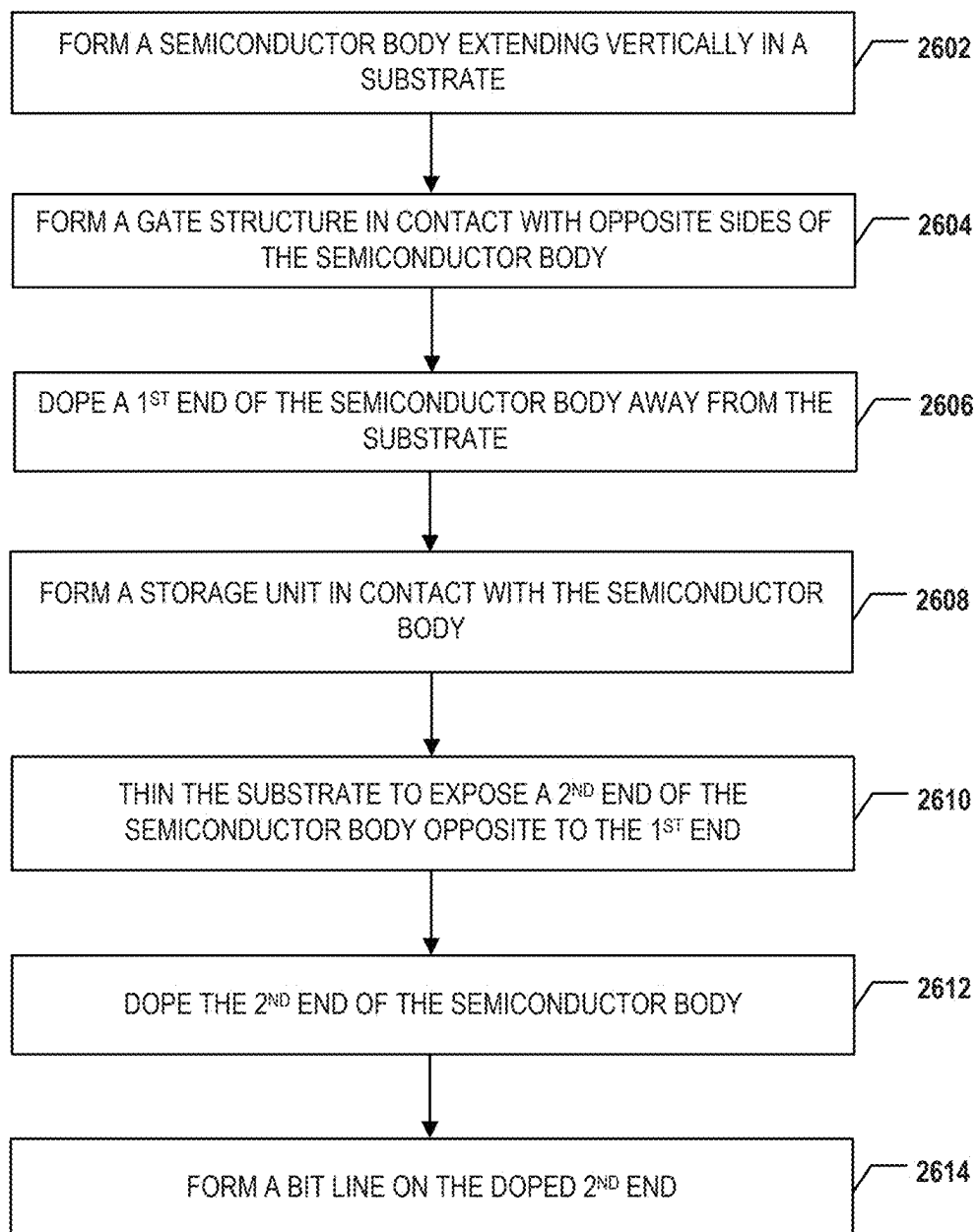


FIG. 26

2700

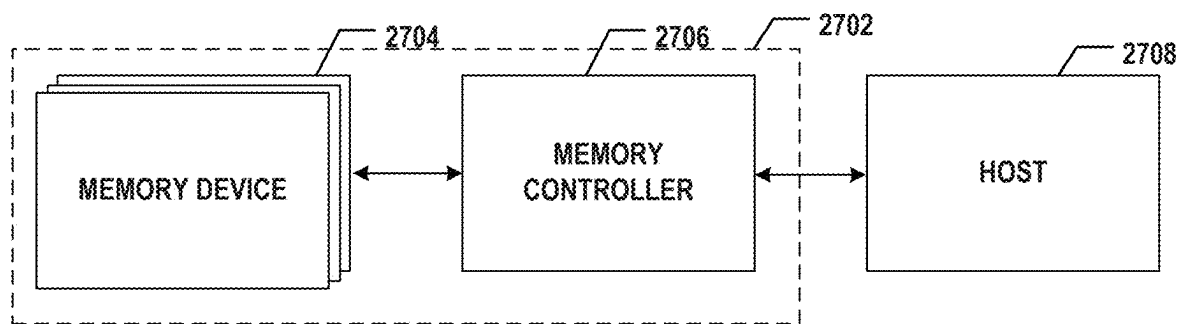


FIG. 27

MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 17/553,776, filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” which is a continuation of International Application No. PCT/CN2021/115775, filed on Aug. 31, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” both of which are hereby incorporated by reference in their entireties. This application is also related to U.S. application Ser. No. 17/553,759 filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/553,763 filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/553,765 filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/553,772 filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/553,773 filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” and U.S. application Ser. No. 17/553,781 filed on Dec. 16, 2021, entitled “memory devices having vertical transistors and methods for forming the same,” all of which are hereby incorporated by reference in their entireties.

BACKGROUND

[0002] The present disclosure relates to memory devices and fabrication methods thereof.

[0003] Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit.

[0004] A three-dimensional (3D) memory architecture can address the density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral circuits for facilitating operations of the memory array.

SUMMARY

[0005] In one aspect, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes an array of memory cells and a plurality of bit lines coupled to the memory cells and each extending in a second direction perpendicular to the first direction. Each of the memory cells

includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction. Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The array of memory cells is coupled to the peripheral circuit across the bonding interface.

[0006] In another aspect, a memory system includes a memory device configured to store data and a memory controller coupled to the memory device. The memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes an array of memory cells and a plurality of bit lines coupled to the memory cells and each extending in a second direction perpendicular to the first direction. Each of the memory cells includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction. Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The array of memory cells is coupled to the peripheral circuit across the bonding interface. The memory controller is configured to control the array of memory cells through the peripheral circuit and the bit lines.

[0007] In still another aspect, a method for forming a 3D memory device is disclosed. A first semiconductor structure including a peripheral circuit is formed. A second semiconductor structure is formed. To form the second semiconductor structure, an array of memory cells is formed, and a plurality of bit lines coupled to the memory cells are formed. Each of the memory cells includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the array of memory cells is coupled to the peripheral circuit across a bonding interface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

[0009] FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device, according to some aspects of the present disclosure.

[0010] FIG. 1B illustrates a schematic view of a cross-section of another 3D memory device, according to some aspects of the present disclosure.

[0011] FIG. 2 illustrates a schematic diagram of a memory device including peripheral circuits and an array of memory

cells each having a vertical transistor, according to some aspects of the present disclosure.

[0012] FIG. 3 illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of dynamic random-access memory (DRAM) cells, according to some aspects of the present disclosure.

[0013] FIG. 4 illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of phase-change memory (PCM) cells, according to some aspects of the present disclosure.

[0014] FIG. 5 illustrates a plan view of an array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

[0015] FIG. 6A illustrates a side view of a cross-section of a 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0016] FIG. 6B illustrates a side view of a cross-section of another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0017] FIG. 6C illustrates a side view of a cross-section of still another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0018] FIG. 6D illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0019] FIG. 6E illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0020] FIG. 7 illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0021] FIG. 8 illustrates a plan view of another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

[0022] FIG. 9 illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0023] FIGS. 10A-10M illustrate a fabrication process for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0024] FIGS. 11A-11I illustrate a fabrication process for forming another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0025] FIGS. 12A-12H illustrate a fabrication process for forming still another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0026] FIGS. 13A-13H illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0027] FIGS. 14A-14E illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0028] FIGS. 15A-15D illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0029] FIG. 16 illustrates a plan view of still another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

[0030] FIG. 17 illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0031] FIG. 18 illustrates a perspective view of an array of vertical transistors, according to some aspects of the present disclosure.

[0032] FIGS. 19A-19M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0033] FIG. 20 illustrates a plan view of yet another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

[0034] FIG. 21 illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0035] FIGS. 22A-22M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0036] FIG. 23 illustrates a flowchart of a method for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure.

[0037] FIG. 24 illustrates a flowchart of a method for forming an array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.

[0038] FIG. 25 illustrates a flowchart of a method for forming another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.

[0039] FIG. 26 illustrates a flowchart of a method for forming still another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.

[0040] FIG. 27 illustrates a block diagram of an exemplary system having a memory device, according to some aspects of the present disclosure.

[0041] The present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

[0042] Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the present disclosure can also be employed in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, and modified with one another and in ways not specifically depicted in the drawings, such that these combinations, adjustments, and modifications are within the scope of the present disclosure.

[0043] In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a

plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

[0044] It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (directly on something).

[0045] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0046] As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

[0047] As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layers thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductors and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

[0048] Transistors are used as the switch or selecting devices in the memory cells of some memory devices, such as DRAM, PCM, and ferroelectric DRAM (FRAM). However, the planar transistors commonly used in existing memory cells usually have a horizontal structure with buried word lines in the substrate and bit lines above the substrate. Since the source and drain of a planar transistor are disposed laterally at different locations, which increases the area

occupied by the transistor. The design of planar transistors also complicates the arrangement of interconnected structures, such as word lines and bit lines, coupled to the memory cells, for example, limiting the pitches of the word lines and/or bit lines, thereby increasing the fabrication complexity and reducing the production yield. Moreover, because the bit lines and the storage units (e.g., capacitors or PCM elements) are arranged on the same side of the planar transistors (above the transistors and substrate), the bit line process margin is limited by the storage units, and the coupling capacitance between the bit lines and storage units, such as capacitors, are increased. Planar transistors may also suffer from a high leakage current as the saturated drain current keeps increasing, which is undesirable for the performance of memory devices.

[0049] On the other hand, the memory cell array and the peripheral circuits for controlling the memory cell array are usually arranged side-by-side in the same plane. As the number of memory cells keeps increasing, to maintain the same chip size, the dimensions of the components in the memory cell array, such as transistors, word lines, and/or bit lines, need to keep decreasing in order not to significantly reduce the memory cell array efficiency.

[0050] To address one or more of the aforementioned issues, the present disclosure introduces a solution in which vertical transistors replace the planar transistors as the switch and selecting devices in a memory cell array of memory devices (e.g., DRAM, PCM, and FRAM). Compared with planar transistors, the vertically arranged transistors (e.g., the drain and source are overlapped in the plan view) can reduce the area of the transistor as well as simplify the layout of the interconnect structures, e.g., metal wiring the word lines and bit lines, which can reduce the fabrication complexity and improve the yield. For example, the pitches of word lines and/or bit lines can be reduced for ease of fabrication. The vertical structures of the transistors also allow the bit lines and storage units, such as capacitors, to be arranged on opposite sides of the transistors in the vertical direction (e.g., one above and on below the transistors), such that the process margin of the bit lines can be increased and the coupling capacitance between the bit lines and the storage units can be decreased.

[0051] Consistent with the scope of the present disclosure, according to some aspects of the present disclosure, the memory cell array having vertical transistors and the peripheral circuits of the memory cell array can be formed on different wafers and bonded together in a face-to-face manner. Thus, the thermal budget of fabricating the memory cell array does not affect the fabrication of the peripheral circuits. The stacked memory cell array and peripheral circuits can also reduce the chip size compared with the side-by-side arrangement, thereby improving the array efficiency. In some implementations, more than one memory cell array is stacked over one another using bonding techniques to further increase the array efficiency. In some implementations, the word lines and bit lines are disposed close to the bonding interface due to the vertically arranged transistors, which can be coupled to the peripheral circuits through a large number (e.g., millions) of parallel bonding contacts across the bonding interface can make direct, short-distance (e.g., micron-level) electrical connections between the memory cell array and peripheral circuits to increase the throughput and input/output (I/O) speed of the memory devices.

[0052] In some implementations, the vertical transistors disclosed herein include multi-gate transistors (e.g., gate-all-around (GAA) transistors, tri-gate transistors, or double-gate transistors), which can have a larger gate control area to achieve better channel control with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current of multi-gate transistors can be significantly reduced as well. Thus, using multi-gate transistors instead of planar transistors can achieve a much better speed (saturated drain current)/leakage current performance.

[0053] In some implementations, the vertical transistors disclosed herein include single-gate transistors (a.k.a. single-side gate transistors) in a mirror-symmetric arrangement with respect to adjacent transistors in the bit line direction as a result of splitting multi-gate transistors (e.g., double-gate transistors) using trench isolations extending along the word line direction. Thus, the memory cell density in the bit line direction can be significantly increased (e.g., doubled) without unduly complicating the fabrication process compared with using processes, such as self-aligned double patterning (SADP). Also, the mirror-symmetric single-gate transistors have a larger process window for word line, bit line, and transistor pitch reduction, compared to either planar transistors or multi-gate vertical transistors, for example, with dual-side or all-around gates.

[0054] FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device **100**, according to some aspects of the present disclosure. 3D memory device **100** represents an example of a bonded chip. The components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) can be formed separately on different substrates and then jointed to form a bonded chip. 3D memory device **100** can include a first semiconductor structure **102** including the peripheral circuits of a memory cell array. 3D memory device **100** can also include a second semiconductor structure **104** including the memory cell array. The peripheral circuits (a.k.a. control and sensing circuits) can include any suitable digital, analog, and/or mixed-signal circuits used for facilitating the operations of the memory cell array. For example, the peripheral circuit can include one or more of a page buffer, a decoder (e.g., a row decoder and a column decoder), a sense amplifier, a driver (e.g., a word line driver), an input/output (I/O) circuit, a charge pump, a voltage source or generator, a current or voltage reference, any portions (e.g., a sub-circuit) of the functional circuits mentioned above, or any active or passive components of the circuit (e.g., transistors, diodes, resistors, or capacitors). The peripheral circuits in first semiconductor structure **102** use complementary metal-oxide-semiconductor (CMOS) technology, e.g., which can be implemented with logic processes (e.g., technology nodes of 90 nm, 65 nm, 60 nm, 45 nm, 32 nm, 28 nm, 22 nm, 20 nm, 16 nm, 14 nm, 10 nm, 7 nm, 5 nm, 3 nm, 2 nm, etc.), according to some implementations.

[0055] As shown in FIG. 1A, 3D memory device **100** can also include first semiconductor structure **104** including an array of memory cells (memory cell array) that can use transistors as the switch and selecting devices. In some implementations, the memory cell array includes an array of DRAM cells. For case of description, a DRAM cell array may be used as an example for describing the memory cell array in the present disclosure. But it is understood that the memory cell array is not limited to DRAM cell array and may include any other suitable types of memory cell arrays

that can use transistors as the switch and selecting devices, such as PCM cell array, static random-access memory (SRAM) cell array, FRAM cell array, resistive memory cell array, magnetic memory cell array, spin transfer torque (STT) memory cell array, to name a few, or any combination thereof.

[0056] Second semiconductor structure **104** can be a DRAM device in which memory cells are provided in the form of an array of DRAM cells. In some embodiments, each DRAM cell includes a capacitor for storing a bit of data as a positive or negative electrical charge as well as one or more transistors (a.k.a. pass transistors) that control (e.g., switch and selecting) access to it. In some implementations, each DRAM cell is a one-transistor, one-capacitor (1T1C) cell. Since transistors always leak a small amount of charge, the capacitors will slowly discharge, causing information stored in them to drain. As such, a DRAM cell has to be refreshed to retain data, for example, by the peripheral circuit in first semiconductor structure **102**, according to some implementation.

[0057] As shown in FIG. 1A, 3D memory device **100** further includes a bonding interface **106** vertically between (in the vertical direction, e.g., the z-direction in FIG. 1A) first semiconductor structure **102** and second semiconductor structure **104**. As described below in detail, first and second semiconductor structures **102** and **104** can be fabricated separately (and in parallel in some implementations) such that the thermal budget of fabricating one of first and second semiconductor structures **102** and **104** does not limit the processes of fabricating another one of first and second semiconductor structures **102** and **104**. Moreover, a large number of interconnects (e.g., bonding contacts) can be formed through bonding interface **106** to make direct, short-distance (e.g., micron-level) electrical connections between first semiconductor structure **102** and second semiconductor structure **104**, as opposed to the long-distance (e.g., millimeter or centimeter-level) chip-to-chip data bus on the circuit board, such as printed circuit board (PCB), thereby eliminating chip interface delay and achieving high-speed I/O throughput with reduced power consumption. Data transfer between the memory cell array in second semiconductor structure **104** and the peripheral circuits in first semiconductor structure **102** can be performed through the interconnects (e.g., bonding contacts) across bonding interface **106**. By vertically integrating first and second semiconductor structures **102** and **104**, the chip size can be reduced, and the memory cell density can be increased.

[0058] It is understood that the relative positions of stacked first and second semiconductor structures **102** and **104** are not limited. FIG. 1B illustrates a schematic view of a cross-section of another exemplary 3D memory device **101**, according to some implementations. Different from 3D memory device **100** in FIG. 1A in which second semiconductor structure **104** including the memory cell array is above first semiconductor structure **102** including the peripheral circuits, in 3D memory device **101** in FIG. 1B, first semiconductor structure **102** including the peripheral circuit is above second semiconductor structure **104** including the memory cell array. Nevertheless, bonding interface **106** is formed vertically between first and second semiconductor structures **102** and **104** in 3D memory device **101**, and first and second semiconductor structures **102** and **104** are jointed vertically through bonding (e.g., hybrid bonding) according to some implementations. Hybrid bonding, also

known as “metal/dielectric hybrid bonding,” is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., copper-to-copper) bonding and dielectric-dielectric (e.g., silicon oxide-to-silicon oxide) bonding simultaneously. Data transfer between the memory cell array in second semiconductor structure **104** and the peripheral circuits in first semiconductor structure **102** can be performed through the interconnects (e.g., bonding contacts) across bonding interface **106**.

[0059] It is noted that x, y, and z axes are included in FIGS. **1A** and **1B** to further illustrate the spatial relationship of the components in 3D memory devices **100** and **101**. The substrate of the 3D memory device includes two lateral surfaces extending laterally in the x-y plane: a top surface on the front side of the wafer on which the semiconductor devices can be formed, and a bottom surface on the backside opposite to the front side of the wafer. The z-axis is perpendicular to both the x and y axes. As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of the 3D memory device is determined relative to the substrate of the 3D memory device in the z-direction (the vertical direction perpendicular to the x-y plane, e.g., the thickness direction of the substrate) when the substrate is positioned in the lowest plane of the 3D memory device in the z-direction. The same notion for describing the spatial relationships is applied throughout the present disclosure.

[0060] FIG. **2** illustrates a schematic diagram of a memory device **200** including peripheral circuits and an array of memory cells each having a vertical transistor, according to some aspects of the present disclosure. Memory device **200** can include a memory cell array **201** and peripheral circuits **202** coupled to memory cell array **201**. 3D memory devices **100** and **101** may be examples of memory device **200** in which memory cell array **201** and peripheral circuits **202** may be included in second and first semiconductor structures **104** and **102**, respectively. Memory cell array **201** can be any suitable memory cell array in which each memory cell **208** includes a vertical transistor **210** and a storage unit **212** coupled to vertical transistor **210**. In some implementations, memory cell array **201** is a DRAM cell array, and storage unit **212** is a capacitor for storing charge as the binary information stored by the respective DRAM cell. In some implementations, memory cell array **201** is a PCM cell array, and storage unit **212** is a PCM element (e.g., including chalcogenide alloys) for storing binary information of the respective PCM cell based on the different resistivities of the PCM element in the amorphous phase and the crystalline phase. In some implementations, memory cell array **201** is a FRAM cell array, and storage unit **212** is a ferroelectric capacitor for storing binary information of the respective FRAM cell based on the switch between two polarization states of ferroelectric materials under an external electric field.

[0061] As shown in FIG. **2**, memory cells **208** can be arranged in a two-dimensional (2D) array having rows and columns. Memory device **200** can include word lines **204** coupling peripheral circuits **202** and memory cell array **201** for controlling the switch of vertical transistors **210** in memory cells **208** located in a row, as well as bit lines **206** coupling peripheral circuits **202** and memory cell array **201** for sending data to and/or receiving data from memory cells **208** located in a column. That is, each word line **204** is

coupled to a respective row of memory cells **208**, and each bit line is coupled to a respective column of memory cells **208**.

[0062] Consistent with the scope of the present disclosure, vertical transistors **210**, such as vertical metal-oxide-semiconductor field-effect transistors (MOSFETs), can replace the planar transistors as the pass transistors of memory cells **208** to reduce the area occupied by the pass transistors, the coupling capacitance, as well as the interconnect routing complexity, as described below in detail. As shown in FIG. **2**, in some implementations, different from planar transistors in which the active regions are formed in the substrates, vertical transistor **210** includes a semiconductor body **214** extending vertically (in the z-direction) above the substrate (not shown). That is, semiconductor body **214** can extend above the top surface of the substrate to allow channels to be formed not only at the top surface of semiconductor body **214**, but also at one or more side surfaces thereof. As shown in FIG. **2**, for example, semiconductor body **214** can have a cuboid shape to expose four sides thereof. It is understood that semiconductor body **214** may have any suitable 3D shape, such as polyhedron shapes or a cylinder shape. That is, the cross-section of semiconductor body **214** in the plan view (e.g., in the x-y plane) can have a square shape, a rectangular shape (or a trapezoidal shape), a circular (or an oval shape), or any other suitable shapes. It is understood that consistent with the scope of the present disclosure, for semiconductor bodies that have a circular or oval shape of their cross-sections in the plan view, the semiconductor bodies may still be considered as having multiple sides, such that the gate structures are in contact with more than one side of the semiconductor bodies. As described below with respect to the fabrication process, semiconductor body **214** can be formed from the substrate (e.g., by etching or epitaxy) and thus, has the same semiconductor material (e.g., silicon crystalline silicon) as the substrate (e.g., a silicon substrate).

[0063] As shown in FIG. **2**, vertical transistor **210** can also include a gate structure **216** in contact with one or more sides of semiconductor body **214**, e.g., in one or more planes of the side surface(s) of the active region. In other words, the active region of vertical transistor **210**, e.g., semiconductor body **214**, can be at least partially surrounded by gate structure **216**. Gate structure **216** can include a gate dielectric **218** over one or more sides of semiconductor body **214**, e.g., in contact with four side surfaces of semiconductor body **214** as shown in FIG. **2**. Gate structure **216** can also include a gate electrode **220** over and in contact with gate dielectric **218**. Gate dielectric **218** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. For example, gate dielectric **218** may include silicon oxide, which is a form of gate oxide. Gate electrode **220** can include any suitable conductive materials, such as polysilicon, metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), metal compounds (e.g., titanium nitride (TiN), tantalum nitride (Ta₂N₅), etc.), or silicides. For example, gate electrode **220** may include doped polysilicon, which is a form of a gate poly. In some implementations, gate electrode **220** includes multiple conductive layers, such as a W layer over a TiN layer. It is understood that gate electrode **220** and word line **204** may be a continuous conductive structure in some examples. In other words, gate electrode **220** may be viewed as part of word line **204** that forms gate structure **216**, or word line **204**

may be viewed as the extension of gate electrode **220** to be coupled to peripheral circuits **202**.

[0064] As shown in FIG. 2, vertical transistor **210** can further include a pair of a source and a drain (S/D, dope regions, a.k.a., source electrode and drain electrode) formed at the two ends of semiconductor body **214** in the vertical direction (the z-direction), respectively. The source and drain can be doped with any suitable P-type dopants, such as boron (B) or Gallium (Ga), or any suitable N-type dopants, such as phosphorus (P) or arsenic (As). The source and drain can be separated by gate structure **216** in the vertical direction (the z-direction). In other words, gate structure **216** is formed vertically between the source and drain. As a result, one or more channels (not shown) of vertical transistor **210** can be formed in semiconductor body **214** vertically between the source and drain when a gate voltage applied to gate electrode **220** of gate structure **216** is above the threshold voltage of vertical transistor **210**. That is, each channel of vertical transistors **210** is also formed in the vertical direction along which semiconductor body **214** extends, according to some implementations.

[0065] In some implementations, as shown in FIG. 2, vertical transistor **210** is a multi-gate transistor. That is, gate structure **216** can be in contact with more than one side of semiconductor body **214** (e.g., four sides in FIG. 2) to form more than one gate, such that more than one channel can be formed between the source and drain in operation. That is, different from the planar transistor that includes only a single planar gate (and resulting in a single planar channel), vertical transistor **210** shown in FIG. 2 can include multiple vertical gates on multiple sides of semiconductor body **214** due to the 3D structure of semiconductor body **214** and gate structure **216** that surrounds the multiple sides of semiconductor body **214**. As a result, compared with planar transistors, vertical transistor **210** shown in FIG. 2 can have a larger gate control area to achieve better channel control with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current (I_{off}) of vertical transistor **210** can be significantly reduced a well. As described below in detail, the multi-gate vertical transistors can include double-gate vertical transistors (e.g., dual-side gate vertical transistors), tri-gate vertical transistors (e.g., tri-side gate vertical transistors), and GAA vertical transistors.

[0066] It is understood that although vertical transistor **210** is shown as a multi-gate transistor in FIG. 2, the vertical transistors disclosed herein may also include single-gate transistors as described below in detail. That is, gate structure **216** may be in contact with a single side of semiconductor body **214**, for example, for the purpose of increasing the transistor and memory cell density. It is also understood that although gate dielectric **218** is shown as being separate (a separate structure) from other gate dielectrics of adjacent vertical transistors (not shown), gate dielectric **218** may be part of a continuous dielectric layer having multiple gate dielectrics of vertical transistors.

[0067] In planar transistors and some lateral multiple-gate transistors (e.g., FinFET), the active regions, such as semiconductor bodies (e.g., Fins), extend laterally (in the x-y plane), and the source and the drain are disposed at different locations in the same lateral plane (the x-y plane). In contrast, in vertical transistor **210**, semiconductor body **214** extends vertically (in the z-direction), and the source and the drain are disposed in the different lateral planes, according to some implementations. In some implementations, the

source and the drain are formed at two ends of semiconductor body **214** in the vertical direction (the z-direction), respectively, thereby being overlapped in the plan view. As a result, the area (in the x-y plane) occupied by vertical transistor **210** can be reduced compared with planar transistor and lateral multiple-gate transistors. Also, the metal wiring coupled to vertical transistors **210** can be simplified as well since the interconnects can be routed in different planes. For example, bit lines **206** and storage units **212** may be formed on opposite sides of vertical transistor **210**. In one example, bit line **206** may be coupled to the source or the drain at the upper end of semiconductor body **214**, while storage unit **212** may be coupled to the other source or the drain at the lower end of semiconductor body **214**.

[0068] As shown in FIG. 2, storage unit **212** can be coupled to the source or the drain of vertical transistor **210**. Storage unit **212** can include any devices that are capable of storing binary data (e.g., 0 and 1), including but not limited to, capacitors for DRAM cells and FRAM cells, and PCM elements for PCM cells. In some implementations, vertical transistor **210** controls the selection and/or the state switch of the respective storage unit **212** coupled to vertical transistor **210**. In some implementations as shown in FIG. 3, each memory cell **208** is a DRAM cell **302** including a transistor **304** (e.g., implementing using vertical transistors **210** in FIG. 2) and a capacitor **306** (e.g., an example of storage unit **212** in FIG. 2). The gate of transistor **304** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **304** may be coupled to bit line **206**, the other one of the source and the drain of transistor **304** may be coupled to one electrode of capacitor **306**, and the other electrode of capacitor **306** may be coupled to the ground. In some implementations as shown in FIG. 4, each memory cell **208** is a PCM cell **402** including a transistor **404** (e.g., implementing using vertical transistors **210** in FIG. 2) and a PCM element **406** (e.g., an example of storage unit **212** in FIG. 2). The gate of transistor **404** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **404** may be coupled to the ground, the other one of the source and the drain of transistor **404** may be coupled to one electrode of PCM element **406**, and the other electrode of PCM element **406** may be coupled to bit line **206**.

[0069] Peripheral circuits **202** can be coupled to memory cell array **201** through bit lines **206**, word lines **204**, and any other suitable metal wirings. As described above, peripheral circuits **202** can include any suitable circuits for facilitating the operations of memory cell array **201** by applying and sensing voltage signals and/or current signals through word lines **204** and bit lines **206** to and from each memory cell **208**. Peripheral circuits **202** can include various types of peripheral circuits formed using CMOS technologies.

[0070] According to some aspects of the present disclosure, the vertical transistors of memory cells in a memory device (e.g., memory device **200**) are multi-gate transistors, and the gate dielectrics of vertical transistors in the word line direction are separate. For example, FIG. 5 illustrates a plan view of an array of memory cells **502** each including a vertical transistor in a memory device **500**, according to some aspects of the present disclosure. As shown in FIG. 5, memory device **500** can include a plurality of word lines **504** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **500**

can also include a plurality of bit lines **506** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **5** does not illustrate a cross-section of memory device **500** in the same lateral plane, and word lines **504** and bit lines **506** may be formed in different lateral planes for ease of routing as described below in detail.

[0071] Memory cells **502** can be formed at the intersections of word lines **504** and bit lines **506**. In some implementations, each memory cell **502** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **508** and a gate structure **510**. Semiconductor body **508** can extend in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a multi-gate transistor in which gate structure **510** is in contact with a plurality of sides (e.g., all 4 sides in FIG. **5**) of semiconductor body **508** (the active region in which channels are formed). As shown in FIG. **5**, the vertical transistor is a GAA transistor in which gate structure **510** fully circumscribes semiconductor body **508** in the plan view. That is, gate structure **510** circumscribes (e.g., surrounding and contacting) all four sides of semiconductor body **508** (having a rectangle or square-shaped cross-section) in the plan view, according to some implementations. Gate structure **510** can include a gate dielectric **512** fully circumscribes semiconductor body **508** in the plan view, and a gate electrode **514** fully circumscribes gate dielectric **512**. In some implementation, gate dielectric **512** is laterally between gate electrode **514** and semiconductor body **508** in the bit line direction and in the word line direction. As described above, gate electrode **514** may be part of word line **504**, and word line **504** may be an extension of gate electrode **514**. In some implementations, gate structures **510** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **5**.

[0072] As shown in FIG. **5**, gate electrodes **514** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **514** and **504**. In contrast, gate dielectrics **512** of adjacent vertical transistors in the word line direction are separate, e.g., not parts of a continuous dielectric layer having gate dielectrics **512**.

[0073] FIG. **6A** illustrates a side view of a cross-section of a 3D memory device **600** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **600** may be one example of memory device **500** including multi-gate vertical transistors in which gate structures fully circumscribes semiconductor bodies in the plan view, e.g., GAA vertical transistors. It is understood that FIG. **6A** is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device **100** described above with respect to FIG. **1A**, 3D memory device **600** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104** stacked over first semiconductor structure **102**. First and second semiconductor structures **102** and **104** are joined at bonding interface **106** therebetween, according to some implementations. As shown in FIG. **6A**, first semiconductor structure **102** can include a substrate **610**, which can include silicon (e.g., single crystalline silicon, c-Si), silicon germa-

nium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon-on-insulator (SOI), or any other suitable materials.

[0074] First semiconductor structure **102** can include peripheral circuits **612** on substrate **610**. In some implementations, peripheral circuits **612** includes a plurality of transistors **614** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **614**) can be formed on or in substrate **610** as well.

[0075] In some implementations, first semiconductor structure **102** further includes an interconnect layer **616** above peripheral circuits **612** to transfer electrical signals to and from peripheral circuits **612**. Interconnect layer **616** can include a plurality of interconnects (also referred to herein as “contacts”), including lateral interconnect lines and vertical interconnect access (VIA) contacts. As used herein, the term “interconnects” can broadly include any suitable types of interconnects, such as middle-end-of-line (MEOL) interconnects and back-end-of-line (BEOL) interconnects. Interconnect layer **616** can further include one or more interlayer dielectric (ILD) layers (also known as “intermetal dielectric (IMD) layers”) in which the interconnect lines and via contacts can form. That is, interconnect layer **616** can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits **612** are coupled to one another through the interconnects in interconnect layer **616**. The interconnects in interconnect layer **616** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

[0076] As shown in FIG. **6A**, first semiconductor structure **102** can further include a bonding layer **618** at bonding interface **106** and above interconnect layer **616** and peripheral circuits **612**. Bonding layer **618** can include a plurality of bonding contacts **619** and dielectrics electrically isolating bonding contacts **619**. Bonding contacts **619** can include conductive materials, such as Cu. The remaining area of bonding layer **618** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **619** and surrounding dielectrics in bonding layer **618** can be used for hybrid bonding. Similarly, as shown in FIG. **6A**, second semiconductor structure **104** can also include a bonding layer **620** at bonding interface **106** and above bonding layer **618** of first semiconductor structure **102**. Bonding layer **620** can include a plurality of bonding contacts **621** and dielectrics electrically isolating bonding contacts **621**. Bonding contacts **621** can include conductive materials, such as Cu. The remaining area of bonding layer **620** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **621** and surrounding dielectrics in bonding layer **620** can be used for hybrid bonding. Bonding contacts **621** are in contact with bonding contacts **619** at bonding interface **106**, according to some implementations. In some implementations, bonding layer **620** includes a dielectric layer opposing DRAM cells **624** with bit line **623** positioned between the dielectric layer and DRAM cells **624**, as shown in FIG. **6A**. The dielectric layer can include bonding interface **106** having bonding contacts **621**.

[0077] Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some implementations,

bonding interface **106** is disposed between bonding layers **620** and **618** as a result of hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface **106** is the place at which bonding layers **620** and **618** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **618** of first semiconductor structure **102** and the bottom surface of bonding layer **620** of second semiconductor structure **104**.

[0078] In some implementations, second semiconductor structure **104** further includes an interconnect layer **622** including bit lines **623** above bonding layer **620** to transfer electrical signals. Interconnect layer **622** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **622** also include local interconnects, such as bit lines **623** (e.g., an example of bit lines **506** in FIG. 5), bit line contacts **625** (which may be omitted in some examples), and word line contacts **627**. Interconnect layer **622** can further include one or more ILD layers in which the interconnect lines and via contacts can form. The interconnects in interconnect layer **622** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits **612** includes a word line driver/row decoder coupled to word line contacts **627** in interconnect layer **622** through bonding contacts **621** and **619** in bonding layers **620** and **618** and interconnect layer **616**. In some implementations, peripheral circuits **612** includes a bit line driver/column decoder coupled to bit lines **623** and bit line contacts **625** in interconnect layer **622** through bonding contacts **621** and **619** in bonding layers **620** and **618** and interconnect layer **616**. In some implementations, bit line **623** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line **623** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, bit line contact **625** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, bit line contact **625** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

[0079] In some implementations, second semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **624** (e.g., an example of memory cells **502** in FIG. 5) above interconnect layer **622** and bonding layer **620**. That is, interconnect layer **622** including bit lines **623** can be disposed between bonding layer **620** and array of DRAM cells **624**. It is understood that the cross-section of 3D memory device **600** in FIG. 6A may be made along the bit line direction (the y-direction), and one bit line **623** in interconnect layer **622** extending laterally in the y-direction may be coupled to a column of DRAM cells **624**.

[0080] Each DRAM cell **624** can include a vertical transistor **626** (e.g., an example of vertical transistors **210** in

FIG. 2) and capacitor **628** (e.g., an example of storage unit **212** in FIG. 2) coupled to the vertical transistor **626**. DRAM cell **624** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **624** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc.

[0081] Vertical transistor **626** can be a MOSFET used to switch a respective DRAM cell **624**. In some implementations, vertical transistor **626** includes a semiconductor body **630** (the active region in which multiple channels can form) extending vertically (in the z-direction), and a gate structure **636** in contact with a plurality of sides of semiconductor body **630**. As described above, as in a GAA vertical transistor, semiconductor body **630** can have a cuboid shape or a cylinder shape, and gate structure **636** can fully circumscribe semiconductor body **630** in the plan view, for example, as shown in FIG. 5. Gate structure **636** includes a gate electrode **634** and a gate dielectric **632** laterally between gate electrode **634** and semiconductor body **630**, according to some implementations. For example, for semiconductor body **630** having a cylinder shape, semiconductor body **630**, gate dielectric **632**, and gate electrode **634** may be disposed radially from the center of vertical transistor **626** in this order. In some implementations, gate dielectric **632** surrounds and contacts semiconductor body **630**, and gate electrode **634** surrounds and contacts gate dielectric **632**.

[0082] As shown in FIG. 6A, in some implementations, semiconductor body **630** has two ends (the upper end and lower end) in the vertical direction (the z-direction), and both ends extend beyond gate structure **636**, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body **630** can have a larger vertical dimension (e.g., the depth) than that of gate structure **636** (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body **630** is flush with the respective end of gate structure **636**. Thus, short circuits between bit lines **623** and word lines/gate electrodes **634** or between word lines/gate electrodes **634** and capacitors **628** can be avoided. In some implementations, the two ILD layers into which semiconductor body **630** extends (e.g., the ILD layer vertically between bit line contacts **625** and word lines **634**, and the ILD layer vertically between word lines **634** and capacitors **628**) include the same dielectric material, such as silicon oxide. Vertical transistor **626** can further include a source and a drain (both referred to as **638** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body **630**, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain **638** (e.g., at the upper end in FIG. 6A) is coupled to capacitor **628**, and the other one of source and drain **638** (e.g., at the lower end in FIG. 6A) is coupled to bit line **623** (e.g., through bit line contact **625** or directly). That is, vertical transistor **626** can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. 6A. In some implementations, a metal bit line (e.g., bit line **623** made of a metal material) is coupled to the second terminal of vertical transistor **626** via an ohmic contact (e.g., bit line contact **625** made of a metal silicide material).

[0083] In some implementations, semiconductor body **630** includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one

example, semiconductor body **630** may include single crystalline silicon. Source and drain **638** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **638** of vertical transistor **626** and bit line **623** as bit line contact **625** or between source/drain **638** of vertical transistor **626** and the first electrode of capacitor **628** as capacitor contact **642** to reduce the contact resistance. In some implementations, gate dielectric **632** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, aluminum oxide (Al_2O_3), hafnium oxide (HfO_2), tantalum oxide (Ta_2O_5), zirconium oxide (ZrO_2), titanium oxide (TiO_2), or any combination thereof. In some implementations, gate electrode **634** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode **634** includes multiple conductive layers, such as a W layer over a TiN layer. In one example, gate structure **636** may be a “gate oxide/gate poly” gate in which gate dielectric **632** includes silicon oxide and gate electrode **634** includes doped polysilicon. In another example, gate structure **636** may be a high-k metal gate (HKMG) in which gate dielectric **632** includes a high-k dielectric and gate electrode **634** includes a metal.

[0084] As described above, since gate electrode **634** may be part of a word line or extend in the word line direction (e.g., the x-direction in FIG. 5) as a word line, although not directly shown in FIG. 6A, second semiconductor structure **104** of 3D memory device **600** can also include a plurality of word lines (e.g., an example of word lines **504** in FIG. 5, referred to as **634** as well) each extending in the word line direction (the x-direction). Each word line **634** can be coupled to a row of DRAM cells **624**. That is, bit line **623** and word line **634** can extend in two perpendicular lateral directions, and semiconductor body **630** of vertical transistor **626** can extend in the vertical direction perpendicular to the two lateral directions in which bit line **623** and word line **634** extend. Word lines **634** are in contact with word line contacts **627**, according to some implementations. In some implementations, word lines **634** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, word line **634** includes multiple conductive layers, such as a W layer over a TiN layer.

[0085] As shown in FIG. 6A, vertical transistor **626** extends vertically through and contacts word lines **634**, and source or drain **638** of vertical transistor **626** at the lower end thereof is in contact with bit line contact **625** or in contact with bit line **623** directly, according to some implementations. Accordingly, word lines **634** and bit lines **623** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **626**, which simplifies the routing of word lines **634** and bit lines **623**. In some implementations, bit lines **623** are disposed vertically between bonding layer **620** and word lines **634**, and word lines **634** are disposed vertically between bit lines **623** and capacitors **628**. Word lines **634** can be coupled to peripheral circuits **612** in first semiconductor structure **102** through word line contacts **627** in interconnect layer **622**, bonding contacts **621** and **619** in bonding layers **620** and **618**, and the interconnects in interconnect layer **616**. Similarly, bit lines **623** in interconnect layer **622** can be coupled to peripheral

circuits **612** in first semiconductor structure **102** through bonding contacts **621** and **619** in bonding layers **620** and **618** and the interconnects in interconnect layer **616**.

[0086] In some implementations, second semiconductor structure **104** further includes a plurality of air gaps **640** each disposed laterally between adjacent word lines **634**. Each air gap **640** can be a trench extending in the word line direction (e.g., the x-direction) in parallel with word lines **634** to separate adjacent rows of vertical transistors **626**. As described below with respect to the fabrication process, air gaps **640** may be formed due to the relatively small pitches of word lines **634** (and rows of DRAM cells **624**) in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps **640** (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between word lines **634** (and rows of DRAM cells **624**) compared with some dielectrics (e.g., silicon oxide).

[0087] As shown in FIG. 6A, in some implementations, capacitor **628** includes a first electrode above and coupled to source or drain **638** of vertical transistor **626**, e.g., the upper end of semiconductor body **630**, via a capacitor contact **642**. In some implementations, capacitor contact **642** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact **642** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor **628** can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor **628** can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **638** of a respective vertical transistor **626** in the same DRAM cell, while all second electrodes are coupled to a common plate **646** coupled to the ground, e.g., a common ground. Capacitor **628** can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in FIG. 6A. In some implementations, the first end of capacitor **628** is coupled to the first terminal of vertical transistor **626** via an ohmic contact (e.g., capacitor contact **642** made of a metal silicide material). It is understood that in some examples, the capacitor contacts described herein (e.g., **642**, **1024**, **1124**, **1224**, **1324**, **1742**, **2142**, and **2228**) may not be included in the memory devices. As shown in FIG. 6A, second semiconductor structure **104** can further include a capacitor contact **647** (e.g., a conductor) in contact with common plate **646** for coupling capacitors **628** to peripheral circuits **612** or to the ground directly. In some implementations, capacitor contact **647** (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer **620** to couple to the second end of capacitor **628** via common plate **646**, as shown in FIG. 6A. In some implementation, the ILD layer in which capacitors **628** are formed has the same dielectric material as the two ILD layers into which semiconductor body **630** extends, such as silicon oxide.

[0088] It is understood that the structure and configuration of capacitor **628** are not limited to the example in FIG. 6A and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a

substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al_2O_3 , HfO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , or any combination thereof. It is understood that in some examples, capacitor 628 may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as lead zirconate titanate (PZT) or strontium bismuth tantalate (SBT). In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

[0089] As shown in FIG. 6A, vertical transistor 626 extends vertically through and contacts word lines 634, source or drain 638 of vertical transistor 626 at the lower end thereof is in contact with bit line contact 625, and source or drain 638 of vertical transistor 626 at the upper end thereof is in contact with capacitor contact 642, according to some implementations. That is, bit line 623 and capacitor 628 can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor 626 of DRAM cell 624 in the vertical direction due to the vertical arrangement of vertical transistor 626. In some implementations, bit line 623 and capacitor 628 are disposed on opposite sides of vertical transistor 626 in the vertical direction, which simplifies the routing of bit lines 623 and reduces the coupling capacitance between bit lines 623 and capacitors 628 compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.

[0090] As shown in FIG. 6A, in some implementations, vertical transistors 626 are disposed vertically between capacitors 628 and bonding interface 106. That is, vertical transistors 626 can be arranged closer to peripheral circuits 614 of first semiconductor structure 102 and bonding interface 106 than capacitors 628. Since bit lines 623 and capacitors 628 are coupled to opposite ends of vertical transistors 626, as described above, bit lines 623 (as part of interconnect layer 622) are disposed vertically between vertical transistors 626 and bonding interface 106, according to some implementations. As a result, interconnect layer 622 including bit lines 623 can be arranged close to bonding interface 106 to reduce the interconnect routing distance and complexity.

[0091] In some implementations, second semiconductor structure 104 further includes a substrate 648 disposed above DRAM cells 624. As described below with respect to the fabrication process, substrate 648 can be part of a carrier wafer. It is understood that in some examples, substrate 648 may not be included in second semiconductor structure 104.

[0092] As shown in FIG. 6A, second semiconductor structure 104 can further include a pad-out interconnect layer 650 above substrate 648 and DRAM cells 624. Pad-out interconnect layer 650 can include interconnects, e.g., contact pads 654, in one or more ILD layers. Pad-out interconnect layer 650 and interconnect layer 622 can be formed on opposite sides of DRAM cells 624. Capacitors 628 are disposed vertically between vertical transistors 626 and pad-out interconnect layer 650, according to some implementations. In some implementations, the interconnects in pad-out interconnect layer 650 can transfer electrical signals between 3D memory device 600 and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure 104 further includes one or more

contacts 652 extending through substrate 648 and part of pad-out interconnect layer 650 to couple pad-out interconnect layer 650 to DRAM cells 624 and interconnect layer 622. As a result, peripheral circuits 612 can be coupled to DRAM cells 624 through interconnect layers 616 and 622 as well as bonding layers 620 and 618, and peripheral circuits 612 and DRAM cells 624 can be coupled to outside circuits through contacts 652 and pad-out interconnect layer 650. Contact pads 654 and contacts 652 can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad 654 may include Al, and contact 652 may include W. In some implementations, contact 652 includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate 648. Depending on the thickness of substrate 648, contact 652 can be an interlayer via (ILV) having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a through substrate via (TSV) having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

[0093] It is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure 104 having DRAM cells 624 as shown in FIG. 6A and may be from first semiconductor structure 102 having peripheral circuit 612. For example, as shown in FIG. 6B, a 3D memory device 601 may include pad-out interconnect layer 650 in first semiconductor structure 102. Pad-out interconnect layer 650 can be disposed above and in contact with substrate 610 of first semiconductor structure 102 on which transistors 614 of peripheral circuit 612 are formed. In some implementations, first semiconductor structure 102 further includes one or more contacts 653 extending vertically through substrate 610. In some implementations, contact 653 couples the interconnects in interconnect layer 616 in first semiconductor structure 102 to contact pads 654 in pad-out interconnect layer 650 to make an electrical connection through substrate 610. Contacts 653 can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact 653 may include W. In some implementations, contact 653 includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate 610. It is understood that in some examples, substrate 610 in FIG. 6B may be a thinned substrate, e.g., compared with substrate 610 in FIG. 6A. Depending on the thickness of substrate 610, contact 653 can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices 600 and 601 are not repeated for case of description. Pad-out from first semiconductor structure 102 including peripheral circuits 612 can reduce the interconnect distance between contact pad 654 and peripheral circuits 612, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device 601.

[0094] It is also understood that the relative vertical positions between the semiconductor body and the respective gate structure and word line are not limited to the example shown in FIG. 6A in which both the upper and lower ends of semiconductor body 630 extend beyond gate structure 636 (and word line 634), respectively, depending on the

various fabrication processes as described below in detail. For example, as shown in FIG. 6C, a 3D memory device 603 may include vertical transistors 626 each having semiconductor body 630 and gate structure 636, and one end of semiconductor body 630 in the vertical direction (the z-direction) may be flush with gate structure 636. In some implementations, the upper or lower end of semiconductor body 630 that is in contact with capacitor contact 642 is flush with the respective end of gate structure 636 and word line 634. That is, one of the upper and lower ends of semiconductor body 630 that is in contact with capacitor 628 does not extend beyond the respective end of gate structure 636 and word line 634, according to some implementations. In some implementations, as shown in FIG. 6C, the other end of semiconductor body 630 in the vertical direction that is in contact with bit line contact 625 still extends beyond the respective end of gate structure 636 and word line 634 into an ILD layer, which is vertically between bit line contacts 625 and word lines 634. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices 600 and 603 are not repeated for ease of description.

[0095] It is further understood that the dielectric materials of the ILD layers into which the semiconductor bodies extend are not limited to the example shown in FIG. 6A in which the ILD layers include silicon oxide, e.g., the same material as the ILD layer in which capacitors 628 are formed, depending on the various fabrication processes as described below in detail. For example, as shown in FIG. 6D, a 3D memory device 605 may include two ILD layers 660 and 662 into which semiconductor body 630 extends. ILD layer 660 is vertically between bit line contacts 625 and word lines 634, and ILD layer 662 is vertically between word lines 634 and capacitor contact 642, according to some implementations. ILD layers 660 and 662 can include a dielectric material that is different from the dielectric material of the ILD layer in which capacitors 628 are formed. In some implementations, ILD layers 660 and 662 include silicon nitride, while the ILD layer of capacitors 628 includes silicon oxide. As shown in FIG. 6D, in some implementations, one end of semiconductor body 630 in the vertical direction (the z-direction) that is in contact with capacitor 628 is flush with the respective end of ILD layer 662. In some implementations, air gap 640 extends vertically through ILD layer 662 to separate ILD layer 662, but does not extend further into ILD layer 660, e.g., being stopped by ILD layer 660. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices 600 and 605 are not repeated for ease of description.

[0096] It is further understood that the air gaps between word lines may be partially or fully filled with dielectrics. For example, as shown in FIG. 6E, a memory device 607 may not include air gaps (e.g., air gaps 640 in FIG. 6A) between adjacent word lines 634. Instead, memory device 607 can include dielectric wall structures 641 (e.g., filled with dielectrics, such as silicon oxide) each formed between adjacent word lines 634. It is understood that in some examples (not shown), air gaps 640 may still exist between word lines 634, but with a smaller vertical dimension (depth) compared with word lines 634. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices 600 and 607 are not repeated for ease of description.

[0097] It is further understood that more than one DRAM cell array may be stacked over one another to vertically scale up the number of DRAM cells. For example, as shown in FIG. 7, a memory device 700 may further include a third semiconductor structure 702 having an array of DRAM cells 624 stacked over second and first semiconductor structures 104 and 102. Third and second semiconductor structures 702 and 104 are joined at another bonding interface 704 therebetween, according to some implementations. Third and second semiconductor structures 702 and 104 may have the same arrays of DRAM cells 624 and interconnect layers 622 and thus, the details of DRAM cells 624 and interconnect layer 622 in third semiconductor structure 702 are not repeated for ease of description.

[0098] Third and second semiconductor structures 702 and 104 can be bonded in a face-to-face manner, such that at least some components (e.g., DRAM cells 624) in third and second semiconductor structures 702 and 104 can be symmetric with respect to bonding interface 704, according to some implementations. Bonding interface 704 can be formed vertically between DRAM cells 624 in third semiconductor structure 702 and DRAM cells 624 in second semiconductor structure 104. As shown in FIG. 7, in some implementations, capacitors 628 in second semiconductor structure 104 are disposed vertically between bonding interface 704 and vertical transistors 626 in second semiconductor structure 104, and capacitors 628 in third semiconductor structure 702 are disposed vertically between bonding interface 704 and vertical transistors 626 in third semiconductor structure 702. That is, capacitors 628 in second semiconductor structure 104 and capacitors 628 in third semiconductor structure 702 can be disposed on opposite sides of bonding interface 704. In some implementations, the second electrodes of capacitors 628 in third semiconductor structure 702 are in contact with common plate 646 at bonding interface 704.

[0099] In some implementations, 3D memory device 700 includes additional interconnects that couple DRAM cells 624 in third semiconductor structure 702 to peripheral circuits 612 across bonding interfaces 704 and 106, such as word line contacts 734 coupling word lines 634 in third semiconductor structure 702 and peripheral circuits 612 in first semiconductor structure 102. As shown in FIG. 7, third semiconductor structure 702, as opposed to first or second semiconductor structure 102 or 104, can include pad-out interconnect layer 650. In some implementations, vertical transistors 626 in third semiconductor structure 702 are disposed vertically between capacitors 628 in third semiconductor structure 702 and pad-out interconnect layer 650. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices 600 and 700 are not repeated for ease of description.

[0100] It is understood that the architecture of multiple memory cell arrays shown in FIG. 7 is not limited to the design of DRAM cells 624 and may be applied to any suitable memory cells disclosed herein. It is also understood that various designs of memory cells disclosed herein may be mixed in the architecture of multiple memory cell arrays shown in FIG. 7. For example, second and third semiconductor structures 104 and 702 may include different designs of memory cells disclosed herein.

[0101] It is further understood that the memory cell array is not limited to the example shown in FIGS. 5, 6A-6D, and

7 in which the vertical transistors are GAA transistors and may be any other suitable multi-gate vertical transistors. For example, FIG. 8 illustrates a plan view of another array of memory cells **802** each including a vertical transistor in a memory device **800**, according to some aspects of the present disclosure. As shown in FIG. 8, memory device **800** can include a plurality of word lines **804** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **800** can also include a plurality of bit lines **806** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. 8 does not illustrate a cross-section of memory device **800** in the same lateral plane, and word lines **804** and bit lines **806** may be formed in different lateral planes for ease of routing as described below in detail.

[0102] Memory cells **802** can be formed at the intersections of word lines **804** and bit lines **806**. In some implementations, each memory cell **802** includes a vertical transistor (e.g., vertical transistor **210** in FIG. 2) having a semiconductor body **808** and a gate structure **810**. The vertical transistor of memory cell **802** in FIG. 8 may be an example of a tri-gate vertical transistor (e.g., tri-side gate vertical transistor). Semiconductor body **808** can extend in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. Gate structure **810** can be in contact with a plurality of sides (e.g., three of all four sides in FIG. 8) of semiconductor body **808** (the active region in which channels are formed). That is, different from the GAA vertical transistor in memory cell **502** in FIG. 5, gate structure **810** of the vertical transistor in memory cell **802** partially circumscribes semiconductor body **808** in the plan view. That is, gate structure **810** circumscribes (e.g., surrounding and contacting) three of all four sides of semiconductor body **808** (having a rectangle or square-shaped cross-section) in the plan view, according to some implementations. Gate structure **810** does not surround and contact at least one side of semiconductor body **808**, according to some implementations. Gate structure **810** can include a gate dielectric **812** partially or fully circumscribes semiconductor body **808** in the plan view, and a gate electrode **814** partially circumscribes gate dielectric **812**. Thus, the vertical transistor having gate structure **810** may be viewed as a “tri-gate” vertical transistor in which gate structure **810** is in contact with two opposite sides of semiconductor body **808** in the word line direction (the x-direction) and one side of semiconductor body **808** in the bit line direction (the y-direction). As described above, gate electrode **814** may be part of word line **804**, and word line **804** may be an extension of gate electrode **814**. For example, as shown in FIG. 8, one edge of each word line **804** may be formed aligned with the same side of each semiconductor body **808**, such that gate electrode **814** may not extend to the side of semiconductor body **808** to form a GAA transistor. By arranging semiconductor bodies **808** of memory cells **802** to be aligned with one side of word lines **804**, the pitch of word lines **804** and/or the pitch of memory cells **802** in the bit line direction (the y-direction) can be further increased to reduce the fabrication complexity and increase the production yield. In some implementations, gate structures **810** of a row of the vertical transistors are continuous in the x-direction, as shown in FIG. 8.

[0103] Similar to memory device **500** in FIG. 5, as shown in FIG. 8, gate electrodes **814** of adjacent vertical transistors

in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **814** and **804**. In contrast, gate dielectrics **812** of adjacent vertical transistors in the word line direction are separate, e.g., not parts of a continuous dielectric layer having gate dielectrics **812**.

[0104] FIG. 9 illustrates a side view of a cross-section of a 3D memory device **900** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **900** may be one example of memory device **800** including multi-gate vertical transistors in which gate structures partially circumscribe semiconductor bodies in the plan view. 3D memory device **900** is similar to 3D memory device **600** in FIG. 6A except for the different structures of multi-gate vertical transistors in DRAM cells **624**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **900** are not repeated for ease of description. Similar to FIG. 6A, the cross-section of 3D memory device **900** in FIG. 9 may be made along the bit line direction (the y-direction).

[0105] Vertical transistor **926** can be a MOSFET used to switch a respective DRAM cell **624**. In some implementations, vertical transistor **926** includes semiconductor body **630** (the active region in which multiple channels can form) extending vertically (in the z-direction), and a gate structure **936** in contact with a plurality of sides of semiconductor body **630**. Semiconductor body **630** can have a cuboid shape or a cylinder shape, and gate structure **936** can partially circumscribe semiconductor body **630** in the plan view, for example, as shown in FIG. 8. As shown in FIG. 9, gate structure **936** does not extend to at least one side of semiconductor body **630**, according to some implementations. Gate structure **936** includes a gate electrode **934** and a gate dielectric **932** laterally between gate electrode **934** and semiconductor body **630**, according to some implementations. As shown in FIG. 9, gate electrode **934** does not extend to at least one side of semiconductor body **630**, according to some implementation. Due to the increased pitch of word lines **934** and/or the pitch of DRAM cells **624** in the bit line direction (the y-direction), the air gaps between word lines **934** may be partially or fully filled with dielectrics.

[0106] It is also understood that the number of gates in multi-gate transistors may vary, e.g., not limited by the GAA vertical transistor example in FIG. 5 and the tri-gate vertical transistor example in FIG. 8. For example, multi-gate vertical transistors may also include double-gate vertical transistors (a.k.a. dual-side gate vertical transistors) in which the gate structure is in contact with two sides of the semiconductor body, such as the two opposite sides in the bit line direction or in the word line direction.

[0107] It is further understood that although storage units are described as capacitors **628** in FIGS. 6A-6D, 7, and 9, storage units may include any other suitable devices, such as PCM elements, as described above with respect to FIG. 4 in some examples. For example, the capacitor dielectric of capacitor **628** may be replaced with a phase-change material layer, such as chalcogenide alloys, vertically sandwiched between the electrodes to form a PCM element. Also, instead of coupling source or drain **638** of vertical transistor **626** or **926** to bit line **623**, the electrode of the PCM element

may be coupled to bit line **623**, while source or drain **638** of vertical transistor **626** or **926** may be coupled to the ground, e.g., a common ground plate.

[0108] According to some aspects of the present disclosure, the vertical transistors of memory cells in a memory device (e.g., memory device **200**) are single-gate transistors, and the gate dielectrics of vertical transistors in the word line direction are continuous. For example, FIG. **16** illustrates a plan view of still another array of memory cells **1602** each including a vertical transistor in a memory device **1600**, according to some aspects of the present disclosure. As shown in FIG. **16**, memory device **1600** can include a plurality of word lines **1604** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **1600** can also include a plurality of bit lines **1606** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **16** does not illustrate a cross-section of memory device **1600** in the same lateral plane, and word lines **1604** and bit lines **1606** may be formed in different lateral planes for ease of routing as described below in detail.

[0109] Memory cells **1602** can be formed at the intersections of word lines **1604** and bit lines **1606**. In some implementations, each memory cell **1602** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **1608** and a gate structure **1610**. Semiconductor body **1608** can extend in a substrate in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a single-gate transistor in which gate structure **1610** is in contact with a single side (e.g., one of four sides in FIG. **16**) of semiconductor body **1608** (the active region in which channels are formed). As shown in FIG. **16**, the vertical transistor is a single-gate transistor in which gate structure **1610** abuts one side of semiconductor body **1608** (having a rectangle or square-shaped cross-section) in the bit line direction (the y-direction) in the plan view. Gate structure **1610** does not surround and contact other three sides of semiconductor body **1608**, according to some implementations. Gate structure **1610** can include a gate dielectric **1612** abuts one side of semiconductor body **1608** in the plan view, and a gate electrode **1614** in contact with gate dielectric **1612**. In some implementations, gate dielectric **1612** is laterally between gate electrode **1614** and semiconductor body **1608** in the bit line direction (the y-direction). As described above, gate electrode **1614** may be part of word line **1604**, and word line **1604** may be an extension of gate electrode **1614**. That is, gate electrodes **1614** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **1614** and **1604**. In some implementations, gate structures **1610** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **16**.

[0110] Different from separate gate dielectrics **512** and **812** in FIGS. **5** and **8**, as shown in FIG. **16**, gate dielectrics **1612** of adjacent vertical transistors in the word line direction are continuous, e.g., parts of a continuous dielectric layer having gate dielectrics **1612** and extending in the word line direction to abut vertical transistors in the same row on the same side. Gate structures **1610** can be thus viewed as parts of a continuous structure extending in the word line

direction at which the continuous structure abut vertical transistors in the same row on the same side.

[0111] As shown in FIG. **16**, two adjacent vertical transistors of memory cells (e.g., **1602A** and **1602B**) in the bit line direction (the y-direction) are mirror-symmetric to one another, according to some implementations. As described below with respect to the fabrication process, semiconductor bodies **1608** of each pair of two adjacent vertical transistors of memory cells (e.g., **1602A** and **1602B**) in the bit line direction (the y-direction) can be formed by separating a semiconductor pillar into two pieces using a trench isolation **1616** extending in the word line direction (the x-direction) and in parallel with word lines **1604**. Trench isolations **1616** and word lines **1604** can be disposed in an interleaved manner in the bit line direction. In some implementations, trench isolation **1616** is formed in the middle of the semiconductor pillars (not shown) such that the resulting pair of semiconductor bodies **1608** are mirror-symmetric to one another with respect to trench isolation **1616**, so are the pair of vertical transistors having semiconductor bodies **1608** when the respective gate structures **1610** are mirror-symmetric to one another with respect to trench isolation **1616** as well.

[0112] It is understood that in some examples, trench isolations **1616** extending in the word line directions may not be formed such that two adjacent semiconductor bodies **1608** separated by a respective trench isolation **1616** may be merged as a single semiconductor body having two opposite sides in the bit line direction in contact with gate structure **1610**. That is, without trench isolations **1616**, the adjacent single-gate vertical transistors may be merged to form a double-gate vertical transistor with increased gate control area and lower leakage current. The gate structure of the double-gate vertical transistor may include two mirror-symmetric gate structures **1610** in FIG. **16**, such that both sides of the merged semiconductor body **1608** in the bit line direction may be in contact with the gate structure in the double-gate vertical transistor. On the other hand, by splitting the double-gate vertical transistors into single-gate vertical transistors using trench isolations **1616**, the number of memory cells **1602** (and the cell density) in the bit line direction can be doubled compared to double-gate vertical transistors without unduly complexing the fabrication process (e.g., compared with using SADP process).

[0113] FIG. **17** illustrates a side view of a cross-section of yet another 3D memory device **1700** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **1700** may be one example of memory device **1600** including single-gate vertical transistors in which gate structures abut a single side of semiconductor bodies in the plan view. It is understood that FIG. **17** is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device **100** described above with respect to FIG. **1A**, 3D memory device **1700** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104** stacked over first semiconductor structure **102**. First and second semiconductor structures **102** and **104** are jointed at bonding interface **106** therebetween, according to some implementations. As shown in FIG. **17**, first semiconductor structure **102** can include a substrate **1710**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials.

[0114] First semiconductor structure **102** can include peripheral circuits **1712** on substrate **1710**. In some implementations, peripheral circuits **1712** include a plurality of transistors **1714** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **1714**) can be formed on or in substrate **1710** as well.

[0115] In some implementations, first semiconductor structure **102** further includes an interconnect layer **1716** above peripheral circuits **1712** to transfer electrical signals to and from peripheral circuits **1712**. Interconnect layer **1716** can include a plurality of interconnects (also referred to herein as “contacts”), including lateral interconnect lines and VIA contacts. Interconnect layer **1716** can further include one or more ILD layers in which the interconnect lines and via contacts can form. That is, interconnect layer **1716** can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits **1712** are coupled to one another through the interconnects in interconnect layer **1716**. The interconnects in interconnect layer **1716** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

[0116] As shown in FIG. 17, first semiconductor structure **102** can further include a bonding layer **1718** at bonding interface **106** and above interconnect layer **1716** and peripheral circuits **1712**. Bonding layer **1718** can include a plurality of bonding contacts **1719** and dielectrics electrically isolating bonding contacts **1719**. Bonding contacts **1719** can include conductive materials, such as Cu. The remaining area of bonding layer **1718** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1719** and surrounding dielectrics in bonding layer **1718** can be used for hybrid bonding. Similarly, as shown in FIG. 17, second semiconductor structure **104** can also include a bonding layer **1720** at bonding interface **106** and above bonding layer **1718** of first semiconductor structure **102**. Bonding layer **1720** can include a plurality of bonding contacts **1721** and dielectrics electrically isolating bonding contacts **1721**. Bonding contacts **1721** can include conductive materials, such as Cu. The remaining area of bonding layer **1720** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1721** and surrounding dielectrics in bonding layer **1720** can be used for hybrid bonding. Bonding contacts **1721** are in contact with bonding contacts **1719** at bonding interface **106**, according to some implementations. In some implementations, bonding layer **1720** includes a dielectric layer opposing DRAM cells **1724** with bit line **1723** positioned between the dielectric layer and DRAM cells **1724**, as shown in FIG. 17. The dielectric layer can include bonding interface **106** having bonding contacts **1721**.

[0117] Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some implementations, bonding interface **106** is disposed between bonding layers **1720** and **1718** as a result of hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhe-

sives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface **106** is the place at which bonding layers **1720** and **1718** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **1718** of first semiconductor structure **102** and the bottom surface of bonding layer **1720** of second semiconductor structure **104**.

[0118] In some implementations, second semiconductor structure **104** further includes an interconnect layer **1722** including bit lines **1723** above bonding layer **1720** to transfer electrical signals. Interconnect layer **1722** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1722** also include local interconnects, such as bit lines **1723** (e.g., an example of bit lines **1606** in FIG. 16) and word line contacts (not shown). Interconnect layer **1722** can further include one or more ILD layers in which the interconnect lines and via contacts can form. The interconnects in interconnect layer **1722** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits **1712** include a word line driver/row decoder coupled to the word line contacts in interconnect layer **1722** through bonding contacts **1721** and **1719** in bonding layers **1720** and **1718** and interconnect layer **1716**. In some implementations, peripheral circuits **1712** include a bit line driver/column decoder coupled to bit lines **1723** and bit line contacts (not shown in FIG. 17, e.g., bit line contacts **625** in FIGS. 6A-6E) in interconnect layer **1722** through bonding contacts **1721** and **1719** in bonding layers **1720** and **1718** and interconnect layer **1716**. In some implementations, bit line **1723** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line **1723** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, the bit line contact (not shown in FIG. 17, e.g., bit line contacts **625** in FIGS. 6A-6E) is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, the bit line contact may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

[0119] In some implementations, second semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **1724** (e.g., an example of memory cells **1602** in FIG. 16) above interconnect layer **1722** and bonding layer **1720**. That is, interconnect layer **1722** including bit lines **1723** can be disposed between bonding layer **1720** and array of DRAM cells **1724**. It is understood that the cross-section of 3D memory device **1700** in FIG. 17 may be made along the bit line direction (the y-direction), and one bit line **1723** in interconnect layer **1722** extending laterally in the y-direction may be coupled to a column of DRAM cells **1724**.

[0120] Each DRAM cell **1724** can include a vertical transistor **1726** (e.g., an example of vertical transistors **210** in FIG. 2) and capacitor **1728** (e.g., an example of storage unit **212** in FIG. 2) coupled to the vertical transistor **1726**. DRAM cell **1724** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell

1724 may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc. To better illustrate vertical transistor 1726, FIG. 18 illustrates a perspective view of an array of vertical transistor 1726, according to some aspects of the present disclosure. FIGS. 17 and 18 will be described together when describing vertical transistors 1726.

[0121] Vertical transistor 1726 can be a MOSFET used to switch a respective DRAM cell 1724. In some implementations, vertical transistor 1726 includes a semiconductor body 1730 (the active region in which a channel can form) extending vertically (in the z-direction), and a gate structure 1736 in contact with one side of semiconductor body 1730 in the bit line direction (the y-direction). As described above, as in a single-gate vertical transistor, semiconductor body 1730 can have a cuboid shape or a cylinder shape, and gate structure 1736 can abut a single side of semiconductor body 1730 in the plan view, for example, as shown in FIGS. 17 and 18. Gate structure 1736 includes a gate electrode 1734 and a gate dielectric 1732 laterally between gate electrode 1734 and semiconductor body 1730 in the bit line direction, according to some implementations. In some implementations, gate dielectric 1732 abuts one side of semiconductor body 1730, and gate electrode 1734 abuts gate dielectric 1732.

[0122] As shown in FIGS. 17 and 18, in some implementations, semiconductor body 1730 has two ends (the upper end and lower end) in the vertical direction (the z-direction), and at least one end (e.g., the lower end in FIGS. 17 and 18) extends beyond gate dielectric 1732 in the vertical direction (the z-direction) into ILD layers. In some implementations, one end (e.g., the upper end in FIGS. 17 and 18) of semiconductor body 1730 is flush with the respective end (e.g., the upper end in FIGS. 17 and 18) of gate dielectric 1732. In some implementations, both ends (the upper end and lower end) of semiconductor body 1730 extend beyond gate electrode 1734, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body 1730 can have a larger vertical dimension (e.g., the depth) than that of gate electrode 1734 (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body 1730 is flush with the respective end of gate electrode 1734. Thus, short circuits between bit lines 1723 and word lines/gate electrodes 1734 or between word lines/gate electrodes 1734 and capacitors 1728 can be avoided. Vertical transistor 1726 can further include a source and a drain (both referred to as 1738 as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body 1730, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain 1738 (e.g., at the upper end in FIGS. 17 and 18) is coupled to capacitor 1728, and the other one of source and drain 1738 (e.g., at the lower end in FIGS. 17 and 18) is coupled to bit line 1723. That is, vertical transistor 1726 can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. 17. In some implementations, a metal bit line (e.g., bit line 1723 made of a metal material) is coupled to the second terminal of vertical transistor 1726 via an ohmic contact (e.g., a bit line contact made of a metal silicide material).

[0123] In some implementations, semiconductor body 1730 includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In

one example, semiconductor body 1730 may include single crystalline silicon. Source and drain 1738 can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain 1738 of vertical transistor 1726 and bit line 1723 as the bit line contact or between source/drain 1738 of vertical transistor 1726 and the first electrode of capacitor 1728 as capacitor contact 1742 to reduce the contact resistance. In some implementations, gate dielectric 1732 includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al_2O_3 , HfO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , or any combination thereof. In some implementations, gate electrode 1734 includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode 1734 includes multiple conductive layers, such as a W layer over a TiN layer, as shown in FIGS. 17 and 18. In one example, gate structure 1736 may be a “gate oxide/gate poly” gate in which gate dielectric 1732 includes silicon oxide and gate electrode 1734 includes doped polysilicon. In another example, gate structure 1736 may be an HKMG in which gate dielectric 1732 includes a high-k dielectric and gate electrode 1734 includes a metal.

[0124] As described above, since gate electrode 1734 may be part of a word line or extend in the word line direction (e.g., the x-direction in FIG. 18) as a word line, as shown in FIG. 18, second semiconductor structure 104 of 3D memory device 1700 can also include a plurality of word lines (e.g., an example of word lines 1604 in FIG. 16, referred to as 1734 as well) each extending in the word line direction (the x-direction). Each word line 1734 can be coupled to a row of DRAM cells 1724. That is, bit line 1723 and word line 1734 can extend in two perpendicular lateral directions, and semiconductor body 1730 of vertical transistor 1726 can extend in the vertical direction perpendicular to the two lateral directions in which bit line 1723 and word line 1734 extend. Word lines 1734 are in contact with word line contacts (not shown), according to some implementations. In some implementations, word lines 1734 include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, word line 1734 includes multiple conductive layers, such as a W layer over a TiN layer, as shown in FIG. 17.

[0125] As shown in FIGS. 17 and 18, vertical transistor 1726 extends vertically through and contacts word lines 1734, and source or drain 1738 of vertical transistor 1726 at the lower end thereof is in contact with bit line 1723 (or bit line contact if any), according to some implementations. Accordingly, word lines 1734 and bit lines 1723 can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor 1726, which simplifies the routing of word lines 1734 and bit lines 1723. In some implementations, bit lines 1723 are disposed vertically between bonding layer 1720 and word lines 1734, and word lines 1734 are disposed vertically between bit lines 1723 and capacitors 1728. Word lines 1734 can be coupled to peripheral circuits 1712 in first semiconductor structure 102 through word line contacts (not shown) in interconnect layer 1722, bonding contacts 1721 and 1719 in bonding layers 1720 and 1718, and the interconnects in interconnect layer 1716. Similarly, bit lines 1723 in interconnect layer

1722 can be coupled to peripheral circuits 1712 in first semiconductor structure 102 through bonding contacts 1721 and 1719 in bonding layers 1720 and 1718 and the interconnects in interconnect layer 1716.

[0126] As described above with respect to FIG. 16, vertical transistors 1726 can be arranged in a mirror-symmetric manner to increase the density of DRAM cells 1724 in the bit line direction (the y-direction). As shown in FIG. 17, two adjacent vertical transistors 1726 in the bit line direction are mirror-symmetric to one another with respect to a trench isolation 1760 (e.g., corresponding to trench isolation 1616 in FIG. 16), according to some implementations. That is, second semiconductor structure 104 can include a plurality of trench isolations 1760 each extending in the word line direction (the x-direction) in parallel with word lines 1734 and disposed between semiconductor bodies 1730 of two adjacent rows of vertical transistors 1726. In some implementations, the rows of vertical transistors 1726 separated by trench isolation 1760 are mirror-symmetric to one another with respect to trench isolation 1760. Trench isolation 1760 can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. It is understood that trench isolation 1760 may include an air gap each disposed laterally between adjacent semiconductor bodies 1730. As described below with respect to the fabrication process, air gaps may be formed due to the relatively small pitches of vertical transistors 1726 in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between vertical transistors 1726 (and rows of DRAM cells 1724) compared with some dielectrics (e.g., silicon oxide). Similarly, in some implementations, air gaps are formed laterally between word lines/gate electrodes 1734 in the bit line direction as well, depending on the pitches of word lines/gate electrodes 1734 in the bit line direction.

[0127] As shown in FIG. 17, in some implementations, capacitor 1728 includes a first electrode above and coupled to source or drain 1738 of vertical transistor 1726, e.g., the upper end of semiconductor body 1730, via a capacitor contact 1742. In some implementations, capacitor contact 1742 is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact 1742 may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor 1728 can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor 1728 can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain 1738 of a respective vertical transistor 1726 in the same DRAM cell, while all second electrodes are coupled to a common plate 1746 coupled to the ground, e.g., a common ground. Capacitor 1728 can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in FIG. 17. In some implementations, the first end of capacitor 1728 is coupled to the first terminal of vertical transistor 1726 via an ohmic contact (e.g., capacitor contact

1742 made of a metal silicide material). As shown in FIG. 17, second semiconductor structure 104 can further include a capacitor contact 1747 (e.g., a conductor) in contact with common plate 1746 for coupling capacitors 1728 to peripheral circuits 1712 or to the ground directly. In some implementations, capacitor contact 1747 (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer 1720 to couple to the second end of capacitor 1728 via common plate 1746, as shown in FIG. 17. In some implementation, the ILD layer in which capacitors 1728 are formed has the same dielectric material as the two ILD layers into which semiconductor body 1730 extends, such as silicon oxide.

[0128] It is understood that the structure and configuration of capacitor 1728 are not limited to the example in FIG. 17 and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al_2O_3 , HfO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , or any combination thereof. It is understood that in some examples, capacitor 1728 may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as PZT or SBT. In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

[0129] As shown in FIG. 17, vertical transistor 1726 extends vertically through and contacts word lines 1734, source or drain 1738 of vertical transistor 1726 at the lower end thereof is in contact with bit line 1723, and source or drain 1738 of vertical transistor 1726 at the upper end thereof is coupled to capacitor 1728, according to some implementations. That is, bit line 1723 and capacitor 1728 can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor 1726 of DRAM cell 1724 in the vertical direction due to the vertical arrangement of vertical transistor 1726. In some implementations, bit line 1723 and capacitor 1728 are disposed on opposite sides of vertical transistor 1726 in the vertical direction, which simplifies the routing of bit lines 1723 and reduces the coupling capacitance between bit lines 1723 and capacitors 1728 compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.

[0130] As shown in FIG. 17, in some implementations, vertical transistors 1726 are disposed vertically between capacitors 1728 and bonding interface 106. That is, vertical transistors 1726 can be arranged closer to peripheral circuits 1714 of first semiconductor structure 102 and bonding interface 106 than capacitors 1728. Since bit lines 1723 and capacitors 1728 are coupled to opposite ends of vertical transistors 1726, as described above, bit lines 1723 (as part of interconnect layer 1722) are disposed vertically between vertical transistors 1726 and bonding interface 106, according to some implementations. As a result, interconnect layer 1722 including bit lines 1723 can be arranged close to bonding interface 106 to reduce the interconnect routing distance and complexity.

[0131] In some implementations, second semiconductor structure 104 further includes a substrate 1748 disposed

above DRAM cells **1724**. As described below with respect to the fabrication process, substrate **1748** can be part of a carrier wafer. It is understood that in some examples, substrate **1748** may not be included in second semiconductor structure **104**.

[0132] As shown in FIG. 17, second semiconductor structure **104** can further include a pad-out interconnect layer **1750** above substrate **1748** and DRAM cells **1724**. Pad-out interconnect layer **1750** can include interconnects, e.g., contact pads **1754**, in one or more ILD layers. Pad-out interconnect layer **1750** and interconnect layer **1722** can be formed on opposite sides of DRAM cells **1724**. Capacitors **1728** are disposed vertically between vertical transistors **1726** and pad-out interconnect layer **1750**, according to some implementations. In some implementations, the interconnects in pad-out interconnect layer **1750** can transfer electrical signals between 3D memory device **1700** and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure **104** further includes one or more contacts **1752** extending through substrate **1748** and part of pad-out interconnect layer **1750** to couple pad-out interconnect layer **1750** to DRAM cells **1724** and interconnect layer **1722**. As a result, peripheral circuits **1712** can be coupled to DRAM cells **1724** through interconnect layers **1716** and **1722** as well as bonding layers **1720** and **1718**, and peripheral circuits **1712** and DRAM cells **1724** can be coupled to outside circuits through contacts **1752** and pad-out interconnect layer **1750**. Contact pads **1754** and contacts **1752** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad **1754** may include Al, and contact **1752** may include W. In some implementations, contact **1752** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **1748**. Depending on the thickness of substrate **1748**, contact **1752** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μ m and 100 μ m).

[0133] Although not shown, it is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having DRAM cells **1724** as shown in FIG. 17 and may be from first semiconductor structure **102** having peripheral circuit **1712** in the similar manner as described above with respect to FIG. 6B. Although not shown, it is also understood that the air gaps between word lines **1734** and/or between semiconductor bodies **1730** may be partially or fully filled with dielectrics in the similar manner as described above with respect to FIG. 6E. Although not shown, it is further understood that more than one array of DRAM cells **1724** may be stacked over one another to vertically scale up the number of DRAM cells **1724** in the similar manner as described above with respect to FIG. 7.

[0134] As described above, in some examples, trench isolations **1616** extending in the word line directions in FIG. 16 may not be formed such that two adjacent semiconductor bodies **1608** separated by a respective trench isolation **1616** may be merged as a single semiconductor body having two opposite sides in the bit line direction in contact with gate structure **1610**. That is, without trench isolations **1616**, the adjacent single-gate vertical transistors may be merged to form a double-gate vertical transistor (e.g., dual-side gate vertical transistor) with increased gate control area and

lower leakage current. For example, FIG. 20 illustrates a plan view of yet another array of memory cells **2002** each including a vertical transistor in a memory device **2000**, according to some aspects of the present disclosure. As shown in FIG. 20, memory device **2000** can include a plurality of word lines **2004** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **2000** can also include a plurality of bit lines **2006** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. 20 does not illustrate a cross-section of memory device **2000** in the same lateral plane, and word lines **2004** and bit lines **2006** may be formed in different lateral planes for ease of routing as described below in detail.

[0135] Memory cells **2002** can be formed at the intersections of word lines **2004** and bit lines **2006**. In some implementations, each memory cell **2002** includes a vertical transistor (e.g., vertical transistor **210** in FIG. 2) having a semiconductor body **2008** and a gate structure **2010**. Semiconductor body **2008** can extend in a substrate in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a double-gate transistor in which gate structure **2010** is in contact with two sides (e.g., two of four sides in FIG. 20) of semiconductor body **2008** (the active region in which channels are formed). As shown in FIG. 20, the vertical transistor is a double-gate transistor in which gate structure **2010** abuts two opposite sides of semiconductor body **1608** (having a rectangle or square-shaped cross-section) in the bit line direction (the y-direction) in the plan view. Gate structure **2010** does not surround and contact the other two sides of semiconductor body **2008** in the word line direction (the x-direction), according to some implementations. That is, gate structure **2010** can partially circumscribes semiconductor body **2008** in the plan view. Gate structure **2010** can include a gate dielectric **2012** abuts two opposite sides of semiconductor body **2008** in the plan view, and a gate electrode **2014** in contact with gate dielectric **2012**. In some implementations, gate dielectric **2012** is laterally between gate electrode **2014** and semiconductor body **2008** in the bit line direction (the y-direction). As described above, gate electrode **2014** may be part of word line **2004**, and word line **2004** may be an extension of gate electrode **2014**. That is, gate electrodes **1614** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **1614** and **1604**. In some implementations, gate structures **2010** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. 20.

[0136] Different from separate gate dielectrics **512** and **812** in FIGS. 5 and 8, as shown in FIG. 20, gate dielectrics **2012** of adjacent vertical transistors in the word line direction are continuous, e.g., parts of a continuous dielectric layer having gate dielectrics **2012** and extending in the word line direction. Gate structures **2010** can be thus viewed as parts of a continuous structure extending in the word line direction at which the continuous structure intersects vertical transistors in the same row.

[0137] FIG. 21 illustrates a side view of a cross-section of yet another 3D memory device **2100** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **2100** may be one example of memory device **2000** including double-gate vertical transis-

tors in which gate structures abut two sides of semiconductor bodies in the plan view. It is understood that FIG. 21 is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device 100 described above with respect to FIG. 1A, 3D memory device 2100 is a bonded chip including first semiconductor structure 102 and second semiconductor structure 104 stacked over first semiconductor structure 102. First and second semiconductor structures 102 and 104 are joined at bonding interface 106 therebetween, according to some implementations. As shown in FIG. 21, first semiconductor structure 102 can include a substrate 2110, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials.

[0138] First semiconductor structure 102 can include peripheral circuits 2112 on substrate 2110. In some implementations, peripheral circuits 2112 include a plurality of transistors 2114 (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors 2114) can be formed on or in substrate 2110 as well.

[0139] In some implementations, first semiconductor structure 102 further includes an interconnect layer 2116 above peripheral circuits 2112 to transfer electrical signals to and from peripheral circuits 2112. Interconnect layer 2116 can include a plurality of interconnects (also referred to herein as “contacts”), including lateral interconnect lines and VIA contacts. Interconnect layer 1716 can further include one or more ILD layers in which the interconnect lines and via contacts can form. That is, interconnect layer 2116 can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits 2112 are coupled to one another through the interconnects in interconnect layer 2116. The interconnects in interconnect layer 2116 can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

[0140] As shown in FIG. 21, first semiconductor structure 102 can further include a bonding layer 2118 at bonding interface 106 and above interconnect layer 2116 and peripheral circuits 2112. Bonding layer 2118 can include a plurality of bonding contacts 2119 and dielectrics electrically isolating bonding contacts 2119. Bonding contacts 2119 can include conductive materials, such as Cu. The remaining area of bonding layer 2118 can be formed with dielectric materials, such as silicon oxide. Bonding contacts 2119 and surrounding dielectrics in bonding layer 2118 can be used for hybrid bonding. Similarly, as shown in FIG. 21, second semiconductor structure 104 can also include a bonding layer 2120 at bonding interface 106 and above bonding layer 2118 of first semiconductor structure 102. Bonding layer 2120 can include a plurality of bonding contacts 2121 and dielectrics electrically isolating bonding contacts 2121. Bonding contacts 2121 can include conductive materials, such as Cu. The remaining area of bonding layer 2120 can be formed with dielectric materials, such as silicon oxide. Bonding contacts 2121 and surrounding dielectrics in bonding layer 2120 can be used for hybrid bonding. Bonding contacts 2121 are in contact with bonding contacts 2119 at

bonding interface 106, according to some implementations. In some implementations, bonding layer 2120 includes a dielectric layer opposing DRAM cells 2124 with bit line 2123 positioned between the dielectric layer and DRAM cells 2124, as shown in FIG. 21. The dielectric layer can include bonding interface 106 having bonding contacts 2121.

[0141] Second semiconductor structure 104 can be bonded on top of first semiconductor structure 102 in a face-to-face manner at bonding interface 106. In some implementations, bonding interface 106 is disposed between bonding layers 2120 and 2118 as a result of hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface 106 is the place at which bonding layers 2120 and 2118 are met and bonded. In practice, bonding interface 106 can be a layer with a certain thickness that includes the top surface of bonding layer 2118 of first semiconductor structure 102 and the bottom surface of bonding layer 2120 of second semiconductor structure 104.

[0142] In some implementations, second semiconductor structure 104 further includes an interconnect layer 2122 including bit lines 2123 above bonding layer 2120 to transfer electrical signals. Interconnect layer 2122 can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer 2122 also include local interconnects, such as bit lines 2123 (e.g., an example of bit lines 2006 in FIG. 20) and word line contacts (not shown). Interconnect layer 2122 can further include one or more ILD layers in which the interconnect lines and via contacts can form. The interconnects in interconnect layer 2122 can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits 2112 include a word line driver/row decoder coupled to the word line contacts in interconnect layer 2122 through bonding contacts 2121 and 2119 in bonding layers 2120 and 2118 and interconnect layer 2116. In some implementations, peripheral circuits 2112 include a bit line driver/column decoder coupled to bit lines 2123 and the bit line contacts in interconnect layer 2122 through bonding contacts 2121 and 2119 in bonding layers 2120 and 2118 and interconnect layer 2116. In some implementations, bit line 2123 is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line 2123 may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, the bit line contact is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, the bit line contact may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

[0143] In some implementations, second semiconductor structure 104 includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells 2124 (e.g., an example of memory cells 2002 in FIG. 20) above interconnect layer 2122 and bonding layer 2120. That

is, interconnect layer **2122** including bit lines **2123** can be disposed between bonding layer **2120** and array of DRAM cells **2124**. It is understood that the cross-section of 3D memory device **2100** in FIG. **21** may be made along the bit line direction (the y-direction), and one bit line **2123** in interconnect layer **2122** extending laterally in the y-direction may be coupled to a column of DRAM cells **2124**.

[0144] Each DRAM cell **2124** can include a vertical transistor **2126** (e.g., an example of vertical transistors **210** in FIG. **2**) and capacitor **2128** (e.g., an example of storage unit **212** in FIG. **2**) coupled to the vertical transistor **2126**. DRAM cell **2124** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **2124** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc.

[0145] Vertical transistor **2126** can be a MOSFET used to switch a respective DRAM cell **2124**. In some implementations, vertical transistor **2126** includes a semiconductor body **2130** (the active region in which channels can form) extending vertically (in the z-direction), and a gate structure **2136** in contact with two opposite sides of semiconductor body **2130** in the bit line direction (the y-direction). As described above, as in a double-gate vertical transistor, semiconductor body **2130** can have a cuboid shape or a cylinder shape, and gate structure **2136** can abut two sides of semiconductor body **2130** in the plan view, for example, as shown in FIG. **21**. Gate structure **2136** includes a gate electrode **2134** and a gate dielectric **2132** laterally between gate electrode **2134** and semiconductor body **2130** in the bit line direction, according to some implementations. In some implementations, gate dielectric **2132** abuts two sides of semiconductor body **2130**, and gate electrode **2134** abuts gate dielectric **2132**.

[0146] As shown in FIG. **21**, in some implementations, semiconductor body **2130** has two ends (the upper end and lower end) in the vertical direction (the z-direction), and at least one end (e.g., the lower end in FIG. **21**) extends beyond gate dielectric **2132** in the vertical direction (the z-direction) into ILD layers. In some implementations, one end (e.g., the upper end in FIG. **21**) of semiconductor body **2130** is flush with the respective end (e.g., the upper end in FIG. **21**) of gate dielectric **2132**. In some implementations, both ends (the upper end and lower end) of semiconductor body **2130** extend beyond gate electrode **2134**, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body **2130** can have a larger vertical dimension (e.g., the depth) than that of gate electrode **2134** (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body **2130** is flush with the respective end of gate electrode **2134**. Thus, short circuits between bit lines **2123** and word lines/gate electrodes **2134** or between word lines/gate electrodes **2134** and capacitors **2128** can be avoided. Vertical transistor **2126** can further include a source and a drain (both referred to as **2138** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body **2130**, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain **2138** (e.g., at the upper end in FIG. **21**) is coupled to capacitor **2128**, and the other one of source and drain **2138** (e.g., at the lower end in FIG. **21**) is coupled to bit line **2123**. That is, vertical transistor **2126** can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. **21**. In some implementations,

a metal bit line (e.g., bit line **2123** made of a metal material) is coupled to the second terminal of vertical transistor **2126** via an ohmic contact (e.g., a bit line contact made of a metal silicide material).

[0147] In some implementations, semiconductor body **2130** includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one example, semiconductor body **2130** may include single crystalline silicon. Source and drain **2138** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **2138** of vertical transistor **2126** and bit line **2123** as the bit line contact or between source/drain **2138** of vertical transistor **2126** and the first electrode of capacitor **2128** as capacitor contact **2142** to reduce the contact resistance. In some implementations, gate dielectric **2132** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al_2O_3 , HfO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , or any combination thereof. In some implementations, gate electrode **2134** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode **2134** includes multiple conductive layers, such as a W layer over a TiN layer. In one example, gate structure **2136** may be a “gate oxide/gate poly” gate in which gate dielectric **2132** includes silicon oxide and gate electrode **2134** includes doped polysilicon. In another example, gate structure **2136** may be an HKMG in which gate dielectric **2132** includes a high-k dielectric and gate electrode **2134** includes a metal.

[0148] As described above, since gate electrode **2134** may be part of a word line or extend in the word line direction as a word line, second semiconductor structure **104** of 3D memory device **2100** can also include a plurality of word lines (e.g., an example of word lines **2004** in FIG. **20**, referred to as **2134** as well) each extending in the word line direction. Each word line **2134** can be coupled to a row of DRAM cells **2124**. That is, bit line **2123** and word line **2134** can extend in two perpendicular lateral directions, and semiconductor body **2130** of vertical transistor **2126** can extend in the vertical direction perpendicular to the two lateral directions in which bit line **2123** and word line **2134** extend. Word lines **2134** are in contact with word line contacts (not shown), according to some implementations. In some implementations, word lines **2134** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, word line **2134** includes multiple conductive layers, such as a W layer over a TiN layer.

[0149] As shown in FIG. **21**, vertical transistor **2126** extends vertically through and contacts word lines **2134**, and source or drain **2138** of vertical transistor **2126** at the lower end thereof is in contact with bit line **2123** (or bit line contact if any), according to some implementations. Accordingly, word lines **2134** and bit lines **2123** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **2126**, which simplifies the routing of word lines **2134** and bit lines **2123**. In some implementations, bit lines **2123** are disposed vertically between bonding layer **2120** and word lines **2134**, and word lines **2134** are disposed vertically between bit lines **2123** and capacitors **2128**. Word lines **2134** can be coupled to periph-

eral circuits **2112** in first semiconductor structure **102** through word line contacts in interconnect layer **2122**, bonding contacts **2121** and **2119** in bonding layers **2120** and **2118**, and the interconnects in interconnect layer **2116**. Similarly, bit lines **2123** in interconnect layer **2122** can be coupled to peripheral circuits **2112** in first semiconductor structure **102** through bonding contacts **2121** and **2119** in bonding layers **2120** and **2118** and the interconnects in interconnect layer **2116**.

[0150] In some implementations, second semiconductor structure **104** further includes a plurality of air gaps **2140** each disposed laterally between adjacent word lines **2134**. Each air gap **2140** can be a trench extending in the word line direction (e.g., the x-direction) in parallel with word lines **2134** to separate adjacent rows of vertical transistors **2126**. As described below with respect to the fabrication process, air gaps **2140** may be formed due to the relatively small pitches of word lines **2134** (and rows of DRAM cells **2124**) in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps **2140** (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between word lines **2134** (and rows of DRAM cells **2124**) compared with some dielectrics (e.g., silicon oxide).

[0151] As shown in FIG. 21, in some implementations, capacitor **2128** includes a first electrode above and coupled to source or drain **2138** of vertical transistor **2126**, e.g., the upper end of semiconductor body **2130**, via a capacitor contact **2142**. In some implementations, capacitor contact **2142** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact **2142** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor **2128** can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor **2128** can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **2138** of a respective vertical transistor **2126** in the same DRAM cell, while all second electrodes are coupled to a common plate **2146** coupled to the ground, e.g., a common ground. Capacitor **2128** can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in FIG. 21. In some implementations, the first end of capacitor **2128** is coupled to the first terminal of vertical transistor **2126** via an ohmic contact (e.g., capacitor contact **2142** made of a metal silicide material). As shown in FIG. 21, second semiconductor structure **104** can further include a capacitor contact **2147** (e.g., a conductor) in contact with common plate **2146** for coupling capacitors **2128** to peripheral circuits **2112** or to the ground directly. In some implementations, capacitor contact **2147** (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer **2120** to couple to the second end of capacitor **2128** via common plate **2146**, as shown in FIG. 21. In some implementation, the ILD layer in which capacitors **2128** are formed has the same dielectric material as the two ILD layers into which semiconductor body **2130** extends, such as silicon oxide.

[0152] It is understood that the structure and configuration of capacitor **2128** are not limited to the example in FIG. 21 and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al_2O_3 , HfO_2 , Ta_2O_5 , ZrO_2 , TiO_2 , or any combination thereof. It is understood that in some examples, capacitor **2128** may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as PZT or SBT. In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

[0153] As shown in FIG. 21, vertical transistor **2126** extends vertically through and contacts word lines **2134**, source or drain **2138** of vertical transistor **2126** at the lower end thereof is in contact with bit line **2123** directly or through the bit line contact, and source or drain **2138** of vertical transistor **2126** at the upper end thereof is coupled to capacitor **2128**, according to some implementations. That is, bit line **2123** and capacitor **2128** can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor **2126** of DRAM cell **2124** in the vertical direction due to the vertical arrangement of vertical transistor **2126**. In some implementations, bit line **2123** and capacitor **2128** are disposed on opposite sides of vertical transistor **2126** in the vertical direction, which simplifies the routing of bit lines **2123** and reduces the coupling capacitance between bit lines **2123** and capacitors **2128** compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.

[0154] As shown in FIG. 21, in some implementations, vertical transistors **2126** are disposed vertically between capacitors **2128** and bonding interface **106**. That is, vertical transistors **2126** can be arranged closer to peripheral circuits **2114** of first semiconductor structure **102** and bonding interface **106** than capacitors **2128**. Since bit lines **2123** and capacitors **2128** are coupled to opposite ends of vertical transistors **2126**, as described above, bit lines **2123** (as part of interconnect layer **2122**) are disposed vertically between vertical transistors **2126** and bonding interface **106**, according to some implementations. As a result, interconnect layer **2122** including bit lines **2123** can be arranged close to bonding interface **106** to reduce the interconnect routing distance and complexity.

[0155] In some implementations, second semiconductor structure **104** further includes a substrate **2148** disposed above DRAM cells **2124**. As described below with respect to the fabrication process, substrate **2148** can be part of a carrier wafer. It is understood that in some examples, substrate **2148** may not be included in second semiconductor structure **104**.

[0156] As shown in FIG. 21, second semiconductor structure **104** can further include a pad-out interconnect layer **2150** above substrate **2148** and DRAM cells **2124**. Pad-out interconnect layer **2150** can include interconnects, e.g., contact pads **2154**, in one or more ILD layers. Pad-out interconnect layer **2150** and interconnect layer **2122** can be formed on opposite sides of DRAM cells **2124**. Capacitors

2128 are disposed vertically between vertical transistors **2126** and pad-out interconnect layer **2150**, according to some implementations. In some implementations, the interconnects in pad-out interconnect layer **2150** can transfer electrical signals between 3D memory device **2100** and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure **104** further includes one or more contacts **2152** extending through substrate **2148** and part of pad-out interconnect layer **2150** to couple pad-out interconnect layer **2150** to DRAM cells **2124** and interconnect layer **2122**. As a result, peripheral circuits **2112** can be coupled to DRAM cells **2124** through interconnect layers **2116** and **2122** as well as bonding layers **2120** and **2118**, and peripheral circuits **2112** and DRAM cells **2124** can be coupled to outside circuits through contacts **2152** and pad-out interconnect layer **2150**. Contact pads **2154** and contacts **2152** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad **2154** may include Al, and contact **2152** may include W. In some implementations, contact **2152** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **2148**. Depending on the thickness of substrate **2148**, contact **2152** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μ m and 100 μ m).

[0157] Although not shown, it is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having DRAM cells **2124** as shown in FIG. 21 and may be from first semiconductor structure **102** having peripheral circuit **2112** in the similar manner as described above with respect to FIG. 6B. Although not shown, it is also understood that the air gaps between word lines **2134** may be partially or fully filled with dielectrics in the similar manner as described above with respect to FIG. 6E. Although not shown, it is further understood that more than one array of DRAM cells **2124** may be stacked over one another to vertically scale up the number of DRAM cells **2124** in the similar manner as described above with respect to FIG. 7.

[0158] FIG. 27 illustrates a block diagram of a system **2700** having a memory device, according to some aspects of the present disclosure. System **2700** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. 27, system **2700** can include a host **2708** and a memory system **2702** having one or more memory devices **2704** and a memory controller **2706**. Host **2708** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **2708** can be configured to send or receive the data to or from memory devices **2704**.

[0159] Memory device **2704** can be any memory devices disclosed herein, such as 3D memory devices **100** and **101**, memory devices **200**, **500**, **800**, **1600**, and **2000**, and 3D memory devices **600**, **601**, **603**, **605**, **607**, **700**, **900**, **1700**, and **2100**. In some implementations, memory device **2704** includes an array of memory cells each including a vertical transistor, as described above in detail.

[0160] Memory controller **2706** is coupled to memory device **2704** and host **2708** and is configured to control memory device **2704**, according to some implementations. Memory controller **2706** can manage the data stored in memory device **2704** and communicate with host **2708**. Memory controller **2706** can be configured to control operations of memory device **2704**, such as read, write, and refresh operations. Memory controller **2706** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **2704** including, but not limited to refresh and timing control, command/request translation, buffer and schedule, and power management. In some implementations, memory controller **2706** is further configured to determine the maximum memory capacity that the computer system can use, the number of memory banks, memory type and speed, memory particle data depth and data width, and other important parameters. Any other suitable functions may be performed by memory controller **2706** as well. Memory controller **2706** can communicate with an external device (e.g., host **2708**) according to a particular communication protocol. For example, memory controller **2706** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

[0161] FIGS. 10A-10M illustrate a fabrication process for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 11A-11I illustrate a fabrication process for forming another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 12A-12H illustrate a fabrication process for forming still another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 13A-13H illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 14A-14E illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 15A-15D illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 19A-19M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. 22A-22M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIG. 23 illustrates a flowchart of a method **2300** for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. 10A-10M include 3D memory devices **600** and **601** depicted in FIGS. 6A and 6B. Examples of the 3D memory devices depicted in FIGS. 11A-11I include 3D memory device **900** depicted in FIG. 9. Examples of the 3D memory devices depicted in FIGS. 12A-12H include 3D memory device **603** depicted in FIG. 6C. Examples of the 3D memory devices depicted in FIGS.

13A-13H include 3D memory device **605** depicted in FIG. 6D. Examples of the 3D memory devices depicted in FIGS. 14A-14E and 15A-15D include 3D memory device **700** depicted in FIG. 7. Examples of the 3D memory devices depicted in FIGS. 19A-19M include 3D memory device **1700** depicted in FIG. 17. Examples of the 3D memory devices depicted in FIGS. 22A-22M include 3D memory device **2100** depicted in FIG. 21. FIGS. 10A-10M, 11A-11I, 12A-12H, 13A-13H, 14A-14E, 15A-15D, 19A-19M, 22A-22M, and **23** will be described together. It is understood that the operations shown in method **2300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **2300**.

[0162] In some implementations, a first semiconductor structure including a peripheral circuit is formed. As depicted in FIG. 10L or 19L, a first semiconductor structure including peripheral circuits is formed. In some implementations, a second semiconductor structure including a first array of memory cells and a plurality of bit lines coupled to the memory cells is formed. Each of the memory cells can include a vertical transistor, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. As depicted in FIG. 10L, 11I, 12H, 13H, 19L, or 22L, a second semiconductor structure including an array of DRAM cells, each of which includes a vertical transistor, and a capacitor coupled to the vertical transistor, is formed. The second semiconductor structure also includes a plurality of bit lines coupled to the memory cells, and a respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. In some implementations, the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. As depicted in FIGS. 10L and 10M, 19L, or 22L, the first and second semiconductor structures are bonded in a face-to-face manner, such that the array of DRAM cells is coupled to the peripheral circuit across a bonding interface.

[0163] Referring to FIG. 23, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.

[0164] As illustrated in FIG. 10L, a plurality of transistors **1042** are formed on a silicon substrate **1038**. Transistors **1042** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, chemical mechanical polishing (CMP), and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1038** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **1042**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1038** by wet/dry etch and thin film deposition. Transistors **1042** can form peripheral circuits **1040** on silicon substrate **1038**.

[0165] As illustrated in FIG. 10L, an interconnect layer **1044** can be formed above peripheral circuits **1040** having

transistors **1042**. Interconnect layer **1044** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **1040**. In some implementations, interconnect layer **1044** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1044** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 10L can be collectively referred to as interconnect layer **1044**.

[0166] Method **2300** proceeds to operation **2304**, as illustrated in FIG. 23, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The first bonding layer can include a first bonding contact. As illustrated in FIG. 10L, a bonding layer **1046** is formed above interconnect layer **1044** and peripheral circuits **1040**. Bonding layer **1046** can include a plurality of bonding contacts **1047** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1044** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1047** then can be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1044** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0167] Method **2300** proceeds to operation **2306**, as illustrated in FIG. 23, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

[0168] For example, FIG. 24 illustrates a flowchart of a method **2400** for forming an array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2402** in FIG. 24, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, three layers having a first dielectric, a second dielectric, and the first dielectric, respectively, are subsequently deposited on the substrate. The first dielectric can include silicon oxide, and the second dielectric can include silicon nitride. The layer having the second dielectric can act as a sacrificial layer sandwiched vertically between two layers having the first dielectric. The sacrificial layer can be removed by selectively etching against the two layers having the first dielectric and replaced with a conductive layer in the later processes.

[0169] As illustrated in FIG. 10A, a stack of a silicon oxide layer 1004, a silicon nitride layer 1006, and a silicon oxide layer 1008 is formed on a silicon substrate 1002. To form the dielectric stack, silicon oxide, silicon nitride, and silicon oxide are subsequently deposited onto silicon substrate 1002 using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer 1004 is formed by oxidizing the top portion of silicon substrate 1002 using dry oxidation and/or wet oxidation, such as in situ steam generation (ISSG) oxidation process. In some implementations, the thickness of silicon oxide layer 1004 (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer 1008 (e.g., CVD silicon oxide). FIG. 10A illustrates both the side view (in the top portion of FIG. 10A) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of FIG. 10A) of a cross-section in the x-y plane (e.g., in the AA plane through silicon nitride layer 1006). The same drawing layout is arranged in FIGS. 10B-10G as well.

[0170] At operation 2404 in FIG. 24, a semiconductor body extending vertically from the substrate through the stack of dielectric layers is formed. In some implementations, to form the semiconductor body, an opening extending through the stack of dielectric layers is etched to expose part of the substrate, and the semiconductor body is epitaxially grown from the exposed part of the substrate in the opening.

[0171] As illustrated in FIG. 10B, an array of openings 1010 is formed, each of which extends vertically (in the z-direction) through the stack of silicon oxide layer 1008, silicon nitride layer 1006, and silicon oxide layer 1004 to silicon substrate 1002. As a result, parts of silicon substrate 1002 can be exposed from openings 1010. In some implementations, a lithography process is performed to pattern the array of openings 1010 using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines and bit lines, and one or more dry etching and/or wet etching processes, such as reactive ion etch (RIE), are performed to etch openings 1010 through silicon oxide layer 1008, silicon nitride layer 1006, and silicon oxide layer 1004 until being stopped by silicon substrate 1002.

[0172] As illustrated in FIG. 10C, an array of semiconductor bodies 1012 are formed in openings 1010. Semiconductor body 1012 can be epitaxially grown from the respective exposed part of silicon substrate 1002 in the respective opening 1010. The fabrication processes for epitaxially growing semiconductor body 1012 can include, but not limited to, vapor-phase epitaxy (VPE), liquid-phase epitaxy (LPE), molecular-beam epitaxy (MBE), or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate 1002 in openings 1010. Semiconductor body 1012 thus can have the same material as silicon substrate 1002, in the form of single crystalline silicon. Depending on the shape of opening 1010, semiconductor body 1012 can have the same shape as opening 1010, such as a cuboid shape or a cylinder shape. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies 1012 beyond the top surface of silicon oxide layer 1008. As a result, an array of semiconductor bodies 1012 (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate 1002 through the stack of silicon oxide layer 1008, silicon nitride

layer 1006, and silicon oxide layer 1004 is formed thereby, according to some implementations.

[0173] At operation 2406 in FIG. 24, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the second dielectric, and the layer having the second dielectric (e.g., the sacrificial layer) is etched away via the trench. In some implementations, the trench is etched between adjacent rows of semiconductor bodies without touching any sides of the semiconductor bodies.

[0174] As illustrated in FIG. 10D, a plurality of trenches 1014 (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon oxide layer 1008 and silicon nitride layer 1006 are formed to expose silicon nitride layer 1006. As a result, parts of silicon nitride layer 1006 can be exposed from trenches 1014. In some implementations, a lithography process is performed to pattern trenches 1014 using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). As shown in FIG. 10D, trench 1014 is patterned to be formed between adjacent rows of semiconductor bodies 1012 without touching any sides of semiconductor bodies 1012, such that semiconductor bodies 1012 are not exposed from any side thereof. In one example, trench 1014 is patterned to be formed in the middle between adjacent rows of semiconductor bodies 1012, e.g., having the same distance two adjacent rows of semiconductor bodies 1012. In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches 1014 through silicon oxide layer 1008, silicon nitride layer 1006, and silicon oxide layer 1004 until being stopped by silicon substrate 1002. It is understood that in some examples, the etching of trenches 1014 may not go all the way to silicon substrate 1002 and may be stopped at silicon oxide layer 1004 so long as silicon nitride layer 1006 is exposed from trenches 1014.

[0175] As illustrated in FIG. 10E, silicon nitride layer 1006 (shown in FIG. 10D) is removed to expose parts of semiconductor bodies 1012 abutting silicon nitride layer 1006. In some implementations, silicon nitride layer 1006 is etched away via trenches 1014. For example, a wet etchant including phosphoric acid may be applied through trenches 1014 to selectively wet etch silicon nitride layer 1006 without etching silicon oxide layers 1004 and 1008 as well as semiconductor bodies 1012 and silicon substrate 1002. As a result, lateral recesses 1016 can be formed vertically between silicon oxide layers 1004 and 1008 thereby, which expose parts of semiconductor bodies 1012. As shown in the plan view, all sides of each semiconductor body 1012 can be exposed from lateral recesses 1016.

[0176] At operation 2408 in FIG. 24, a gate structure in contact with a plurality of sides of the exposed part of the semiconductor body is formed. In some implementations, to form the gate structure, a gate dielectric is formed over the exposed part of the semiconductor body, a conductive layer is deposited over the gate dielectric, and the conductive layer is patterned to form a gate electrode over the gate dielectric.

[0177] As illustrated in FIG. 10F, a gate dielectric 1018 is formed over the exposed part of each semiconductor body 1012, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body 1012. As shown in the plan view, gate dielectric 1018 can fully circumscribe a

respective semiconductor body **1012**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1012** (e.g., single crystalline silicon) as gate dielectric **1018**. In some implementations, gate dielectric **1018** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1012** through trenches **1014** and lateral recesses **1016** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1016** and trenches **1014**.

[0178] As illustrated in FIG. 10G, a conductive layer **1020** is formed over gate dielectrics **1018** in lateral recesses **1016** (shown in FIG. 10F) through trenches **1014**. In some implementations, conductive layer **1020** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1018** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1014** to fill lateral recesses **1016**. In one example, the deposition of conductive layer **1020** is controlled not to fill trenches **1014**. It is understood that in some examples, the deposition of conductive layer **1020** may fill trenches **1014** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1020** over the top surface of silicon oxide layer **1008**, and conductive layer **1020** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1014** filled with conductive layer **1020** may be patterned and etched again to separate conductive layers **1020** between adjacent rows of semiconductor bodies **1012** and gate dielectrics **1018**. As described above, a lithography process can be performed to pattern trenches **1014** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches).

[0179] As a result, patterned conductive layers **1020** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1014**, and parts of patterned conductive layers **1020** that are over gate dielectrics **1018** (e.g., fully circumscribes a respective gate dielectric **1018** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1018** over the exposed part of semiconductor body **1012** and a respective gate electrode (part of conductive layer **1020**) over gate dielectric **1018** can be formed thereby. Since conductive layer **1020** remains on all sides of semiconductor body **1012** (and gate dielectric **1018** thereover) when patterning conductive layer **1020** (etching trenches **1014**), the gate structure is in contact with all sides of semiconductor body **1012**, according to some implementations, as shown in FIG. 10G. As shown in the plan view, the gate structure (having gate dielectric **1018** and the gate electrode) can fully circumscribe a respective semiconductor body **1012**, and all sides of each semiconductor body **1012** can be surrounded and contacted by the respective gate structure. Comparing FIG. 10G with FIG. 10A, silicon nitride layer **1006** (sacrificial layer) in FIG. 10A is eventually replaced with conductive layer **1020** in FIG. 10G, according to some implementations.

[0180] At operation **2410** in FIG. 24, a first end of the semiconductor body away from the substrate is doped. As illustrated in FIG. 10G, the exposed upper end of each semiconductor body **1012**, e.g., one of the two ends of

semiconductor body **1012** in the vertical direction (the z-direction) that is away from silicon substrate **1002**, is doped to form a source/drain **1021** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** to form sources/drains **1021**. In some implementations, a silicide layer is formed on source/drain **1021** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1012**.

[0181] At operation **2412** in FIG. 24, a storage unit in contact with the semiconductor body, e.g., the doped first end thereof, is formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

[0182] As illustrated in FIG. 10H, one or more ILD layers are formed over the top surface of silicon oxide layer **1008**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that in some examples in which the ILD layers include silicon oxide, the same material as silicon oxide layer **1008**, the boundary and interface between the ILD layer and silicon oxide layer **1008** may become indistinguishable after the deposition. Depending on the lateral dimensions of trenches **1014** (shown in FIG. 10G), trenches **1014** may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps **1022** between adjacent word lines (patterned conductive layers **1020**). It is understood that in some examples, when the lateral dimensions of trenches **1014** are sufficiently large, dielectrics may fully fill trenches **1014** during the formation of the ILD layers, thereby eliminating air gaps **1022**.

[0183] As illustrated in FIG. 10H, capacitor contacts **1024**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1026**, and common plate **1028** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1012**. In some implementations, each capacitor contact **1024** is formed on a respective source/drain **1021**, e.g., the doped upper end of a respective semiconductor body **1012** by patterning and etching an electrode hole aligned with respective source/drain **1021** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1028** is formed on the second electrodes of capacitors **1026** by patterning and etching an electrode trench aligned with capacitors **1026** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[0184] At operation **2414** in FIG. 24, the substrate is removed to expose a second end of the semiconductor body opposite to the first end. As illustrated in FIG. 10I, a carrier substrate **1030** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1002** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1002** become above carrier substrate **1030**.

[0185] As illustrated in FIG. 10J, silicon substrate 1002 (shown in FIG. 10I) is removed to expose the undoped upper ends of semiconductor bodies 1012 (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate 1002 until being stopped by silicon oxide layer 1004 and the upper ends of semiconductor bodies 1012.

[0186] At operation 2416 in FIG. 24, the exposed second end of the semiconductor body is doped. As illustrated in FIG. 10J, the exposed upper end of each semiconductor body 1012, e.g., one of the two ends of semiconductor body 1012 in the vertical direction (the z-direction) that is away from carrier substrate 1030, is doped to form another source/drain 1023 (e.g., a source terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies 1012 to form sources/drains 1023. In some implementations, a silicide layer is formed on source/drain 1023 by performing a silicidation process at the exposed upper ends of semiconductor bodies 1012. As a result, vertical transistors having semiconductor body 1012, sources/drains 1021 and 1023, gate dielectric 1018, and the gate electrode (part of conductive layer 1020) are formed thereby, as shown in FIG. 10J, according to some implementations. As described above, capacitors 1026 each having the first and second electrodes and the capacitor dielectric are thereby formed as well, and DRAM cells 1080 each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. 10J, according to some implementations.

[0187] Referring back to FIG. 23, method 2300 proceeds to operation 2308, as illustrated in FIG. 23, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. 10K, an interconnect layer 1032 can be formed above DRAM cells 1080. Interconnect layer 1032 can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells 1080. In some implementations, interconnect layer 1032 includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers 1032 can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer 1004 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 10K can be collectively referred to as interconnect layer 1032.

[0188] As shown in FIG. 24, at operation 2418, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. 10K, bit line 1034 can be formed on sources/drains 1023 by patterning and etching a trench aligned with respective source/drain 1023 using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line 1034 includes

depositing a metal layer onto the exposed end of semiconductor body 1012. As a result, bit line 1034 and capacitor 1026 can be formed on opposite sides of semiconductor body 1012 and coupled to opposite ends of semiconductor body 1012. It is understood that additional local interconnects, such as word line contacts 1039, capacitor contacts 1083 (e.g., a conductor), and bit line contacts 1041 (e.g., a metal silicide contact) may be similarly formed as well. In some implementations, bit line contact 1041 (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body 1012, and bit line 1034 is formed on bit line contact 1041.

[0189] Method 2300 proceeds to operation 2310, as illustrated in FIG. 23, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. 10K, a bonding layer 1036 is formed above interconnect layer 1032 and DRAM cells 1080. Bonding layer 1036 can include a plurality of bonding contacts 1037 surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer 1032 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts 1037 can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer 1032 by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0190] Method 2300 proceeds to operation 2312, as illustrated in FIG. 23, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

[0191] As illustrated in FIG. 10L, carrier substrate 1030 and components formed thereon (e.g., DRAM cells 1080) are flipped upside down. As illustrated in FIG. 10M, bonding layer 1036 facing down is bonded with bonding layer 1046 facing up, e.g., in a face-to-face manner, thereby forming a bonding interface 1050. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIGS. 10L and 10M, silicon substrate 1038 and components formed thereon (e.g., peripheral circuits 1040) can be flipped upside down, and bonding layer 1046 facing down can be bonded with bonding layer 1036 facing up, e.g., in a face-to-face manner, thereby forming bonding interface 1050. After the bonding, bonding contacts 1037 in bonding layer 1036 and bonding contacts 1047 in bonding layer 1046 are aligned and in contact with one another, such that DRAM cells 1080 can be electrically connected to peripheral circuits 1040

across bonding interface **1050**. It is understood that in the bonded chip, DRAM cells **1080** may be either above or below peripheral circuits **1040**. Nevertheless, bonding interface **1050** can be formed vertically between peripheral circuits **1040** and DRAM cells **1080** after the bonding.

[0192] Method **2300** proceeds to operation **2314**, as illustrated in FIG. **23**, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. **10M**, a pad-out interconnect layer **1052** is formed above on the backside of carrier substrate **1030**. Pad-out interconnect layer **1052** can include interconnects, such as pad contacts **1054**, formed in one or more ILD layers. Pad contacts **1054** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts **1056** are formed extending vertically through carrier substrate **1030**, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts **1056** can be in contact with the interconnects in pad-out interconnect layer **1052**. It is understood that in some examples, carrier substrate **1030** may be thinned or removed after bonding and prior to forming pad-out interconnect layer **1052** and contacts **1056**, for example, using planarization processes and/or etching processes.

[0193] Although not shown, it is understood that in some examples, pad-out interconnect layer **1052** may be formed above on the backside of silicon substrate **1038**, and contacts **1056** may be formed extending vertically through silicon substrate **1038**. Silicon substrate **1038** may be thinned prior to forming pad-out interconnect layer **1052** and contacts **1056**, for example, using planarization processes and/or etching processes.

[0194] As described above, FIGS. **10A-10M** illustrates a fabrication process of forming a DRAM array having a vertical transistor in which the gate structure is in contact with all sides of the semiconductor body, in the form of a GAA transistor. In some implementations as shown in FIG. **11A-11I**, by changing the layout of word line trenches, a DRAM array having a vertical transistor in which the gate structure is in contact with some sides (e.g., three of the four sides) of the semiconductor body are formed with a relatively larger pitch of word lines and reduced fabrication complexity.

[0195] As illustrated in FIG. **11A**, a stack of a silicon oxide layer **1104**, a silicon nitride layer **1106**, and a silicon oxide layer **1108** is formed on a silicon substrate **1102**. To form the dielectric stack, silicon oxide, silicon nitride, and silicon oxide are subsequently deposited onto silicon substrate **1102** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1104** is formed by oxidizing the top portion of silicon substrate **1102** using dry oxidation and/or wet oxidation, such as ISSG oxidation process. In some implementations, the thickness of silicon oxide layer **1104** (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer **1108** (e.g., CVD silicon oxide). Besides the side view of the cross-section along the y-direction (e.g., the bit line direction) shown in the top portion of FIG. **11A**, the plan view of the cross-section in the x-y plane through silicon nitride layer

1106 is also shown in the bottom portion of FIG. **11A**. The same drawing layout is arranged in FIGS. **11B-11E** as well.

[0196] As illustrated in FIG. **11B**, an array of semiconductor bodies **1112** each extending vertically through the stack of silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** are formed. Semiconductor body **1112** can be epitaxially grown from the respective exposed part of silicon substrate **1102** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1112** can include, but not limited to, VPE, LPE, MPE, or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1102** in the openings. Semiconductor body **1112** thus can have the same material as silicon substrate **1102**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1112** beyond the top surface of silicon oxide layer **1108**. As a result, an array of semiconductor bodies **1112** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1102** through the stack of silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** is formed thereby, according to some implementation.

[0197] In some implementations, at operation **2406** in FIG. **24**, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the second dielectric, and the layer having the second dielectric (e.g., the sacrificial layer) is etched away via the trench. In some implementations, the trench is etched aligned with one side of the semiconductor body to expose the semiconductor body from the side.

[0198] As illustrated in FIG. **11C**, a plurality of trenches **1114** (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon oxide layer **1108** and silicon nitride layer **1106** are formed to expose silicon nitride layer **1106**. As a result, parts of silicon nitride layer **1106** can be exposed from trenches **1114**. In some implementations, a lithography process is performed to pattern trenches **1114** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). As shown in FIG. **11C**, trench **1114** is patterned to be formed between adjacent rows of semiconductor bodies **1112** and aligned with one side of semiconductor bodies **1112** to expose the semiconductor bodies **1112** from the side, according to some implementations. That is, trench **1114** can be patterned to touch one side of semiconductor bodies **1112**, such that semiconductor bodies **1112** are exposed from the side. In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1114** through silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** until being stopped by silicon substrate **1102**. It is understood that in some examples, the etching of trenches **1114** may not go all the way to silicon substrate **1102** and may be stopped at silicon oxide layer **1104** so long as silicon nitride layer **1106** is exposed from trenches **1114**.

[0199] As illustrated in FIG. **11D**, silicon nitride layer **1106** (shown in FIG. **11C**) is removed to expose parts of semiconductor bodies **1112** abutting silicon nitride layer

1106. In some implementations, silicon nitride layer **1106** is etched away via trenches **1114**. For example, a wet etchant including phosphoric acid may be applied through trenches **1114** to selectively wet etch silicon nitride layer **1106** without etching silicon oxide layers **1104** and **1108** as well as semiconductor bodies **1112** and silicon substrate **1102**. As a result, lateral recesses **1116** can be formed vertically between silicon oxide layers **1104** and **1108** thereby, which expose parts of semiconductor bodies **1112**.

[0200] As illustrated in FIG. **11D**, a gate dielectric **1118** is formed over the exposed part of each semiconductor body **1112**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1112**. As shown in the plan view, gate dielectric **1118** can fully circumscribe a respective semiconductor body **1112**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1112** (e.g., single crystalline silicon) as gate dielectric **1118**. In some implementations, gate dielectric **1118** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1112** through trenches **1114** and lateral recesses **1116** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1116** and trenches **1114**.

[0201] Different from FIG. **10F** in which gate dielectric **1118** has a uniform vertical dimension (thickness in the z-direction) because all sides of semiconductor body **1112** is surrounded by lateral recess **1016** having the same vertical dimension, in FIG. **11D**, because one side of semiconductor body **1112** is aligned with and touches trench **1114** having a greater vertical dimension than that of lateral recess **1116**, part of gate dielectric **1118** formed on that side of semiconductor body **1112** (referred to as elongated gate dielectric part **1119**) can have a greater vertical dimension than the remainder of gate dielectric **1118** formed on other sides of semiconductor body **1112** touching lateral recess **1116**, as shown in the side view of FIG. **11D**.

[0202] As illustrated in FIG. **11E**, a conductive layer **1120** is formed over gate dielectric **1118** in lateral recesses **1116** (shown in FIG. **11D**) through trenches **1114**, but not over elongated gate dielectric part **1119**. In some implementations, conductive layer **1120** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1118** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1114** to fill lateral recesses **1116**. In one example, the deposition of conductive layer **1120** is controlled not to fill trenches **1114** (and not over elongated gate dielectric part **1119**). It is understood that in some examples, the deposition of conductive layer **1120** may fill trenches **1114** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1120** over the top surface of silicon oxide layer **1108**, and conductive layer **1120** may be patterned to form a gate electrode over only gate dielectric **1118**, but not elongated gate dielectric part **1119**. For example, trenches **1114** filled with conductive layer **1120** may be patterned and etched again to separate conductive layers **1120** between adjacent rows of semiconductor bodies **1112** and gate dielectrics **1118**. As described above, a lithography process can be performed to pattern

trenches **1114** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches).

[0203] As a result, patterned conductive layers **1120** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1114**, and parts of patterned conductive layers **1120** that are over gate dielectrics **1118**, but not elongated gate dielectric part **1119**, can become gate electrodes. Gate structures each including a respective gate dielectric **1118** over the exposed part semiconductor body **1112** and a respective gate electrode (part of conductive layer **1120**) over gate dielectric **1118** can be formed thereby. Since conductive layer **1120** remains on only some sides of semiconductor body **1012** (and gate dielectric **1018** thereover) when patterning conductive layer **1120** (etching trenches **1114**), the gate structure is in contact with some, but not all, sides of semiconductor body **1012**, according to some implementations, as shown in FIG. **11E**. As shown in the plan view, the gate structure (having gate dielectric **1118** and the gate electrode) can partially circumscribe a respective semiconductor body **1112**, and not all sides of each semiconductor body **1112** can be surrounded and contacted by the respective gate structure. Compared with the pitch of word lines **1020** in the example in FIG. **10G**, the pitch of word lines **1120** in FIG. **11E** may be increased to reduce the fabrication complexity.

[0204] As illustrated in FIG. **11E**, the exposed upper end of each semiconductor body **1112**, e.g., one of the two ends of semiconductor body **1112** in the vertical direction (the z-direction) that is away from silicon substrate **1102**, is doped to form a source/drain **1121** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1112** to form sources/drains **1021**.

[0205] As illustrated in FIG. **11F**, one or more ILD layers are formed over the top surface of silicon oxide layer **1108**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that in some examples in which the ILD layers include silicon oxide, the same material as silicon oxide layer **1108**, the boundary and interface between the ILD layer and silicon oxide layer **1108** may become indistinguishable after the deposition. Due to the relatively larger lateral dimensions of trenches **1114** (shown in FIG. **11E**) compared with trenches **1014** (as a result of a larger pitch of word lines **1120**), trenches **1114** may be fully filled or at least partially with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, eliminating air gaps **1022** or at least reducing air gaps **1022** between adjacent word lines (patterned conductive layers **1120**).

[0206] As illustrated in FIG. **11F**, capacitor contacts **1124**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1126**, and common plate **1128** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1112**. In some implementations, each capacitor contact **1124** is formed on a respective source/drain **1121**, e.g., the doped upper end of a respective semiconductor body **1112** by patterning and etching an electrode hole aligned with respective source/drain **1121** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In

some implementations, common plate **1128** is formed on the second electrodes of capacitors **1126** by patterning and etching an electrode trench aligned with capacitors **1126** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[**0207**] As illustrated in FIG. **11G**, a carrier substrate **1130** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1102** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1102** become above carrier substrate **1130**.

[**0208**] As illustrated in FIG. **11H**, silicon substrate **1102** (shown in FIG. **11G**) is removed to expose the undoped upper ends of semiconductor bodies **1112** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1102** until being stopped by silicon oxide layer **1104** and the upper ends of semiconductor bodies **1112**.

[**0209**] As illustrated in FIG. **11H**, the exposed upper end of each semiconductor body **1112**, e.g., one of the two ends of semiconductor body **1112** in the vertical direction (the z-direction) that is away from carrier substrate **1130**, is doped to form another source/drain **1123** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1112** to form sources/drains **1123**. As a result, multi-gate vertical transistors having semiconductor body **1112**, sources/drains **1121** and **1123**, gate dielectric **1118** (not including elongated gate dielectric part **1119**), and the gate electrode (part of conductive layer **1120**) are formed thereby, as shown in FIG. **11H**, according to some implementations. As described above, capacitors **1126** are thereby formed as well, and DRAM cells **1180** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. **11H**, according to some implementations.

[**0210**] As illustrated in FIG. **11I**, an interconnect layer **1132** can be formed above DRAM cells **1180**. Interconnect layer **1132** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1180**. In some implementations, interconnect layer **1132** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1132** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1104** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **11I** can be collectively referred to as interconnect layer **1132**. As illustrated in FIG. **11I**, bit line **1134** can be formed on sources/drains **1123** by patterning and etching a trench aligned with respective source/drain **1123** using lithography

and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1134** includes depositing a metal layer onto the exposed end of semiconductor body **1112**.

[**0211**] As illustrated in FIG. **11I**, a bonding layer **1136** is formed above interconnect layer **1132** and DRAM cells **1180**. Bonding layer **1136** can include a plurality of bonding contacts **1137** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1132** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1137** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1132** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[**0212**] As described above, FIGS. **10A-10M** illustrates a fabrication process of forming a DRAM cell array from a three-layer dielectric stack having a sacrificial layer (e.g., silicon nitride layer **1006**) sandwiched between two dielectric layers (e.g., silicon oxide layers **1004** and **1008**). It is understood that the configuration of the dielectric stack from which the DRAM cell array is formed can vary in other examples, resulting DRAM cells with different structures, such as in 3D memory devices **603** and **605** in FIGS. **6C** and **6D**. In some implementations as shown in FIGS. **12A-12H**, a DRAM cell array is formed from a two-layer dielectric stack having a sacrificial layer on a dielectric layer.

[**0213**] At operation **2402** in FIG. **24**, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, two layers having a first dielectric and a second dielectric, respectively, are subsequently deposited on the substrate. The first dielectric can include silicon oxide, and the second dielectric can include silicon nitride. The layer having the second dielectric can act as a sacrificial layer on the layer having the first dielectric. The sacrificial layer can be removed by selectively etching against the other layer having the first dielectric and replaced with a conductive layer in the later processes.

[**0214**] As illustrated in FIG. **12A**, a stack of a silicon oxide layer **1204** and a silicon nitride layer **1206** is formed on a silicon substrate **1202**. To form the dielectric stack, silicon oxide and silicon nitride are subsequently deposited onto silicon substrate **1202** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1204** is formed by oxidizing the top portion of silicon substrate **1202** using dry oxidation and/or wet oxidation, such as in situ steam generation (ISSG) oxidation process.

[**0215**] As illustrated in FIG. **12B**, an array of semiconductor bodies **1212** each extending vertically through the stack of silicon nitride layer **1206** and silicon oxide layer **1204** are formed. Semiconductor body **1212** can be epitaxially grown from the respective exposed part of silicon substrate **1202** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1212** can include, but not limited to, VPE, LPE, MPE,

or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1202** in the openings. Semiconductor body **1212** thus can have the same material as silicon substrate **1202**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1212** beyond the top surface of silicon nitride layer **1206**. As a result, an array of semiconductor bodies **1212** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1202** through the stack of silicon nitride layer **1206** and silicon oxide layer **1204** is formed thereby, according to some implementation.

[0216] As illustrated in FIG. 12C, a plurality of trenches **1214** (slits/openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon nitride layer **1206** are formed to expose silicon nitride layer **1206**. As a result, parts of silicon nitride layer **1206** can be exposed from trenches **1214**. In some implementations, a lithography process is performed to pattern trenches **1214** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1214** through silicon nitride layer **1206** and silicon oxide layer **1204** until being stopped by silicon substrate **1202**. It is understood that in some examples, the etching of trenches **1214** may not go all the way to silicon substrate **1202** and may be stopped at silicon oxide layer **1204** so long as silicon nitride layer **1206** is exposed from trenches **1214**.

[0217] As illustrated in FIG. 12D, silicon nitride layer **1206** (shown in FIG. 12C) is removed to expose parts of semiconductor bodies **1212** abutting silicon nitride layer **1206**. In some implementations, silicon nitride layer **1206** is etched away via trenches **1214**. For example, a wet etchant including phosphoric acid may be applied through trenches **1214** to selectively wet etch silicon nitride layer **1206** without etching silicon oxide layer **1204** as well as semiconductor bodies **1212** and silicon substrate **1202**. As a result, lateral recesses **1216** can be formed thereby, which expose parts of semiconductor bodies **1212**. It is understood that in some examples, the top surface of silicon nitride layer **1206** may be exposed, such that trenches **1214** may not be needed to remove silicon nitride layer **1206**. Dry etching and/or wet etching processes may be applied directly on silicon nitride layer **1206** to remove silicon nitride layer **1206** (from FIG. 12B to FIG. 12D directly without going through FIG. 12C).

[0218] As illustrated in FIG. 12D, a gate dielectric **1218** is formed over the exposed part of each semiconductor body **1212**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1212**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1212** (e.g., single crystalline silicon) as gate dielectric **1218**. In some implementations, gate dielectric **1218** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1212** through trenches **1214** and lateral recesses **1216** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1216** and trenches **1214**. Due to the omission of silicon

oxide layer **1008**, the upper end of gate dielectric **1218** can be flush with the upper end of semiconductor body **1212** as shown in FIG. 12D, while the upper end of gate dielectric **1018** is below the upper end of semiconductor body **1012** in FIG. 10F.

[0219] As illustrated in FIG. 12E, a conductive layer **1220** is formed over gate dielectrics **1218** in lateral recesses **1216** (shown in FIG. 12D) through trenches **1214**. In some implementations, conductive layer **1220** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1218** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1214** to fill lateral recesses **1216**. In one example, the deposition of conductive layer **1220** is controlled not to fill trenches **1214**. It is understood that in some examples, the deposition of conductive layer **1220** may fill trenches **1214** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1220** to expose the upper ends of semiconductor bodies **1212**, and conductive layer **1220** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1214** filled with conductive layer **1220** may be patterned and etched again to separate conductive layers **1220** between adjacent rows of semiconductor bodies **1212** and gate dielectrics **1218**. As described above, a lithography process can be performed to pattern trenches **1214** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). Due to the omission of silicon oxide layer **1008**, the top surface of conductive layer **1220** (including gate electrodes and word line) can be flush with the upper end of semiconductor body **1212** as shown in FIG. 12E, while the top surface of word line **1020** is below the upper end of semiconductor body **1012** in FIG. 10G.

[0220] As a result, patterned conductive layers **1220** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1214**, and parts of patterned conductive layers **1220** that are over gate dielectrics **1218** (e.g., fully circumscribes a respective gate dielectric **1218** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1218** over the exposed part semiconductor body **1212** and a respective gate electrode (part of conductive layer **1220**) over gate dielectric **1218** can be formed thereby. Comparing FIG. 12E with FIG. 12A, silicon nitride layer **1206** (sacrificial layer) in FIG. 12A is eventually replaced with conductive layer **1220** in FIG. 12E, according to some implementations.

[0221] As illustrated in FIG. 12E, the exposed upper end of each semiconductor body **1212**, e.g., one of the two ends of semiconductor body **1212** in the vertical direction (the z-direction) that is away from silicon substrate **1202**, is doped to form a source/drain **1221**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1212** to form sources/drains **1221**.

[0222] As illustrated in FIG. 12F, one or more ILD layers are formed over the top surface of conductive layer **1220**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Depending on the lateral dimensions of trenches **1214** (shown in FIG.

12E), trenches 1214 may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps 1222 between adjacent word lines (patterned conductive layers 1220). It is understood that in some examples, when the lateral dimensions of trenches 1214 are sufficiently large, dielectrics may fully fill trenches 1214 during the formation of the ILD layers, thereby eliminating air gaps 1222.

[0223] As illustrated in FIG. 12F, capacitor contacts 1224, first electrodes, capacitor dielectrics, and second electrodes of capacitors 1226, and a common plate 1228 are subsequently formed in the ILD layers to be coupled to semiconductor bodies 1212. In some implementations, each capacitor contact 1224 is formed on a respective source/drain 1221, e.g., the doped upper end of a respective semiconductor body 1212 by patterning and etching an electrode hole aligned with respective source/drain 1221 using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate 1228 is formed on the second electrodes of capacitors 1226 by patterning and etching an electrode trench aligned with capacitors 1226 using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[0224] As illustrated in FIG. 12G, a carrier substrate (a.k.a. a handle substrate) 1230 is bonded onto the front side of silicon substrate 1202 on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate 1202 become above carrier substrate 1230.

[0225] As illustrated in FIG. 12G, silicon substrate 1202 (shown in FIG. 12F) is removed to expose the undoped upper ends of semiconductor bodies 1212 (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate 1202 until being stopped by silicon oxide layer 1204 and the upper ends of semiconductor bodies 1212.

[0226] As illustrated in FIG. 12G, the exposed upper end of each semiconductor body 1212, e.g., one of the two ends of semiconductor body 1212 in the vertical direction (the z-direction) that is away from carrier substrate 1230, is doped to form another source/drain 1223. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies 1212 to form sources/drains 1223. As a result, multi-gate vertical transistors having semiconductor body 1212, sources/drains 1221 and 1223, gate dielectric 1218, and the gate electrode (part of conductive layer 1220) are formed thereby, as shown in FIG. 12G, according to some implementations. As described above, capacitors each having first and second electrodes 1224 and 1228 and capacitor dielectric 1226 are thereby formed as well, and DRAM cells 1280 each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. 12G, according to some implementations.

[0227] As illustrated in FIG. 12H, an interconnect layer 1232 can be formed above DRAM cells 1280. Interconnect layer 1232 can include interconnects of MEOL and/or

BEOL in a plurality of ILD layers to make electrical connections with DRAM cells 1280. In some implementations, interconnect layer 1232 includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers 1232 can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer 1204 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 12H can be collectively referred to as interconnect layer 1232. As illustrated in FIG. 12H, bit line 1234 can be formed on sources/drains 1223 by patterning and etching a trench aligned with respective source/drain 1223 using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line 1234 includes depositing a metal layer onto the exposed end of semiconductor body 1212.

[0228] As illustrated in FIG. 12H, a bonding layer 1236 is formed above interconnect layer 1232 and DRAM cells 1280. Bonding layer 1236 can include a plurality of bonding contacts 1237 surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer 1232 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts 1237 can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer 1232 by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0229] In some implementations as shown in FIGS. 13A-13H, a DRAM cell array is formed from a four-layer dielectric stack having a sacrificial layer (e.g., a silicon oxide layer) sandwiched between two dielectric layers (e.g., silicon nitride layers) on a pad layer (e.g., a silicon oxide layer).

[0230] At operation 2402 in FIG. 24, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, four layers having a first dielectric, a second dielectric, a third dielectric, and the second dielectric, respectively, are subsequently deposited on the substrate. The second dielectric can include silicon nitride, and the third dielectric can include silicon oxide. The layer having the third dielectric can act as a sacrificial layer vertically sandwiched between the two layers having the second dielectric. The sacrificial layer can be removed by selectively etching against the other layer having the second dielectric and replaced with a conductive layer in the later processes.

[0231] As illustrated in FIG. 13A, a stack of a silicon oxide layer 1304, a silicon nitride layer 1306, a silicon oxide layer 1308, and a silicon nitride layer 1309 is formed on a silicon substrate 1302. To form the dielectric stack, silicon oxide and silicon nitride are subsequently and alternatively

deposited onto silicon substrate **1202** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1304** (a pad layer) is formed by oxidizing the top portion of silicon substrate **1302** using dry oxidation and/or wet oxidation, such as ISSG oxidation process. In some implementations, the thickness of silicon oxide layer **1304** (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer **1308** (e.g., CVD silicon oxide).

[0232] As illustrated in FIG. 13B, an array of semiconductor bodies **1312** each extending vertically through the stack of silicon oxide layer **1304**, silicon nitride layer **1306**, silicon oxide layer **1308**, and silicon nitride layer **1309** are formed. Semiconductor body **1312** can be epitaxially grown from the respective exposed part of silicon substrate **1302** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1312** can include, but not limited to, VPE, LPE, MPE, or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1302** in the openings. Semiconductor body **1312** thus can have the same material as silicon substrate **1302**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1312** beyond the top surface of silicon nitride layer **1309**. As a result, an array of semiconductor bodies **1312** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1302** through the stack of silicon oxide layer **1304**, silicon nitride layer **1306**, silicon oxide layer **1308**, and silicon nitride layer **1309** is formed thereby, according to some implementation.

[0233] At operation **2406** in FIG. 24, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the third dielectric, and the layer having the third dielectric (e.g., the sacrificial layer) is etched away via the trench.

[0234] As illustrated in FIG. 13C, a plurality of trenches **1314** (slits/openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon nitride layer **1309** and silicon oxide layer **1308** are formed to expose silicon oxide layer **1308**. As a result, parts of silicon oxide layer **1308** can be exposed from trenches **1314**. In some implementations, a lithography process is performed to pattern trenches **1314** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1314** through silicon nitride layer **1309** and silicon oxide layer **1308** until being stopped by silicon nitride layer **1306**. It is understood that in some examples, the etching of trenches **1314** may go further into silicon nitride layer **1306**, but not into silicon oxide layer **1304**.

[0235] As illustrated in FIG. 13D, silicon oxide layer **1308** (shown in FIG. 13C) is removed to expose parts of semiconductor bodies **1312** abutting silicon oxide layer **1308**. In some implementations, silicon oxide layer **1308** is etched away via trenches **1214**. For example, a wet etchant including hydrofluoric acid may be applied through trenches **1314**

to selectively wet etch silicon oxide layer **1308** without etching silicon nitride layers **1309** and **1306** as well as semiconductor bodies **1312**. As a result, lateral recesses **1316** can be formed thereby, which expose parts of semiconductor bodies **1312**.

[0236] As illustrated in FIG. 13E, a gate dielectric **1318** is formed over the exposed part of each semiconductor body **1312**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1312**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1312** (e.g., single crystalline silicon) as gate dielectric **1318**. In some implementations, gate dielectric **1318** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1312** through trenches **1314** and lateral recesses **1316** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1316** and trenches **1314**. Due to the existence of silicon nitride layer **1309**, the upper end of gate dielectric **1318** can be below the upper end of semiconductor body **1312** in FIG. 13E.

[0237] As illustrated in FIG. 13F, a conductive layer **1320** is formed over gate dielectrics **1318** in lateral recesses **1316** (shown in FIG. 13E) through trenches **1314**. In some implementations, conductive layer **1320** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1318** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1314** to fill lateral recesses **1316**. In one example, the deposition of conductive layer **1320** is controlled not to fill trenches **1314**. It is understood that in some examples, the deposition of conductive layer **1320** may fill trenches **1314** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1320** to expose the upper ends of semiconductor bodies **1312**, and conductive layer **1320** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1314** filled with conductive layer **1320** may be patterned and etched again to separate conductive layers **1320** between adjacent rows of semiconductor bodies **1312** and gate dielectrics **1318**. As described above, a lithography process can be performed to pattern trenches **1314** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). Due to the existence of silicon nitride layer **1309**, the top surface of conductive layer **1320** (including gate electrodes and word line) can be below the upper end of semiconductor body **1312** in FIG. 13F.

[0238] As a result, patterned conductive layers **1320** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1314**, and parts of patterned conductive layers **1320** that are over gate dielectrics **1318** (e.g., fully circumscribes a respective gate dielectric **1318** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1318** over the exposed part semiconductor body **1312** and a respective gate electrode (part of conductive layer **1320**) over gate dielectric **1318** can be formed thereby. Comparing FIG. 13F with FIG. 13A, silicon oxide layer

1308 (sacrificial layer) in FIG. 13A is eventually replaced with conductive layer **1320** in FIG. 13F, according to some implementations.

[0239] As illustrated in FIG. 13F, the exposed upper end of each semiconductor body **1312**, e.g., one of the two ends of semiconductor body **1312** in the vertical direction (the z-direction) that is away from silicon substrate **1302**, is doped to form a source/drain **1321** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1312** to form sources/drains **1321**.

[0240] As illustrated in FIG. 13G, one or more ILD layers are formed over the top surface of silicon nitride layer **1309**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Depending on the lateral dimensions of trenches **1314** (shown in FIG. 13F), trenches **1314** may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps **1322** between adjacent word lines (patterned conductive layers **1320**). It is understood that in some examples, when the lateral dimensions of trenches **1314** are sufficiently large, dielectrics may fully fill trenches **1314** during the formation of the ILD layers, thereby eliminating air gaps **1322**.

[0241] As illustrated in FIG. 13G, capacitor contacts **1324**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1326**, and a common plate **1328** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1312**. In some implementations, each capacitor contact **1324** is formed on a respective source/drain **1321**, e.g., the doped upper end of a respective semiconductor body **1312** by patterning and etching an electrode hole aligned with respective source/drain **1321** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1328** is formed on capacitors **1326** by patterning and etching an electrode trench aligned with capacitors **1326** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[0242] As illustrated in FIG. 13H, a carrier substrate **1330** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1302** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1302** become above carrier substrate **1330**.

[0243] As illustrated in FIG. 13H, silicon substrate **1302** (shown in FIG. 13G) is removed to expose the undoped upper ends of semiconductor bodies **1312** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1302** until being stopped by silicon oxide layer **1304** and the upper ends of semiconductor bodies **1312**.

[0244] As illustrated in FIG. 13H, the exposed upper end of each semiconductor body **1312**, e.g., one of the two ends of semiconductor body **1312** in the vertical direction (the z-direction) that is away from carrier substrate **1330**, is

doped to form another source/drain **1323** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1312** to form sources/drains **1323**. As a result, vertical transistors having semiconductor body **1312**, sources/drains **1321** and **1323**, gate dielectric **1318**, and the gate electrode (part of conductive layer **1320**) are formed thereby, as shown in FIG. 13H, according to some implementations. As described above, capacitors each having first and second electrodes **1324** and **1328** and capacitor dielectric **1326** are thereby formed as well, and DRAM cells **1380** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. 13H, according to some implementations.

[0245] As illustrated in FIG. 13H, an interconnect layer **1332** can be formed above DRAM cells **1380**. Interconnect layer **1332** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1380**. In some implementations, interconnect layer **1332** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1332** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1304** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 13H can be collectively referred to as interconnect layer **1332**. As illustrated in FIG. 13H, bit line **1334** can be formed on sources/drains **1323** by patterning and etching a trench aligned with respective source/drain **1323** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1334** includes depositing a metal layer onto the exposed end of semiconductor body **1312**.

[0246] As illustrated in FIG. 13H, a bonding layer **1336** is formed above interconnect layer **1332** and DRAM cells **1380**. Bonding layer **1336** can include a plurality of bonding contacts **1337** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1332** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1337** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1332** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0247] A third semiconductor structure including a second array of memory cells can be formed. Each of the memory cells can also include a vertical transistor, and a storage unit coupled to the vertical transistor. The second semiconductor

structure and the third semiconductor structure can be bonded in a face-to-face manner. In some implementations, the second and third semiconductor structures are bonded prior to bonding the first and second semiconductor structures. For example, as shown in FIG. 23, the second and third semiconductor structures may be bonded prior to operation 2312, e.g., between operation 2306 and operation 2308.

[0248] As illustrated in FIG. 14A, two semiconductor structures 1000 and 1400 are formed separately (e.g., in parallel) using any suitable fabrication processes disclosed herein (e.g., in FIGS. 10A-10H). For ease of description, the fabrication process of forming semiconductor structure 1400 is not repeated and is the same as that of forming semiconductor structure 1000. Thus, two semiconductor structures 1000 and 1400 may have the same devices therein.

[0249] As illustrated in FIG. 14A, semiconductor structure 1400 is flipped upside down. As illustrated in FIG. 14B, semiconductor structure 1400 facing down is bonded with semiconductor structure 1000 facing up, e.g., in a face-to-face manner, thereby forming a bonding interface 1402, using any suitable substrate/wafer bonding processes including, for example, hybrid bonding (as described above in detail), anodic bonding, and fusion (direct) bonding. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface 1402. For example, silicon oxide layers may be formed on the top surfaces of semiconductor structures 1000 and 1400 to allow SiO₂—SiO₂ bonding using fusion bonding. In some implementations, common plate 1028 of semiconductor structure 1400 are in contact with common plate of semiconductor structure 1000 at bonding interface 1402 and thus, may be viewed as a common electrode (e.g., common ground plate) of both semiconductor structures 1000 and 1400.

[0250] As illustrated in FIG. 14B, silicon substrate 1002 (shown in FIG. 14A) of semiconductor structure 1400 (on top of semiconductor structure 1000 after bonding) is removed to expose the undoped upper ends of semiconductor bodies 1012 (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate 1002 of semiconductor structure 1400 until being stopped by silicon oxide layer 1004 and the upper ends of semiconductor bodies 1012 of semiconductor structure 1400.

[0251] As illustrated in FIG. 14B, the exposed upper end of each semiconductor body 1012 of semiconductor structure 1400, e.g., one of the two ends of semiconductor body 1012 in the vertical direction (the z-direction) that is away from semiconductor structure 1000, is doped to form another source/drain 1023. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies 1012 of semiconductor structure 1400 to form sources/drains 1023. As a result, multi-gate vertical transistors having semiconductor body 1012, sources/drains 1021 and 1023, gate dielectric

1018, and the gate electrode (part of conductive layer 1020) are formed thereby in semiconductor structure 1400, as shown in FIG. 14B, according to some implementations. As described above, capacitors 1026 are thereby formed as well, and DRAM cells 1080 each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure 1400, as shown in FIG. 14B, according to some implementations.

[0252] As illustrated in FIG. 14C, an interconnect layer 1032 can be formed above DRAM cells 1080. Interconnect layer 1032 can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells 1080. In some implementations, interconnect layer 1032 includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers 1032 can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer 1004 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 14C can be collectively referred to as interconnect layer 1032.

[0253] As illustrated in FIG. 14D, a carrier substrate 1030 (a.k.a. a handle substrate) is bonded onto the front side of semiconductor structure 1400 on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that semiconductor structure 1000 become above carrier substrate 1030 (not shown in FIG. 14D).

[0254] As illustrated in FIG. 14D, silicon substrate 1002 of semiconductor structure 1000 (shown in FIG. 14C) is removed to expose the undoped upper ends of semiconductor bodies 1012 of semiconductor structure 1000 (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate 1002 of semiconductor structure 1000 until being stopped by silicon oxide layer 1004 and the upper ends of semiconductor bodies 1012 of semiconductor structure 1000.

[0255] As illustrated in FIG. 14D, the exposed upper end of each semiconductor body 1012 of semiconductor structure 1000, e.g., one of the two ends of semiconductor body 1012 in the vertical direction (the z-direction) that is away from semiconductor structure 1400, is doped to form another source/drain 1023. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies 1012 of semiconductor structure 1000 to form sources/drains 1023. As a result, vertical transistors having semiconductor body 1012, sources/drains 1021 and 1023, gate dielectric 1018, and the gate electrode (part of conductive layer 1020) are formed thereby in semiconductor structure 1000, as shown in FIG. 14D, according to some implementations. As described above, capacitors 1026 are thereby formed as well, and

DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure **1000**, as shown in FIG. **14D**, according to some implementations.

[0256] As illustrated in FIG. **14E**, an interconnect layer **1032** can be formed above DRAM cells **1080** in semiconductor structure **1000**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080** in semiconductor structure **1000**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **14E** can be collectively referred to as interconnect layer **1032**.

[0257] As illustrated in FIG. **14E**, a bonding layer **1036** is formed above interconnect layer **1032** and DRAM cells **1080** in semiconductor structure **1000**. Bonding layer **1036** can include a plurality of bonding contacts **1037** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1032** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1037** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1032** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0258] The bonded structure shown in FIG. **14E** then can be bonded with a semiconductor structure including peripheral circuits in a face-to-face manner, as described above in detail with respect to operation **2312** in FIG. **23** and FIGS. **10L** and **10M**.

[0259] In some implementations, the second and third semiconductor structures are bonded after bonding the first and second semiconductor structures. For example, as shown in FIG. **23**, the second and third semiconductor structures may be bonded after operation **2312**, e.g., between operation **2312** and operation **2314**.

[0260] As illustrated in FIG. **15A**, a bonded semiconductor structure **1500** is formed after the fabrication process shown in FIG. **10L** by removing carrier substrate **1030**. A semiconductor structure **1000** is formed separately (e.g., in parallel) using any suitable fabrication processes disclosed herein (e.g., in FIGS. **10A-10H**). For ease of description, the fabrication processes of forming semiconductor structures **1000** and **1500** are not repeated.

[0261] As illustrated in FIG. **15A**, semiconductor structure **1000** is flipped upside down. As illustrated in FIG. **15B**, semiconductor structure **1000** facing down is bonded with semiconductor structure **1500** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **1502**, using any suitable substrate/wafer bonding processes including, for example, hybrid bonding (as described above in detail), anodic bonding, and fusion (direct) bonding. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **1502**. For example, silicon oxide layers may be formed on the top surfaces of semiconductor structures **1000** and **1500** to allow SiO₂—SiO₂ bonding using fusion bonding. In some implementations, common plate **1028** of semiconductor structure **1000** are in contact with common plate **1028** of semiconductor structure **1500** at bonding interface **1502** and thus, may be viewed as a common electrode (e.g., common ground plate) of both semiconductor structures **1000** and **1500**.

[0262] As illustrated in FIG. **15C**, silicon substrate **1002** (shown in FIG. **15B**) of semiconductor structure **1000** (on top of semiconductor structure **1500** after bonding) is removed to expose the undoped upper ends of semiconductor bodies **1012** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1002** of semiconductor structure **1000** until being stopped by silicon oxide layer **1004** and the upper ends of semiconductor bodies **1012** of semiconductor structure **1000**.

[0263] As illustrated in FIG. **15C**, the exposed upper end of each semiconductor body **1012** of semiconductor structure **1000**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from semiconductor structure **1500**, is doped to form another source/drain **1023**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** of semiconductor structure **1000** to form sources/drains **1023**. As a result, multi-gate vertical transistors having semiconductor body **1012**, sources/drains **1021** and **1023**, gate dielectric **1018**, and the gate electrode (part of conductive layer **1020**) are formed thereby in semiconductor structure **1000**, as shown in FIG. **15C**, according to some implementations. As described above, capacitors **1026** are thereby formed as well, and DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure **1400**, as shown in FIG. **15C**, according to some implementations.

[0264] As illustrated in FIG. **15D**, an interconnect layer **1032** can be formed above DRAM cells **1080** in semiconductor structure **1000**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects therein

formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **15D** can be collectively referred to as interconnect layer **1032**.

[0265] A pad-out interconnect layer then can be formed on the bonded structure shown in FIG. **15D** as described above in detail with respect to operation **2314** in FIG. **23** and FIG. **10M**.

[0266] Method **2300** may also be implemented by the fabrication process described in FIGS. **19A-19M** and **22** to form 3D memory device **1700** depicted in FIG. **17** having single-gate vertical transistors, as opposed to multiple-gate vertical transistors. Referring to FIG. **23**, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.

[0267] As illustrated in FIG. **19L**, a plurality of transistors **1948** are formed on a silicon substrate **1944**. Transistors **1948** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1944** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **1948**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1944** by wet/dry etch and thin film deposition. Transistors **1948** can form peripheral circuits **1946** on silicon substrate **1944**.

[0268] As illustrated in FIG. **19L**, an interconnect layer **1950** can be formed above peripheral circuits **1946** having transistors **1948**. Interconnect layer **1950** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **1946**. In some implementations, interconnect layer **1950** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1950** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19L** can be collectively referred to as interconnect layer **1950**.

[0269] Method **2300** proceeds to operation **2304**, as illustrated in FIG. **23**, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The

first bonding layer can include a first bonding contact. As illustrated in FIG. **19L**, a bonding layer **1952** is formed above interconnect layer **1950** and peripheral circuits **1946**. Bonding layer **1952** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1950** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts then can be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1950** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0270] Method **2300** proceeds to operation **2306**, as illustrated in FIG. **23**, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

[0271] For example, FIG. **25** illustrates a flowchart of a method **2500** for forming another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2502** in FIG. **25**, a semiconductor pillar extending vertically in a substrate is formed. The substrate can be a silicon substrate. In some implementations, to form the semiconductor pillar, the substrate is etched in a first lateral direction to form a plurality of first trenches, a dielectric is deposited to fill the first trenches to form second trench isolations, and the substrate and the second trench isolations are etched in a second lateral direction to form a plurality of second trenches and the semiconductor pillar surrounded by the second trenches and the second trench isolations. In some implementations, a dielectric is deposited to partially fill the second trenches.

[0272] As illustrated in FIG. **19A**, a plurality of parallel trenches **1904** are formed in the y-direction (e.g., the bit line direction) to form a plurality of parallel semiconductor walls **1905** in the y-direction. In some implementations, a lithography process is performed to pattern trenches **1904** and semiconductor walls **1905** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1904** in a silicon substrate **1902**. Thus, semiconductor wall **1905** extending vertically in silicon substrate **1902** can be formed. The bottom of semiconductor wall **1905** can be below the top surface of silicon substrate **1902**. Since semiconductor walls **1905** are formed by etching silicon substrate **1902**, semiconductor walls **1905** can have the same material as silicon substrate **1902**, such as single crystalline silicon. FIG. **19A** illustrates both the side view (in the top portion of FIG. **19A**) of a cross-section along the x-direction (the word line direction, e.g., in the BB plane) and the plan view (in the bottom portion of FIG. **19A**) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor walls **1905**). The same drawing layout is arranged in FIG. **19B** as well.

[0273] As illustrated in FIG. 19B, trench isolations 1908 (e.g., STIs) are formed in trenches 1904. In some implementations, a dielectric, such as silicon oxide, is deposited to fully fill trenches 1904 using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, a planarization process, such as CMP, is performed to remove excess dielectric deposited beyond the top surface of silicon substrate 1902. As a result, parallel semiconductor walls 1905 can be separated by trench isolations 1908.

[0274] As illustrated in FIG. 19C, a plurality of parallel trenches 1910 are formed in the x-direction (e.g., the word line direction) to form an array of semiconductor pillars 1906 each extending vertically in silicon substrate 1902. In some implementations, a lithography process is performed to pattern trenches 1910 to be perpendicular to trench isolations 1908 using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed on silicon substrate 1902 and trench isolation 1908 to etch trenches 1910 in silicon substrate 1902. As a result, semiconductor walls 1905 (shown in FIG. 19B) can be cut by trenches 1910 to form an array of semiconductor pillars 1906 each extending vertically in silicon substrate 1902. The bottom of semiconductor pillar 1906 can be below the top surface of silicon substrate 1902. Since semiconductor pillars 1906 are formed by etching silicon substrate 1902, semiconductor pillars 1906 can have the same material as silicon substrate 1902, such as single crystalline silicon. FIG. 19C illustrates both the side view (in the top portion of FIG. 19C) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of FIG. 19C) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor pillars 1906). The same drawing layout is arranged in FIGS. 19C-19G as well.

[0275] As illustrated in FIG. 19C, a dielectric layer 1912 is formed at the bottom of trench 1910, for example, by depositing a dielectric, such as silicon oxide, to partially fill trench 1910, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The deposition conditions, such as deposition rate and/or time, can be controlled to control the thickness of dielectric layer 1912 and avoid fully filling trench 1910. As a result, the bottom surface of trenches 1910 can be elevated to be above the bottom surface of semiconductor pillars 1906. As shown in the plan view, the two opposite sides of semiconductor pillar 1906 in the y-direction are exposed by trenches 1910, and the other two opposite sides of semiconductor pillar 1906 in the x-direction are in contact with trench isolation 1908. In other words, semiconductor pillar 1906 is surrounded by trenches 1910 and trench isolations 1908.

[0276] At operation 2504 in FIG. 25, gate structures in contact with opposite sides of the semiconductor pillar are formed. In some implementations, to form the gate structures, gate dielectrics are formed over the opposite sides of the semiconductor pillar, and gate electrodes are formed over the gate dielectrics. In some implementations, to form the gate electrodes, conductive layers are deposited over the gate dielectrics, and the conductive layers are etched back.

[0277] As illustrated in FIG. 19D, gate dielectrics 1914 are formed over the two opposite sides of semiconductor pillars 1906 in the bit line direction (the y-direction) exposed from

trenches 1910. As shown in the plan view, gate dielectrics 1914 can be parts of a continuous dielectric layer formed over sidewalls of each row of semiconductor pillars 1906 and trench isolations 1908. In some implementations, gate dielectric 1914 is formed by depositing a layer of dielectric, such as silicon oxide, over the sidewalls of trenches 1910 using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without fully filling trenches 1910. It is understood that in some examples, gate dielectrics 1914 may not be parts of a continuous dielectric layer. For example, a wet oxidation and/or a dry oxidation process, such as in situ steam generation (ISSG) oxidation, is performed to form native oxide (e.g., silicon oxide) on semiconductor pillar 1906 (e.g., single crystalline silicon) as gate dielectric 1914.

[0278] As illustrated in FIG. 19D, conductive layers 1916 are formed over gate dielectrics 1914 in trenches 1910. In some implementations, conductive layers 1916 are formed by depositing one or more conductive materials, such as metal and/or metal compounds (e.g., W and TiN), over gate dielectrics 1914 using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, to partially fill trenches 1910. For example, layers of TiN and W may be sequentially deposited to form conductive layer 1916. A planarization process, e.g., CMP, can be performed to remove the excess conductive materials over the top surface of silicon substrate 1902.

[0279] As illustrated in FIG. 19E, in some implementations, conductive layers 1916 are etched back, for example, using dry etch and/or wet etch (e.g., RIE), to form dents, such that the upper ends of conductive layers 1916 are below the top surface of semiconductor pillars 1906. In some implementations, as gate dielectrics 1914 are not etched back, the upper ends of conductive layers 1916 are below the upper ends of gate dielectrics 1914 as well, which are flush with the top surface of semiconductor pillars 1906. As a result, etched-back conductive layers 1916 can become word lines each extending in the word line direction (the x-direction), and parts of etched-back conductive layers 1916 that are facing semiconductor pillars 1906 can become gate electrodes. Gate structures each including a respective gate dielectric 1914 over the exposed side of semiconductor pillar 1906 and a respective gate electrode (part of conductive layer 1916) over gate dielectric 1914 can be formed thereby. In some implementations, as shown in FIG. 19E, a dielectric layer 1918 is formed in the remaining space of trenches 1910 as well as the dents (not shown) resulting from etching back of conductive layers 1916, for example, by depositing a dielectric, such as silicon oxide, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that depending on the pitches of the word lines (the dimension of trenches 1910), air gaps may be formed in dielectric layer 1918.

[0280] At operation 2506 in FIG. 25, a first trench isolation extending vertically through the semiconductor pillar is formed to separate the semiconductor pillar into semiconductor bodies each in contact with a respective one of the gate structures. In some implementations, to form the first trench, the semiconductor pillar is etched in the second lateral direction to form a third trench, and a dielectric is deposited to fill the third trench.

[0281] As illustrated in FIG. 19F, a plurality of parallel trenches 1922 in the x-direction (e.g., the word line direc-

tion) are formed to form an array of semiconductor bodies **1920** each extending vertically in silicon substrate **1902**. In some implementations, a lithography process is performed to pattern trenches **1922** on semiconductor pillars **1906** (shown in FIG. **19E**) using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, are performed on semiconductor pillars **1906** and trench isolation **1908** to etch trenches **1922**. The etching can be controlled such that bottom of trenches **1922** is flush with or below the bottom surface of semiconductor pillars **1906**. As a result, each semiconductor pillar **1906** can be separated by a respective trench **1922** into two semiconductor bodies **1920** in the y-direction. Since semiconductor bodies **1920** are formed by etching silicon substrate **1902**, semiconductor bodies **1920** can have the same material as silicon substrate **1902**, such as single crystalline silicon. As shown in the plan view, each semiconductor body **1920** can be in contact with a gate structure having gate dielectric **1914** and gate electrode **1916** on one side of semiconductor body **1920** in the y-direction. The opposite side of semiconductor body **1920** can be exposed by trench **1922**. In some implementations, a mirror-symmetric arrangement of two semiconductor bodies **1920** and two gate structures thereof is achieved by forming trench **1922** across the middle of a respective semiconductor pillar **1906**.

[0282] As illustrated in FIG. **19G**, a trench isolation **1926** is formed in trench **1922** (shown in FIG. **19F**), for example, by depositing a dielectric, such as silicon oxide, to fill trench **1922**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A planarization process can be performed to remove excess dielectric over the top surface of silicon substrate **1902**. It is understood that depending on the pitches of the semiconductor bodies **1920** (the dimension of trenches **1922**), air gaps may be formed in trench isolation **1926**. As shown in the plan view, parallel trench isolations **1926** each extending in the x-direction can form an array of semiconductor bodies **1920** in which a single side is in contact with a gate structure having gate dielectric **1914** and gate electrode **1916**.

[0283] At operation **2508** in FIG. **25**, first ends of the semiconductor bodies away from the substrate are doped. As illustrated in FIG. **19G**, the exposed upper end of each semiconductor body **1920**, e.g., one of the two ends of semiconductor body **1920** in the vertical direction (the z-direction) that is away from silicon substrate **1902**, is doped to form a source/drain **1924** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1920** to form sources/drains **1924**. In some implementations, a silicide layer is formed on source/drain **1924** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1920**.

[0284] At operation **2510** in FIG. **25**, storage units in contact with the semiconductor bodies, e.g., the doped first ends thereof, are formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

[0285] As illustrated in FIG. **19H**, one or more ILD layers are formed over the top surface of silicon substrate **1902**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Capacitor contacts **1928**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1930**, and a common plate **1932** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1920**. In some implementations, capacitor contact **1928** is formed on a respective source/drain **1924**, e.g., the doped upper end of a respective semiconductor body **1920** by patterning and etching an electrode hole aligned with respective source/drain **1924** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1932** is formed on the second electrodes of capacitors **1930** by patterning and etching an electrode trench aligned with capacitors **1930** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[0286] At operation **2512** in FIG. **25**, the substrate is thinned to expose second ends of the semiconductor bodies opposite to the first end. As illustrated in FIG. **19I**, a carrier substrate **1934** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1902** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1902** become above carrier substrate **1934**.

[0287] As illustrated in FIG. **19J**, silicon substrate **1902** (shown in FIG. **19I**) is thinned to expose the undoped upper ends of semiconductor bodies **1920** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to thin silicon substrate **1902** until being stopped by dielectric layer **1918** and the upper ends of semiconductor bodies **1920**.

[0288] At operation **2514** in FIG. **25**, the exposed second ends of the semiconductor bodies are doped. As illustrated in FIG. **19J**, the exposed upper end of each semiconductor body **1920**, e.g., one of the two ends of semiconductor body **1920** in the vertical direction (the z-direction) that is away from carrier substrate **1934**, is doped to form another source/drain **1936** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1920** to form sources/drains **1936**. In some implementations, a silicide layer is formed on source/drain **1936** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1920**. As a result, vertical transistors having semiconductor body **1920**, sources/drains **1924** and **1936**, gate dielectric **1914**, and the gate electrode (part of conductive layer **1916**) are formed thereby, as shown in FIG. **19J**, according to some implementations. As described above, capacitors **1930** each having the first and second electrodes and the capacitor dielectric are thereby formed as well, and DRAM cells **1980** each having a single-gate vertical transistor and a capacitor coupled to the single-gate vertical transistor are thereby formed, as shown in FIG. **19J**, according to some implementations.

[0289] Referring back to FIG. 23, method 2300 proceeds to operation 2308, as illustrated in FIG. 23, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. 19K, an interconnect layer 1940 can be formed above DRAM cells 1980. Interconnect layer 1940 can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells 1980. In some implementations, interconnect layer 1940 includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers 1940 can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 19K can be collectively referred to as interconnect layer 1940.

[0290] As shown in FIG. 25, at operation 2516, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. 19K, bit line 1938 can be formed on sources/drains 1936 by patterning and etching a trench aligned with respective source/drain 1936 using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line 1938 includes depositing a metal layer onto the exposed end of semiconductor body 1920. As a result, bit line 1938 and capacitor 1930 can be formed on opposite sides of semiconductor body 1920 and coupled to opposite ends of semiconductor body 1920. It is understood that additional local interconnects, such as word line contacts, capacitor contacts (e.g., a conductor), and bit line contacts (not shown in FIG. 19K, e.g., a metal silicide contact) may be similarly formed as well. In some implementations, the bit line contact (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body 1820, and bit line 1938 is formed on the bit line contact.

[0291] Method 2300 proceeds to operation 2310, as illustrated in FIG. 23, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. 19K, a bonding layer 1942 is formed above interconnect layer 1940 and DRAM cells 1980. Bonding layer 1942 can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer 1940 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer 1940 by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0292] Method 2300 proceeds to operation 2312, as illustrated in FIG. 23, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

[0293] As illustrated in FIG. 19L, carrier substrate 1934 and components formed thereon (e.g., DRAM cells 1980) are flipped upside down. As illustrated in FIG. 19L, bonding layer 1942 facing down is bonded with bonding layer 1952 facing up, e.g., in a face-to-face manner, thereby forming a bonding interface 1954. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIG. 19L, silicon substrate 1944 and components formed thereon (e.g., peripheral circuits 1946) can be flipped upside down, and bonding layer 1952 facing down can be bonded with bonding layer 1942 facing up, e.g., in a face-to-face manner, thereby forming bonding interface 1954. After the bonding, the bonding contacts in bonding layer 1942 and the bonding contacts in bonding layer 1952 are aligned and in contact with one another, such that DRAM cells 1980 can be electrically connected to peripheral circuits 1946 across bonding interface 1954. It is understood that in the bonded chip, DRAM cells 1980 may be either above or below peripheral circuits 1946. Nevertheless, bonding interface 1954 can be formed vertically between peripheral circuits 1946 and DRAM cells 1980 after the bonding.

[0294] Method 2300 proceeds to operation 2314, as illustrated in FIG. 23, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. 19M, a pad-out interconnect layer 1956 is formed above on the backside of carrier substrate 1934. Pad-out interconnect layer 1956 can include interconnects, such as pad contacts 1958, formed in one or more ILD layers. Pad contacts 1958 can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts 1960 are formed extending vertically through carrier substrate 1934, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts 1960 can be in contact with the interconnects in pad-out interconnect layer 1956. It is understood that in some examples, carrier substrate 1934 may be thinned or removed after bonding and prior to forming pad-out interconnect layer 1956 and contacts 1960, for example, using planarization processes and/or etching processes.

[0295] Although not shown, it is understood that in some examples, pad-out interconnect layer 1956 may be formed above on the backside of silicon substrate 1944, and contacts 1960 may be formed extending vertically through silicon substrate 1944. Silicon substrate 1944 may be thinned prior

to forming pad-out interconnect layer **1956** and contacts **1960**, for example, using planarization processes and/or etching processes. Although not shown, it is further understood that in some examples, the fabrication processes described with respect to FIGS. **14A-14E** and **15A-15D** may be applied to form another array of DRAM cells **1980** in another semiconductor structure bonded to the semiconductor structure including DRAM cells **1980** described above with respect to FIGS. **19A-19M**.

[**0296**] Method **2300** may further be implemented by the fabrication process described in FIGS. **22A-22M** and **26** to form 3D memory device **2100** depicted in FIG. **21** having double-gate vertical transistors, as opposed to single-gate vertical transistors. Referring to FIG. **23**, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.

[**0297**] As illustrated in FIG. **22L**, a plurality of transistors **2248** are formed on a silicon substrate **2244**. Transistors **2248** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2244** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **2248**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2244** by wet/dry etch and thin film deposition. Transistors **2248** can form peripheral circuits **2246** on silicon substrate **2244**.

[**0298**] As illustrated in FIG. **22L**, an interconnect layer **2250** can be formed above peripheral circuits **2246** having transistors **2248**. Interconnect layer **2250** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **2246**. In some implementations, interconnect layer **2250** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **2250** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **22L** can be collectively referred to as interconnect layer **2250**.

[**0299**] Method **2300** proceeds to operation **2304**, as illustrated in FIG. **23**, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The first bonding layer can include a first bonding contact. As illustrated in FIG. **22L**, a bonding layer **2252** is formed above interconnect layer **2250** and peripheral circuits **2246**. Bonding layer **2252** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **2250** by one or more thin film deposition processes including, but not limited to,

CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2250** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[**0300**] Method **2300** proceeds to operation **2306**, as illustrated in FIG. **23**, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

[**0301**] For example, FIG. **26** illustrates a flowchart of a method **2600** for forming still another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2602** in FIG. **26**, a semiconductor body extending vertically in a substrate is formed. The substrate can be an SOI substrate including a handle layer, a buried oxide layer, and a device layer. In some implementations, to form the semiconductor body, the handle layer is etched in a first lateral direction to form first trenches, and the handle layer is etched in a second lateral direction to form second trenches, such that the two opposite sides of the semiconductor body is exposed by the second trenches. In some implementations, a dielectric is deposited to partially fill the second trenches.

[**0302**] As illustrated in FIG. **22A**, a plurality of parallel trenches **2204** are formed in the y-direction (e.g., the bit line direction) to form a plurality of parallel semiconductor walls **2205** in the y-direction. In some implementations, a lithography process is performed to pattern trenches **2204** and semiconductor walls **2205** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1904** in an SOI substrate **2201**. Thus, semiconductor wall **1905** extending vertically in SOI substrate **2201** can be formed. As shown in FIG. **22A**, SOI substrate **2201** can include a handle layer **2202**, a buried oxide layer **2203** on handle layer **2202**, and a device layer **2209** on buried oxide layer **2203**. In some implementations, buried oxide layer **2203** includes silicon oxide, and device layer **2209** includes single crystalline silicon. In some implementations, to form trenches **2204**, device layer **2209** is etched using RIE, stopped at buried oxide layer **2203**. That is, buried oxide layer **2203** can serve as the etch stop layer. It is understood that in some examples, device layer **2209** may not be part of an SOI substrate, but transferred and bonded onto buried oxide layer **2203** from another silicon substrate (not shown, e.g., an SOI substrate). It is also understood that in some examples, SOI substrate **2201** may be replaced with a silicon substrate, such as silicon substrate **1902** in FIG. **19A**; the etching of trenches **2204** may not be stopped by buried oxide layer **2203**, but by controlling the etching rate and/or duration, for example, as shown in FIG. **19A**.

[**0303**] Nevertheless, the bottom of semiconductor wall **2205** can be below the top surface of SOI substrate **2201**. Since semiconductor walls **2205** are formed by etching

device layer **2209** of SOI substrate **2201**, semiconductor walls **2205** can have the same material as device layer **2209** of SOI substrate **2201**, such as single crystalline silicon. FIG. **22A** illustrates both the side view (in the top portion of FIG. **22A**) of a cross-section along the x-direction (the word line direction, e.g., in the BB plane) and the plan view (in the bottom portion of FIG. **22A**) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor walls **2205**).

[0304] As illustrated in FIG. **22B**, a plurality of parallel trenches **2210** are formed in the x-direction (e.g., the word line direction) to form an array of semiconductor bodies **2206** each extending vertically in SOI substrate **2201**. In some implementations, a lithography process is performed to pattern trenches **2210** to be perpendicular to trenches **2204** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **2210** in device layer **2209** of SOI substrate **2201**. As a result, semiconductor walls **2205** (shown in FIG. **22A**) can be cut by trenches **2210** to form an array of semiconductor bodies **2206** each extending vertically in SOI substrate **2201**. The bottom of semiconductor body **2206** can be below the top surface of SOI substrate **2201**. Since semiconductor bodies **2206** are formed by etching device layer **2209** of SOI substrate **2201**, semiconductor bodies **2206** can have the same material as device layer **2209** of SOI substrate **2201**, such as single crystalline silicon. FIG. **22B** illustrates both the side view (in the top portion of FIG. **22B**) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of FIG. **22B**) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor bodies **2206**). The same drawing layout is arranged in FIGS. **22C-22G** as well.

[0305] It is understood that in some examples, trenches **2204** and **2210** may be formed in the same process, as opposed to two consecutive processes. For example, the same lithography process may be used to pattern and trenches **2204** and **2210**, followed by the same etching process. It is also understood that in some examples, trenches **2210** in the word line direction may be formed prior to the formation of trenches **2204** in the bit line direction. Nevertheless, after the formation of trenches **2204** and **2210**, semiconductor body **2206** can be formed, and all four sides of semiconductor body **2206** can be exposed by trenches **2204** and **2210**. In some implementations, two opposite sides of semiconductor body **2206** in the word line direction are exposed by trenches **2204**, and two opposite sides of semiconductor body **2206** in the bit line direction are exposed by trenches **2210**. As shown in the plan view. In other words, semiconductor body **2206** can be surrounded by trenches **2204** and **2210**.

[0306] As illustrated in FIG. **22C**, a dielectric layer **2212** is formed at the bottom of trench **2210** (and trench **2204** in some examples), for example, by depositing a dielectric, such as silicon oxide, to partially fill trench **2210**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The deposition conditions, such as deposition rate and/or time, can be controlled to control the thickness of dielectric layer **2212** and avoid fully filling trench **2210**. As a result, the bottom surface of trenches **2210** can be elevated to be above the bottom surface of semiconductor bodies **2206**.

[0307] At operation **2604** in FIG. **26**, a gate structure in contact with opposite sides of the semiconductor body is formed. In some implementations, to form the gate structure, a gate dielectric is formed over the opposite sides of the semiconductor body, and a gate electrode is formed over the gate dielectric. In some implementations, to form the gate electrode, a conductive layer is deposited over the gate dielectric, and the conductive layer is etched back.

[0308] As illustrated in FIG. **22D**, a gate dielectric **2214** is formed over the two opposite sides of semiconductor body **2206** in the bit line direction (the y-direction) exposed from trenches **2210**. As shown in the plan view, gate dielectrics **2214** can be parts of a continuous dielectric layer formed over sidewalls of each row of semiconductor bodies **2206**. In some implementations, gate dielectric **2214** is formed by depositing a layer of dielectric, such as silicon oxide, over the sidewalls and top surfaces of semiconductor bodies **2206** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without fully filling trenches **2210**. It is understood that in some examples, gate dielectrics **2214** may not be parts of a continuous dielectric layer. For example, a wet oxidation and/or a dry oxidation process, such as ISSG oxidation, is performed to form native oxide (e.g., silicon oxide) on semiconductor bodies **2206** (e.g., single crystalline silicon) as gate dielectric **2214**.

[0309] As illustrated in FIG. **22D**, a conductive layer **2216** is formed over gate dielectrics **2214**. In some implementations, conductive layers **2216** are formed by depositing one or more conductive materials, such as metal and/or metal compounds (e.g., W and TiN), over gate dielectrics **2214** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, to partially fill trenches **2210**. For example, layers of TiN and W may be sequentially deposited to form conductive layer **2216**. As shown in the side view, conductive layers **2216** can be a continuous layer in the bit line direction as the conductive materials can be deposited over the top surfaces of semiconductor bodies **2206** and the bottom surfaces of trenches **2210**.

[0310] As illustrated in FIG. **22E**, in some implementations, parts of conductive layers **2216** at the bottom surfaces of trenches **2210** are removed to separate the continuous conductive layers **2216** into discrete pieces in the bit line direction, for example, using dry etch and/or wet etch (e.g., RIE) to form cuts **2211** on the bottom surfaces of trenches **2210**. In some implementations, parts of conductive layers **2216** at the top surfaces of semiconductor bodies **2206** are removed as well by the same etching process to expose gate dielectrics **2214** at the top surfaces of semiconductor bodies **2206**.

[0311] As illustrated in FIG. **22F**, in some implementations, trench isolations **2218** are formed in trench **2210** (shown in FIG. **22E**), for example, by depositing a dielectric, such as silicon oxide, to fill trench **2210**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A planarization process (e.g., CMP and/or etching process) can be performed to remove excess dielectric over the top surface of semiconductor bodies **2206**. In some implementations, the planarization process removes parts of gate dielectrics **2214** over the top surfaces of semiconductor bodies **2206** as well to expose the top surfaces of semiconductor bodies **2206**. It is understood that depending on the pitches of the

semiconductor bodies **2206** (the dimension of trenches **2210**), air gaps may be formed in trench isolation **2218**. As shown in the plan view, the deposition of the dielectric may fill the remaining spaces of trenches **2204** (shown in FIG. **22E**) as well to form isolations **2219** between adjacent semiconductor bodies **2206** in the word line direction (e.g., in the same row).

[**0312**] As illustrated in FIG. **22G**, in some implementations, conductive layers **2216** are etched back, for example, using dry etch and/or wet etch (e.g., RIE), to form dents, such that the upper ends of conductive layers **2216** are below the top surface of semiconductor bodies **1906**. In some implementations, as gate dielectrics **2214** are not etched back, the upper ends of conductive layers **2216** are below the upper ends of gate dielectrics **2214** as well, which are flush with the top surface of semiconductor bodies **2206**. As a result, etched-back conductive layers **2216** can become word lines each extending in the word line direction (the x-direction), and parts of etched-back conductive layers **2216** that are facing semiconductor bodies **2206** can become gate electrodes. Gate structures each including a respective gate dielectric **2214** over the exposed two opposite sides (in the bit line direction) of semiconductor body **2206** and a respective gate electrode (part of conductive layer **2216**) over gate dielectric **2214** can be formed thereby.

[**0313**] At operation **2608** in FIG. **26**, a first end of the semiconductor body away from the substrate is doped. As illustrated in FIG. **22G**, the exposed upper end (top surface) of each semiconductor body **2206**, e.g., one of the two ends of semiconductor body **2206** in the vertical direction (the z-direction) that is away from handle layer **2202** of SOI substrate **2201**, is doped to form a source/drain **2224** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **2206** to form sources/drains **2224**. In some implementations, a silicide layer is formed on source/drain **2224** by performing a silicidation process at the exposed upper ends of semiconductor bodies **2206**.

[**0314**] At operation **2608** in FIG. **26**, a storage unit in contact with the semiconductor body, e.g., the doped first end thereof, is formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

[**0315**] As illustrated in FIG. **22H**, one or more ILD layers are formed over the top surface of semiconductor bodies **2206**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Capacitor contacts **2228**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **2230**, and a common plate **2232** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **2206**. In some implementations, each capacitor contact **2228** is formed on a respective source/drain **2224**, e.g., the doped upper end of a respective semiconductor body **2206** by patterning and etching an electrode hole aligned with respective source/drain **2224** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementa-

tions, common plate **2232** is formed on capacitors **2230** by patterning and etching an electrode trench aligned with capacitors dielectrics **2230** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[**0316**] At operation **2610** in FIG. **26**, the substrate is thinned to expose a second end of the semiconductor body opposite to the first end. As illustrated in FIG. **22**, a carrier substrate **2234** (a.k.a. a handle substrate) is bonded onto the front side of SOI substrate **2201** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that handle layer **2202** of SOI substrate **2201** become above carrier substrate **2234**.

[**0317**] As illustrated in FIG. **22J**, SOI substrate **2201** is thinned to expose the undoped upper ends of semiconductor bodies **2206** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove handle layer **2202** and buried oxide layer **2203** (shown in FIG. **22G**) of SOI substrate **2201** until being stopped by the upper ends of semiconductor bodies **2206**.

[**0318**] At operation **2612** in FIG. **26**, the exposed second end of the semiconductor body is doped. As illustrated in FIG. **22J**, the exposed upper end of each semiconductor body **2206**, e.g., one of the two ends of semiconductor body **2206** in the vertical direction (the z-direction) that is away from carrier substrate **2234**, is doped to form another source/drain **2236** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **2206** to form sources/drains **2236**. In some implementations, a silicide layer is formed on source/drain **2236** by performing a silicidation process at the exposed upper ends of semiconductor bodies **2206**. As a result, vertical transistors having semiconductor body **2206**, sources/drains **2224** and **2236**, gate dielectric **2214**, and the gate electrode (part of conductive layer **2216**) are formed thereby, as shown in FIG. **22J**, according to some implementations. As described above, capacitors **2230** are thereby formed as well, and DRAM cells **2280** each having a double-gate vertical transistor and a capacitor coupled to the double-gate vertical transistor are thereby formed, as shown in FIG. **22J**, according to some implementations.

[**0319**] Referring back to FIG. **23**, method **2300** proceeds to operation **2308**, as illustrated in FIG. **23**, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. **22K**, an interconnect layer **2240** can be formed above DRAM cells **2280**. Interconnect layer **2240** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **2280**. In some implementations, interconnect layer **2240** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **2240** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electrodeless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited using one or

more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 22K can be collectively referred to as interconnect layer 2240.

[0320] As shown in FIG. 26, at operation 2614, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. 22K, bit line 2238 can be formed on sources/drains 2236 by patterning and etching a trench aligned with respective source/drain 2236 using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line 2238 includes depositing a metal layer onto the exposed end of semiconductor body 2220. As a result, bit line 2238 and capacitor 2230 can be formed on opposite sides of semiconductor body 2206 and coupled to opposite ends of semiconductor body 2206. It is understood that additional local interconnects, such as word line contacts, capacitor contacts (e.g., a conductor), and bit line contacts (e.g., a metal silicide contact) may be similarly formed as well. In some implementations, the bit line contact (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body 2220, and bit line 2238 is formed on the bit line contact.

[0321] Method 2300 proceeds to operation 2310, as illustrated in FIG. 23, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. 22K, a bonding layer 2242 is formed above interconnect layer 2240 and DRAM cells 2280. Bonding layer 2242 can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer 2240 by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer 2240 by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

[0322] Method 2300 proceeds to operation 2312, as illustrated in FIG. 23, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

[0323] As illustrated in FIG. 22L, carrier substrate 2234 and components formed thereon (e.g., DRAM cells 2280) are flipped upside down. As illustrated in FIG. 22L, bonding layer 2242 facing down is bonded with bonding layer 2252 facing up, e.g., in a face-to-face manner, thereby forming a bonding interface 2254. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment,

and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIG. 22L, silicon substrate 2244 and components formed thereon (e.g., peripheral circuits 2246) can be flipped upside down, and bonding layer 2252 facing down can be bonded with bonding layer 2242 facing up, e.g., in a face-to-face manner, thereby forming bonding interface 2254. After the bonding, the bonding contacts in bonding layer 2242 and the bonding contacts in bonding layer 2252 are aligned and in contact with one another, such that DRAM cells 2280 can be electrically connected to peripheral circuits 2246 across bonding interface 2254. It is understood that in the bonded chip, DRAM cells 2280 may be either above or below peripheral circuits 2246. Nevertheless, bonding interface 2254 can be formed vertically between peripheral circuits 2246 and DRAM cells 2280 after the bonding.

[0324] Method 2300 proceeds to operation 2314, as illustrated in FIG. 23, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. 22M, a pad-out interconnect layer 2256 is formed above on the backside of carrier substrate 2234. Pad-out interconnect layer 2256 can include interconnects, such as pad contacts 2258, formed in one or more ILD layers. Pad contacts 2258 can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts 2260 are formed extending vertically through carrier substrate 2234, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts 2260 can be in contact with the interconnects in pad-out interconnect layer 2256. It is understood that in some examples, carrier substrate 2234 may be thinned or removed after bonding and prior to forming pad-out interconnect layer 2256 and contacts 2260, for example, using planarization processes and/or etching processes.

[0325] Although not shown, it is understood that in some examples, pad-out interconnect layer 2256 may be formed above on the backside of silicon substrate 2244, and contacts 2260 may be formed extending vertically through silicon substrate 2244. Silicon substrate 2244 may be thinned prior to forming pad-out interconnect layer 2256 and contacts 2260, for example, using planarization processes and/or etching processes. Although not shown, it is further understood that in some examples, the fabrication processes described with respect to FIGS. 14A-14E and 15A-15D may be applied to form another array of DRAM cells 2280 in another semiconductor structure bonded to the semiconductor structure including DRAM cells 2280 described above with respect to FIGS. 22A-22M.

[0326] According to one aspect of the present disclosure, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes an array of memory cells and a plurality of bit lines coupled to the memory cells and each extending in a second direction perpendicular to the first direction. Each of the memory cells includes a vertical transistor extending in a first direction, and a storage unit

coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction. Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The array of memory cells is coupled to the peripheral circuit across the bonding interface.

[0327] In some implementations, the vertical transistor includes a semiconductor body extending in the first direction, and a gate structure in contact with one side of the semiconductor body in the second direction.

[0328] In some implementations, the second semiconductor structure further includes a plurality of word lines each extending in a third direction perpendicular to the first direction and the second direction.

[0329] In some implementations, the second semiconductor structure further includes a trench isolation extending in the third direction and disposed between the two adjacent vertical transistors in the second direction.

[0330] In some implementations, the gate structure includes a gate electrode, and a gate dielectric between the gate electrode and the semiconductor body in the second direction.

[0331] In some implementations, the gate dielectrics of two adjacent vertical transistors of the vertical transistors in the third direction are continuous.

[0332] In some implementations, the vertical transistor further includes a source and a drain disposed at two ends of the semiconductor body, respectively, in the first direction.

[0333] In some implementations, one of the source and the drain of the vertical transistor is coupled to the storage unit in a respective memory cell.

[0334] In some implementations, another one of the source and the drain of the vertical transistor is coupled to the respective bit line.

[0335] In some implementations, the bit lines are disposed between the vertical transistors and the bonding interface.

[0336] In some implementations, two ends of the semiconductor body in the first direction extend beyond the gate structure, respectively.

[0337] In some implementations, the second semiconductor structure further includes a pad-out interconnect layer, and the storage units are disposed between the vertical transistors and the pad-out interconnect layer.

[0338] In some implementations, the first semiconductor structure further includes a pad-out interconnect layer, and the peripheral circuit is disposed between the bonding interface and the pad-out interconnect layer.

[0339] According to another aspect of the present disclosure, a memory system includes a memory device configured to store data and a memory controller coupled to the memory device. The memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes an array of memory cells and a plurality of bit lines coupled to the memory cells and each extending in a second direction perpendicular to the first direction. Each of the memory cells includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction.

Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The array of memory cells is coupled to the peripheral circuit across the bonding interface. The memory controller is configured to control the array of memory cells through the peripheral circuit and the bit lines.

[0340] In some implementations, the memory system further includes a host coupled to the memory controller and configured to send or receive the data to or from the memory device.

[0341] In some implementations, the memory cells comprise at least a DRAM cell, a PCM cell, or a FRAM cell.

[0342] According to still another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. A first semiconductor structure including a peripheral circuit is formed. A second semiconductor structure is formed. To form the second semiconductor structure, an array of memory cells is formed, and a plurality of bit lines coupled to the memory cells are formed. Each of the memory cells includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. Two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another. The first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the array of memory cells is coupled to the peripheral circuit across a bonding interface.

[0343] In some implementations a pad-out interconnect layer is formed on a backside of the first semiconductor structure or the second semiconductor structure after the bonding.

[0344] In some implementations, the bonding includes hybrid bonding.

[0345] In some implementations, to form the array of memory cells, a semiconductor pillar extending vertically is formed in a substrate, gate structures in contact with opposite sides of the semiconductor pillar are formed, a first trench isolation extending vertically through the semiconductor pillar is formed to separate the semiconductor pillar into semiconductor bodies each in contact with a respective one of the gate structures, and storage units in contact with the semiconductor bodies are formed.

[0346] In some implementations, to form the semiconductor pillar, the substrate is etched in a first lateral direction to form a plurality of first trenches, a dielectric is deposited to fill the first trenches to form second trench isolations, and the substrate and the second trench isolations are etched in a second lateral direction to form a plurality of second trenches and the semiconductor pillar surrounded by the second trenches and the second trench isolations.

[0347] In some implementations, to form the semiconductor pillar, a dielectric is deposited to partially fill the second trenches.

[0348] In some implementations, to form the gate structure, gate dielectrics are formed over the opposite sides of the semiconductor pillar, and gate electrodes are formed over the gate dielectrics.

[0349] In some implementations, to form the gate electrodes, conductive layers are deposited over the gate dielectrics, and the conductive layers are etched back.

[0350] In some implementations, to form the first trench isolation, the semiconductor pillar is etched in the second lateral direction to form a third trench, and a dielectric is deposited to fill the third trench.

[0351] In some implementations, to form the array of memory cells, a first end of the semiconductor body away from the substrate is doped prior to forming the storage unit, the substrate is thinned to expose a second end of the semiconductor body opposite to the first end after forming the storage unit, and the exposed second end of the semiconductor body is doped.

[0352] In some implementations, to form the storage unit, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

[0353] In some implementations, to form the bit lines, a respective one of the bit lines is formed on the doped second end of the semiconductor body.

[0354] The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.

[0355] The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

What is claimed is:

1. A memory device, comprising:

a bit line extending in a first direction;

a first transistor comprising a first semiconductor body extending in a second direction perpendicular to the first direction and a first gate structure located at one side of the first semiconductor body in the first direction, the first semiconductor body coupled to the bit line;

a second transistor comprising a second semiconductor body extending in the second direction and a second gate structure located at one side of the second semiconductor body in the first direction, the second semiconductor body coupled to the bit line; and

a first isolation structure located between the first semiconductor body and the second semiconductor body in the first direction,

wherein the first semiconductor body is in contact with the first isolation structure, and the second semiconductor body is in contact with the first isolation structure.

2. The memory device of claim 1, wherein

the first semiconductor body is between the first isolation structure and the first gate structure in the first direction; and

the second semiconductor body is between the first isolation structure and the second gate structure in the first direction.

3. The memory device of claim 1, wherein

the first gate structure comprises a first gate electrode and a first gate dielectric between the first gate electrode and the first semiconductor body; and

the second gate structure comprises a second gate electrode and a second gate dielectric between the second gate electrode and the second semiconductor body.

4. The memory device of claim 3, wherein

the first gate dielectric is in contact with the first semiconductor body located between the first isolation structure and the first gate dielectric in the first direction; and

the second gate dielectric is in contact with the second semiconductor body which is located between the first isolation structure and the second gate dielectric in the first direction.

5. The memory device of claim 1, wherein

the first isolation structure comprises an air gap located between the first semiconductor body and the second semiconductor body in the first direction.

6. The memory device of claim 1, further comprising:

a third transistor comprising a third semiconductor body extending in the second direction, the third semiconductor body coupling to the bit line; and

a second isolation structure located between the second semiconductor body and the third semiconductor body in the first direction.

7. The memory device of claim 6, wherein

the third transistor comprising a third gate structure located at one side of the third semiconductor body in the first direction;

the second gate structure is between the second semiconductor body and the second isolation structure; and the third gate structure is between the third semiconductor body and the second isolation structure.

8. The memory device of claim 7, wherein

the third gate structure comprises a third gate electrode and a third gate dielectric between the third gate electrode and the third semiconductor body; and the third gate electrode is in contact with the second isolation structure.

9. The memory device of claim 7, wherein

the second isolation structure comprises an air gap located between the second gate structure and the third gate structure in the first direction.

10. The memory device of claim 1, wherein

both the first gate structure and the second gate structure extend in a third direction perpendicular to the first direction and the second direction.

11. The memory device of claim 1, further comprising a first storage unit and a second storage unit;

the first storage unit and the bit line are coupled to opposite ends of the first semiconductor body in the second direction; and

the second storage unit and the bit line are coupled to opposite ends of the second semiconductor body in the second direction.

12. A memory device, comprising:

a first semiconductor structure comprising:

bit lines extending in a first direction; and

an array of memory cells coupled to the bit lines, each of the memory cells comprising a transistor extending in a second direction perpendicular to the first direction, and a storage unit coupled to the transistor,

wherein a respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the transistors in the second direction; and

one of two adjacent transistors in the first direction comprising a first semiconductor body, another one of the two adjacent transistors comprising a second semi-

conductor body, an air gap is located between the first semiconductor body and the second semiconductor body.

13. The memory device of claim **12**, wherein the first semiconductor structure further comprises an isolation structure located between the first semiconductor body and the second semiconductor body in the first direction; and

the air gap is located in the isolation structure.

14. The memory device of claim **12**, wherein

the one of the two adjacent transistors further comprises a first gate structure located at one side of the first semiconductor body in the first direction; and

the another one of the two adjacent transistors further comprises a second gate structure located at one side of the second semiconductor body in the first direction.

15. The memory device of claim **14**, wherein

the first semiconductor body is between the air gap and the first gate structure in the first direction; and

the second semiconductor body is between the air gap and the second gate structure in the first direction.

16. The memory device of claim **14**, wherein

the first gate structure is between the air gap and the first semiconductor body in the first direction; and

the second gate structure is between the air gap and the second semiconductor body in the first direction.

17. The memory device of claim **12**, further comprising: a second semiconductor structure comprising a peripheral circuit, wherein the second semiconductor structure is bonded with the first semiconductor structure and the array of memory cells is coupled to the peripheral circuit.

18. A method for forming a memory device, comprising: forming a bit line extending in a first direction;

forming a first transistor comprising a first semiconductor body extending in a second direction perpendicular to the first direction and a first gate structure located at one side of the first semiconductor body in the first direction, the first semiconductor body coupled to the bit line;

forming a second transistor comprising a second semiconductor body extending in the second direction and a second gate structure located at one side of the second semiconductor body in the first direction, the second semiconductor body coupled to the bit line; and

forming a first isolation structure located between the first semiconductor body and the second semiconductor body in the first direction,

wherein the first semiconductor body is in contact with the first isolation structure, and the second semiconductor body is in contact with the first isolation structure.

19. The method of claim **18**, wherein forming the first isolation structure comprises:

forming a semiconductor pillar extending in the second direction in a substrate; and

forming a trench extending in the second direction through the semiconductor pillar to separate the semiconductor pillar into semiconductor bodies.

20. The method of claim **19**, wherein forming the first isolation structure further comprises:

depositing a dielectric to fill the trench, wherein the dielectric in the trench is in contact with the first semiconductor body and the second semiconductor body.

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